Observation and Recovery in Arithmetic and Probability

Prime Flows, Rare Events, and Infinite Mixtures

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Abstract

We study what arithmetic and probabilistic observations reveal about hidden infinite-dimensional structure. Quantitative recurrence of weighted prime flows leads to finite-height boundary zeros and sharp Euler-edge locking. Weighted Weil forms localize simple critical-line zeros in prime phase regions, including shrinking targets, and connect selected phases to conditional Gibbs profiles and joint edge laws. In weighted random series, macroscopic joint rare events recover the microscopic law, while microscopic fluctuations recover a discrete coefficient spectrum. Controlled thinning and interior probes then identify unknown dimensions, power ratios, and correlated laws, with explicit finite-data obstructions and stability estimates. For infinite mixtures we give sparse partial-sum identification criteria, construct analytic families with arbitrary replication thresholds despite independence of all finite parameter jets, and show that a finite-dimensional one-view defect becomes supported on finitely many atoms after two observations. A localized multiplicative-smoothing argument proves the prescribed-moment assertion of Bürgin, Epperlein, and Wirth and yields an exact classification of integer moment coincidence sets on positive compact intervals, realizable by strictly log-concave Gevrey densities. The final results separate exact identification from uniform statistical recovery through sharp regularity boundaries, logarithmic rates, and arbitrarily slow examples. This manuscript consolidates eleven source papers into five connected parts, retaining their distinct proofs while replacing repeated arguments by internal references. Conditional inputs, classical ingredients, and remaining quantitative and priority questions are stated explicitly.

Introduction and organization

An observation can retain a surprisingly detailed description of an infinite system, or conceal an entire family of alternatives. The distinction depends on more than the smoothness of the observation map. It depends on which coordinates are tested, how several observations are coupled, and which limiting operations are allowed. This manuscript develops that issue in arithmetic dynamics, rare-event probability, and mixture identification.

The arithmetic starting point is a weighted metric on an infinite prime torus. A small metric ball imposes simultaneous phase coherence at many primes. Its probability controls an entropy scale, while the actual logarithms of primes introduce an additional arithmetic cost. Near the extremal edges of Euler products, analytic defects become positive weighted phase energies, allowing recurrence estimates to control the location and spacing of zeros and critical points. Weil-form localization provides a different observation mechanism: it selects critical-line zeros with prescribed phase constraints.

The probabilistic middle of the manuscript asks what is retained by the resulting continuum and conditional limits. Joint rare-event surfaces admit Mellin inversions that recover a local law, whereas centered fluctuation series retain a discrete spectrum. Once the law and the observation architecture are both unknown, extra channels, controlled Bernoulli gates, and interior probes give exact recovery criteria. Counterexamples show why finite marginal summaries, resonances, and flat endpoint jets can fail.

The final part places these observations in the language of mixing operators. If P_θ is a known component family and μ is an unknown mixing law, then

$$A_q\mu = \int P_\theta^{\otimes q}\,\mu(d\theta)$$

describes q observations sharing a latent parameter. Products of observable functions enlarge the collection of accessible moments. This makes the number of repeated observations an algebraic property of the observable function space. Sparse moments, finite derivative tests, and closure of model classes then explain why finite mixtures, arbitrary mixing laws, and uniformly estimable subclasses can have different thresholds.

Five parts. Part I develops quantitative prime-flow recurrence, boundary geometry, and the difference between entropy and arithmetic time scales. Part II establishes phase localization at zeta zeros and develops shrinking targets, simultaneous edge approximation, and conditional phase laws. Part III derives inverses and fluctuation laws for weighted series and analyzes minimal multichannel observation. Part IV uses thinning and interior probes for blind identification, including finite noisy observation guarantees and exact obstructions. Part V treats partial-sum marginals, finite differential barriers, replication thresholds, prescribed moment coincidences, and statistical recovery.

Three kinds of conclusion. Throughout, exact recovery from an entire response function is distinguished from stable recovery from noisy finite data. A theorem about recurrence on a prime torus is also distinguished from its arithmetic transfer to a particular Euler product. Finally, an explicit construction with strong regularity is an existence theorem; it need not supply an efficient numerical representation of the constructed object. The hypotheses attached to each transfer and inverse theorem are part of its statement.

Scope of the consolidation. The eleven source manuscripts are listed in Section 78. Their substantive results and proof arguments are incorporated here. Where a later manuscript proves a

stronger version of an earlier result, the integrated exposition uses the stronger proof and records the earlier specialization or its reference. Questions resolved by a later part are no longer listed as open. Independent results that use similar tools but concern different objects remain separate.

The integration unifies notation, numbering, references, and dependencies. It does not turn an explicitly conditional statement into an unconditional one, and it does not establish historical priority or publication-level significance. The source drafts contained several independent proof checks; the present editorial pass adds coverage and cross-reference checks rather than a new external peer review of every assertion.

Notation. The symbols $\mathbb{R}, \mathbb{C}, \mathbb{N}, \mathbb{Z}$ denote the usual number systems and $\mathbb{T}$ denotes a circle group, with its period specified when first used in a part. The shift between period 1 and period 2π is kept explicit. For a signed measure, $\|\nu\|_{\mathrm{TV}}$ denotes its total variation norm. For probabilities, $\mathrm{TV}(P, Q) = \sup_B |P(B) - Q(B)|$ is half the variation norm of $P - Q$. Wasserstein metrics refer to the underlying parameter metric specified in the relevant section. The symbols A, E denote the two Euler-edge locations in the arithmetic parts and positive power exponents in the weighted-series parts; those sections state the intended parameters explicitly. Likewise N can denote an observation count or a spectral counting function with an argument. No identification of these different uses is assumed.

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Part I
Prime flows and quantitative recurrence

The first part develops a common quantitative framework for weighted prime-phase flows and analytic boundary points. The geometry of a small phase ball determines its Haar probability; the arithmetic of the prime frequencies determines the additional cost of reaching it uniformly. A quadratic real-part estimate then transports phase recurrence into zeros or critical points near a vertical boundary. These three steps will also supply the phase geometry used in the subsequent parts.

The qualitative prime-zeta boundary and almost-periodic transport of zeros are classical [7, 2, 3, 5, 1]. For Euler products, the right boundaries of $\zeta(s) = 1$ and $\zeta'(s) = 0$, and their qualitative correspondence, are due to Arias de Reyna and van de Lune [8]. The quantitative claims developed here concern shrinking targets, uniform maximal gaps, and sharp local moduli between these two boundaries. The phase-hull viewpoint motivating the prime-zeta construction is also present in [4]; none of its density or correlation asymptotics is assumed in the proofs below.

Finite-dimensional small-divisor covering estimates and infinite-dimensional product-measure nonresonance arguments are established methods [9, 10, 13]. Here their growing-dimensional costs are tracked explicitly. The contrast between probability-one recurrence and residual slowness is related to classical exceptional-rotation phenomena [11]. The Fejér coefficient inequality used in the localization optimization is likewise classical [12]; its weighted prime-budget consequences are proved in Section 11.

The results in this part concern the half-plane of absolute convergence. They do not assert the Riemann hypothesis or a deterministic lower first-waiting bound at the origin for ordinary prime zeta or ordinary ζ'. Maximal gaps, recurrence of long empty intervals, and almost-sure first waiting times have different quantifiers throughout.

1 Weighted prime flows and a finite Fourier certificate

Let $\mathcal{P}$ be the set of primes, with finitely many primes possibly removed, and let $\Omega = \mathbb{T}^{\mathcal{P}}$ carry normalized product Haar measure μ. Here $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$, parametrized by $z = e^{i\theta}$ with θ of period 2π. For positive summable weights $w = (w_p)$, put

$$\mathcal{E}_w(\eta) = \sum_{p\in\mathcal{P}} w_p(1 - \operatorname{Re}\eta_p), \qquad d_w(\eta,\zeta)^2 = \sum_p w_p|\eta_p - \zeta_p|^2. \tag{1}$$

Thus $d_w(\eta, 1)^2 = 2\mathcal{E}_w(\eta)$. The metric d_w induces the product topology. For a real frequency vector λ, write $\Phi_t\eta = (\eta_p e^{-it\lambda_p})_p$. This flow is an isometry. We use the uniform covering time

$$\mathcal{R}_{w,\lambda}(\varepsilon) = \inf\{T > 0 : \text{for every } \eta \in \Omega \text{ there is } t \in [0, T] \text{ with } \mathcal{E}_w(\Phi_t\eta) < \varepsilon\}. \tag{2}$$

Translation in the group makes this the same as covering every target, from every initial phase, with squared-distance tolerance 2ε. All estimates below are also uniform in the initial time of an interval. The fixed-origin waiting time is $\tau_\eta(\varepsilon) = \inf\{t \ge 0 : \mathcal{E}_w(\Phi_t\eta) < \varepsilon\}$.

Our main weight class is

$$c_0 p^{-\sigma}(\log p)^\kappa \le w_p \le C_0 p^{-\sigma}(\log p)^\kappa, \qquad \sigma > 1, \quad \kappa \in \mathbb{R}. \tag{3}$$

Throughout, implicit constants may depend on these parameters and on a fixed finite set of omitted primes. Set

$$\beta = \frac{1}{\sigma - 1}, \qquad \ell = \log(1/\varepsilon), \qquad X_{\sigma,\kappa}(\varepsilon) = \varepsilon^{-\beta}\ell^{(\kappa-1)\beta}, \qquad Y_{\sigma,\kappa}(\varepsilon) = \frac{X_{\sigma,\kappa}(\varepsilon)}{\ell}. \tag{4}$$

We restrict to $0 < \varepsilon < e^{-1}$. Multiplying ε by a fixed positive constant changes X and Y by bounded factors. If $x \asymp X(\varepsilon)$, then

$$x^{1-\sigma}(\log x)^{\kappa-1} \asymp \varepsilon, \qquad x/\log x \asymp Y(\varepsilon). \tag{5}$$

Lemma 1.1 (Prime sums and logarithmic budgets)**.** *For $a < 1$, $u > 1$, and real b, as $x \to \infty$,*

$$\sum_{p \le x} p^{-a}(\log p)^b \asymp x^{1-a}(\log x)^{b-1}, \tag{6}$$

$$\sum_{p > x} p^{-u}(\log p)^b \asymp x^{1-u}(\log x)^{b-1}. \tag{7}$$

In particular, for fixed $u > 1$ and $0 \le v < 1$,

$$\sum_{p>x} p^{-u} \asymp_u \frac{x^{1-u}}{\log x}, \qquad \sum_{3 \le p \le x} p^{-v} \asymp_v \frac{x^{1-v}}{\log x}, \tag{8}$$

$$\sum_{3 \le p \le x} p^{-v}\log p \asymp_v x^{1-v}, \qquad \#\{p : x \le p \le 2x\} \asymp \frac{x}{\log x}. \tag{9}$$

Moreover,

$$\sum_{p \le x} \log(x/p) = O(x/\log x), \tag{10}$$

$$\sum_{p \le x} \log(\log x/\log p) = O(x/(\log x)^2). \tag{11}$$

Deleting finitely many primes changes none of these orders.

Proof. Partial summation from $\pi(x) \sim x/\log x$ and $\sum_{p \le x} \log p \sim x$ gives the power-weight estimates and $\pi(2x) - \pi(x) \asymp x/\log x$. For the logarithmic budgets,

$$\sum_{p \le x} \log(x/p) = \int_2^x \frac{\pi(t)}{t}\, dt, \qquad \sum_{p \le x} \log(\log x/\log p) = \int_2^x \frac{\pi(t)}{t \log t}\, dt.$$

The prime number theorem gives the stated bounds. These are the only asymptotic facts about prime distribution used in the recurrence arguments; see [6]. □

For a finite trigonometric polynomial $P = \sum_k \widehat{P}(k)\eta^k$, write $\|P\|_A = \sum_k |\widehat{P}(k)|$. Finite multi-indices are understood throughout. Direct integration of each character gives

$$\left| \frac{1}{T}\int_0^T P(\Phi_t\eta)\, dt - \int P\, d\mu \right| \le \frac{2\|P\|_A}{T\Delta}, \tag{12}$$

provided every nonconstant frequency occurring in P satisfies $|k \cdot \lambda| \ge \Delta > 0$. This bound is uniform in η.

Lemma 1.2 (Positive Fourier certificate)**.** *Let $Q \subset \mathcal{P}$ be finite, and let $K \geq 0$ be a trigonometric polynomial depending only on Q, with $\int K\, d\mu = 1$. Write*

$$\mathcal{E}_Q = \sum_{p \in Q} w_p(1 - \operatorname{Re} \eta_p), \qquad m = \int \mathcal{E}_Q K\, d\mu, \qquad W = \sum_p w_p.$$

Choose $a > m$, and suppose the nonzero frequencies of $P = (a - \mathcal{E}_Q)K$ have absolute value at least Δ. If

$$T > \frac{2(a + 2W)\|K\|_A}{\Delta(a - m)}, \tag{13}$$

then every orbit has a time in $[0, T]$ at which $\mathcal{E}_w < a + 2\sum_{p \notin Q} w_p$.

Proof. The mean of P is $a - m > 0$, and $\|P\|_A \leq (a + 2W)\|K\|_A$. Equation (12) makes its time average positive. At a time where $P > 0$, nonnegativity of K forces $\mathcal{E}_Q < a$. The remaining coordinates contribute at most twice their total weight. □

Haar integration on one circle is normalized to total mass one. For an integer $m \geq 1$, define

$$F_m(\theta) = \frac{1}{m}\left|\sum_{j=0}^{m-1} e^{ij\theta}\right|^2, \quad A_m = \frac{2m^2 + 1}{3m}, \quad J_m = F_m^2/A_m. \tag{14}$$

Equivalently writing $J_m(e^{i\theta}) = J_m(\theta)$, we use the Fourier coefficient norm $\|Q\|_{\mathcal{A}} = \sum_k |\widehat{Q}(k)|$ for a trigonometric polynomial; the analogous definition applies to several variables. Thus $\|Q\|_{\mathcal{A}} = \|Q\|_A$.

Lemma 1.3 (Squared Fejér identities)**.** *The polynomial J_m is nonnegative, has degree $2m - 2$, and has Haar mean one. All its Fourier coefficients are nonnegative. Moreover,*

$$\int (1 - \cos\theta) J_m(\theta)\, \mathrm{d}\theta = \frac{3}{2m^2 + 1}, \tag{15}$$

$$\|J_m\|_{\mathcal{A}} = J_m(0) = \frac{3m^3}{2m^2 + 1} \leq \frac{3}{2}m. \tag{16}$$

Here and below the normalized Haar factor is included in $\mathrm{d}\theta$ when integration is over a circle.

Proof. Write

$$F_m(\theta) = \sum_{k \in \mathbb{Z}} a_k e^{ik\theta}, \qquad a_k = (1 - |k|/m)_+.$$

Convolution shows positivity of every Fourier coefficient of F_m^2. Its constant coefficient is

$$\sum_k a_k^2 = 1 + 2\sum_{k=1}^{m-1}(1 - k/m)^2 = \frac{2m^2 + 1}{3m} = A_m.$$

If b_0, b_1 are the zeroth and first coefficients of F_m^2, then

$$b_0 - b_1 = \frac{1}{2}\sum_{k \in \mathbb{Z}}(a_k - a_{k-1})^2 = \frac{1}{m}.$$

There are $2m$ nonzero differences, each of magnitude $1/m$. Division by A_m proves (15). Nonnegative Fourier coefficients make the coefficient norm equal to the value at zero. Since $F_m(0) = m$, this proves (16). □

The gain m^{-2} in (15) is the quantitative reason for squaring the kernel. An unweighted, equal-precision treatment of all prime phases loses the exponent needed below.

For $K = \prod_{p\in Q} J_{m_p}$, its localized mean is consequently

$$m = \sum_{p\in Q} \frac{3w_p}{2m_p^2+1}. \tag{17}$$

2 Small balls, arithmetic recurrence, and recurrent holes

Theorem 2.1 (Small-ball scale)**.** *Under (3), for every $\sigma > 1$ and $\kappa \in \mathbb{R}$, there are positive constants c, C such that*

$$e^{-CY(\varepsilon)} \le \mu\{\mathcal{E}_w \le \varepsilon\} \le e^{-cY(\varepsilon)} \tag{18}$$

for all sufficiently small ε. If $L = \sum_p w_p|\lambda_p| < \infty$, then also

$$\mu\{\min_{0\le t\le T} \mathcal{E}_w(\Phi_t\eta) \le \varepsilon\} \le C(1+T/\varepsilon)e^{-cY(\varepsilon)}. \tag{19}$$

In particular $\mathcal{R}_{w,\lambda}(\varepsilon) \ge e^{cY(\varepsilon)}$.

Proof. For the upper small-ball bound take $x = bX(\varepsilon)$ with $b > 0$ a sufficiently small fixed constant. The block $x \le p \le 2x$ contains $N \asymp x/\log x$ primes, each of weight comparable to $v = x^{-\sigma}(\log x)^{\kappa}$. As $Nv \asymp b^{1-\sigma}\varepsilon$, the event $\mathcal{E}_w \le \varepsilon$ forces the average of N independent copies of $U = 1-\cos\Theta$ to be at most a fixed $q < 1 = \mathbb{E}U$. For small fixed $s > 0$, $e^{sq}\mathbb{E}e^{-sU} < 1$. Markov's inequality thus bounds the probability by e^{-cN}.

For the lower bound take $x = BX(\varepsilon)$, with B large enough that $2\sum_{p>x} w_p \le \varepsilon/2$. On the coordinates $p \le x$, require

$$|\arg\eta_p| \le \min\left\{1, b_0(p/x)^{\sigma/2}(\log x/\log p)^{\kappa/2}\right\}.$$

Each of their energy contributions is at most $Cb_0^2 x^{-\sigma}(\log x)^{\kappa}$. Their sum is at most $\varepsilon/2$ after fixing b_0 small. The negative logarithm of the probability of this box is bounded by a constant times

$$\pi(x) + \sum_{p\le x}\log(x/p) + \sum_{p\le x}\log(\log x/\log p) = O(x/\log x).$$

This proves (18).

The energy along any orbit is L-Lipschitz. A mesh of spacing $\varepsilon/(L+1)$ converts a hit below ε into a mesh value below 2ε. Haar invariance, the union bound, and (18) give (19). Taking $T = e^{c_1 Y(\varepsilon)}$ with c_1 small makes the last probability less than one. Here $Y(\varepsilon)/\log(1/\varepsilon) \to \infty$, so the polynomial mesh factor is harmless. □

There is also a metric interpretation of Y. If $N_w(r)$ denotes the least number of d_w-balls of radius r covering Ω, then

$$\log N_w(\sqrt{\varepsilon}) \asymp Y(\varepsilon). \tag{20}$$

The lower estimate follows by measuring a cover. For the upper estimate, take a maximal $\sqrt{\varepsilon}$-separated set. Its half-radius balls are disjoint, each of the same Haar measure, and the full-radius balls cover. Compactness makes this a finite argument.

Theorem 2.2 (Arithmetic prime frequencies)**.** *Let* $\lambda_p = \log p$*. Under* (3) *with* $1 < \sigma < 3$*,*

$$\exp(cY_{\sigma,\kappa}(\varepsilon)) \le \mathcal{R}_{w,\lambda}(\varepsilon) \le \exp(CX_{\sigma,\kappa}(\varepsilon)). \tag{21}$$

The same conclusion holds for $\lambda_p = \log(p/p_0)$ *with a fixed prime* p_0 *omitted from* $\mathcal{P}$*.*

Proof. Unique factorization shows that each nonzero finite integer vector k has $\sum k_p \log p = \log(A/B) \neq 0$, where A, B are positive integers at most $\exp(\sum |k_p| \log p)$. Integer separation gives

$$\left|\sum k_p \log p\right| \ge \exp\left(-\sum |k_p| \log p\right). \tag{22}$$

For relative frequencies the denominator cost is at most a fixed multiple of this cost, since the additional prime is p_0.

Take $Q = \{p \le x\}$ and

$$m_p = \left\lceil \max\left\{1, (x/p)^{\sigma/3} (\log p/\log x)^{\kappa/3}\right\}\right\rceil.$$

Equations (6)–(17) give

$$m \le Cx^{-2\sigma/3}(\log x)^{2\kappa/3} \sum_{p\le x} p^{-\sigma/3}(\log p)^{\kappa/3} \le Cx^{1-\sigma}(\log x)^{\kappa-1},$$

$$B_x := \sum_{p\le x} (2m_p - 1)\log p \le Cx, \qquad \log \|K\|_A \le Cx/\log x.$$

The estimate for B_x uses $\sigma/3 < 1$; the norm estimate uses (10) and (11). Choose x a sufficiently large constant multiple of $X(\varepsilon)$, so that $m \le \varepsilon/4$ and $2\sum_{p>x} w_p \le \varepsilon/4$. The polynomial $(\varepsilon/2 - \mathcal{E}_Q)K$ has coordinate degrees at most $2m_p - 1$. Equations (22) and (13) give a covering time at most e^{Cx}. The lower estimate is Theorem 2.1, since $\sum w_p|\lambda_p| < \infty$. □

Proposition 2.3 (Robust holes and first waiting times)**.** *Suppose* (3) *holds,* $\sum w_p|\lambda_p| < \infty$*, and* $\mathcal{R}_{w,\lambda}(\varepsilon) \le U(\varepsilon) < \infty$*. There are constants* $c, b > 0$ *with the following property. For every sufficiently small* ε*, every orbit, and every* $A \in \mathbb{R}$*, a time* $t_0 \in [A, A + 2U(b\varepsilon)]$ *begins an interval of length* $e^{cY(\varepsilon)}$ *on which* $\mathcal{E}_w(\Phi_t\eta) > \varepsilon$*. Moreover, for Haar-almost every* η*,*

$$\tau_\eta(\varepsilon) \ge e^{cY(\varepsilon)} \quad \text{for all sufficiently small } \varepsilon. \tag{23}$$

If $U(\varepsilon) = e^{CX(\varepsilon)}$*, the double-logarithmic waiting-time exponent is* β *almost surely. If* $U(\varepsilon) = e^{CY(\varepsilon)}$*, then* $\log \tau_\eta(\varepsilon) \asymp Y(\varepsilon)$ *almost surely.*

Proof. Apply (19) at threshold 16ε and time $H = e^{cY(\varepsilon)}$ to obtain a phase η^* satisfying $\mathcal{E}_w(\Phi_t\eta^*) > 16\varepsilon$ throughout $[0, H]$. Every phase within d_w-distance $\sqrt{\varepsilon}$ of η^* still has energy greater than ε on that interval: use the triangle inequality and the isometry property. Covering at tolerance $b\varepsilon$ brings any prescribed orbit into that ball in time $2U(b\varepsilon)$, also after any initial time A. This proves the robust-hole assertion.

For the almost-sure assertion, take $\varepsilon_n = e^{-n}$. With $T_n = e^{cY(\varepsilon_n)}$ and c sufficiently small, the probabilities in (19) are summable. Borel–Cantelli excludes a hit before T_n for all sufficiently large n, almost surely. Monotonicity of τ and comparability of $Y(\varepsilon_n)$ at consecutive indices give (23) for all small ε after reducing c. The asserted upper bounds follow from uniform covering. □

The power-weight specialization and its explicit certificate are given next. The endpoint $\sigma = 3$ is resolved in Section 11; the frequency-perturbation theorem in Section 9 has no endpoint restriction.

3 Power weights and an explicit arithmetic certificate

Specialize the preceding framework to the relative frequencies and write

$$\Omega = \prod_{p\geq 3} \mathbb{T}, \qquad \mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}, \qquad (\Phi_t \eta)_p = \eta_p e^{-it\lambda_p}, \quad \lambda_p = \log(p/2).$$

Equip Ω with product Haar probability μ. Fix $\sigma > 1$ and positive weights $w = (w_p)_{p\geq 3}$ satisfying

$$c_0 p^{-\sigma} \leq w_p \leq C_0 p^{-\sigma} \qquad (p \geq 3) \tag{24}$$

for two fixed positive constants. Define

$$E_w(\eta) = \sum_{p\geq 3} w_p(1 - \operatorname{Re}\eta_p), \qquad d_w(\eta, \xi)^2 = \sum_{p\geq 3} w_p |\eta_p - \xi_p|^2. \tag{25}$$

Here $E_w = \mathcal{E}_w$ in the notation of (1).

The covering time is

$$\mathcal{R}_w(\varepsilon) = \inf\left\{T > 0 : \text{for all } \eta, \xi \in \Omega \text{ there is } t \in [0, T] \text{ with } d_w(\Phi_t \eta, \xi)^2 \leq 2\varepsilon\right\}. \tag{26}$$

Multiplication by $\bar{\xi}$ reduces the target to 1. All logarithms are natural. For $0 < \varepsilon < e^{-1}$ put

$$\beta_\sigma = \frac{1}{\sigma - 1}, \qquad \ell_\varepsilon = \log(1/\varepsilon), \qquad X_\sigma(\varepsilon) = \varepsilon^{-\beta_\sigma} \ell_\varepsilon^{-\beta_\sigma}, \qquad Y_\sigma(\varepsilon) = \frac{X_\sigma(\varepsilon)}{\ell_\varepsilon}. \tag{27}$$

Constants may depend on the fixed weights through σ, c_0, C_0, but never on the phases or the initial time.

Theorem 3.1 (Weighted recurrence and small balls)**.** *Assume* (24) *and* $1 < \sigma < 3$. *There are constants* $c, C, \varepsilon_0 > 0$ *such that, for* $0 < \varepsilon < \varepsilon_0$,

$$e^{cY_\sigma(\varepsilon)} \leq \mathcal{R}_w(\varepsilon) \leq e^{CX_\sigma(\varepsilon)}, \tag{28}$$

$$e^{-CY_\sigma(\varepsilon)} \leq \mu\{E_w \leq \varepsilon\} \leq e^{-cY_\sigma(\varepsilon)}. \tag{29}$$

The upper recurrence bound holds on every interval of the stated length. In particular,

$$\lim_{\varepsilon \downarrow 0} \frac{\log\log \mathcal{R}_w(\varepsilon)}{\log(1/\varepsilon)} = \frac{1}{\sigma - 1}. \tag{30}$$

Moreover, if $\tau_\eta(\varepsilon) = \inf\{t \geq 0 : E_w(\Phi_t \eta) \leq \varepsilon\}$, *then for* μ*-almost every* η,

$$\lim_{\varepsilon \downarrow 0} \frac{\log\log(e^e + \tau_\eta(\varepsilon))}{\log(1/\varepsilon)} = \frac{1}{\sigma - 1}. \tag{31}$$

The theorem is the $\kappa = 0$ specialization of the preceding general results. Its exponent is exact, while its upper and lower logarithms differ by a factor $\log(1/\varepsilon)$. Section 11 extends the exact double-logarithmic exponent to $\sigma = 3$.

Lemma 3.2 (Arithmetic separation of frequencies)**.** *Let* $\mathcal{Q}$ *be a finite set of primes at least* 3, *let* $d_p \geq 1$ *be integers, and put* $B = \sum_{p\in\mathcal{Q}} d_p \log p$. *If* $k \neq 0$ *and* $|k_p| \leq d_p$, *then*

$$\left|\sum_{p\in\mathcal{Q}} k_p \log(p/2)\right| \geq e^{-2B}. \tag{32}$$

Consequently, for any trigonometric polynomial Q with these coordinate degrees, any initial phase η, and any real interval I of length T,

$$\left|\frac{1}{T}\int_I Q(\Phi_t\eta)\,\mathrm{d}t - \int Q\,\mathrm{d}\mu\right| \le \frac{2e^{2B}}{T}\,\|Q\|_{\mathcal{A}}\,. \tag{33}$$

Proof. Exponentiating the frequency gives a positive rational A/D in lowest terms. Unique factorization at every prime $p \ge 3$ implies $A \ne D$. Both integers are at most

$$\exp\left(\sum |k_p|\log p + \left|\sum k_p\right|\log 2\right) \le e^{2B}.$$

For distinct positive integers, $|\log A - \log D| \ge 1/\max(A,D)$, proving (32). The average of $e^{-it\nu}$ on an interval of length T has magnitude at most $2/(T|\nu|)$. Expand Q and apply this to each nonconstant coefficient. The phase and the initial endpoint have modulus one and do not change the bound. □

Proposition 3.3 (Finite certificate)**.** *Fix a nonempty finite set $\mathcal{Q}$, positive w_p, and integers $m_p \ge 1$. Put*

$$W_{\mathcal{Q}} = \sum_{p\in\mathcal{Q}} w_p, \qquad M = \sum_{p\in\mathcal{Q}} \frac{3w_p}{2m_p^2+1},$$
$$B = \sum_{p\in\mathcal{Q}} (2m_p-1)\log p, \qquad K_0 = \prod_{p\in\mathcal{Q}} \frac{3m_p^3}{2m_p^2+1}.$$

Every interval of length

$$T \ge 8e^{2B}\frac{(M+W_{\mathcal{Q}})K_0}{M} \tag{34}$$

contains a t for which

$$\sum_{p\in\mathcal{Q}} w_p(1-\mathrm{Re}(\Phi_t\eta)_p) < 2M. \tag{35}$$

The estimate is uniform in η and remains true for any target phase after translation.

Proof. Let $E_{\mathcal{Q}}$ denote the sum in (35), and set $K = \prod_{p\in\mathcal{Q}} J_{m_p}$. Lemma 1.3 gives $\int K = 1$, $\int E_{\mathcal{Q}}K = M$, and $\|K\|_{\mathcal{A}} = K_0$. The real polynomial

$$Q = (2M - E_{\mathcal{Q}})K$$

has mean $M > 0$, coordinate degree at most $2m_p - 1$, and

$$\|Q\|_{\mathcal{A}} \le (2M + 2W_{\mathcal{Q}})K_0.$$

By (33) and (34), its average on the interval is at least $M/2$. Hence Q is positive somewhere. Since $K \ge 0$, positivity implies (35). □

Proof of the upper bound in Theorem 3.1. Take $\mathcal{Q} = \{3 \le p \le x\}$ and allocate the degrees by

$$m_p = \left\lceil (x/p)^{\sigma/3} \right\rceil. \tag{36}$$

Because $\sigma/3 < 1$, Lemma 1.1 gives

$$M \le \frac{3}{2} C_0 x^{-2\sigma/3} \sum_{p \le x} p^{-\sigma/3} \ll \frac{x^{1-\sigma}}{\log x}, \tag{37}$$

$$B \ll x^{\sigma/3} \sum_{p \le x} p^{-\sigma/3} \log p \ll x, \tag{38}$$

$$\log K_0 \le \sum_{p \le x} \log(3m_p/2) \ll \pi(x) + \sum_{p \le x} \log(x/p) \ll \frac{x}{\log x}. \tag{39}$$

The single term $p = 3$ also gives $M \gg x^{-2\sigma/3}$, so $\log(1/M) = O(\log x)$. Thus (34) is at most $\exp(Cx)$ for large x. At the resulting time, the unprescribed tail has the deterministic bound

$$E_w(\Phi_t \eta) \le 2M + 2 \sum_{p > x} w_p \ll \frac{x^{1-\sigma}}{\log x}.$$

Choose $x = K X_\sigma(\varepsilon)$ with a sufficiently large fixed K. The scale identity

$$\frac{X_\sigma(\varepsilon)^{1-\sigma}}{\log X_\sigma(\varepsilon)} \asymp_\sigma \varepsilon \tag{40}$$

shows that the last bound is at most ε. This proves the upper bound on every initial interval, for every phase and target. □

Remark 3.4 (Why the degrees are anisotropic). Ignoring integer rounding and slow logarithmic factors, minimizing $\sum m_p$ subject to $\sum w_p/m_p^2 \le \varepsilon$ leads to $m_p \propto w_p^{1/3}$. Formula (36) implements that allocation at the truncation scale. We do not claim exact optimization of constants or logarithmic factors.

Lemma 3.5 (Power-weight small balls). *For every $\sigma > 1$ under (24),*

$$e^{-CY_\sigma(\varepsilon)} \le \mu\{E_w \le \varepsilon\} \le e^{-cY_\sigma(\varepsilon)}.$$

Proof. This is Theorem 2.1 with $\kappa = 0$. Its lower bound may in this specialization use the explicit event

$$|\arg \eta_p| \le (p/x)^{\sigma/2} \quad (3 \le p \le x), \qquad x = K X_\sigma(\varepsilon). \tag{41}$$

Its energy is at most $Cx^{-\sigma}\pi(x) + 2\sum_{p>x} w_p \ll x^{1-\sigma}/\log x$, and its negative log probability is $(\pi(x) - 1)\log \pi + (\sigma/2)\sum_{3 \le p \le x} \log(x/p) = O(x/\log x)$. Choosing K large gives the assertion. □

Lemma 3.6 (Finite-time avoidance estimate). *For weights satisfying (24), there are $c, C > 0$ such that for all small ε and all $T \ge 0$,*

$$\mu\left\{\eta : \min_{0 \le t \le T} E_w(\Phi_t \eta) \le \varepsilon\right\} \le C(1 + T/\varepsilon) e^{-cY_\sigma(\varepsilon)}. \tag{42}$$

In particular, for a sufficiently small fixed $c_1 > 0$, some η^* satisfies

$$E_w(\Phi_t \eta^*) > \varepsilon \quad \text{for every } 0 \le t \le e^{c_1 Y_\sigma(\varepsilon)}. \tag{43}$$

Proof. The phase energy is uniformly Lipschitz:

$$\left|\frac{\mathrm{d}}{\mathrm{d}t}E_w(\Phi_t\eta)\right| \le L_w := \sum_{p\ge 3} w_p\lambda_p < \infty. \tag{44}$$

Apply Theorem 2.1 with $\kappa = 0$ and relative prime frequencies. Its time-mesh bound is exactly (42); its nonempty complement at $T = \exp(c_1 Y_\sigma(\varepsilon))$ gives (43). No independence between different times is used. □

Completion of Theorem 3.1. The bad phase in Lemma 3.6 proves the lower bound for $\mathcal{R}_w$. The small-ball bounds are Theorem 2.1 with $\kappa = 0$. Taking two logarithms in (28) yields

$$\beta_\sigma \ell_\varepsilon - (\beta_\sigma + 1)\log \ell_\varepsilon + O(1) \le \log\log \mathcal{R}_w(\varepsilon) \le \beta_\sigma \ell_\varepsilon - \beta_\sigma \log \ell_\varepsilon + O(1),$$

which proves (30).

For the almost-sure assertion set $\varepsilon_n = e^{-n}$ and $T_n = \exp(c_1 Y_\sigma(\varepsilon_n))$. By (42), decreasing c_1 if needed,

$$\sum_{n=1}^{\infty} \mu\{\tau_\eta(\varepsilon_n) \le T_n\} < \infty.$$

The first Borel–Cantelli lemma implies $\tau_\eta(\varepsilon_n) > T_n$ eventually for almost every η. The uniform upper bound applies to every η. If $e^{-(n+1)} < \varepsilon \le e^{-n}$, monotonicity sandwiches $\tau_\eta(\varepsilon)$ between its two adjacent values. Since $(n+1)/n \to 1$, the two logarithms give (31). □

Remark 3.7 (Entropy and arithmetic cost). The number of active primes is $\asymp X_\sigma(\varepsilon)/\ell_\varepsilon$. This determines the small-ball probability. The separation bound for prime logarithms pays the larger cost $\sum m_p \log p = O(X_\sigma(\varepsilon))$. The remaining logarithmic gap between the upper and lower recurrence bounds occurs exactly here. We have not identified the true leading order of $\log \mathcal{R}_w$.

4 Zeros of prime Dirichlet series in shrinking edge strips

In this section $\lambda_p = \log(p/2)$ and $\Omega = \prod_{p\ge 3} \mathbb{T}$. Let a_p be real and satisfy

$$0 < a_- \le a_p \le a_+ < \infty, \qquad F_\omega(s) = \sum_p a_p \omega_p p^{-s}, \quad |\omega_p| = 1. \tag{45}$$

The series and all its derivatives converge locally uniformly on $\operatorname{Re} s > 1$. There is a unique $S > 1$ satisfying

$$a_2 2^{-S} = \sum_{p\ge 3} a_p p^{-S}. \tag{46}$$

Indeed, after division by $a_2 2^{-S}$ the right side is strictly decreasing from infinity to zero. In the results that follow we assume

$$1 < S < 3. \tag{47}$$

No such restriction is needed for the elementary zero-free half-plane.

Write $\mathcal{Z}_\omega(\varepsilon)$ for the set of ordinates of zeros in

$$\mathcal{S}_\varepsilon = \{s : S - \varepsilon < \operatorname{Re} s \le S\}.$$

Define the maximal vertical gap and the first waiting time by

$$G_\omega(\varepsilon) = \sup\{b-a : a<b, (a,b)\cap \mathcal{Z}_\omega(\varepsilon)=\varnothing\}, \tag{48}$$
$$W_\omega(\varepsilon) = \inf(\mathcal{Z}_\omega(\varepsilon)\cap[0,\infty)). \tag{49}$$

As usual the infimum of the empty set is infinity. The theorem shows it is finite here. The definition of a gap ignores multiplicity.

Theorem 4.1 (Finite-height edge zeros and recurrent long gaps)**.** *Assume (45)–(47). There are constants $c, C, \varepsilon_0 > 0$, independent of ω, such that for $0 < \varepsilon < \varepsilon_0$ the following hold.*

(i) *Every real interval of length $U(\varepsilon) = \exp(CX_S(\varepsilon))$ contains the ordinate of a simple zero in $\mathcal{S}_\varepsilon$. Every zero in this strip is simple.*

(ii) *For every $A \in \mathbb{R}$, some $b \in [A, A+U(\varepsilon)]$ has*

$$[b, b+L(\varepsilon)]\cap \mathcal{Z}_\omega(\varepsilon)=\varnothing, \qquad L(\varepsilon)=\exp(cY_S(\varepsilon)).$$

Thus the long empty intervals occur arbitrarily high for every deterministic twist, with an explicit bound on the waiting time for their starting points.

(iii) *For every twist,*

$$e^{cY_S(\varepsilon)} \le G_\omega(\varepsilon) \le e^{CX_S(\varepsilon)}, \qquad \lim_{\varepsilon\downarrow 0}\frac{\log\log G_\omega(\varepsilon)}{\log(1/\varepsilon)} = \frac{1}{S-1}. \tag{50}$$

(iv) *For product-Haar-almost every twist,*

$$\lim_{\varepsilon\downarrow 0}\frac{\log\log(e^e + W_\omega(\varepsilon))}{\log(1/\varepsilon)} = \frac{1}{S-1}. \tag{51}$$

The upper bound in (i) and the lower bound in (ii) concern different lengths. They are compatible: $L(\varepsilon)$ is much smaller than $U(\varepsilon)$. The long gaps in (ii) do not imply a corresponding lower bound for the first zero above a fixed origin of a prescribed twist.

Corollary 4.2 (Ordinary prime zeta)**.** *Let $\mathcal{P}(s) = \sum_p p^{-s}$ and let S be the solution of $\mathcal{P}(S) = 2^{1-S}$. Then $1 < S < 2$, the line $\operatorname{Re} s = S$ is zero-free, and the maximal gaps between ordinates of zeros in $S-\varepsilon < \operatorname{Re} s < S$ satisfy (50). Their exact double-logarithmic exponent is $1/(S-1)$. For orientation,*

$$S = 1.7795446535\ldots, \qquad \frac{1}{S-1} = 1.2828001519\ldots.$$

These decimals are illustrative; the theorem uses the exact edge equation.

5 The edge germ and quadratic horizontal control

Assume (45) and let S satisfy (46). Normalize the positive weights by

$$b_p = \frac{a_p}{a_2}(2/p)^S \quad (p\ge 3), \qquad \sum_{p\ge 3} b_p = 1, \qquad h = \sum_{p\ge 3} b_p\lambda_p > 0. \tag{52}$$

The weights b_p satisfy (24) with $\sigma = S$. For an arbitrary phase $\eta \in \Omega$ put

$$g_\eta(z) = 1 - \sum_{p\geq 3} b_p \eta_p e^{-\lambda_p z}, \qquad H(z) = g_1(z). \tag{53}$$

They are analytic for $\operatorname{Re} z > 1 - S$. We have $H(0) = 0$, $H'(0) = h$, and

$$\operatorname{Re} g_\eta(0) = E_b(\eta). \tag{54}$$

For the original Dirichlet series define

$$\eta_p = -\frac{\omega_p}{\omega_2}. \tag{55}$$

Then the exact identity

$$\frac{F_\omega(S + it + z)}{a_2\omega_2 2^{-(S+it+z)}} = g_{\Phi_t\eta}(z) \tag{56}$$

holds throughout the domain.

Lemma 5.1 (One-sided boundary and quadratic zero transfer)**.** *For all twists, F_ω has no zero in $\operatorname{Re} s > S$. There are $C, \eta_0 > 0$ such that whenever $E = E_b(\eta) \leq \eta_0$, g_η has a simple zero z_η satisfying*

$$-CE \leq \operatorname{Re} z_\eta \leq 0, \qquad |\operatorname{Im} z_\eta| \leq C\sqrt{E}, \qquad z_\eta = -\frac{g_\eta(0)}{h} + O(E). \tag{57}$$

All constants are uniform in the infinitely many phases.

Proof. For real $\sigma > S$, $a_2 2^{-\sigma} > \sum_{p\geq 3} a_p p^{-\sigma}$, so the first term cannot be canceled. Equivalently, g_η has no zero with positive real part.

Choose a fixed small disk $|z| \leq r$ with $S - 2r > 1$, on which H has only its simple zero at zero and $|H'(z) - h| < h/4$. Cauchy–Schwarz gives, for $j = 0, 1, 2$,

$$\sup_{|z|\leq r} |(g_\eta - H)^{(j)}(z)| \leq \sqrt{2E}\left(\sum_{p\geq 3} b_p \lambda_p^{2j} e^{2r\lambda_p}\right)^{1/2} \leq C_j\sqrt{E}. \tag{58}$$

Each sum is finite by $S - 2r > 1$. Rouché's theorem on the boundary circle gives a unique zero in the disk when E is small, counted with multiplicity. The derivative there is bounded below by $h/2$, so this zero is simple. Shrinking the disk if necessary, $|H(z)| \geq c|z|$ throughout it. At the zero, (58) therefore implies $|z_\eta| \leq C\sqrt{E}$.

Taylor's formula at zero, with a uniform bound for the second derivative, gives

$$0 = g_\eta(0) + hz_\eta + (g_\eta'(0) - h)z_\eta + O(|z_\eta|^2) = g_\eta(0) + hz_\eta + O(E).$$

This proves the last estimate in (57). Its real part, together with (54), is $O(E)$ rather than merely $O(\sqrt{E})$. The zero-free half-plane supplies the upper sign restriction. The other bounds follow. □

The distinction between horizontal and vertical displacement is decisive: a direct uniform-error estimate without (54) would need phase energy of order ε^2 and would lose the principal exponent in Theorem 4.1.

Lemma 5.2 (Necessary energy at every edge zero)**.** *There are $B, \varepsilon_0 > 0$ such that every zero $S - \varepsilon \leq \sigma \leq S$, $F_\omega(\sigma + it) = 0$, satisfies*

$$E_b(\Phi_t\eta) \leq B\varepsilon, \tag{59}$$

where (55) defines η. Every zero with $S - \varepsilon_0 \leq \operatorname{Re} s \leq S$ is simple.

Proof. Put $\delta = S - \sigma \geq 0$ and $q = \Phi_t \eta$. At the zero,

$$\sum_{p \geq 3} b_p e^{\lambda_p \delta} q_p = 1. \tag{60}$$

Taking real parts gives the exact defect formula

$$\sum_{p \geq 3} b_p e^{\lambda_p \delta}(1 - \operatorname{Re} q_p) = A(\delta) - 1, \quad A(\delta) := \sum_{p \geq 3} b_p e^{\lambda_p \delta}. \tag{61}$$

For a fixed sufficiently small $\delta_0 < S - 1$, this is an analytic function with $A(\delta) = 1 + h\delta + O(\delta^2)$, uniformly on $[0, \delta_0]$. Since $e^{\lambda_p \delta} \geq 1$ and each summand is nonnegative, (61) implies (59).

Finally,

$$\begin{aligned} |g_q'(-\delta) - h| &\leq \sum b_p \lambda_p (e^{\lambda_p \delta} - 1) + \left| \sum b_p \lambda_p e^{\lambda_p \delta}(q_p - 1) \right| \\ &\leq O(\delta) + \sqrt{2(A(\delta) - 1)} \left(\sum b_p \lambda_p^2 e^{\lambda_p \delta} \right)^{1/2} = O(\sqrt{\delta}). \end{aligned}$$

For small δ_0 this derivative is nonzero. By (56), the original zero is also simple. At $\delta = 0$, the defect vanishes and the derivative is exactly h. □

5.1 Curvature transverse to the vertical flow

The preceding lemmas require no coordinatewise small-angle assumption. For smooth one-parameter phase perturbations, the geometry can be identified more precisely.

Proposition 5.3 (Exact quadratic form)**.** *Let v_p be real and satisfy $|v_p| \leq C_v \lambda_p$. Set $\eta_p(u) = e^{iuv_p}$ for real u near zero. The zero from Lemma 5.1 is an analytic function $z(u)$ with $z(0) = 0$. If $V = \sum b_p v_p$, then*

$$\operatorname{Re} z(u) = -\frac{u^2}{2h} \sum_{p \geq 3} b_p \left(v_p - \frac{V}{h} \lambda_p \right)^2 + O(u^4). \tag{62}$$

The quadratic form vanishes exactly when $v_p = c\lambda_p$ for every p. In that case $z(u) = icu$ identically near zero.

Proof. The growth condition ensures local analyticity of

$$G(z, u) = 1 - \sum b_p \exp(iuv_p - \lambda_p z)$$

in both complex variables, by normal convergence. The implicit function theorem applies since $G_z(0, 0) = h > 0$. Write

$$C_1 = \sum b_p \lambda_p v_p, \quad V_2 = \sum b_p v_p^2, \quad H_2 = \sum b_p \lambda_p^2.$$

Differentiation gives $z'(0) = iV/h$ and

$$z''(0) = -\frac{1}{h} \left(V_2 - \frac{2VC_1}{h} + \frac{V^2 H_2}{h^2} \right).$$

The bracket is the sum of squares in (62). For real u, uniqueness gives $z(-u) = \overline{z(u)}$, so the real part is even and its remainder after the quadratic term is $O(u^4)$. All b_p are positive, hence the sum vanishes precisely when $v_p = (V/h)\lambda_p$. Direct substitution then gives the exact root $z(u) = i(V/h)u$. □

Thus the only direction with vanishing quadratic inward displacement is the actual vertical-translation direction. The $O(E)$ control in Lemma 5.1 is consistent with this more precise local geometry, while remaining valid for arbitrary tail phases.

6 Proof of the finite-height and gap theorems

6.1 Uniform simple zeros in every long interval

Apply the upper bound of Theorem 3.1 with weights b_p and energy tolerance $c_2\varepsilon$, where c_2 is a sufficiently small fixed constant. In every interval of length $\exp(CX_S(\varepsilon))$ there is a time t with $E_b(\Phi_t\eta) \le c_2\varepsilon$. By Lemma 5.1, the original function has a simple zero

$$\rho = S + it + z, \qquad -\varepsilon/2 \le \operatorname{Re} z \le 0, \quad |\operatorname{Im} z| \le C\sqrt{\varepsilon}.$$

For small ε, the last displacement is less than one. Given an interval $[A, A+T+2]$, run recurrence on $[A+1, A+T+1]$; the zero lies in the required height interval. Increasing the fixed constant in the exponential absorbs the additive two. Lemma 5.2 proves the simplicity of every zero in the strip. This proves Theorem 4.1(i), and also

$$W_\omega(\varepsilon) \le \exp(CX_S(\varepsilon)). \tag{63}$$

6.2 Robust empty intervals and their recurrence

Let B be the constant in (59). Apply Lemma 3.6 at tolerance $16B\varepsilon$. It gives a phase η^* and a length

$$L = \exp(cY_S(\varepsilon))$$

such that

$$E_b(\Phi_u\eta^*) > 16B\varepsilon \qquad (0 \le u \le L). \tag{64}$$

Changing the tolerance by a fixed factor changes only the constants in this length. The strict inequality holds simultaneously on the whole compact interval.

The metric triangle inequality and isometry of the flow imply

$$\left|\sqrt{E_b(\Phi_u\xi)} - \sqrt{E_b(\Phi_u\eta^*)}\right| \le \frac{1}{\sqrt{2}} d_b(\xi, \eta^*) \qquad (u \in \mathbb{R}). \tag{65}$$

Therefore every phase ξ satisfying $d_b(\xi, \eta^*)^2 \le 2B\varepsilon$ has

$$E_b(\Phi_u\xi) > 9B\varepsilon > B\varepsilon \qquad (0 \le u \le L).$$

By (59), its associated Dirichlet series has no zero in the edge strip at any of these heights.

Now fix any deterministic twist ω and any initial height A. The covering theorem, with target η^* and tolerance $B\varepsilon$, provides a

$$b \in [A, A + \exp(CX_S(\varepsilon))] \quad \text{such that} \quad d_b(\Phi_b\eta, \eta^*)^2 \le 2B\varepsilon.$$

The entire segment $[b, b+L]$ is consequently zero-free in the strip. This proves (ii). It also shows why a probabilistically constructed bad phase gives long holes for *every* fixed twist: the same weighted isometry preserves the exclusion for the entire segment, and the covering theorem transports its starting point effectively.

Part (iii) follows at once. For the upper bound on a supremum of open zero-free intervals, if an interval were longer than $U(\varepsilon)$, it would contain a closed subinterval of that length and hence a zero by (i). Taking two logarithms as in Theorem 3.1 gives the limit in (50).

6.3 Almost-sure first waiting time

For independent Haar phases ω_p, the relative phases $\eta_p = -\omega_p/\omega_2$, $p \geq 3$, have product Haar law. One may condition on ω_2 to verify this. If an edge zero occurs at some $t \in [0, T]$, Lemma 5.2 implies $E_b(\Phi_t \eta) \leq B\varepsilon$. Hence

$$\mathbb{P}\{W_\omega(\varepsilon) < T\} \leq C(1 + T/\varepsilon)e^{-cY_S(\varepsilon)}. \tag{66}$$

The use of $<$ avoids any issue of an infimum at the open left boundary of the strip; using $T + 1$ yields the same bound for $\leq T$ after changing constants.

Apply this estimate with $\varepsilon_n = e^{-n}$ and $T_n = \exp(c_3 Y_S(\varepsilon_n))$, $c_3 > 0$ sufficiently small. The resulting probabilities are summable. Borel–Cantelli, (63), and monotonicity in ε give part (iv). Independence between times or between the events indexed by n is not required.

7 Consequences for prime zeta and boundary-attaining twists

Lemma 7.1 (Boundary classification)**.** *The function F_ω has a zero at $S + it$ if and only if*

$$-\frac{\omega_p}{\omega_2} e^{-it \log(p/2)} = 1 \qquad (p \geq 3). \tag{67}$$

Each twist has at most one such ordinate. The untwisted prime zeta function has none.

Proof. At $\delta = 0$, the defect identity (61) is a sum of nonnegative terms equal to zero, and every weight is positive. This is equivalent to (67); the converse follows from the edge equation. If two distinct ordinates satisfied the condition, their difference $t_0 \neq 0$ would have both $t_0 \log(3/2)$ and $t_0 \log(5/2)$ in $2\pi\mathbb{Z}$. This would make $\log(3/2)/\log(5/2)$ rational. Unique factorization rules that out: an equality $(3/2)^m = (5/2)^n$ for integers forces $m = n = 0$. For the untwisted function, (67) would instead make these two products odd multiples of π, giving the same contradiction. At $t = 0$ the condition also fails. □

Proof of Corollary 4.2. For $a_p = 1$, divergence of $\sum p^{-\sigma}$ as $\sigma \downarrow 1$ gives $S > 1$. At $\sigma = 2$,

$$\sum_{p \geq 3} (2/p)^2 \leq 4 \sum_{\substack{n \geq 3 \\ n \text{ odd}}} n^{-2} = \frac{\pi^2}{2} - 4 < 1,$$

so $S < 2 < 3$. Theorem 4.1 applies, and Lemma 7.1 removes the edge line from the strip without changing its zero set. □

Corollary 7.2 (Strict strips for every twist)**.** *The maximal-gap exponent in Theorem 4.1 is unchanged if one uses $S - \varepsilon < \operatorname{Re} s < S$ instead of $\mathcal{S}_\varepsilon$, for any twist. A uniform upper waiting bound of the same order also holds.*

Proof. At most one ordinate is removed by Lemma 7.1. Two disjoint closed intervals of length $U(\varepsilon)$, separated by one unit, each contain an edge-strip zero. They have different ordinates, so at least one zero is strictly to the left of S. This gives an upper bound of length $2U(\varepsilon) + 1$, absorbed in the exponential constant. The long holes already exclude the larger strip, so their lower bound persists. □

Corollary 7.3 (An explicit approach rate at finite height)**.** *For every twist and all sufficiently large T, a simple zero has $0 \leq \operatorname{Im} \rho \leq T$ and*

$$0 \leq S - \operatorname{Re} \rho \leq \frac{C}{(\log T)^{S-1} \log\log T}. \tag{68}$$

The zero can be chosen strictly to the left of S, with a changed constant.

Proof. Put $L = \log T$ and choose $\varepsilon = KL^{-(S-1)}(\log L)^{-1}$. Then

$$\varepsilon \log(1/\varepsilon) = K(S-1)L^{-(S-1)}(1+o(1)),$$

so $X_S(\varepsilon) = (K(S-1))^{-1/(S-1)}L(1+o(1))$. Choose K large enough that the upper bound in (63) is less than T. Apply Theorem 4.1(i), or Corollary 7.2 for the strict version. □

Remark 7.4 (Uniformity versus practical search). Equation (34) is a finite, verifiable certificate once the weights and degrees are supplied. Nevertheless, the heights in the general bound are enormous. These results quantify existence and recurrence; they do not provide a practical numerical search method for extremely thin edge strips.

7.1 Quantifiers in the prime-zeta conclusions

The same exponent appears in three logically different statements. The following table records their quantifiers.

Quantity	Phase quantifier	Conclusion
Weighted torus covering time	All starts and targets	Exact exponent $1/(\sigma-1)$
Maximal shrinking-edge zero gap	Every fixed twist	Exact exponent $1/(S-1)$
Fixed-origin first zero	Haar-almost every twist	Exact exponent $1/(S-1)$
Fixed-origin first zero of $\mathcal{P}$	One specified twist	Uniform upper bound only

Here "exact exponent" always means a limit of the form $\log\log(\text{time})/\log(1/\varepsilon)$, not a full asymptotic for the time itself.

7.2 Proof assumptions for the recurrence-to-zero transfer

The recurrence-to-zero arguments above depend on the following distinctions.

1. **Growing dimension.** The number of retained primes tends to infinity. All Fourier norms and frequency-separation constants are tracked explicitly in (34); a fixed-dimensional equidistribution theorem alone is insufficient.

2. **Relative frequencies.** Removing the common phase replaces $\log p$ by $\log(p/2)$. The resulting powers of 2 are included in Lemma 3.2; unique factorization still rules out nontrivial resonances.

3. **Energy versus amplitude.** Small energy gives a complex function error of order $\sqrt{E}$. The exact real-part identity and one Newton expansion are what improve the horizontal zero error to $O(E)$.

4. **The unobserved tail.** It is bounded for all phases by $2\sum_{p>x} w_p$. No finite phase box is treated as though it controlled infinitely many coordinates.

5. **No mixing.** The avoidance estimate uses one-time Haar laws and a finite union bound. The almost-sure step uses the first Borel–Cantelli lemma, which requires no independence.

6. **Holes for a fixed function.** A positive-measure family of bad initial phases alone gives only a worst-twist lower bound. The isometry argument (65) and the uniform covering theorem are explicitly used to transfer an entire bad segment to every twist.

7. **Height endpoints.** The generated zero changes its ordinate by $O(\sqrt{\varepsilon})$. Unit buffers are inserted before the upper waiting statement is made.

8. **Sharpness.** The power of ε after two logarithms is determined. The single-logarithmic asymptotic and its constant are not determined.

8 Euler products: critical points and quantitative locking

For a sequence $\omega = (\omega_p)_p \in \prod_p \mathbb{T}$, define

$$Z_\omega(s) = \prod_p (1 - \omega_p p^{-s})^{-1}, \qquad Q_\omega(s) = -\frac{Z'_\omega(s)}{Z_\omega(s)} = \sum_p \frac{(\log p)\omega_p p^{-s}}{1 - \omega_p p^{-s}}. \tag{69}$$

These expressions and their derivatives converge locally uniformly in $\operatorname{Re} s > 1$, where Z_ω never vanishes. The phases of prime powers in an Euler product are coupled; they cannot be replaced by independent coordinates. The relevant flow here uses the full frequencies $\log p$, including $p = 2$.

Let $E_* > 1$ be the unique solution of

$$\frac{\log 2}{2^{E_*} + 1} = \sum_{p \geq 3} \frac{\log p}{p^{E_*} - 1}, \qquad \beta_* = (E_* - 1)^{-1}. \tag{70}$$

For $\omega_p = 1$, this is the classical right boundary of the zeros of ζ'. Arias de Reyna and van de Lune [8, Theorem 5.1] give $E_* = 2.813014020252898\ldots$, in particular $1 < E_* < 3$. Their Theorem 6.1 also proves qualitative freezing of individual prime phases along sequences approaching the boundary. The quantitative energy estimates below strengthen that coordinatewise conclusion.

Write $\mathcal{C}_\omega(\varepsilon)$ for the ordinates of critical points of Z_ω in $E_* - \varepsilon < \operatorname{Re} s \leq E_*$, and define

$$G_\omega^{\mathrm{crit}}(\varepsilon) = \sup\{b - a : (a, b) \cap \mathcal{C}_\omega(\varepsilon) = \varnothing\}.$$

Theorem 8.1 (Critical-point gaps for every Euler-product twist)**.** *There are positive constants c, C, ε_0, independent of ω, such that, for $0 < \varepsilon < \varepsilon_0$, the following assertions hold. Put*

$$U_*(\varepsilon) = \exp(C\varepsilon^{-\beta_*}), \qquad L_*(\varepsilon) = \exp\left(\frac{c\varepsilon^{-\beta_*}}{\log(1/\varepsilon)}\right).$$

Every real interval of length $U_*(\varepsilon)$ contains an ordinate in $\mathcal{C}_\omega(\varepsilon)$, and all the critical points in this strip are simple zeros of Z'_ω. For every real a, there is $b \in [a, a + U_*(\varepsilon)]$ such that

$$[b, b + L_*(\varepsilon)] \cap \mathcal{C}_\omega(\varepsilon) = \varnothing.$$

Consequently

$$L_*(\varepsilon) \leq G_\omega^{\mathrm{crit}}(\varepsilon) \leq U_*(\varepsilon), \qquad \lim_{\varepsilon \downarrow 0} \frac{\log\log G_\omega^{\mathrm{crit}}(\varepsilon)}{\log(1/\varepsilon)} = \frac{1}{E_* - 1}. \tag{71}$$

If $W_\omega^{\mathrm{crit}}(\varepsilon)$ is the first nonnegative ordinate in $\mathcal{C}_\omega(\varepsilon)$, then for product-Haar-almost every ω,

$$\lim_{\varepsilon \downarrow 0} \frac{\log\log(e^e + W_\omega^{\mathrm{crit}}(\varepsilon))}{\log(1/\varepsilon)} = \frac{1}{E_* - 1}. \tag{72}$$

8.1 A circle identity and a positive boundary defect

The frozen phase is

$$z_2^* = -1, \qquad z_p^* = 1 \quad (p \geq 3).$$

Define its weighted squared-distance defect by

$$D_E(z) = \sum_p (\log p) p^{-E_*} |z_p - z_p^*|^2. \tag{73}$$

There is no common-rotation freedom in this definition.

Lemma 8.2 (Quantitative rigidity of critical points)**.** *For every twist, Q_ω has no zeros in* $\operatorname{Re} s > E_*$. *If $Q_\omega(\sigma + it) = 0$ and $E_* - \varepsilon < \sigma \leq E_*$, then*

$$D_E((\omega_p p^{-it})_p) \leq C\varepsilon \tag{74}$$

for sufficiently small ε, uniformly in the twist and the ordinate. At $\sigma = E_*$ the phases equal z^* exactly.

Proof. For real $\sigma > 1$, put $r_p = p^{-\sigma}$ and

$$c_p = \frac{(\log p) r_p^2}{1 - r_p^2}, \qquad R_p = \frac{(\log p) r_p}{1 - r_p^2}, \qquad \xi_p = \frac{z_p - r_p}{1 - r_p z_p}.$$

For $|z_p| = 1$, one has $|\xi_p| = 1$ and the exact identity

$$\frac{(\log p) r_p z_p}{1 - r_p z_p} = c_p + R_p \xi_p. \tag{75}$$

Set

$$C_0 = \sum_p c_p, \qquad b_0 = R_2, \qquad R_0 = \sum_{p \geq 3} R_p, \qquad \Delta(\sigma) = C_0 + R_0 - b_0.$$

Then

$$\Delta(\sigma) = \sum_{p \geq 3} \frac{\log p}{p^\sigma - 1} - \frac{\log 2}{2^\sigma + 1}. \tag{76}$$

Divide the first sum by $h(\sigma) = (\log 2)/(2^\sigma + 1)$. Each summand of the resulting ratio has logarithmic derivative

$$\frac{(\log 2) 2^\sigma}{2^\sigma + 1} - \frac{(\log p) p^\sigma}{p^\sigma - 1} < 0.$$

The ratio decreases from infinity to zero. Thus Δ has the unique zero E_* and is negative to its right.

At a zero of Q, take $z_p = \omega_p p^{-it}$. Equation (75) gives

$$C_0 + b_0 \xi_2 = -\sum_{p \geq 3} R_p \xi_p, \qquad |C_0 + b_0 \xi_2| \leq R_0. \tag{77}$$

If $\sigma > E_*$, then $b_0 > C_0 + R_0$, contradicting this inequality. Near E_* from the left, $b_0 - C_0$ is positive and C_0, b_0, R_0 stay bounded above and below by positive constants. Squaring (77) yields

$$C_0 b_0 |\xi_2 + 1|^2 \leq R_0^2 - (b_0 - C_0)^2 = \Delta(\sigma)(R_0 + b_0 - C_0). \tag{78}$$

Taking real parts in the same identity gives

$$\sum_{p\ge 3} R_p|\xi_p-1|^2 = 2\Delta(\sigma) + b_0|\xi_2+1|^2. \tag{79}$$

Since $\Delta(\sigma) = O(E_*-\sigma)$, both defects are $O(\varepsilon)$. Finally,

$$|\xi-1| = \frac{(1+r)|z-1|}{|1-rz|}, \qquad |\xi+1| = \frac{(1-r)|z+1|}{|1-rz|}.$$

These are uniform comparisons because all r_p are bounded away from 1, and $R_p \ge (\log p)p^{-E_*}$ for $\sigma \le E_*$. This proves (74). At the boundary both right sides in (78)–(79) vanish, which forces precisely $z = z^*$. □

Lemma 8.3 (Realization with a quadratic real-part loss)**.** *There are constants $d_0, K > 0$ such that $D = D_E(z) < d_0$ implies that Q_z has a simple zero $E_* + h_z$ with*

$$-KD \le \operatorname{Re} h_z \le 0, \qquad |\operatorname{Im} h_z| \le K\sqrt{D}. \tag{80}$$

Furthermore every zero in a sufficiently thin right-edge strip is simple.

Proof. The frozen function is $Q_{z^*}(s) = \Delta(s)$. Put $a_E = \Delta'(E_*)$. If $h = (\log 2)/(2^{E_*}+1)$ and $t_p = (\log p)/(p^{E_*}-1)$, then $\sum_{p\ge 3} t_p = h$ and

$$a_E = h\frac{(\log 2)2^{E_*}}{2^{E_*}+1} - \sum_{p\ge 3} t_p \frac{(\log p)p^{E_*}}{p^{E_*}-1} < h(\log 2 - \log 3) < 0.$$

For $F_z(u) = Q_z(E_*+u)$, the circle identity at $\sigma = E_*$, followed by Cauchy–Schwarz, gives

$$|F_z(0)| \le K\sqrt{D}, \qquad |\operatorname{Re} F_z(0)| \le KD, \qquad |F_z'(0) - a_E| \le K\sqrt{D}. \tag{81}$$

For the real estimate each tail summand has real-part change $-R_p|\xi_p-1|^2/2$, and the $p=2$ summand has change $R_2|\xi_2+1|^2/2$. For the derivative estimate the local difference is at most $K(\log p)^2 p^{-E_*}|z_p - z_p^*|$; its Cauchy–Schwarz companion sum is $\sum_p (\log p)^3 p^{-E_*} < \infty$.

Choose a fixed $r_0 > 0$ with $E_* - 2r_0 > 1$. On $|u| \le r_0$ the second derivatives are uniformly bounded, and hence

$$F_z(u) = F_z(0) + a_E u + O(\sqrt{D}\,|u| + |u|^2). \tag{82}$$

Rouché's theorem on $|u| = K_0\sqrt{D}$, with K_0 sufficiently large and then D sufficiently small, gives one zero h_z in that disk. Its derivative is $a_E + O(\sqrt{D})$, so it is simple. Taking real parts in (82) proves $|\operatorname{Re} h_z| \le KD$; Lemma 8.2 supplies the sign. The case $D = 0$ is the frozen zero itself.

Conversely, translate any zero in the edge strip to its own ordinate. Lemma 8.2 gives $D = O(\varepsilon)$, so its derivative is $a_E + O(\sqrt{\varepsilon}) \ne 0$. At a zero of Q_z one has $Z_z'' = -Z_z Q_z'$, proving the asserted simplicity for Z_z'. □

8.2 The finite-height consequences

Proof of Theorem 8.1. Apply Theorem 2.2 to full prime frequencies and weights $(\log p)p^{-E_*}$. This is the logarithmic-weight case: the truncation and entropy scales are respectively $\varepsilon^{-\beta_*}$ and $\varepsilon^{-\beta_*}/\log(1/\varepsilon)$. The energy in (73) differs by a factor of two from the energy used there, after translating the target to z^*. Thus every orbit reaches $D_E \le c_0\varepsilon$ on every interval of length

$\exp(C\varepsilon^{-\beta_*})$. Lemma 8.3 produces a critical point in the required strip and moves its ordinate by at most $K\sqrt{\varepsilon}$. Padding the initial interval and enlarging C gives the stated upper bound.

By Lemma 8.2, the ordinate of every critical point in the strip is a visit to $D_E \le K\varepsilon$. The robust-hole assertion of Proposition 2.3, applied to this enlarged target, therefore gives the asserted empty intervals and their recurrence windows. This proves (71).

For completeness, the almost-sure lower waiting estimate requires no independence between times. The weighted energy is uniformly Lipschitz along the flow, since $\sum_p (\log p)^2 p^{-E_*} < \infty$. A grid of mesh comparable to ε and the upper small-ball bound in Theorem 2.1 give

$$\mathbb{P}\Big\{\min_{0\le t\le T} D_E(\Phi_t\omega) \le K\varepsilon\Big\} \le C(1+T/\varepsilon)\exp\left(-\frac{c\varepsilon^{-\beta_*}}{\log(1/\varepsilon)}\right).$$

Take $\varepsilon_n = e^{-n}$ and $T_n = \exp(c_1\varepsilon_n^{-\beta_*}/n)$ with c_1 sufficiently small. The displayed probabilities are summable. Borel–Cantelli, Lemma 8.2, monotone interpolation between consecutive ε_n, and the deterministic upper bound prove (72). □

Corollary 8.4 (Characters of an odd modulus)**.** *Let q be odd, and let $E_q > 1$ solve*

$$\frac{\log 2}{2^{E_q}+1} = \sum_{\substack{p\ge 3\\ p\nmid q}} \frac{\log p}{p^{E_q}-1}.$$

Then $1 < E_q \le E_* < 3$. For every Dirichlet character χ modulo q, the deterministic assertions of Theorem 8.1 hold for the critical points of $L(s,\chi)$ with E_q replacing E_* and constants depending on q. In particular their maximal-gap exponent is $(E_q-1)^{-1}$, independently of the character.

Proof. Omit the finitely many primes dividing q in every sum and product. The smallest remaining prime is 2, and the decreasing-ratio proof gives the stated boundary. Omitting positive tail terms can only lower it. All circle identities and local estimates remain valid. The prime-flow and small-ball estimates are unchanged in scale by finitely many omissions. Finally $\chi(p)$ supplies a fixed unimodular twist on the remaining primes. This proves the deterministic assertions; an almost-sure statement, if desired, concerns additional Haar twists, not probability on the finite set of characters. □

8.3 A sharp quantitative relation between 1-points and critical points

Let $A_* > 1$ be the unique solution of

$$\sum_{\substack{n\ge 3\\ n \text{ odd}}} n^{-A_*} = 2^{-A_*}, \quad \text{equivalently} \quad \zeta(A_*) = \frac{2^{A_*}+1}{2^{A_*}-1}. \tag{83}$$

Its classical value is $A_* = 1.940101683743625\ldots$ [8, Theorem 4.1]. Put

$$\theta = \frac{A_*-1}{E_*-1} \in (0,1), \qquad \Psi(\delta) = \frac{\delta^\theta}{\log(1/\delta)}, \qquad D_A(z) = \sum_p p^{-A_*}|z_p - z_p^*|^2. \tag{84}$$

Theorem 6.2 of [8] establishes a qualitative relation between the two kinds of almost-extremal points. The next result gives an explicit rate, including its logarithmic factor, and proves that this rate cannot be uniformly improved.

Theorem 8.5 (Quantitative locking and its sharpness)**.** *There are constants $K, \delta_0 > 0$, independent of the twist, with the following properties.*

1. *If $Z'_\omega(E_* - \delta + i\gamma) = 0$, where $0 < \delta < \delta_0$, then Z_ω has a 1-point s satisfying*
$$0 \le A_* - \operatorname{Re} s \le K\Psi(\delta), \qquad |\operatorname{Im} s - \gamma| \le K\sqrt{\delta}.$$

2. *If $Z_\omega(A_* - \varepsilon + i\gamma) = 1$, where $0 < \varepsilon < \delta_0$, then it has a critical point ρ satisfying*
$$0 \le E_* - \operatorname{Re}\rho \le K\varepsilon, \qquad |\operatorname{Im}\rho - \gamma| \le K\sqrt{\varepsilon}.$$

3. *The real-part rate Ψ in the first assertion is sharp up to constant factors, both uniformly among twists and along sequences for the ordinary zeta function. More precisely, there are critical points ρ_n of ζ and 1-points s_n such that, with $\delta_n = E_* - \operatorname{Re}\rho_n \downarrow 0$ after passage to a subsequence,*
$$A_* - \operatorname{Re} s_n \asymp \Psi(\delta_n), \qquad |\operatorname{Im} s_n - \operatorname{Im}\rho_n| \longrightarrow 0.$$
The 1-point s_n is the unique 1-point in a fixed sufficiently small disk centered at $A_* + i\operatorname{Im}\rho_n$. Thus the sharpness concerns the local counterpart supplied by the first assertion.

Lemma 8.6 (Rigidity and realization for 1-points)**.** *Every twist has no 1-points in $\operatorname{Re} s > A_*$. A 1-point $\sigma + it$ with $A_* - \varepsilon < \sigma \le A_*$ satisfies*
$$D_A((\omega_p p^{-it})_p) \le K\varepsilon.$$
Conversely, if $D = D_A(z)$ is sufficiently small, there is a simple 1-point $A_* + u_z$ with $-KD \le \operatorname{Re} u_z \le 0$ and $|\operatorname{Im} u_z| \le K\sqrt{D}$.

Proof. For prime phases z, write $z(n) = \prod_p z_p^{v_p(n)}$, and set $M(\sigma) = \sum_{n\ge3,\, n \text{ odd}} n^{-\sigma}$. The equation $Z_z(\sigma) = 1$ is equivalent to
$$\sum_{\substack{n\ge3\\ n \text{ odd}}} z(n)n^{-\sigma} = -2^{-\sigma}z_2.$$
The triangle inequality implies $2^{-\sigma} \le M(\sigma)$. Since $M(\sigma)/2^{-\sigma} = \sum_{n\ge3,\, n \text{ odd}}(2/n)^\sigma$ is strictly decreasing, this proves the boundary assertion. Put $\tau = -z_2$. The same equation gives the exact identity
$$\sum_{\substack{n\ge3\\ n \text{ odd}}} n^{-\sigma}|z(n) - \tau|^2 = 2(M(\sigma) - 2^{-\sigma}) = O(\varepsilon). \tag{85}$$
The terms $n = 3, 9$ show that $|z_3 - \tau| + |z_3^2 - \tau| = O(\sqrt{\varepsilon})$; hence $|\tau - 1| = |\tau^2 - \tau| = O(\sqrt{\varepsilon})$. The prime terms in (85), the inequality $p^{-A_*} \le p^{-\sigma}$, and summability of $\sum_p p^{-A_*}$ now give $D_A(z) = O(\varepsilon)$.

For the converse, the frozen product is
$$Z_{z^*}(s) = \frac{1 - 2^{-s}}{1 + 2^{-s}}\zeta(s), \qquad Z_{z^*}(s) - 1 = \frac{M(s) - 2^{-s}}{1 + 2^{-s}}.$$
Its derivative a_A at A_* is strictly negative, since $M(s)/2^{-s}$ decreases strictly through 1 there. Use the absolutely convergent logarithm $\mathcal{L}_z(s) = -\sum_p \log(1 - z_p p^{-s})$, defined by its power series. At $s = A_*$ the frozen logarithm is 0. Each local real part has a stationary extremum at its frozen phase, so
$$|\mathcal{L}_z(A_*)| = O(\sqrt{D}), \qquad |\operatorname{Re}\mathcal{L}_z(A_*)| = O(D).$$
These estimates follow directly from $|1 - rz|^2 = (1 - r)^2 + r|z - 1|^2$ at $z^* = 1$, and $|1 - rz|^2 = (1 + r)^2 - r|z + 1|^2$ at $z^* = -1$, followed by Cauchy–Schwarz. Exponentiating preserves the bounds $|Z_z(A_*) - 1| = O(\sqrt{D})$ and $|\operatorname{Re}(Z_z(A_*) - 1)| = O(D)$. The derivative differs from a_A by $O(\sqrt{D})$ and the second derivative is uniformly bounded in a fixed disk: the necessary sums $\sum_p (\log p)^j p^{-A_* + 2r_0}$ converge after choosing $A_* - 2r_0 > 1$. The Rouché and real-part argument of Lemma 8.3 proves the assertion. □

Proof of Theorem 8.5. First $D_E(z) \le K D_A(z)$, because $(\log p)p^{-(E_*-A_*)}$ is bounded. In the reverse direction, for sufficiently large x,

$$D_A(z) \le K\frac{x^{E_*-A_*}}{\log x} D_E(z) + K\frac{x^{1-A_*}}{\log x}. \tag{86}$$

Indeed the ratio of the two weights for $p \le x$ is bounded by $Kx^{E_*-A_*}/\log x$, and the tail uses $|z_p - z_p^*|^2 \le 4$ and the prime-number estimate for $\sum_{p>x} p^{-A_*}$. For a critical point with real-part gap δ, Lemma 8.2 gives $D_E \le K\delta$ at its own ordinate. Taking $x = \delta^{-1/(E_*-1)}$ in (86) gives $D_A \le K\Psi(\delta)$. The sharper ordinate estimate uses

$$2A_* - E_* > 1. \tag{87}$$

The certified enclosures in Lemma 8.9 give $2A_* - E_* > 1.04$. Choose a fixed disk radius $r_0 > 0$ such that $2A_* - E_* - 2r_0 > 1$. Cauchy–Schwarz against the D_E weights bounds the logarithmic Euler germ and its first derivative in that disk by $O(\sqrt{D_E})$: the companion series are bounded by

$$\sum_p \frac{p^{E_*-2A_*+2r_0}}{\log p}, \qquad \sum_p (\log p)p^{E_*-2A_*+2r_0},$$

both convergent. Thus $Z_z(A_* + u) - 1$ differs in C^1 from the frozen germ by $O(\sqrt{\delta})$. Rouché gives a zero with $|u| = O(\sqrt{\delta})$. At $u = 0$ its real part remains $O(D_A)$, by the proof of Lemma 8.6. Taylor's formula therefore gives $|\operatorname{Re} u| = O(D_A + \delta) = O(\Psi(\delta))$. The global 1-point boundary supplies its sign. This proves the first assertion, including the sharper ordinate bound. The reverse assertion follows from that lemma, $D_E \le K D_A$, and Lemma 8.3.

To prove sharpness, let $x \to \infty$ and take the real twist

$$z_2^{(x)} = -1, \quad z_p^{(x)} = 1 \ (3 \le p \le x), \quad z_p^{(x)} = -1 \ (p > x).$$

These products converge locally uniformly, with all derivatives, to Z_{z^*} on $\operatorname{Re} s > 1$. At the critical boundary,

$$Q_{z^{(x)}}(E_*) = -2\sum_{p>x} \frac{(\log p)p^{-E_*}}{1 - p^{-2E_*}} \sim -\frac{2}{E_* - 1} x^{1-E_*}. \tag{88}$$

At the 1-point boundary,

$$Z_{z^{(x)}}(A_*) = \prod_{p>x} \frac{1 - p^{-A_*}}{1 + p^{-A_*}}, \qquad Z_{z^{(x)}}(A_*) - 1 \sim -\frac{2}{A_* - 1}\frac{x^{1-A_*}}{\log x}. \tag{89}$$

The prime number theorem gives both comparisons. Because the two frozen derivatives are real and strictly negative, the unique nearby zeros are real, lie to the left, and have gaps

$$\delta_x \asymp x^{1-E_*}, \qquad \varepsilon_x \asymp \frac{x^{1-A_*}}{\log x} \asymp \Psi(\delta_x). \tag{90}$$

More precisely, if $c_E = 2/((E_* - 1)|a_E|)$ and $c_A = 2/((A_* - 1)|a_A|)$, then $\delta_x \sim c_E x^{1-E_*}$ and $\varepsilon_x \sim c_A x^{1-A_*}/\log x$, whence $\varepsilon_x/\Psi(\delta_x) \to c_A(E_* - 1)c_E^{-\theta} > 0$. Reality follows alternatively from conjugation and local uniqueness. Choose fixed small disks around A_* and E_* containing just the respective frozen zeros. For all sufficiently large x, the nearby zeros in these disks remain unique, by Rouché's theorem.

Finally every fixed Euler-product twist is a locally uniform limit of vertical translates of ζ in $\operatorname{Re} s > 1$. To check this directly, truncate the logarithmic Euler series so that its tail and the

required derivative tails are uniformly small on the two disks, and use the rational independence of the finitely many $\log p$ to approximate the remaining prime phases. Such approximations occur at arbitrarily large ordinates by Kronecker's theorem. For each member of a sequence $x_n \to \infty$, approximate $Z_{z^{(x_n)}}$ and its derivative closely enough that both local zeros move by $o(\delta_{x_n})$. The derivatives at these zeros stay bounded away from zero, so this follows from Rouché's theorem, or the local inverse estimate. The resulting ordinary-zeta points retain (90), their ordinate difference tends to zero, and the local 1-point is unique in a fixed smaller disk. Passing to a subsequence makes the critical gaps decrease strictly to zero. This proves the final assertion. □

Remark 8.7 (What the sharpness assertion measures). The construction above concerns the real-part transfer $\delta \mapsto \delta^\theta/\log(1/\delta)$. Proposition 8.8 below treats the other rates. The almost-sure assertion (72) does not provide a deterministic first-occurrence lower bound for ordinary ζ'.

8.4 Optimal ordinate estimates and the reverse horizontal rate

The preceding tail construction concerns the forward horizontal modulus. A different deformation proves sharpness of the remaining three estimates.

Proposition 8.8 (One-prime sharpness)**.** *The square-root ordinate exponents in both directions of Theorem 8.5, and the linear horizontal bound from a* 1*-point to a critical point, cannot be improved uniformly. Each assertion is sharp along sequences for ordinary ζ, with the counterparts chosen as the unique roots in fixed local disks.*

Proof. Fix a sufficiently large odd prime P. Starting from z^*, change only z_P to e^{iu} for real u near zero. The holomorphic implicit-function theorem gives analytic functions $h_A(u), h_E(u)$ such that $A_* + h_A(u)$ is a 1-point and $E_* + h_E(u)$ is a critical point of the perturbed product, with $h_A(0) = h_E(0) = 0$. Let $a_A < 0$ and $a_E < 0$ be the frozen derivatives from the preceding proofs. Differentiating in u gives

$$h_A'(0) = \frac{-i}{a_A(P^{A_*} - 1)}, \qquad h_E'(0) = \frac{-i(\log P)P^{E_*}}{a_E(P^{E_*} - 1)^2}. \tag{91}$$

Write their positive imaginary parts as $b_A(P)$ and $b_E(P)$. Since $E_* > A_*$,

$$b_A(P) \sim P^{-A_*}/|a_A|, \qquad b_E(P) \sim (\log P)P^{-E_*}/|a_E| = o(b_A(P)).$$

Thus $b = b_A(P) - b_E(P) > 0$ after fixing P large enough. Conjugation gives $h_j(-u) = \overline{h_j(u)}$, so the real parts are even and the imaginary parts odd.

Put $\delta_j(u) = -\operatorname{Re} h_j(u)$ and $t_j(u) = \operatorname{Im} h_j(u)$ for $j = A, E$. The realization lemmas give $0 \le \delta_j(u) = O(u^2)$. For the reverse bound, recenter at $t_E(u)$. The prime-2 phase is $-e^{-it_E(u)\log 2}$, whose contribution to the critical energy is

$$(\log 2)2^{-E_*}|1 - e^{-it_E(u)\log 2}|^2 \sim (\log 2)^3 2^{-E_*} b_E(P)^2 u^2.$$

The necessary-energy inequality bounds this by $C\delta_E(u)$. The identical argument with the 1-point energy at height $t_A(u)$ gives $\delta_A(u) \ge cu^2$. At zero deficit the necessary identities still hold with zero right side, so there is no exception in this argument. Analyticity and evenness now give positive constants k_A, k_E such that

$$\delta_A(u) = k_A u^2 + O(u^4), \qquad \delta_E(u) = k_E u^2 + O(u^4), \qquad t_A(u) - t_E(u) = bu + O(u^3).$$

Therefore the ordinate difference is comparable to the square root of either deficit, and the two deficits are comparable to each other.

To realize these asymptotics for ordinary ζ, take $u_n \to 0$ and approximate each fixed twist by a vertical translate of ζ on slightly enlarged fixed disks around A_* and E_*, with error $o(u_n^2)$. Finite-coordinate Kronecker approximation and absolute convergence permit this at arbitrarily large heights, exactly as in the preceding sharpness proof. Cauchy's formula also controls the derivatives. Uniform simple-root stability moves the two roots by $o(u_n^2)$, preserving all three comparisons and local uniqueness. □

8.5 A finite certificate for the cross-weight inequality

Lemma 8.9. *The edges defined in (70) and (83) satisfy*

$$1.93 < A_* < 2 < E_* < 2.82, \qquad 2A_* - E_* > 1.04. \tag{92}$$

Proof. For $a = 193/100$, the finite positive sum gives

$$\sum_{\substack{3\le n\le 499\\ n \text{ odd}}} (2/n)^a > 1.0054767 > 1.$$

The decreasing-ratio characterization of A_* implies $A_* > a$. At $s = 2$, the odd sum $M(2) = \pi^2/8 - 1 < 1/4$, since $\pi^2 < 10$; hence $A_* < 2$. The terms $p = 3, 5$ alone give $\Delta(2) > 0$ and therefore $E_* > 2$.

For the remaining upper bound put $s = 141/50$ and $u = s - 1$. Direct outward enclosures of the following finite expressions give

$$\begin{aligned} P_{200} &:= \sum_{3\le p\le 200} \frac{\log p}{p^s - 1} < 0.085425810, \\ T_{200} &:= \frac{200^{1-s}}{1 - 201^{-s}} \left(\frac{\log 200}{u} + \frac{1}{u^2} \right) < 0.000208471, \\ H_2 &:= \frac{\log 2}{2^s + 1} > 0.085981085. \end{aligned}$$

The positive decreasing function $(\log x)x^{-s}$ and the integral test show

$$\sum_{p>200} \frac{\log p}{p^s - 1} \le \frac{1}{1 - 201^{-s}} \int_{200}^{\infty} (\log x) x^{-s}\, dx = T_{200}.$$

Thus $\Delta(s) < -0.000346804 < 0$, proving $E_* < 2.82$.

The displayed numerical inequalities are only coarse finite certificates, not assumed high-precision values of either edge. They can be verified using rational interval arithmetic as follows. Reduce logarithms by powers of 2 to arguments in $[1, 2]$ and use

$$\log y = 2 \sum_{j=0}^{N} \frac{t^{2j+1}}{2j+1} + R_N, \quad t = \frac{y-1}{y+1}, \quad 0 \le R_N \le \frac{2t^{2N+3}}{(2N+3)(1-t^2)}.$$

Taking $N = 80$ suffices, since $0 \le t \le 1/3$. For exponentials of nonnegative arguments $v \le 18$, use the Taylor polynomial through degree 256 and the tail bound

$$0 \le e^v - \sum_{j=0}^{256} \frac{v^j}{j!} \le \frac{v^{257}}{257!} \frac{1}{1 - v/258}.$$

Use reciprocals for negative arguments and propagate endpoints outward. These errors are much smaller than the stated decimal margins. Finally $2(1.93) - 2.82 = 1.04$, proving the last inequality. □

9 Small random perturbations and the matching recurrence scale

Let $\mathbb{P}$ denote the full set of primes. Fix $A > 0$ and $0 < \rho < \log 2/2$. Consider the compact product box

$$\mathcal{B} = \prod_{p\in\mathbb{P}} [\log p - \rho p^{-A}, \log p + \rho p^{-A}], \qquad \lambda_p = \log p + \xi_p,$$

with its product topology and the product probability law under which the ξ_p are independent and uniform on $[-\rho p^{-A}, \rho p^{-A}]$. The radius ρ may be arbitrarily small. We use either the full frequencies $\nu_p = \lambda_p$, $p \in \mathbb{P}$, or the relative frequencies $\nu_p = \lambda_p - \lambda_2$, $p \geq 3$. The conclusions below hold simultaneously for both models. Sums and products below range over the coordinates of the chosen model. Write $\mathcal{R}_{w,\nu}$ for the covering time associated with $\mathcal{E}_w$ and these frequencies.

Theorem 9.1 (Matching scale on one full-measure frequency set)**.** *There is a set $\mathcal{B}_{\mathrm{good}}$ of full product probability such that the following assertion holds for every $\lambda \in \mathcal{B}_{\mathrm{good}}$, simultaneously for every $\sigma > 1$, every $\kappa \in \mathbb{R}$, and every positive weight sequence satisfying*

$$w_p \asymp p^{-\sigma}(\log p)^{\kappa}.$$

There are $c, C > 0$ such that, for all sufficiently small ε,

$$\exp\bigl(cY_{\sigma,\kappa}(\varepsilon)\bigr) \leq \mathcal{R}_{w,\nu}(\varepsilon) \leq \exp\bigl(CY_{\sigma,\kappa}(\varepsilon)\bigr). \tag{93}$$

The upper bound is uniform in the starting phase, target phase, and initial time. In particular, with $\beta = (\sigma-1)^{-1}$ and $\ell_\varepsilon = \log(1/\varepsilon)$,

$$\log\log \mathcal{R}_{w,\nu}(\varepsilon) = \beta\ell_\varepsilon + \bigl((\kappa-1)\beta - 1\bigr)\log \ell_\varepsilon + O(1). \tag{94}$$

Constants and the small-ε threshold may depend on the fixed weights and frequencies. No assertion of a limiting leading constant for $\log \mathcal{R}_{w,\nu}$ is made. The same estimates hold for the supremum of energy-avoidance interval lengths along every individual orbit.

The phrase "simultaneously" is material: the exceptional frequency set is chosen before the real parameters σ, κ or the weight sequence. This permits applications in which the weight exponent itself depends on the realized frequencies.

Lemma 9.2 (A countable family of frequency boxes)**.** *There is a full-probability set of frequencies with the following property. For each integer $M \geq 2$, there is $D_M > 0$ such that, at every sufficiently large dyadic x, every nonzero integer vector supported on the relevant primes $p \leq x$ and satisfying*

$$|k_p| \leq 2\left\lceil (x/p)^{M/2}(\log x/\log p)^{M/2}\right\rceil + 1$$

obeys

$$|k\cdot\nu| \geq \exp(-D_M x/\log x). \tag{95}$$

Both frequency models satisfy this property on the same set.

Proof. The elementary prime estimates needed for counting these vectors are

$$\sum_{p\leq x} \log(x/p) = O(x/\log x), \tag{96}$$

$$\sum_{p\leq x} \log(\log x/\log p) = \int_2^x \frac{\pi(t)}{t\log t}\,\mathrm{d}t = O(x/\log^2 x). \tag{97}$$

The equalities follow by summing the relevant integrals; the estimates follow from the prime number theorem. Removing the prime 2 changes neither order estimate. Consequently the number of admitted vectors is at most $\exp(C_M x/\log x)$.

For any one nonzero vector, choose a coordinate p with $k_p \neq 0$. Condition on every random coordinate except ξ_p. The linear form $k \cdot \nu$ is then uniform on an interval of length $2|k_p|\rho p^{-A}$. This remains true in the relative model: $p \geq 3$ there, and the common coordinate ξ_2 is among those conditioned upon. Thus, for $u > 0$,

$$\mathbb{P}\{|k \cdot \nu| < u\} \leq \rho^{-1} x^A u.$$

A union bound makes the probability of failure of (95) at most

$$\rho^{-1} x^A \exp(-(D_M - C_M)x/\log x).$$

Choose $D_M > C_M + 1$. These probabilities are summable for $x = 2^n$. The first Borel–Cantelli lemma, followed by intersection over the countably many integers M and the two models, proves the assertion. No independence between distinct linear forms is required. □

Proof of Theorem 9.1. Fix frequencies furnished by Lemma 9.2. At a dyadic truncation x choose squared Fejér kernels with degrees allocated through

$$m_p = \left\lceil (x/p)^{\sigma/2} (\log p/\log x)^{\kappa/2} \right\rceil, \qquad K = \prod_{p \leq x} J_{m_p}. \tag{98}$$

The quantity inside the ceiling is allowed to be smaller than one. The identities for J_m give

$$\begin{aligned} m := \int \mathcal{E}_{w,\leq x} K &= \sum_{p \leq x} \frac{3w_p}{2m_p^2 + 1} \\ &\ll x^{-\sigma} (\log x)^{\kappa} \pi(x) \ll x^{1-\sigma} (\log x)^{\kappa - 1}. \end{aligned}$$

Also $\log \|K\|_A = O_{\sigma,\kappa}(x/\log x)$. Indeed, if h_p denotes the quantity rounded in (98), then

$$\log\lceil h_p \rceil \leq \log 2 + (\log h_p)_+ \leq \log 2 + \frac{\sigma}{2} \log(x/p) + \frac{|\kappa|}{2} \log(\log x/\log p),$$

and (96)–(97) apply.

Choose an integer $M \geq \max\{2, \sigma, |\kappa|\}$. Every frequency of $(2m - \mathcal{E}_{w,\leq x})K$ lies in the box of Lemma 9.2. Its nonzero frequencies therefore have separation at least $\exp(-D_M x/\log x)$. The finite-frequency certificate, Lemma 1.2, shows that every interval of length $\exp(Cx/\log x)$ contains a time with $\mathcal{E}_{w,\leq x} < 2m$. To see that the factor $1/m$ in the certificate causes no loss, a single fixed prime gives $m \gg x^{-\sigma}(\log x)^{\kappa}$ for large x, so $\log(1/m) = O(\log x)$.

The uncontrolled tail contributes at most

$$2 \sum_{p > x} w_p \ll x^{1-\sigma} (\log x)^{\kappa - 1}.$$

Take the least dyadic $x \geq K_1 X_{\sigma,\kappa}(\varepsilon)$, with a sufficiently large fixed K_1. Since x lies between $K_1 X$ and $2K_1 X$, the total energy is less than ε and the elapsed time is at most $\exp(CY_{\sigma,\kappa}(\varepsilon))$. Target translation changes Fourier coefficients only by unimodular factors. This proves the upper bound with all the stated uniformities.

For the lower bound, every frequency vector in $\mathcal{B}$ satisfies

$$\sum_p w_p|\nu_p| < \infty.$$

Thus $\mathcal{E}_w(\Phi_t\eta)$ is uniformly Lipschitz in t. A time mesh of size proportional to ε, Haar invariance, and the upper small-ball estimate in Theorem 2.1 give

$$\mu\left\{\min_{0\le t\le T}\mathcal{E}_w(\Phi_t\eta)\le\varepsilon\right\}\le C(1+T/\varepsilon)\exp\bigl(-cY_{\sigma,\kappa}(\varepsilon)\bigr).$$

For $T=\exp(c_1Y_{\sigma,\kappa}(\varepsilon))$ with c_1 small, this probability is less than one. Some starting phase therefore avoids the target for that long, proving the lower covering bound. Taking two logarithms proves (94). For completeness, the proved uniform covering bound makes every orbit dense. The bad segment just constructed has a strict energy margin on its compact time interval. Continuity and the flow isometry preserve it in an open neighborhood of its starting phase. Every orbit visits that neighborhood, proving the same lower bound for each orbit's maximal avoidance interval; the upper bound follows from the uniform covering estimate. □

Remark 9.3. The deterministic prime-flow estimate paid the arithmetic cost $\sum m_p\log p$. Here the almost-sure separation estimate pays the logarithm of the number of Fourier modes, of order $x/\log x$. This is why the bounds now match and why there is no restriction $\sigma<3$. It is not a proof of the corresponding improvement for the unperturbed prime logarithms.

10 Resonance obstructions and topologically typical slowness

Probability-one behavior and residual behavior in the same frequency box are very different. The obstruction is already visible in one integer character of a two-coordinate torus.

Lemma 10.1 (Periodic exclusion intervals from one character)**.** *Let $w_p>0$ be summable, let $k\neq 0$ be a finite integer vector, and put*

$$Q_w(k)^2=\sum_p\frac{k_p^2}{w_p},\qquad v=k\cdot\nu,\qquad \chi_k(\eta)=\prod_p\eta_p^{k_p}.$$

Suppose $v\neq 0$ and $\varepsilon Q_w(k)^2\le 1/8$. Every orbit has closed intervals of length $\pi/|v|$ on which $\mathcal{E}_w>\varepsilon$. These intervals repeat with period $2\pi/|v|$. In particular,

$$\mathcal{R}_{w,\nu}(\varepsilon)\ge\frac{\pi}{|k\cdot\nu|}. \tag{99}$$

Proof. Choose principal arguments $\theta_p\in[-\pi,\pi]$. Since $\theta_p^2\le(\pi^2/2)(1-\cos\theta_p)$, Cauchy–Schwarz gives

$$\left|\sum_p k_p\theta_p\right|\le\frac{\pi}{\sqrt{2}}\sqrt{\mathcal{E}_w(\eta)}\,Q_w(k).$$

Hence $\mathcal{E}_w\le\varepsilon$ forces $\chi_k(\eta)$ into the closed arc with argument between $-\pi/4$ and $\pi/4$. Along an orbit this character rotates with angular speed v: $\chi_k(\Phi_t\eta)=\chi_k(\eta)e^{-itv}$. Its passage through the opposite closed half-circle takes time $\pi/|v|$ and excludes the target throughout. Choosing a starting phase at the beginning of that passage proves (99). □

Theorem 10.2 (Residual slowness beyond any prescribed rate)**.** *Let $\Psi : (0,1) \longrightarrow [e,\infty)$ be any prescribed finite-valued function. There is a dense G_δ subset $\mathcal{B}_{\mathrm{slow}}(\Psi)$ of $\mathcal{B}$, consisting of rationally independent frequency vectors, such that for every λ in this set and every positive summable weight sequence,*

$$\limsup_{\varepsilon\downarrow 0} \frac{\log \mathcal{R}_{w,\nu}(\varepsilon)}{\log \Psi(\varepsilon)} = \infty. \tag{100}$$

The same assertion holds when covering time is replaced by the supremum of energy-avoidance interval lengths along any one orbit. Both assertions hold for both the full and relative frequency models. Each covering time at a fixed positive tolerance is nevertheless finite.

Proof. First fix one of the two models and two of its coordinates: use ν_2, ν_3 in the full model and ν_3, ν_5 in the relative model. Denote them by α, γ. For integers $q \geq 1$ and $p \in \mathbb{Z}$, put

$$N_{p,q} = p^2 + q^2, \qquad e_{p,q} = N_{p,q}^{-2}.$$

For each integer $n \geq 2$, let U_n be the union, over $N_{p,q} \geq n$, of the relatively open slabs

$$|q\alpha - p\gamma| < \frac{\pi}{\Psi(e_{p,q})^n}. \tag{101}$$

Every U_n is dense in $\mathcal{B}$. Indeed, every nonempty basic product neighborhood contains nondegenerate interior intervals for the two chosen coordinates. In the relative model first hold λ_2 fixed at an interior value and vary λ_3, λ_5. This again supplies a rectangle of relative frequencies. Choose a smaller rectangle on which $\gamma \neq 0$. Rational slopes p/q with arbitrarily large denominators are dense, so a line $q\alpha = p\gamma$ crosses the rectangle. Its open slab in (101) intersects the neighborhood, however small its width. This argument also covers neighborhoods touching endpoints of the original closed intervals.

The compact metrizable product $\mathcal{B}$ is a Baire space. Thus $\bigcap_{n\geq 2} U_n$ is dense G_δ. Intersect the sets obtained for the two models and remove every nontrivial finite integer resonance hyperplane for the full frequencies. The complements of these countably many hyperplanes are open dense. The resulting set is the required $\mathcal{B}_{\mathrm{slow}}(\Psi)$; full rational independence implies relative rational independence as well.

For a frequency vector in this set, choose (p_n, q_n) witnessing membership in U_n. Then $N_n \geq n \to \infty$, $e_n = N_n^{-2} \to 0$, and $v_n = q_n\alpha - p_n\gamma \neq 0$. For any fixed positive weights on the two chosen coordinates,

$$e_n Q_w(k_n)^2 \leq \frac{\max\{w_\alpha^{-1}, w_\gamma^{-1}\}}{N_n} \longrightarrow 0,$$

where $k_n = (q_n, -p_n)$ on those coordinates. Here the subscripts α, γ refer to the chosen coordinate indices. Lemma 10.1 therefore applies eventually and gives

$$\mathcal{R}_{w,\nu}(e_n) \geq \frac{\pi}{|v_n|} > \Psi(e_n)^n.$$

It supplies intervals of the same length for every orbit. These facts prove the limsup assertions. The frequency set was chosen without reference to the weights; only the initial index from which the dual condition holds depends on them.

Finally, rational independence makes every finite-frequency minimum strictly positive. At any fixed tolerance choose a finite truncation with sufficiently small tail, then choose sufficiently large kernel degrees. Lemma 1.2 yields a finite covering time. Thus the theorem describes arbitrarily slow dense motion, not a loss of density. □

For each prescribed Ψ the residual set can be chosen separately; the theorem does not assert one set that outgrows every conceivable function. Countably many prescribed rates can be imposed at once. The underlying Liouville mechanism is classical in almost-periodic dynamics. What is contrasted here is its residual strength with the matching infinite-dimensional probability-one scale in Theorem 9.1.

11 Optimal positive kernels and the limitation of a height estimate

The almost-sure small-divisor estimate removes a restriction that cannot be removed merely by choosing a better positive localization kernel. We now make that distinction precise. The optimization below concerns a Fourier *certificate cost*; it does not identify the covering time of the exact prime flow when the weights decay faster than p^{-3}.

Throughout this section primes include 2, and $w_p > 0$ with $\sum_p w_p < \infty$. For a finitely supported vector of nonnegative integers $D = (D_p)$ put

$$B(D) = \sum_p D_p \log p, \qquad c(D_p) = 1 - \cos \frac{\pi}{D_p + 2}, \qquad M_w(D) = \sum_p w_p c(D_p).$$

In particular, $c(0) = 1$: a coordinate on which a kernel does not depend has a uniform marginal, and contributes its full mean energy w_p. Let $\mathcal{K}(D)$ be the nonnegative trigonometric polynomials of Haar mean one whose degree in coordinate p is at most D_p. Define

$$\Psi_w(B) = \inf_{B(D) \le B} M_w(D). \tag{102}$$

The infimum can be taken over a finite set for each fixed B: a positive D_p requires $p \le e^B$, and every D_p is bounded by $B/\log 2$.

For clarity, $B(D)$ is a *rectangular* Fourier budget. If D_p are the coordinate degrees of a polynomial K, its actual support height is

$$H(K) = \max_{k \in \operatorname{supp} \widehat{K}} \sum_p |k_p| \log p \le B(D).$$

Equality need not hold for an arbitrary multivariate polynomial. It does hold for the product extremizers constructed below. We will not identify these two costs in a lower bound.

11.1 Exact localization at prescribed coordinate degrees

Proposition 11.1 (Rectangular localization optimum)**.** *For every finitely supported bandwidth vector D,*

$$\inf_{K \in \mathcal{K}(D)} \int \mathcal{E}_w K \, d\mu = \sum_p w_p \left(1 - \cos \frac{\pi}{D_p + 2}\right) = M_w(D). \tag{103}$$

A product of one-variable extremizers attains the infimum. Each factor has nonnegative Fourier coefficients and Fourier coefficient norm at most $D_p + 1$.

Proof. The one-variable coefficient inequality used here is classical Fejér extremality; see [12]. We include its short proof. A nonnegative trigonometric polynomial J of degree at most d and mean one has, by Fejér–Riesz factorization, a representation

$$J(\theta) = \left| \sum_{j=0}^{d} a_j e^{ij\theta} \right|^2, \qquad \sum_{j=0}^{d} |a_j|^2 = 1.$$

Consequently

$$\int \cos\theta\, J(\theta)\, d\theta = \operatorname{Re} \sum_{j=0}^{d-1} a_{j+1}\overline{a_j} \le \cos\frac{\pi}{d+2}.$$

The last inequality is the Rayleigh bound for the $(d+1)$-vertex path matrix with entries $1/2$ immediately above and below the diagonal. Its largest eigenvalue is $\cos(\pi/(d+2))$, with normalized eigenvector

$$a_j = \sqrt{\frac{2}{d+2}} \sin\frac{(j+1)\pi}{d+2}, \qquad 0 \le j \le d.$$

These positive coefficients give an extremizer J_d. All Fourier coefficients of J_d are nonnegative, and Cauchy–Schwarz gives

$$\|J_d\|_A = J_d(0) = \left(\sum_{j=0}^{d} a_j\right)^2 \le d+1.$$

For $d = 0$, take $J_0 = 1$.

For a multivariate K, integrate out every coordinate except p. The resulting marginal is nonnegative, has mean one, and has degree at most D_p. Applying the preceding inequality to each marginal and summing the nonnegative energy terms proves the lower bound in (103). The product $\prod_{D_p>0} J_{D_p}$ attains every marginal bound simultaneously. □

Thus allowing correlations among the coordinates cannot improve this mean-energy objective when the rectangular degree bounds are fixed. This observation concerns arbitrary nonnegative kernels, not just tensor products or squared Fejér kernels.

Lemma 11.2 (Budget inequality)**.** *For a finite set of primes Q set*

$$L_Q = \sum_{p\in Q} w_p^{1/3} (\log p)^{2/3}, \qquad \vartheta(Q) = \sum_{p\in Q} \log p.$$

If $B(D) \le B$, then

$$M_w(D) \ge \frac{2L_Q^3}{(B + 2\vartheta(Q))^2}. \tag{104}$$

Proof. For $d \ge 0$, elementary cosine bounds give

$$\frac{2}{(d+2)^2} \le c(d) \le \frac{\pi^2}{2(d+2)^2}. \tag{105}$$

Hölder's inequality, applied to

$$w_p^{1/3}(\log p)^{2/3} = \left(\frac{w_p}{(D_p+2)^2}\right)^{1/3} \big((D_p+2)\log p\big)^{2/3},$$

shows that

$$L_Q^3 \le \left(\sum_{p\in Q} \frac{w_p}{(D_p+2)^2}\right) \left(\sum_{p\in Q} (D_p+2)\log p\right)^2.$$

Now use (105) and the budget constraint. □

11.2 The optimal prime budget and its transition

Theorem 11.3 (Sharp rectangular cost)**.** *Suppose $w_p \asymp p^{-\sigma}$ for a fixed $\sigma > 1$. As $B \to \infty$,*

$$\Psi_w(B) \asymp \begin{cases} B^{1-\sigma}/\log B, & 1 < \sigma < 3, \\ B^{-2}(\log B)^2, & \sigma = 3, \\ B^{-2}, & \sigma > 3. \end{cases} \tag{106}$$

More generally, if

$$L = \sum_p w_p^{1/3} (\log p)^{2/3} < \infty, \tag{107}$$

then the exact limit is

$$\lim_{B\to\infty} B^2 \Psi_w(B) = \frac{\pi^2}{2} L^3. \tag{108}$$

In particular, (108) applies to every $\sigma > 3$.

Proof. First consider the lower estimates in (106). For $Q = \{p : p \le B\}$, prime partial summation gives $\vartheta(Q) = O(B)$ and

$$L_Q \asymp \begin{cases} B^{1-\sigma/3}(\log B)^{-1/3}, & 1 < \sigma < 3, \\ (\log B)^{2/3}, & \sigma = 3. \end{cases}$$

Lemma 11.2 gives the first two lower bounds. For $\sigma > 3$, its application to $Q = \{2\}$ already gives $\Psi_w(B) \gg B^{-2}$.

For the upper bound when $1 < \sigma < 3$, choose $x = cB$ with a sufficiently small fixed $c > 0$, and set

$$D_p = \left\lceil (x/p)^{\sigma/3} \right\rceil \quad (p \le x), \qquad D_p = 0 \quad (p > x).$$

Partial summation shows

$$B(D) \ll x^{\sigma/3} \sum_{p\le x} p^{-\sigma/3} \log p + \vartheta(x) \ll x.$$

Thus $B(D) \le B$ after fixing c. Also,

$$M_w(D) \ll x^{-2\sigma/3} \sum_{p\le x} p^{-\sigma/3} + \sum_{p>x} p^{-\sigma} \ll \frac{x^{1-\sigma}}{\log x},$$

which gives the required upper estimate.

At $\sigma = 3$, put

$$A = \frac{B}{C(\log B)^{2/3}}, \qquad D_p = \left\lfloor \frac{A}{p(\log p)^{1/3}} \right\rfloor,$$

where C is sufficiently large. The nonzero degrees occur for $p \le x$, where $x(\log x)^{1/3} = A$; hence $x \asymp B/\log B$. The budget satisfies

$$B(D) \le A \sum_{p\le x} \frac{(\log p)^{2/3}}{p} \ll A(\log x)^{2/3} \le B.$$

Since $D_p + 1 \geq A/[p(\log p)^{1/3}]$ on the active coordinates,

$$M_w(D) \ll A^{-2} \sum_{p \leq x} \frac{(\log p)^{2/3}}{p} + \sum_{p > x} p^{-3}$$
$$\ll A^{-2} (\log x)^{2/3} + \frac{x^{-2}}{\log x} \ll \frac{(\log B)^2}{B^2}.$$

The inactive tail is only $O(B^{-2} \log B)$.

It remains to prove (108), which also supplies the last upper bound. For its lower bound fix a finite set Q. Along any sequence with $B^2 M_w(D)$ bounded, every D_p with $p \in Q$ tends to infinity. Otherwise a fixed positive summand of $M_w(D)$ would remain. Thus, uniformly on this fixed set,

$$c(D_p) = \frac{\pi^2/2 + o(1)}{(D_p + 2)^2}.$$

The Hölder argument of Lemma 11.2 gives

$$\liminf_{B \to \infty} B^2 M_w(D) \geq \frac{\pi^2}{2} L_Q^3.$$

Sequences with unbounded scaled energy do not weaken the lower bound. Letting Q increase to all primes proves the desired lower limit.

For the matching construction set

$$t_p = w_p^{1/3} (\log p)^{-1/3}, \qquad D_p = \left\lfloor \frac{B t_p}{L} \right\rfloor.$$

Since $t_p \to 0$, this vector has finite support, and $B(D) \leq B$. For each fixed prime,

$$B^2 w_p c(D_p) \longrightarrow \frac{\pi^2}{2} L^2 w_p^{1/3} (\log p)^{2/3}.$$

Furthermore, $D_p + 1 \geq B t_p / L$ and (105) imply

$$B^2 w_p c(D_p) \leq C L^2 w_p^{1/3} (\log p)^{2/3}.$$

The right side is summable by (107). Dominated convergence therefore gives the upper limit $(\pi^2/2) L^3$. □

11.3 Consequences for the unperturbed prime flow

The exact localization optimum can be inserted into the same elementary arithmetic averaging argument. Here $\mathcal{R}_{w,\log p}$ denotes the covering time for frequencies $\lambda_p = \log p$.

Corollary 11.4 (Endpoint and faster-decay upper bounds)**.** *For $w_p \asymp p^{-\sigma}$, $\sigma > 1$, put $\ell = \log(1/\varepsilon)$. For all sufficiently small ε,*

$$\log \mathcal{R}_{w,\log p}(\varepsilon) \ll \begin{cases} \varepsilon^{-1/(\sigma-1)} \ell^{-1/(\sigma-1)}, & 1 < \sigma < 3, \\ \varepsilon^{-1/2} \ell, & \sigma = 3, \\ \varepsilon^{-1/2}, & \sigma > 3. \end{cases} \tag{109}$$

In particular, at $\sigma = 3$,

$$e^{c\varepsilon^{-1/2} \ell^{-3/2}} \leq \mathcal{R}_{w,\log p}(\varepsilon) \leq e^{C\varepsilon^{-1/2} \ell}, \qquad \lim_{\varepsilon \downarrow 0} \frac{\log \log \mathcal{R}_{w,\log p}(\varepsilon)}{\ell} = \frac{1}{2}. \tag{110}$$

The same conclusions hold for the relative frequencies $\log(p/2)$, with the prime 2 omitted.

Proof. Choose D with $M_w(D) \le \varepsilon/4$ and let $Q = \{p : D_p > 0\}$. For small ε, this set is nonempty. Use the product extremizer K from Proposition 11.1. Its truncated energy mean is at most $\varepsilon/4$, while

$$\sum_{p \notin Q} w_p \le M_w(D) \le \varepsilon/4.$$

The omitted coordinates therefore contribute at most $\varepsilon/2$ pointwise. The certificate polynomial

$$P = (\varepsilon/2 - \mathcal{E}_Q)K$$

has mean at least $\varepsilon/4$ and coordinate degrees at most $D_p + 1$. Its arithmetic degree budget is at most

$$\sum_{p \in Q} (D_p + 1) \log p \le 2B(D),$$

since each active $D_p \ge 1$. Moreover

$$\log \|K\|_A \le \sum_{p \in Q} \log(D_p + 1) \le \sum_p D_p \le B(D)/\log 2.$$

A nonzero frequency $\sum k_p \log p$ with these coordinate bounds is the logarithm of a nonunit positive rational whose numerator and denominator are at most $\exp(2B(D))$. Its absolute value is therefore at least $\exp(-2B(D))$. The averaging certificate of Lemma 1.2, or direct integration of each nonconstant Fourier term, now gives a positive average of P on every interval of length $\exp(CB(D))$. Here $\log(1/\varepsilon)$ is absorbed into $B(D)$; indeed its single-prime localization bound already forces $B(D) \gg \varepsilon^{-1/2}$. Positivity of P gives $\mathcal{E}_Q < \varepsilon/2$, and hence $\mathcal{E}_w < \varepsilon$.

Inverting the three orders in Theorem 11.3 proves (109). The lower bound in (110) is the small-ball avoidance consequence of Theorem 2.1 at $\sigma = 3$; taking two logarithms gives the limit. For relative frequencies, clearing the additional power of 2 changes the elementary frequency estimate by only a fixed factor in its exponent. Deleting one coordinate changes none of the orders. □

The local analytic transfer of phase energy to an edge zero needs only $S > 1$. Consequently, if an unperturbed Dirichlet edge occurs at $S = 3$, (110) supplies the recurrence endpoint, with its corresponding zero-gap exponent $1/2$.

11.4 What is, and is not, obstructed

For the linear energy certificate $(a - \mathcal{E}_Q)K$, Theorem 11.3 shows that a better positive polynomial cannot reduce the *rectangular localization budget* below the stated orders. In particular, for $\sigma > 3$, a proof that pays $\exp(CB(D))$ through the elementary arithmetic height estimate cannot reach the entropy exponent $1/(\sigma - 1) < 1/2$ merely by changing its positive kernel. At $\sigma = 3$, the factor $\log(1/\varepsilon)$ in the optimal rectangular budget is also unavoidable within that optimization.

There is a weaker obstruction using actual joint support height, which requires no rectangular comparison. If $K \ge 0$, $\int K = 1$, and $\int \mathcal{E}_w K \le \varepsilon$, its prime-2 marginal must have degree $d \gg \varepsilon^{-1/2}$ by the univariate extremum. A nonzero top coefficient of that marginal is a pure Fourier mode $k = de_2$ of K. Thus

$$H(K) \ge d \log 2 \gg \varepsilon^{-1/2}. \tag{111}$$

Accordingly, using only a worst-case separation estimate of the form $\exp(-CH(K))$ retains this power barrier even without replacing the actual support by a rectangle. Equation (111) does *not* establish the stronger logarithmic factor at $\sigma = 3$ for arbitrary joint supports.

None of these statements is a dynamical lower bound of exponent $1/2$ when $\sigma > 3$. The true lower bound for the exact prime flow remains the entropy bound with exponent $1/(\sigma - 1)$. A better arithmetic estimate, a more selective Fourier averaging argument, or another recurrence mechanism could improve the upper bound. The generic frequency theorem does precisely the first of these: it controls the small divisors by the number of modes, rather than by their elementary rational height.

12 Scope and further questions

The sharp locking theorem compares two different boundary defects of one Euler product. Its two constructions separate the obstructions: changing a distant prime tail controls the optimal forward horizontal modulus, whereas changing one fixed prime controls the optimal ordinate exponents and the reverse horizontal rate. The ordinary zeta function inherits both obstructions through its vertical approximation of the phase hull.

The remaining recurrence questions are arithmetic. For exact prime frequencies and $1 < \sigma < 3$, the established bounds are

$$cY_{\sigma,\kappa}(\varepsilon) \le \log \mathcal{R}_{w,\log p}(\varepsilon) \le CX_{\sigma,\kappa}(\varepsilon).$$

The ratio of these two scales is $\log(1/\varepsilon)$. Closing this gap would sharpen the critical-point gap theorem beyond its present double-logarithmic exponent. The almost-sure perturbation theorem shows that the weighted geometry does not require that loss; it gives no improved small-divisor estimate at the exact prime vector.

For $w_p \asymp p^{-\sigma}$ with $\sigma > 3$, the present unperturbed bounds only imply

$$\frac{1}{\sigma - 1} \le \liminf_{\varepsilon \downarrow 0} \frac{\log\log \mathcal{R}_{w,\log p}(\varepsilon)}{\log(1/\varepsilon)} \le \limsup_{\varepsilon \downarrow 0} \frac{\log\log \mathcal{R}_{w,\log p}(\varepsilon)}{\log(1/\varepsilon)} \le \frac{1}{2}.$$

The exact kernel optimization explains why the specified arithmetic certificate stops at this upper exponent. Determining the actual exponent requires an improvement in the arithmetic analysis or a different recurrence argument. At $\sigma = 3$ the two power exponents coincide, although logarithmic factors remain separated.

Finally, maximal gaps and first visits from one prescribed origin are different quantities. The long-hole construction locates large gaps on every deterministic orbit, but it does not force such a gap to begin at ordinate zero. The almost-sure first-waiting result therefore does not establish a lower waiting exponent for the untwisted zeta function at a fixed origin. This specific problem remains open in the present work.

Part II
Phase-localized zeta zeros and Euler-edge laws

The recurrence theory of the preceding part concerns the prime flow at arbitrary heights. We now restrict those heights to ordinates of simple critical-line zeros. The first step transports a Weil-form counting certificate to fixed phase regions; the second makes that transport uniform for shrinking boxes and for a growing prime set. The resulting phase measures support both quantitative Euler-edge companions and conditional laws for their shape. The final sections treat inverse zero detection and the separate matrix constraints on higher-moment amplification.

13 Fixed phase regions and the counting constant

Write $N(T, 2T)$ for the number of nontrivial zeta zeros with $T < \operatorname{Im}\rho \le 2T$, counted with multiplicity. For a finite prime set S, let $F_S(t) = (p^{-it})_{p\in S}$ and let μ_S be normalized Haar measure on $\mathbb{T}^S$. Put

$$c_{\mathrm{MT}} = \frac{3}{2} - \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}}.$$

Theorem 13.1 (Prime-phase localization)**.** *For every fixed finite S and every Haar-Jordan-measurable $B \subset \mathbb{T}^S$,*

$$\liminf_{T\to\infty} \frac{\#\{\gamma \in (T, 2T] : \zeta(\frac{1}{2} + i\gamma) = 0,\ \zeta'(\frac{1}{2} + i\gamma) \neq 0,\ F_S(\gamma) \in B\}}{N(T, 2T)} \ge c_{\mathrm{MT}}\mu_S(B).$$

No Riemann hypothesis or fixed zero-free half-plane is assumed. The set S and the region B are fixed before T tends to infinity.

The assertion is a lower domination statement for the simple critical-line submeasure. It neither identifies its limiting density nor asserts equidistribution of all simple zeros. Classical all-zero phase distribution is discussed in [19, 18, 24]; the subsequence equidistribution results of Çiçek–Gonek [20] use RH and a spacing hypothesis. The rank–inertia viewpoint and the unconditioned Montgomery–Taylor certificate are from Alpöge–Furman [14]. Lamzouri [15] gives another proof, and Knausgård [16] reports a larger global constant. We use the specified second-moment constant throughout; the present localization does not assert a new global percentage.

The fixed-region result is proved first without a zero-free half-plane. Its shrinking-region extensions are established in Sections 18–19. The simultaneous growing-prime limit in Section 21 is distinct from the iterated height-then-tolerance limits used for the Gibbs laws. The H_ϑ hypothesis occurs only in the labeled half-plane transfer of Section 26; the Gaussian inverse and Vinogradov–Korobov arguments do not use it.

14 The rank–inertia counting inequality

For a Hermitian matrix H, let $n_+(H)$ count its strictly positive eigenvalues. The following elementary version will be used for localized zero counts.

Lemma 14.1. *Suppose $G = P+Q$, where $P \succeq 0$, $\operatorname{rank} P \le s$, $\operatorname{tr} P \le s$, $n_+(Q) \le b$, and $s+2b \le M$. Then*

$$s \ge 4\operatorname{tr} G - \operatorname{tr} G^2 - 2M.$$

Additive $o(N)$ errors in the count and trace constraints change the conclusion by only $o(N)$.

Proof. Let $\lambda_1 \ge \cdots \ge \lambda_r > 0$ be the positive eigenvalues of G. Since $G \preceq P + Q_+$ and $\operatorname{rank} Q_+ \le b$, min–max gives $\lambda_{b+j}(G) \le \lambda_j(P)$. Thus the tail after the first b positive eigenvalues has at most s terms and sum at most s. For $x \le 0$, $4x - x^2 \le 0$; for all x, $4x - x^2 \le 4$; and $4x - x^2 \le 2x + 1$. Consequently

$$4 \operatorname{tr} G - \operatorname{tr} G^2 \le 4b + 3s \le 2M + s.$$

The same calculation tracks additive errors. □

15 Prime-phase weights in the Weil second moment

Fix a finite set $\mathcal{P}$ of primes and write

$$\Phi_{\mathcal{P}}(t) = (p^{-it})_{p \in \mathcal{P}}, \qquad \mu_{\mathcal{P}} = \text{Haar probability measure on } \mathbb{T}^{\mathcal{P}}.$$

The arithmetic observation behind our localization result is that a nontrivial rational twist of the prime–prime term has no bulk diagonal. Indeed, if two prime powers have a ratio supported on $\mathcal{P}$, then either their ratio is one or both prime bases belong to $\mathcal{P}$. We give the details, including the passage from Fourier weights to arbitrary continuous phase weights.

Choose a nonzero, nonnegative, even function $f \in C_c^\infty((-1/2, 1/2))$ and put

$$\psi = f^2, \quad a = \int_{\mathbb{R}} \psi(u)\, du, \quad L = \log T - D \log\log T, \quad X = e^L, \quad \phi(u) = f(u/L), \tag{112}$$

where $D > 4$ is fixed. Our Fourier convention in this section is $\widehat{h}(t) = \int_{\mathbb{R}} h(u) e^{-itu}\, du$. Let $I = [T, 2T]$, $\alpha_k = T + 2\pi k/L$, and retain the integers k for which $\alpha_k \in I$. Define the real symmetric Weil compression

$$G_{k\ell} = \frac{1}{aL^2} \int_{\mathbb{R}} \widehat{\phi}(t - \alpha_k) \widehat{\phi}(t - \alpha_\ell) \nu_X(t)\, dt. \tag{113}$$

Here the explicit-formula density is

$$\begin{aligned} \nu_X(t) &= \mu(t) + \Pi_X(t) + P_X(t), \\ \mu(t) &= \frac{1}{2\pi} \operatorname{Re} \frac{\Gamma'}{\Gamma}\left(\frac{1}{4} + \frac{it}{2}\right) - \frac{\log \pi}{2\pi}, \\ \Pi_X(t) &= \frac{1}{\pi} \operatorname{Re} \frac{X^{1/2+it}}{1/2 + it}, \qquad P_X(t) = -\frac{1}{\pi} \sum_{n \le X} \frac{\Lambda(n)}{\sqrt{n}} \cos(t \log n). \end{aligned}$$

The compact support in (112) makes (113) an exact compression of the zero-side Weil form. We use its standard explicit-formula interpretation; the weighted estimates below are proved directly on the real side. Set

$$R_\psi = \frac{\int \psi(u)^2\, du + \iint |u - v| \psi(u)\psi(v)\, du\, dv}{a^2}, \qquad N_I = N(2T) - N(T). \tag{114}$$

Proposition 15.1 (Phase-weighted first and second moments)**.** *For every fixed continuous real function $K \in C(\mathbb{T}^{\mathcal{P}})$, put $k(t) = K(\Phi_{\mathcal{P}}(t))$ and $\overline{K} = \int K\, d\mu_{\mathcal{P}}$. Then*

$$\sum_k k(\alpha_k) G_{kk} = (\overline{K} + o(1)) N_I, \qquad \sum_k k(\alpha_k) \sum_\ell G_{k\ell}^2 = (R_\psi \overline{K} + o(1)) N_I. \tag{115}$$

In particular, if $0 \le K \le 1$ and $D_K = \operatorname{diag}(\sqrt{k(\alpha_k)})$, then

$$\operatorname{tr}(D_K G D_K) = (\overline{K} + o(1)) N_I, \qquad \|D_K G D_K\|_{\mathrm{HS}}^2 \le (R_\psi \overline{K} + o(1)) N_I. \tag{116}$$

All limits are taken with $\mathcal{P}, K, f, D$ fixed.

Proof. We separate the boundary estimates from the arithmetic computation. Put

$$u_k(t) = \frac{\widehat{\phi}(t-\alpha_k)}{\sqrt{a}L} \quad (k \in \mathbb{Z}),$$

and let $U : \ell^2(\mathbb{Z}) \longrightarrow L^2(\mathbb{R})$ be their synthesis operator. Parseval's formula and the Fourier-series Bessel inequality give

$$\|U\|^2 \ll_f L^{-1}. \tag{117}$$

Moreover, the Schwartz bounds for $\widehat{f}$ imply, for each fixed $A > 0$,

$$\sum_{\alpha_k \in I} |u_k(t)|^2 \ll_{A,f} (1 + L\,\mathrm{dist}(t, I))^{-A} \qquad (t \notin I), \tag{118}$$

$$\sum_{\alpha_k \notin I} |u_k(t)|^2 \ll_{A,f} (1 + L\,\mathrm{dist}(t, \mathbb{R} \setminus I))^{-A} \qquad (t \in I). \tag{119}$$

The integrals of these tails are $O_f(L^{-1})$. On $|t| \le 4T$ we have $|\nu_X(t)| \ll L + \sqrt{X}$; outside this range its growth is at most logarithmic in $|t|$, apart from the bounded prime polynomial. The remote tails are therefore harmless. If V is any subfamily of U and J a measurable set, the elementary operator estimate

$$\|V^* M_{\nu_X 1_J} V\|_{\mathrm{HS}}^2 \le \|V\|^2 \int_J |\nu_X(t)|^2 \sum_k |v_k(t)|^2\, dt$$

shows that restricting the integral defining G to I changes G by a matrix of squared Hilbert–Schmidt norm $O_f((L^2+X)/L^2) = o(N_I)$. Likewise, for the infinite matrix $H = U^* M_{\nu_X 1_I} U$, the squared Hilbert–Schmidt mass in rows or columns whose centers lie outside I is $O_f((L^2+X)/L^2)$. Its full squared norm is $O_f(N_I)$ independently: the exact Poisson identity gives $\sum_k |u_k(t)|^2 = 1$, while the standard Dirichlet-polynomial mean-value bound gives $\int_I |\nu_X(t)|^2 dt \ll TL^2$; applying the preceding operator estimate and (117) yields $\|H\|_{\mathrm{HS}}^2 \ll TL$. Cauchy–Schwarz thus makes every resulting cross-error $o(N_I)$. The linear trace boundary error is $O_f((L+\sqrt{X})/L) = o(N_I)$ by the same tail bounds. We may consequently compute with all centers and the real integral restricted to I.

First take $k(t) = q^{it}$, where q is a fixed positive rational supported on $\mathcal{P}$, and put $\lambda = \log q$. Poisson summation gives the exact identity

$$\sum_{k \in \mathbb{Z}} e^{i\lambda\alpha_k} \widehat{\phi}(t-\alpha_k)\widehat{\phi}(t'-\alpha_k) = L e^{i\lambda(t+t')/2} \Psi_\lambda(t-t'), \tag{120}$$

where

$$\Psi_\lambda(s) = \int_{\mathbb{R}} \phi(u-\lambda/2)\phi(u+\lambda/2) e^{-isu}\, du.$$

There are no nonzero aliases: $\mathrm{supp}\, f$ is a compact subset of $(-1/2, 1/2)$ and λ is fixed. Write $\Phi = \Psi_0 = \widehat{\phi^2}$ and $J_\lambda = \Psi_\lambda \Phi$. The weighted row energy is exactly

$$\frac{1}{a^2L^2} \iint_{I^2} e^{i\lambda(t+t')/2} J_\lambda(t-t') \nu_X(t) \nu_X(t')\, dt\, dt'. \tag{121}$$

Uniformly for fixed λ,

$$\int |J_\lambda(s)|\, ds \ll L, \qquad \int |s J_\lambda(s)|\, ds \ll 1, \qquad \sup_v \left| \int J_\lambda(s) e^{ivs}\, ds \right| \ll L. \tag{122}$$

These follow by scaling and the Schwartz bounds for f.

Let $\ell = \log T$. Uniformly on I,

$$\nu_X(t) = \frac{\ell}{2\pi} + P_X(t) + O(1 + \sqrt{X}/T).$$

The standard Dirichlet-polynomial mean-value inequality gives $\int_I |P_X(t)|^2 dt \ll TL^2 + XL^2$. Consequently replacing ν_X by $\ell/(2\pi) + P_X$ in (121) costs $O(TL^2) = o(TL^3)$ before normalization. In midpoint and lag coordinates, the midpoint interval is I with $|s|/2$ removed at each end. By (122), replacing it by I costs $O(L^2 + X) = o(TL^3)$; the long-lag tail is smaller by Schwartz decay.

The constant–constant term is lower order if $q \neq 1$. At $q = 1$ it equals

$$\frac{T\ell^2}{2\pi} L \int \psi(u)^2 \, du$$

before normalization. For the prime–prime term, put $a_n = \Lambda(n)/\sqrt{n}$. The midpoint frequencies are

$$\log q + \varepsilon \log n + \eta \log m, \qquad \varepsilon, \eta \in \{-1, 1\}.$$

After coincident frequencies have been combined, the two frequency lists in each product have separation at least c_q/X. The Montgomery–Vaughan Hilbert inequality [21], applied to their union and polarized, bounds all unequal-frequency terms by

$$O_q\left(X \sum_{n \leq X} a_n^2\right) = O_q(XL^2),$$

uniformly in the lag s. The coefficient phases introduced by s have modulus one. Integration against J_λ therefore gives $O_q(XL^3) = o(TL^3)$. The separation assertion is elementary: after writing $q = b/c$, a nonzero difference $bn - cm$ is an integer of absolute value at least one; the same-sign cases are no worse.

If $q = 1$, the exact resonances are $n = m$ and $\varepsilon = -\eta$. Their contribution is

$$\frac{T}{\pi} \sum_{n \leq X} \frac{\Lambda(n)^2}{n} (\phi^2 * \phi^2)(\log n). \tag{123}$$

Partial summation, using $\sum_{n \leq y} \Lambda(n)^2/n = \frac{1}{2} \log^2 y + O(\log y)$, turns the sum in (123) into

$$L^3 \int_0^1 v(\psi * \psi)(v) \, dv + O_f(L^2).$$

If $q \neq 1$, write a resonant pair as $n = p^j, m = r^h$. Every prime base must belong to $\mathcal{P}$: a prime outside $\mathcal{P}$ can cancel only when $p = r$ and the signed exponents cancel, which would force $q = 1$. Hence

$$\sum_{\substack{q n^\varepsilon m^\eta = 1 \\ \Lambda(n)\Lambda(m) \neq 0}} \frac{\Lambda(n)\Lambda(m)}{\sqrt{nm}} \leq \left(\sum_{p \in \mathcal{P}} \sum_{j \geq 1} (\log p) p^{-j/2} \right)^2 \ll_{\mathcal{P}} 1. \tag{124}$$

Together with (122), this bounds all nontrivial rational resonances by $O_{\mathcal{P}}(TL)$.

The constant–prime terms can resonate only when q or q^{-1} is a prime power supported on $\mathcal{P}$. Their resonant part is $O_{\mathcal{P}}(TL^2)$. For the remaining terms, $|\log q \pm \log n| \geq \frac{1}{2} \log n$ once n exceeds a constant depending on q. Direct integration in the midpoint thus bounds them by $O_q(L^2(1 + \sqrt{X})) = o(TL^3)$.

It follows that (121) is $o(N_I)$ for $q \neq 1$. For $q = 1$ it is

$$\frac{N_I}{a^2}\left\{\frac{\ell}{L}\int \psi^2 + \frac{2L}{\ell}\int_0^1 v(\psi * \psi)(v)\,dv\right\} + o(N_I) = (R_\psi + o(1))N_I.$$

Here $L/\ell \to 1$, and the evenness of ψ identifies the convolution expression with that in (114). This sharpens the rough unweighted Hilbert–Schmidt bound used in the boundary reduction.

For the first moment, setting $t = t'$ in (120) yields

$$\sum_k q^{i\alpha_k} H_{kk} = \frac{b_\lambda}{aL}\int_I q^{it}\nu_X(t)\,dt, \qquad b_\lambda = \int \phi(u - \lambda/2)\phi(u + \lambda/2)\,du = aL + O_\lambda(L^{-1}).$$

The $q = 1$ main term is $N_I + o(N_I)$. For $q \neq 1$ the archimedean contribution is $O_q(L)$, any prime resonance contributes $O_q(T)$, and the remaining prime terms contribute $O_q(\sqrt{X})$. All are $o(N_I)$. We have proved both assertions for every fixed trigonometric polynomial on the prime torus.

Finally, approximate K uniformly by such a polynomial F with $\|K - F\|_\infty \le \delta$. The error in the weighted row energy is at most $\delta\|G\|_{\mathrm{HS}}^2 = O(\delta N_I)$, and the trace error is at most $\delta\sqrt{d}\,\|G\|_{\mathrm{HS}} = O(\delta N_I)$, where $d \sim N_I$ is the matrix dimension. First let $T \to \infty$ and then $\delta \to 0$. This proves (115). The last assertion follows directly from

$$\|D_K G D_K\|_{\mathrm{HS}}^2 = \sum_{k,\ell} k(\alpha_k)k(\alpha_\ell)G_{k\ell}^2 \le \sum_k k(\alpha_k)\sum_\ell G_{k\ell}^2.$$

□

Remark 15.2. The proposition takes the phase set and its approximating Fourier polynomial to be fixed before T tends to infinity. It supplies no uniform estimate for a phase ball shrinking with T. The reduction $X = T/(\log T)^D$ absorbs the elementary twisted mean-value error while leaving the limiting window functional unchanged.

16 Localization on the zero side

The arithmetic moment calculation must be combined with a localization argument before it gives a proportion of zeros in a phase set. A positive weight does not weight the positive index of a Hermitian form. We use the weight on the sampling coordinates, retain the relatively few zeros far from the critical line, and estimate the discarded near-line vectors in Hilbert–Schmidt norm.

Fix a finite nonempty set of primes $\mathcal{P}$, and write

$$\Phi_{\mathcal{P}}(t) = (p^{-it})_{p\in\mathcal{P}}, \qquad \mu_{\mathcal{P}} = \text{normalized Haar measure on } \mathbb{T}^{\mathcal{P}}.$$

A phase set below is an open subset B of this torus whose boundary has Haar measure zero. Let $N_B(T, 2T)$ count all nontrivial zeta zeros with ordinate in $(T, 2T]$ and phase in B, with multiplicity, and let $N_{0,B}^s(T, 2T)$ count the simple zeros on the critical line satisfying the same conditions. Put $N = N(T, 2T)$.

We use two classical facts about zero ordinates:

$$N(t, t+1) \ll \log(|t| + 3), \tag{125}$$

$$N(\sigma, T) \ll T^{1-(\sigma-1/2)/4}\log T, \qquad \tfrac{1}{2} \le \sigma \le 1. \tag{126}$$

Here $N(\sigma, T)$ counts zeros with $|\operatorname{Im}\rho| \le T$ and $\operatorname{Re}\rho \ge \sigma$, with multiplicity. The second is a uniform form of Selberg's density estimate; see [18, Lemma 2]. We also use the joint form of the Hlawka equidistribution theorem,

$$N_B(T, 2T) = \big(\mu_{\mathcal{P}}(B) + o(1)\big)N. \tag{127}$$

Indeed Weyl's criterion reduces this assertion to the uniform distribution of $\gamma \log r/(2\pi)$ for a fixed positive rational $r \ne 1$. Unique factorization ensures that each nontrivial torus character gives such an r. The one-dimensional assertion follows from Landau's formula and the classical density estimate; see [19]. Subtracting the asymptotic at T from that at $2T$ gives (127).

Here is the normalization of the finite compression needed in this section. Fix a real even function $f \in C_c^\infty((-1/2, 1/2))$, $f \not\equiv 0$, and let

$$L = \log T - D \log\log T, \quad D > 4, \quad \phi(u) = f(u/L), \quad a = \int_{\mathbb{R}} f(u)^2\, du,$$

$$\alpha_k = T + \frac{2\pi k}{L} \in [T, 2T].$$

For a zero $\rho = \beta + i\gamma$ put $z_\rho = \gamma + i(1/2 - \beta)$ and

$$u_\rho(k) = \frac{\widehat{\phi}(z_\rho - \alpha_k)}{\sqrt{a}\, L}. \tag{128}$$

By the explicit formula, G is the sum of $m_\rho u_\rho u_\rho^{\mathsf{T}}$ over all nontrivial zeros. The Fourier support of a product of two test transforms lies in $(-L, L)$. On that interval the transform of the pole density is $2\cosh(u/2)$, which accounts for both pole evaluations; the other terms are the prime and gamma terms of Weil's formula. Rapid decay and the unit-interval zero count make the zero sum convergent, as in [14]. Let G' be the sum restricted to the extended height window $I' = [T - \sqrt{T}, 2T + \sqrt{T}]$. Then $\|G - G'\|_1 = o(1)$, where $\|\cdot\|_1$ denotes trace norm. Indeed, repeated integration by parts bounds each normalized coordinate by $C_M T^{1/4}(1 + L|\gamma - \alpha_k|)^{-M} L^M$ for every zero. Sum its square over the centers and over zeros at distance at least $\sqrt{T}$ from the center interval, using the unit-interval count. The result is $O_M(NT^{1-M}\log T) = o(1)$ for large fixed M. The trace norm of the sum of the rank-one bilinear terms is at most this sum of squared vector norms. Coordinate multiplication by a contraction preserves the estimate.

Choose $K \in C^\infty(\mathbb{T}^{\mathcal{P}})$ with $0 \le K \le 1$ and $\operatorname{supp} K \subset B$, and put

$$D_K = \operatorname{diag}\big(K(\Phi_{\mathcal{P}}(\alpha_k))^{1/2}\big), \qquad G_K = D_K G D_K, \quad F_{\text{out}} = D_K(G - G')D_K.$$

Only boundedness of D_K is needed on the zero side; the square root need not be smooth. The arithmetic input from Proposition 15.1 is

$$\operatorname{tr} G_K = \big(\mu_{\mathcal{P}}(K) + o(1)\big)N, \tag{129}$$

$$\|G_K\|_{\mathrm{HS}}^2 \le \big(R_\psi \mu_{\mathcal{P}}(K) + o(1)\big)N. \tag{130}$$

For the second inequality one uses $\|D_K G D_K\|_{\mathrm{HS}}^2 \le \operatorname{tr}(D_K^2 G^2)$, since G is Hermitian and $0 \le D_K^2 \le I$.

Lemma 16.1 (Discarding shallow zeros outside a phase neighborhood)**.** *Fix $\eta > 0$, and let B_η be the open η-neighborhood of B in the Euclidean metric inherited by the torus. Put*

$$d_0 = 16\frac{\log\log T}{\log T}.$$

Let E be the contribution to G_K from zeros in I' for which $|\beta - 1/2| \le d_0$ and $\Phi_{\mathcal{P}}(\gamma) \notin B_\eta$. Then

$$|\operatorname{tr} E| = o(N), \qquad \|E\|_{\mathrm{HS}} = o(\sqrt{N}).$$

The total multiplicity in I' of zeros with $|\beta - 1/2| > d_0$ is $o(N)$.

Proof. The last assertion follows from (126), reflection about the critical line, and $N \asymp T \log T$: the relevant count is $O(N(\log T)^{-4})$.

The map $\Phi_{\mathcal{P}}$ is Lipschitz with constant $C_{\mathcal{P}} = (\sum_{p\in\mathcal{P}} (\log p)^2)^{1/2}$. Consequently every discarded zero is separated from every sampling center with nonzero weight by at least $r_\eta = \eta / C_{\mathcal{P}} > 0$ in real height. Repeated integration by parts, for any fixed integer M, gives

$$|u_\rho(k)| \ll_M e^{L|\beta-1/2|/2} (1 + L|\beta - 1/2|)^M (1 + L|\gamma - \alpha_k|)^{-M}.$$

For $|\beta - 1/2| \le d_0$ and sufficiently large T this implies

$$|u_\rho(k)| \ll_M (\log T)^9 (1 + L|\gamma - \alpha_k|)^{-M}. \tag{131}$$

The factor $(\log\log T)^M$ has been absorbed in one power of $\log T$. Constants may depend on the fixed M.

Form a rectangular matrix U with columns $D_K u_\rho$ for the discarded zeros, repeating a column according to its multiplicity. The grid spacing and (125) give

$$\sup_k \sum_\rho |U_{k\rho}| \ll (\log T)^{10}, \qquad \sup_\rho \sum_k |U_{k\rho}| \ll (\log T)^9.$$

Schur's test yields $\|U\|^2 \ll (\log T)^{19}$. The separation $|\gamma - \alpha_k| \ge r_\eta$ additionally gives

$$\sum_k |U_{k\rho}|^2 \ll (\log T)^{18} (L r_\eta)^{1-2M}, \qquad \|U\|_{\mathrm{HS}}^2 \ll N (\log T)^{19-2M}.$$

Here and below the constants can depend on η. Since $E = UU^{\mathsf{T}}$, standard singular-value inequalities imply

$$|\operatorname{tr} E| \le \|U\|_{\mathrm{HS}}^2, \qquad \|E\|_{\mathrm{HS}}^2 \le \|U\|^2 \|U\|_{\mathrm{HS}}^2 \ll N (\log T)^{38-2M}.$$

Taking $M = 24$ proves both assertions. Conjugate zero pairs show that E is Hermitian, although its factorization uses a transpose rather than an adjoint. □

Theorem 16.2 (The localization implication)**.** *Suppose the weighted moment estimates* (129)–(130) *hold for every fixed smooth K with $0 \le K \le 1$. Then, for every phase set B as above,*

$$\liminf_{T\to\infty} \frac{N^s_{0,B}(T, 2T)}{N(T, 2T)} \ge (2 - R_\psi) \mu_{\mathcal{P}}(B). \tag{132}$$

In particular the same proportion constant furnished by the global second-moment certificate is valid after restriction to any fixed finite-prime phase set.

Proof. First fix K and $\eta > 0$, choosing η so that $\mu_{\mathcal{P}}(\partial B_\eta) = 0$. Such choices exist arbitrarily close to zero, since disjoint level sets of the distance function have positive measure for at most countably many levels. Retain all deep zeros $|\beta - 1/2| > d_0$ in I', independently of their phase, and all shallow zeros whose phase belongs to B_η. By Lemma 16.1 and $\|F_{\mathrm{out}}\|_1 = o(1)$, the exact retained matrix $H = G_K - F_{\mathrm{out}} - E$ satisfies

$$\operatorname{tr} H = \mu_{\mathcal{P}}(K) N + o(N), \qquad \|H\|_{\mathrm{HS}}^2 \le R_\psi \mu_{\mathcal{P}}(K) N + o(N).$$

Let P_1 be its contribution from simple on-line zeros, and put $Q = H - P_1$. Each such zero contributes a positive matrix of rank at most one. Its trace is at most one, by the full-grid sampling identity and $\|D_K\| \le 1$. Each distinct multiple on-line zero costs at most one positive direction, and each

off-line reflected pair likewise costs at most one. The retained deep zeros have total multiplicity $o(N)$. It follows that

$$\operatorname{rank} P_1 \le N^s_{0,B_\eta}(I'),$$
$$N^s_{0,B_\eta}(I') + 2n_+(Q) \le N_{B_\eta}(I') + o(N).$$

Together with $\operatorname{tr} P_1 \le N^s_{0,B_\eta}(I')$, Lemma 14.1 now gives

$$\begin{aligned} N^s_{0,B_\eta}(I') &\ge 4\operatorname{tr} H - 2N_{B_\eta}(I') - \|H\|^2_{\mathrm{HS}} - o(N) \\ &\ge \big((4 - R_\psi)\mu_{\mathcal{P}}(K) - 2\mu_{\mathcal{P}}(B_\eta)\big)N - o(N). \end{aligned}$$

We used (127); the two added height intervals contribute $O(\sqrt{T}\log T) = o(N)$ by (125).

To obtain the result for B itself, first choose an inner open phase set C with closure contained in B and Haar-null boundary. Choose K supported in C, with $\mu_{\mathcal{P}}(K)$ arbitrarily close to $\mu_{\mathcal{P}}(C)$, and let $\eta \downarrow 0$ through the admissible values, always keeping $C_\eta \subset B$. The preceding inequality gives a lower bound approaching $(2 - R_\psi)\mu_{\mathcal{P}}(C)$. Inner regularity and the Haar-null boundary of B allow $\mu_{\mathcal{P}}(C) \uparrow \mu_{\mathcal{P}}(B)$. All phase sets, weights, and buffers are fixed before T tends to infinity, so no unproved uniformity for shrinking targets is used. □

Remark 16.3. Neither a zero-free half-plane nor an independence assumption between phase conditions and critical-line zeros enters this implication. The small family of deep zeros is charged by its count rather than estimated by the norms of its evaluation vectors. This distinction is what permits localization on a fixed physical height scale.

Proof of Theorem 13.1. Proposition 15.1 supplies the hypotheses of Theorem 16.2. Smooth compactly supported nonnegative even functions f can approximate the square root of

$$\psi_{\mathrm{MT}}(u) = \cos(\sqrt{2}u)\,\mathbf{1}_{[-1/2,1/2]}(u).$$

Both integrals defining R_ψ converge under this approximation. Direct integration gives $R_{\psi_{\mathrm{MT}}} = 1/2 + 2^{-1/2}\cot(2^{-1/2})$. Let the approximation parameter tend to zero after T tends to infinity. For a Jordan-measurable B, its interior has the same Haar measure, so the open-set assertion implies the stated result. The empty prime set reduces to the unconditioned case. □

17 Uniform Weil arithmetic at variable bandwidth

For shrinking phase sets we use the same compression (113), but now allow $\ell/2 \le L \le \ell$, where $\ell = \log T$ and $X = e^L$. The window f remains fixed, nonzero, even and smooth, with support strictly inside $(-1/2, 1/2)$; again $a = \int f^2$ and $\psi = f^2$. The exact zero-side representation and the full-grid normalization are unchanged. A reflected off-line pair with vectors $u, \bar{u}$ contributes $uu^{\mathsf{T}} + \bar{u}\bar{u}^{\mathsf{T}}$; writing $u = v + iw$ makes this $2vv^{\mathsf{T}} - 2ww^{\mathsf{T}}$, with at most one positive direction. Thus the single rank–inertia certificate, Lemma 14.1, applies at this bandwidth as well. We record the variable-bandwidth estimates explicitly because their errors must be small relative to the Haar mass.

Here are the analytic identities used in the uniform arithmetic estimate below. For the full-grid synthesis operator with functions $u_k(t) = \widehat{\phi}(t - \alpha_k)/(\sqrt{a}L)$, Parseval gives $\|U\|^2 \ll_f L^{-1}$ and $\sum_k |u_k(t)|^2 = 1$. Schwartz decay controls the mass outside the center interval by $(1 + L\operatorname{dist}(t, I))^{-M}$, and the analogous inside-boundary tail has integral $O_f(L^{-1})$. The Montgomery–Vaughan mean-value inequality [21] gives $\int_I |\nu_X(t)|^2 dt \ll_f TL^2$. The operator inequality

$$\|V^* M_{\nu_X 1_J} V\|^2_{\mathrm{HS}} \le \|V\|^2 \int_J |\nu_X(t)|^2 \sum_k |v_k(t)|^2 dt$$

then bounds the squared matrix error caused by a boundary reduction by $O_f(1 + X/L^2)$ and the full squared norm by $O_f(TL)$.

For $\lambda = \log q$ small enough to avoid aliases, Poisson summation gives

$$\sum_{k\in\mathbb{Z}} e^{i\lambda\alpha_k}\widehat{\phi}(t-\alpha_k)\widehat{\phi}(t'-\alpha_k) = Le^{i\lambda(t+t')/2}\Psi_\lambda(t-t'),$$

where $\Psi_\lambda(s) = \int \phi(u-\lambda/2)\phi(u+\lambda/2)e^{-isu}du$. Put $J_\lambda = \Psi_\lambda\Psi_0$. After the boundary reduction, the weighted row energy is

$$\frac{1}{a^2L^2}\int_{I^2} e^{i\lambda(t+t')/2}J_\lambda(t-t')\nu_X(t)\nu_X(t')\,dt\,dt'.$$

For the zero character, its prime diagonal before division by a^2L^2 is

$$\frac{T}{\pi}\sum_{n\le X}\frac{\Lambda(n)^2}{n}(\phi^2 * \phi^2)(\log n).$$

The elementary prime-number-theorem consequence $\sum_{n\le y}\Lambda(n)^2/n = \frac{1}{2}\log^2 y + O(\log y)$ and partial summation give $L^3\int_0^1 v(\psi*\psi)(v)dv + O_f(L^2)$ for its sum. The gamma–gamma term supplies $T\ell^2L\int\psi^2/(2\pi)$ before the same division. Together they give the functional $R_f(L/\ell)$ below. We also write $R_\psi = R_f(1)$. Approximating $\psi_{\mathrm{MT}}(u) = \cos(\sqrt{2}u)1_{[-1/2,1/2]}(u)$ by squares of fixed smooth windows gives $R_{\psi_{\mathrm{MT}}} = 1/2 + 2^{-1/2}\cot(2^{-1/2})$.

17.1 Uniform arithmetic with a small Haar mass

Proposition 17.1 (Uniform rational arithmetic)**.** *Fix a nonzero even real $f \in C_c^\infty((-1/2,1/2))$ and choose $\sigma_f < 1$ with* $\operatorname{supp} f \subset [-\sigma_f/2, \sigma_f/2]$. *Let $\ell = \log T$, $\ell/2 \le L \le \ell$, $X = e^L$, and form the Weil matrix G with this bandwidth as in the preceding construction. Let S be a finite set of primes, allowed to vary, and let*

$$F(z) = \sum_{\mathbf{n}} c_{\mathbf{n}} z^{\mathbf{n}}, \qquad m = c_{\mathbf{0}} \ge 0, \qquad |c_{\mathbf{n}}| \le m, \qquad A = \sum_{\mathbf{n}} |c_{\mathbf{n}}|.$$

The Fourier sum is finite. For each character write its associated rational in lowest terms as $q = b/c$ and suppose $b, c \le H$, with

$$\log H \le \kappa L, \qquad 0 < \kappa < 1 - \sigma_f.$$

Put

$$B_S = \sum_{p\in S}\frac{\log p}{\sqrt{p}-1}, \qquad R_f(\alpha) = \frac{\alpha^{-1}\int f^4 + \alpha\iint |u-v|f(u)^2f(v)^2\,du\,dv}{(\int f^2)^2}.$$

Then, uniformly in S, F, H, L subject to these conditions,

$$\sum_k F(F_S(\alpha_k))G_{kk} = mN + O_f\left(mN\frac{1+B_S}{L} + AN\mathcal{E}\right),$$

$$\sum_k F(F_S(\alpha_k))\sum_j G_{kj}^2 = mR_f(L/\ell)N + O_{f,\kappa}\left[mN\left(\frac{1+B_S}{L} + \frac{B_S^2}{L^2}\right) + AN\mathcal{E}\right],$$

where $N = N(T, 2T)$ and one may take

$$\mathcal{E} = \frac{HX}{T} + \frac{1}{\sqrt{TL}} + \frac{\sqrt{X/T}}{L^{3/2}} + \frac{\sqrt{X}}{TL}.$$

The constants do not depend on the number of primes in S. The coefficient condition holds, in particular, for any Fourier truncation of a nonnegative integrable phase weight, with m its Haar integral.

Proof. The boundary and full-grid reduction uses only $|q^{it}| = 1$ for individual characters. The squared Hilbert–Schmidt error is $O_f(1 + X/L^2)$, whereas the full squared norm is $O_f(TL)$. Cauchy–Schwarz therefore gives, after division by N, the second and third terms of $\mathcal{E}$. Linear trace boundary errors are smaller. Summing over characters costs A.

For $\lambda = \log q$, the weighted Poisson formula is exact and has no aliases, because $|\lambda| \le \kappa L$ and $\kappa + \sigma_f < 1$. The lag kernels J_λ obey

$$\int |J_\lambda(s)|\, ds \ll_f L, \qquad \int |sJ_\lambda(s)|\, ds \ll_f 1$$

uniformly in this entire range of λ. Indeed, after scaling, the two translated arguments of f range over a fixed compact set and every derivative has a uniform bound.

The rational-frequency separation used in the mean-value estimate is uniformly $\gg (HX)^{-1}$. For opposite signs this follows from $|bn - cm| \ge 1$ whenever $bn \ne cm$. For equal signs, a small $|\log(qnm)|$ implies $bnm \asymp c \le H$, so the same lower bound is more than sufficient. Frequencies internal to either list are separated by $\gg X^{-1}$.

It is essential not to replace the archimedean density by a constant in every Fourier mode. Keep

$$\mu(t) = \frac{1}{2\pi} \operatorname{Re} \frac{\Gamma'}{\Gamma}(1/4 + it/2) - \frac{\log \pi}{2\pi}$$

exact in all nontrivial modes. For $|s| \le T/2$, as functions of the midpoint $\tau \in [T, 2T]$, each $\mu(\tau \pm s/2)$ has supremum plus total variation $O(\ell)$; their product has the corresponding bound $O(\ell^2)$. Abel summation therefore extends the Montgomery–Vaughan Hilbert inequality to these weights. All nonresonant prime–prime and mixed terms have midpoint integral $O(HX\ell^2)$, uniformly in the lag. The constant–constant nontrivial mode is $O(H\ell^2)$ by direct integration. Integration against the lag kernel and division by a^2L^2 give $O(NHX/T)$ per unit Fourier ℓ^1 mass. The large-lag Schwartz tails and the lag-boundary correction are smaller.

On $[T, 2T]$, the pole density has L^2 norm $O(\sqrt{X/T})$. The remaining density has L^2 norm $O(\sqrt{T}L)$. Its insertion in the lag integral thus contributes $O_f(\sqrt{X})$ after normalization, yielding the final term of $\mathcal{E}$.

For the zero character the usual diagonal computation gives $R_f(L/\ell)N + O_f(N/L)$. Here only this character uses $\mu = \ell/(2\pi) + O(1)$, so its error is multiplied by m. The nontrivial resonances retain the same factor. If $qn^\varepsilon m_1^\eta = 1$, $q \ne 1$, and both von Mangoldt factors are nonzero, both prime bases must lie in S. Summing over all nontrivial q in the Fourier support therefore gives

$$\sum_{q \ne 1} |c_q| \sum_{qn^\varepsilon m_1^\eta = 1} \frac{\Lambda(n)\Lambda(m_1)}{\sqrt{nm_1}} \le 4mB_S^2.$$

Similarly the total mixed-resonance amplitude is at most $2mB_S$. Their normalized contributions are respectively $O_f(mNB_S^2/L^2)$ and $O_f(mNB_S/L)$.

For the trace, the diagonal Poisson formula reduces the calculation to a single midpoint integral with a prefactor bounded by one. The zero-character integral is $N + O_f(\sqrt{X} + \ell)$. Nontrivial resonances sum to $O(mTB_S)$, and nonresonant prime integrals are $O(AHXL)$ by Cauchy–Schwarz and the frequency separation. The archimedean nontrivial terms are $O(AH\ell)$. These bounds give the asserted first-moment estimate. □

Remark 17.2 (A useful choice of bandwidth). If $\log H = o(\log T)$, choose

$$L = \log T - \log H - D \log\log T.$$

Then $HX/T = (\log T)^{-D}$ and $L/\log T \to 1$. For fixed S and a smooth phase bump of width $r = (\log T)^{-a}$, $a < 1$, Fourier truncation at degree $r^{-1}(\log T)^{\epsilon}$, $a+\epsilon < 1$, has $\log H = O_S((\log T)^{a+\epsilon})$, uniformly bounded Fourier ℓ^1 norm, and coefficients bounded by its Haar mass. The arithmetic moments consequently remain asymptotic relative to that mass. This arithmetic observation alone does not establish a shrinking-target theorem: the zero-side inertia budget must also be localized with a uniform relative error.

18 Shrinking prime-phase boxes

Throughout this section the prime set $S = \{p_1, \dots, p_d\}$ is fixed. We identify $\mathbb{T}^d$ with $(\mathbb{R}/\mathbb{Z})^d$, and write

$$\Theta_S(t) = \left(-\frac{t \log p_j}{2\pi} \pmod 1 \right)_{j=1}^{d}.$$

For a box B let $N_B(T, 2T)$ count all nontrivial zeros, with multiplicity, whose ordinates lie in $(T, 2T]$ and whose phase is in B. The analogous count restricted to simple critical-line zeros is $N^s_{0,B}(T, 2T)$.

Theorem 18.1 (Anisotropic shrinking targets)**.** *Fix $a_1, \dots, a_d \geq 0$ and $c_1, \dots, c_d \in (0, 1/4]$. Put*

$$A = \sum_{j=1}^{d} a_j, \qquad a_* = \max_j a_j, \qquad r_j(T) = c_j (\log T)^{-a_j}.$$

Suppose

$$a_* < 1. \tag{133}$$

For any choice of centers $\theta(T) \in \mathbb{T}^d$, let $B_T = \prod_{j=1}^{d} I_j(T)$, where $I_j(T)$ is the circular interval of length $r_j(T)$ centered at $\theta_j(T)$. Then, uniformly in the choice of centers,

$$N_{B_T}(T, 2T) = (1 + o(1))\mu(B_T) N(T, 2T), \tag{134}$$

$$\liminf_{T\to\infty} \inf_{\theta \in \mathbb{T}^d} \frac{N^s_{0,B_T}(T, 2T)}{\mu(B_T) N(T, 2T)} \geq \frac{3}{2} - \frac{1}{\sqrt{2}} \cot \frac{1}{\sqrt{2}}. \tag{135}$$

In particular, equal side lengths $(\log T)^{-a}$, up to fixed positive factors, are allowed whenever $a < 1$.

The strict inequality is a proved sufficient range, not a proposed optimal threshold. The dimension and the primes remain fixed. The centers can depend arbitrarily on T; no uniformity in a growing prime set is asserted.

18.1 A paired quantitative Landau estimate

Set $\ell = \log T$ and $N = N(T, 2T)$. For $n \in \mathbb{Z}^d$ put $q_n = \prod_j p_j^{n_j}$ and $\lambda_n = \log q_n$. If $\|n\|_\infty \leq H$, then the numerator and denominator of q_n in lowest terms are at most $\exp(C_S H)$. In particular, $|\lambda_n| \leq C_S H$, and, for $n \neq 0$, $|\lambda_n| \geq \exp(-C_S H)$ after increasing C_S.

Lemma 18.2. *Uniformly for $H = o(\ell)$ and $0 < \|n\|_\infty \leq H$, let $x_n = \max(q_n, q_n^{-1}) > 1$. Then*

$$\sum_{T < \gamma \leq 2T} e^{i\lambda_n \gamma} = -\frac{T}{2\pi} \frac{\Lambda(x_n)}{\sqrt{x_n}} + O_S\left(N \frac{\lambda_n^2}{\ell^2} + e^{C_S H} \ell^2 \right), \tag{136}$$

where $\Lambda(x) = 0$ unless x is an integral prime power, and the sum counts multiplicities. The same estimate holds with the opposite sign in the exponential.

Proof. The uniform Landau formula in [18, Lemma 1], applied at T and $2T$, gives, for $x > 1$,

$$\sum_{T<\gamma\le 2T} x^{\rho} = -\frac{\Lambda(n_x)}{2\pi}\frac{e^{2iT\log(x/n_x)} - e^{iT\log(x/n_x)}}{i\log(x/n_x)} + O\left(x\log^2(2xT) + \frac{\log(2T)}{\log x}\right),$$

where n_x is a nearest prime power and the quotient is interpreted as T when $x = n_x$. For $x = x_n$, rational separation shows that all nonresonant terms here are $O_S(e^{C_S H}\ell^2)$, after increasing C_S: a nonzero numerator of $x_n - n_x$ is an integer, whereas all relevant numerators and denominators are at most exponential in H. Dividing by $\sqrt{x_n}$ therefore gives the asserted main term for the sum with weight $x_n^{\beta-1/2}$.

It remains to remove that real weight. Horizontal reflection $\beta + i\gamma \mapsto 1 - \beta + i\gamma$, preserving multiplicity, cancels its linear variation. Writing $u_\rho = |\beta - 1/2|$ and $\lambda = \log x_n$, the absolute replacement error is bounded by

$$\sum_{T<\gamma\le 2T} (\cosh(\lambda u_\rho) - 1).$$

The uniform Selberg estimate gives

$$\#\{\rho : T < \gamma \le 2T,\ u_\rho > v\} \ll Ne^{-v\ell/4} \qquad (0 \le v \le 1/2).$$

By layer integration, for $\lambda \le \ell/8$ the preceding error is at most

$$CN\int_0^{1/2} \lambda\sinh(\lambda v)e^{-v\ell/4}\,dv \ll N\lambda^2\int_0^{\infty} ve^{-v\ell/8}\,dv \ll N\frac{\lambda^2}{\ell^2}.$$

The condition $H = o(\ell)$ ensures $\lambda \le \ell/8$ eventually. Complex conjugation handles a negative λ_n. □

Pairing before taking absolute values is essential: a first-order estimate would give $N|\lambda|/\ell$ and a smaller shrinking range.

18.2 Smooth boxes and the all-zero budget

Let $\tau = 1/\log\ell$ and $r = \min_j r_j$. By convolving circular intervals with nonnegative smooth mollifiers of widths comparable to τr_j, we obtain $K_T^-, K_T^+ \in C^\infty(\mathbb{T}^d)$ with

$$0 \le K_T^- \le 1_{B_T} \le K_T^+ \le 1, \qquad \int K_T^{\pm}\,d\mu = (1 + O_d(\tau))\mu(B_T). \tag{137}$$

The support of K_T^- is separated from the complement of B_T by at least $c\tau r$ in the torus metric. This can be arranged by shrinking the initial intervals slightly more before convolution. The construction and all following estimates are invariant under translation of the center.

Writing either Fourier series as $K = \sum_n c_n e^{2\pi i n\cdot\theta}$, put $m = \int K\,d\mu$. Nonnegativity gives $|c_n| \le m$. Standard scaling estimates for the mollifiers give

$$\sum_n |c_n| \ll_d \tau^{-d}, \tag{138}$$

$$\sum_n |c_n||\lambda_n|^2 \ll_S \tau^{-d-2}r^{-2}, \tag{139}$$

$$\sum_{\|n\|_\infty > H} |c_n| \ll_{S,M} \tau^{-d}(H\tau r)^{-M} \quad (H\tau r \ge 1), \tag{140}$$

for every fixed M, with a harmless change of M in the derivative estimate. For example, the one-dimensional coefficients are bounded by $C_M r_j(1+\tau r_j|n_j|)^{-M}$, and summation of the product estimates proves these assertions.

Choose b with $a_* < b < 1$ and set $H = \ell^b$. For sufficiently large fixed M, the last error in (140) is $o(\mu(B_T))$. Apply Lemma 18.2 to the truncated series. The prime-power main terms satisfy the sharper bound

$$\sum_{n\neq 0} |c_n| \frac{\Lambda(x_n)}{\sqrt{x_n}} \le 2m \sum_{p\in S}\sum_{k\ge 1} (\log p) p^{-k/2} \ll_S m.$$

Indeed an integral prime power supported on S has just one nonzero coordinate in its exponent vector. Thus

$$\sum_{T<\gamma\le 2T} K(\Theta_S(\gamma)) = mN + O_S\left(\frac{mN}{\ell} + \frac{N\tau^{-d-2}}{\ell^2 r^2} + \tau^{-d} e^{C_S H}\ell^2\right) + o(\mu(B_T)N). \tag{141}$$

Since $\mu(B_T) \asymp \ell^{-A}$ and $r \asymp \ell^{-a_*}$, the stronger restriction $A + 2a_* < 2$ makes every error $o(\mu(B_T)N)$. Sandwiching by (137) proves (134) in that smaller range. We next remove this restriction.

18.3 Analytic box bounds in a thin horizontal strip

The loss in the elementary Fourier argument can be removed by bounding the phase box directly at complex zero coordinates. This avoids replacing each Fourier weight by its value on the critical line separately.

Put $v = \log \ell$, $\tau = v^{-1}$, and choose Gaussian widths $s_j = \tau r_j / v = r_j / v^2$. Let B^+ and B^- denote boxes obtained by increasing or decreasing every side of B_T by $4\tau r_j$. For large T the latter box is nonempty. On the complexified torus define the periodized Gaussian and its convolutions

$$g_s(z) = \left(\prod_{j=1}^{d} \frac{1}{\sqrt{2\pi} s_j}\right) \sum_{k\in\mathbb{Z}^d} \exp\left(-\sum_{j=1}^{d} \frac{(z_j+k_j)^2}{2s_j^2}\right), \qquad F^{\pm}(z) = \int_{B^{\pm}} g_s(z-u)\,du.$$

Both functions are entire and periodic in every real coordinate, and

$$\int_{\mathbb{T}^d} g_s(z-u)\,du = 1.$$

Their Fourier coefficients are

$$\widehat{F^{\pm}}(n) = \widehat{1_{B^{\pm}}}(n) \exp\left(-2\pi^2 \sum_j s_j^2 n_j^2\right).$$

In particular $|\widehat{F^{\pm}}(n)| \le \mu(B^{\pm})$ and

$$\sum_n |\widehat{F^{\pm}}(n)| \ll_d \mu(B^{\pm}) \prod_j s_j^{-1} \ll_d v^{2d}. \tag{142}$$

Choose a sufficiently large fixed B_0 and put $d_0 = 4B_0 \log \ell / \ell$. A zero with $|\beta - 1/2| \le d_0$ gives a complex phase

$$z_\rho = \Theta_S(\gamma) + i y_\rho, \qquad (y_\rho)_j = (\beta - 1/2)\frac{\log p_j}{2\pi}.$$

For all these phases,

$$h := \sup_{\rho} \left(\sum_j \frac{|(y_\rho)_j|^2}{s_j^2} \right)^{1/2} \ll_{S,B_0} \ell^{a_*-1} v^3 = o(1).$$

The absolute value of the complex Gaussian is exactly its real Gaussian times $\exp(\|y/s\|^2/2)$. Its negative real part can occur only when, in at least one lattice term,

$$\left| \sum_j \frac{(x_j + k_j) y_j}{s_j^2} \right| \geq \pi/2.$$

Cauchy–Schwarz forces $\|(x+k)/s\| \geq \pi/(2\|y/s\|)$. Gaussian tail summation therefore gives, uniformly in real x and $\|y/s\| \leq h$,

$$\operatorname{Re} g_s(x+iy) \geq -C_d \left(\prod_j s_j^{-1} \right) e^{-c_d/h^2}. \tag{143}$$

The assertion is interpreted by continuity when $y = 0$.

The real distance, in each relevant normal direction, from B_T to the complement of B^+ is at least $2\tau r_j = 2v s_j$. The identity for the full Gaussian integral and the Gaussian tail bound imply

$$\operatorname{Re} F^+(x+iy) \geq 1 - O_d(e^{-c_d v^2}) \quad (x \in B_T).$$

The analogous separation of B^- from the complement of B_T gives

$$|F^-(x+iy)| \ll_d e^{-c_d v^2} \quad (x \notin B_T).$$

Globally $\operatorname{Re} F^- \leq e^{h^2/2} = 1 + o(1)$, whereas (143) gives $\operatorname{Re} F^+ \geq -O_d((\prod_j s_j^{-1}) e^{-c_d/h^2})$. Because $a_* < 1$, all the displayed exponentially small errors are $o(\ell^{-C})$ for every fixed $C > 0$. Consequently there is $\epsilon_T = o(\ell^{-C})$ for every fixed C such that, at all shallow zero phases,

$$(1+o(1))^{-1} \operatorname{Re} F^-(z_\rho) - \epsilon_T \leq 1_{B_T}(\Theta_S(\gamma)) \leq (1+o(1)) \operatorname{Re} F^+(z_\rho) + \epsilon_T. \tag{144}$$

The multiplicative errors are independent of the phase center.

Truncate $F^\pm$ to $\|n\|_\infty \leq H = \ell^b$, with $a_* < b < 1$, and call the resulting polynomials $P^\pm$. The relation $d_0/s_{\min}^2 = o(H)$ allows the growing imaginary factor in the Fourier tail to be absorbed by Gaussian decay. At shallow phases the truncation error is

$$O_{S,d}\left(v^{2d} \exp(-c s_{\min}^2 H^2 + C_S d_0 H) \right) = o(\ell^{-C})$$

for every fixed C. Apply the uniform Landau formula directly to $P^\pm(z_\rho)$, without removing the real zero weights. Horizontal reflection or complex conjugation gives the required sign convention for the frequencies. The prime-power terms are $O_S(\mu(B_T)N/\ell)$ by the coefficient bound and the convergent one-prime series used above. The nonresonant terms are $O_S(v^{2d} e^{C_S H} \ell^2) = o(\mu(B_T)N)$. Thus, with all zeros included,

$$\sum_{T < \gamma \leq 2T} P^\pm(z_\rho) = (\mu(B^\pm) + o(\mu(B_T)))N. \tag{145}$$

The contribution of deep zeros must be removed with their complex weights, not merely by counting them. By (142),

$$|P^{\pm}(z_\rho)| \ll_d v^{2d} e^{C_S H|\beta-1/2|}.$$

Layer integration of Selberg's estimate, using $C_S H = o(\ell)$, therefore gives

$$\sum_{\substack{T<\gamma\le 2T\\ |\beta-1/2|>d_0}} |P^{\pm}(z_\rho)| \ll_{S,d} N v^{2d} \exp(-d_0\ell/8) = O_{S,d}(N v^{2d}\ell^{-B_0/2}).$$

Choose $B_0 > 2A+4$. This quantity, and the unweighted count of deep zeros, are $o(\mu(B_T)N)$. Combining (144)–(145) and $\mu(B^{\pm}) = (1+O_d(\tau))\mu(B_T)$ proves (134) throughout the full range $a_* < 1$.

18.4 Relative moments and zero-side localization

Fix a nonnegative even $f \in C_c^\infty((-1/2,1/2))$, and retain the notation $a = \int f^2$, $\psi = f^2$ and R_ψ for the Weil compression. Use the variable bandwidth

$$L = \ell - C_S H - D\log\ell, \qquad X = e^L, \tag{146}$$

where C_S is sufficiently large and D is a sufficiently large fixed constant. Then $L/\ell \to 1$. The support margin of the fixed f ensures that the weighted Poisson identities have no aliases for all $\|n\|_\infty \le H$.

Proposition 17.1 gives, for $K = K_T^-$ and $D_K = \mathrm{diag}(K(\Theta_S(\alpha_k))^{1/2})$,

$$\mathrm{tr}(D_K G D_K) = (m + o(\mu(B_T)))N, \qquad \|D_K G D_K\|_{\mathrm{HS}}^2 \le (R_\psi m + o(\mu(B_T)))N. \tag{147}$$

Here the rational height costs $\exp(O_S(H))$ are absorbed by (146). In checking the relative error, the nonzero resonances must be summed using $|c_n| \le m$: their total is bounded by m times

$$\left(\sum_{p\in S}\sum_{k\ge1}(\log p)p^{-k/2}\right)^2,$$

or its one-prime analogue. A bound using only $\sum|c_n|$ would unnecessarily lose the factor m. Nonresonant archimedean factors are kept as slowly varying functions and handled by partial summation; replacing them by a constant with a global absolute error would likewise be insufficient for a relative estimate. Fourier truncation costs $o(\mu(B_T)N)$ by (140) and the unweighted trace and Hilbert–Schmidt bounds. These observations give the stated uniform extension of the weighted moment proof.

We detail the additional zero-side uniformity. Choose $B_0 > A+2$ and put

$$d_0 = 4B_0\frac{\log\ell}{\ell}, \qquad I' = [T-\sqrt{T}, 2T+\sqrt{T}].$$

The total multiplicity of zeros in I' with $|\beta-1/2| > d_0$ is $O(N\ell^{-B_0}) = o(\mu(B_T)N)$. Retain all these deep zeros, regardless of their phase. Among the remaining zeros retain precisely those whose phase lies in B_T. Let E be the matrix contribution of the discarded shallow zeros. The normalized vectors satisfy, for every fixed M,

$$|u_\rho(k)| \ll_M \ell^{2B_0+1}(1+L|\gamma-\alpha_k|)^{-M}.$$

Every discarded zero is separated from every nonzero weighted sampling center by at least $\eta = c_S \tau r$ in real height, because Θ_S is Lipschitz. For the matrix U whose columns are the corresponding weighted vectors, repeated according to multiplicity, the grid spacing and unit-interval zero count give

$$\|U\|^2 \ll \ell^{4B_0+3}, \qquad \|U\|_{\mathrm{HS}}^2 \ll N\ell^{4B_0+2}(L\eta)^{1-2M}.$$

Since $E = UU^{\mathsf{T}}$,

$$|\operatorname{tr} E| \le \|U\|_{\mathrm{HS}}^2,$$
$$\|E\|_{\mathrm{HS}}^2 \le \|U\|^2\|U\|_{\mathrm{HS}}^2 \ll N\ell^{8B_0+6-a_*-2M(1-a_*)}(\log \ell)^{2M-1}.$$

The condition (133) implies $a_* < 1$. Consequently a sufficiently large fixed M makes both errors $o(\mu(B_T)N)$, where for the Hilbert–Schmidt statement this means $\|E\|_{\mathrm{HS}}^2 = o(\mu(B_T)N)$.

Subtract also the exact contribution F_{out} of zeros outside I'. The full-grid decay gives $\|F_{\mathrm{out}}\|_1 = O(T^{-10})$ after choosing sufficiently many integrations by parts. Thus the exact retained matrix

$$H_K = D_K G D_K - F_{\mathrm{out}} - E$$

still satisfies (147). This exact subtraction is necessary: a small-norm matrix need not have small positive index.

Let $s = N^s_{0,B_T}(I')$ and decompose $H_K = P + Q$, where P is the sum over the retained simple critical-line zeros. Full-grid normalization gives $\operatorname{rank} P \le s$ and $\operatorname{tr} P \le s$. Every reflected off-line pair costs at most one positive direction; every multiple on-line zero costs at most one. Hence

$$s + 2n_+(Q) \le N_{B_T}(I') + o(\mu(B_T)N).$$

The two added height intervals contain only $O(\sqrt{T}\log T)$ zeros, uniformly in the phase condition. Applying the rank–inertia inequality, (134), and (147), gives

$$N^s_{0,B_T}(T, 2T) \ge (2 - R_\psi - o(1))\mu(B_T)N.$$

All constants and errors above are uniform in the box center. Finally fix f successively closer to the Montgomery–Taylor window. For every fixed such f the displayed lower bound holds; taking the supremum over this family after the T limit yields (135). No varying-window estimate is required.

Corollary 18.3 (A covering statement at simple critical zeros)**.** *Fix S with $|S| = d$ and any $0 < a < 1$. For all sufficiently large T, every phase box with side lengths $(\log T)^{-a}$ contains the phase of a simple critical-line zero with ordinate in $(T, 2T]$. In fact each such box contains at least $(c_{\mathrm{MT}} - o(1))(\log T)^{-ad}N(T, 2T)$ such ordinates, uniformly in its center.*

This is a quantitative covering conclusion for a fixed finite prime torus. It does not, by itself, give a rate for approximating an infinite Euler product: such a conclusion would also require estimates uniform as the prime set grows.

19 Near-logarithmic phase resolution

The polynomial shrinking theorem can be sharpened by choosing Gevrey windows. This replaces a fixed inverse power of the sampling distance by a stretched exponential, which is needed when the phase buffer is only a power of $\log\log T$ sampling spacings.

Theorem 19.1 (Near-logarithmic boxes)**.** *Fix a finite prime set $S = \{p_1, \dots, p_d\}$, a constant $c_0 > 0$, and $C > 2$. Put $\ell = \log T$ and $v = \log\log T$. Let B_T be any phase box, with arbitrary center and side lengths satisfying*

$$c_0 \frac{v^C}{\ell} \le r_j(T) \le \frac{1}{4} \qquad (1 \le j \le d). \tag{148}$$

Uniformly in the centers and all these side lengths,

$$N_{B_T}(T, 2T) = (1 + o(1))\mu(B_T)N(T, 2T),$$

and

$$\liminf_{T\to\infty} \inf_{B_T} \frac{N^s_{0,B_T}(T, 2T)}{\mu(B_T)N(T, 2T)} \ge c_{\mathrm{MT}}.$$

Thus every such box contains many simple critical-line zero phases in every sufficiently large dyadic height interval.

The theorem concerns a fixed prime set. It does not reach side length $1/\log T$, and the power $C > 2$ is a sufficient condition of this argument, not an optimality assertion.

Lemma 19.2 (A complex Gevrey estimate)**.** *Suppose $f \in C^\infty_c([-a, a])$ satisfies*

$$\|f^{(m)}\|_{L^1} \le AB^m(m!)^2 \qquad (m \ge 0)$$

with fixed $A, B > 0$. Then there are constants $c, C > 0$, depending only on these data, such that

$$|\widehat{f}(x + iy)| \le C \exp\left(a|y| + C\sqrt{|y|} - c\sqrt{|x|}\right) \qquad (x, y \in \mathbb{R}). \tag{149}$$

There exist nonnegative even functions of this class, supported strictly inside $(-1/2, 1/2)$, whose squares approximate the Montgomery–Taylor window in the integrals defining R_ψ.

Proof. Leibniz's formula, with j derivatives falling on e^{yu}, gives

$$\begin{aligned}
\|(f(u)e^{yu})^{(m)}\|_1 &\le Ae^{a|y|}B^m(m!)^2 \sum_{j=0}^{m} \frac{(|y|/B)^j(m-j)!}{j!m!} \\
&\le Ae^{a|y|}B^m(m!)^2 \sum_{j=0}^{\infty} \frac{(|y|/B)^j}{(j!)^2} \\
&\le A \exp\left(a|y| + 2\sqrt{|y|/B}\right) B^m(m!)^2.
\end{aligned}$$

The last inequality follows by comparing the series with $\exp(2\sqrt{|y|/B})$, using $\binom{2j}{j} \le 4^j$. Repeated integration by parts bounds the Fourier transform by this expression divided by $|x|^m$. Taking m to be a sufficiently small fixed multiple of $\sqrt{|x|}$ and using Stirling's bound proves (149); bounded $|x|$ is absorbed into the constant.

For existence, the standard bump $b(u) = \exp(-1/(1-u^2))$ for $|u| < 1$ and $b(u) = 0$ otherwise is Gevrey of order two. One direct verification applies Cauchy's estimate on a complex circle of radius a small constant times $1 - |u|$, on which the real part of $1/(1-z^2)$ is bounded below by $c/(1-|u|)$. Maximizing $(1-|u|)^{-m}\exp(-c/(1-|u|))$ gives a bound $C^{m+1}(m!)^2$. Translates, dilates, products, and convolution preserve this class, with constants allowed to depend on the chosen functions. Smooth cutoffs formed from b can therefore approximate the interval indicator from within. Multiplying their even square roots in the form $f = \chi\sqrt{\cos(\sqrt{2}u)}$, where χ itself is a nonnegative even Gevrey cutoff, gives $f^2 \to \cos(\sqrt{2}u)1_{[-1/2,1/2]}$ in the required integrals. The analytic square root is harmless since the cosine is strictly positive on this interval. □

Proof of Theorem 19.1. Choose fixed $u, k > 0$ such that

$$u + k < C - 2, \tag{150}$$

and put

$$\eta = v^{-u}, \qquad H = \ell/v^k, \qquad L = \ell - C_S H - Dv.$$

Here C_S and D are sufficiently large fixed constants. We use a fixed nonnegative even Gevrey window f from Lemma 19.2. All the uniform arithmetic moment bounds apply because $H = o(\ell)$ and $L/\ell \to 1$.

We first record the phase weights. Convolve slightly shrunken interval indicators with dilates of a fixed nonnegative Gevrey-two mollifier, with smoothing widths comparable to ηr_j. The resulting compact inner cutoff K satisfies

$$0 \le K \le 1_{B_T}, \quad m := \int K = (1 + O_d(\eta))\mu(B_T), \quad \operatorname{dist}(\operatorname{supp} K, \mathbb{T}^d \setminus B_T) \gg_d \eta r_{\min}.$$

Its Fourier coefficients c_n obey $|c_n| \le m$ and

$$\sum_n |c_n| \ll_d \eta^{-d},$$
$$\sum_{\|n\|_\infty > H} |c_n| \ll_d \eta^{-d} \exp(-c\sqrt{H\eta r_{\min}}).$$

Polynomial factors in the tail summation have been absorbed by reducing c. These estimates follow by multiplying the interval coefficient bound by the Gevrey Fourier decay of the mollifier and summing the resulting product. By (148) and (150),

$$H\eta r_{\min} \gg v^{C-u-k}, \qquad C - u - k > 2.$$

The Fourier tail is consequently $o(\ell^{-M})$ for every fixed $M > 0$. Since $\mu(B_T) \gg \ell^{-d}$, Fourier truncation and the uniform rational-character arithmetic give relative first and second moment estimates for this K, exactly as in (147).

We next prove the required all-zero budget. Use the Gaussian majorants and minorants from the preceding section, now expanding or contracting each side by $4\eta r_j$ and taking widths

$$s_j = \frac{\eta r_j}{\sqrt{K_0 v}},$$

where the fixed constant K_0 is sufficiently large in terms of d. Their Fourier coefficient mass is $O_{d,K_0}(v^{d(u+1/2)})$. For shallow zeros with $|\beta - 1/2| \le d_0 = 4B_0 v/\ell$,

$$h := \|y/s\| \ll_{S,B_0,K_0} v^{3/2+u-C}.$$

The complementary Gaussian tails on the target box and outside the contracted box are $O_d(e^{-c_d K_0 v})$. Choose K_0 so these are $o(\ell^{-d-2})$. The global negative part of the upper analytic majorant is at most

$$C_d \left(\prod_j s_j^{-1} \right) \exp(-c/h^2) \ll \ell^d v^{O_d(1)} \exp(-cv^{2C-3-2u}).$$

Since $C - u > 2$, the exponent $2C - 3 - 2u$ exceeds one, so this error is $o(\ell^{-M})$ for every fixed M. The lower analytic bound's global upper bound remains $e^{h^2/2} = 1 + o(1)$. The Gaussian Fourier truncation at H is also super-polynomially accurate in ℓ, since

$$s_{\min} H \gg v^{C-u-k-1/2}, \qquad C - u - k > 2, \qquad d_0/s_{\min}^2 = o(H).$$

The last relation ensures that the growing imaginary factor in the tail is absorbed by Gaussian decay.

Apply Landau's formula to the truncated analytic polynomials at the complex zero coordinates. Exact prime-power resonances have total contribution $O_S(\mu(B_T)N/\ell)$, and the rational-height error is $T^{o(1)}v^{O_d(1)} = o(\mu(B_T)N)$. The weighted contribution of deep zeros is at most

$$Nv^{O_d(1)}\exp(-d_0\ell/8) = Nv^{O_d(1)}\ell^{-B_0/2}.$$

Taking $B_0 > 2d+4$ makes it $o(\mu(B_T)N)$. The analytic sandwich therefore proves the asserted all-zero asymptotic, uniformly in the box.

It remains to localize the zero-side inertia with the smaller buffer. Set $\delta_\rho = \beta - 1/2$. Lemma 19.2 gives for the normalized sampling vector

$$|u_\rho(k)| \ll_f \exp\left(C_f L|\delta_\rho| - c_f\sqrt{L|\gamma-\alpha_k|}\right).$$

An increase in the constant covers bounded $L|\delta_\rho|$. For a discarded shallow zero, the real separation from a weighted sampling center is at least $c_S\eta r_{\min}$. Thus

$$L|\gamma-\alpha_k| \gg v^{C-u}.$$

The unit-interval zero count and the grid spacing give Schur bounds for the matrix U of discarded vectors of the form $\|U\|^2 \ll \ell^{C_fB_0+C_f}$. Summing the stretched-exponential tail on the grid gives

$$\|U\|_{\mathrm{HS}}^2 \ll N\ell^{C_fB_0+C_f}\exp(-c_fv^{(C-u)/2}).$$

Because $(C-u)/2 > 1$, both $|\operatorname{tr}(UU^{\mathsf{T}})|$ and $\|UU^{\mathsf{T}}\|_{\mathrm{HS}}^2$ are $o(N\ell^{-M})$ for every fixed M. They are in particular $o(\mu(B_T)N)$. The exact outer-height tail is subtracted as before, and the deep zeros cost $o(\mu(B_T)N)$ in the count budget.

The rank–inertia argument now gives

$$N^s_{0,B_T}(T,2T) \geq (2 - R_{f^2} - o(1))\mu(B_T)N(T,2T).$$

Finally optimize through the fixed Gevrey windows supplied by Lemma 19.2, after taking the T limit. This recovers c_{MT} without requiring any estimate uniform in the approximating window. □

Corollary 19.3 (A quantitatively equidistributed selection)**.** *Fix a finite prime set S, $0 < c < c_{\mathrm{MT}}$, and $C > 2$. For every sufficiently large T one can select a set $\mathcal{Z}_T$ of simple critical-line zero ordinates in $(T,2T]$ with $|\mathcal{Z}_T| = (c+o(1))N(T,2T)$ such that its normalized phase distribution has box discrepancy*

$$\sup_{B \text{ a phase box}}\left|\frac{\#\{\gamma\in\mathcal{Z}_T : F_S(\gamma)\in B\}}{|\mathcal{Z}_T|} - \mu(B)\right| \ll_{S,c,C} \frac{(\log\log T)^C}{\log T}.$$

This is a selected subset; equidistribution of all simple critical-line zeros is not asserted.

Proof. Partition the torus into m^d equal boxes, where $m \asymp \log T/(\log\log T)^C$. Theorem 19.1 ensures each cell contains more than cN/m^d suitable ordinates. Select exactly $\lfloor cN/m^d\rfloor$ in every cell, using increasing height to make the selection definite. There are $(c+o(1))N$ selected ordinates. Any phase box meets at most $O_d(m^{d-1})$ boundary cells. Fully included cells carry exactly their Haar fraction of the selected measure, and the boundary cells give discrepancy $O_d(m^{-1})$. □

20 From phase-localized zeros to Euler-edge triples

Let $\Omega = \prod_p \mathbb{T}$ with product Haar measure μ. For $z \in \Omega$ put

$$Z_z(s) = \prod_p (1 - z_p p^{-s})^{-1}, \qquad Q_z(s) = -\frac{Z_z'(s)}{Z_z(s)} = \sum_p \frac{(\log p) z_p p^{-s}}{1 - z_p p^{-s}}.$$

All these expressions converge normally in $\operatorname{Re} s > 1$. The frozen configuration is $z_2^* = -1$, $z_p^* = 1$ for odd p. The classical constants A_* and E_* are characterized by

$$\sum_{\substack{n \geq 3 \\ n \text{ odd}}} n^{-A_*} = 2^{-A_*}, \qquad A_* = 1.940101\ldots,$$

$$\sum_{p \geq 3} \frac{\log p}{p^{E_*} - 1} = \frac{\log 2}{2^{E_*} + 1}, \qquad E_* = 2.813014\ldots.$$

These are the right boundaries for $Z_z(s) = 1$ and $Q_z(s) = 0$; see [8]. In particular no such point has real part larger than its boundary. Their identification is not a new claim. The frozen functions have simple roots at these boundaries: $a_E = Q_{z^*}'(E_*) < 0$ and $a_A = Z_{z^*}'(A_*) < 0$. The inequalities also follow directly by differentiating the defining positive sums. Define

$$D_E(z) = \sum_p (\log p) p^{-E_*} |z_p - z_p^*|^2, \qquad D_A(z) = \sum_p p^{-A_*} |z_p - z_p^*|^2,$$

and put $\beta_E = (E_* - 1)^{-1}$, $\tau = (A_* - 1)/(E_* - 1)$ and $\Psi(\delta) = \delta^\tau / \log(1/\delta)$.

Lemma 20.1 (Local edge estimates)**.** *For sufficiently small $D = D_E(z) > 0$, there are simple roots $Q_z(E_* + u_E) = 0$ and $Z_z(A_* + u_A) = 1$, unique in fixed small disks about the respective boundary points, such that*

$$0 \leq -\operatorname{Re} u_E \leq CD, \qquad |\operatorname{Im} u_E| \leq C\sqrt{D},$$
$$0 \leq -\operatorname{Re} u_A \leq C\Psi(D), \qquad |\operatorname{Im} u_A| \leq C\sqrt{D}.$$

Conversely, a critical point at $E_* - \delta$ in a twist implies $D_E(z) \leq C\delta$, and a 1-point at $A_* - \varepsilon$ implies $D_A(z) \leq C\varepsilon$, when the deficits are sufficiently small. Here the phases are evaluated at the ordinate of the point.

Proof. The converse assertions are Lemmas 8.2 and 8.6. Critical-root realization, including its quadratic real-part estimate, is Lemma 8.3. For completeness, the passage from a phase defect to the other local root uses the estimates already established in the proof of Theorem 8.5. In (86) take $x = D^{-1/(E_*-1)}$ to obtain $D_A \ll \Psi(D)$. The inequality $2A_* - E_* > 1$, certified in Lemma 8.9, gives $\|Z_z - Z_{z^*}\|_{C^1} = O(\sqrt{D})$ on a fixed small disk about A_*. At its center, $\operatorname{Re}(Z_z(A_*) - 1) = O(D_A)$, so Taylor's formula gives

$$u_A = -\frac{Z_z(A_*) - 1}{a_A} + O(D), \qquad |u_A| \ll \sqrt{D}, \qquad |\operatorname{Re} u_A| \ll D_A + D \ll \Psi(D).$$

The two global boundary inequalities give the signs. Choose the fixed disks to contain only the simple frozen roots. The same C^1 estimates at E_* and A_* imply uniform convergence to these frozen germs as $D \to 0$; Rouché's theorem gives exactly one root in each disk, and convergence of their derivatives proves simplicity. Thus all the local estimates have the same fixed-disk interpretation used below. □

Proposition 20.2 (Infinite phase regions)**.** *Let $U \subset \Omega$ be open. The lower density of simple critical-line zero ordinates whose full phase vector belongs to U is at least $c_{\mathrm{MT}}\mu(U)$, relative to $N(T, 2T)$. For a continuity set of product Haar measure, the density of all zero ordinates is its Haar measure.*

Proof. Finite cylinder functions are uniformly dense in $C(\Omega)$. The finite-dimensional all-zero equidistribution thus implies weak convergence of the full all-zero phase measures to μ. Every open set is the increasing union of finite unions of open cylinders with Haar-null boundaries, up to an arbitrarily small loss of measure. Apply Theorem 13.1 to each such finite union and then use inner regularity. The assertion for all zeros follows from weak convergence on continuity sets. □

Corollary 20.3 (Euler-product patterns along simple critical zeros)**.** *Let $\omega \in \Omega$, let K be a compact subset of $\operatorname{Re} s > 1$, let $m \geq 0$ be an integer and let $\varepsilon > 0$. Then a positive lower proportion, relative to all zeros in $(T, 2T]$, of simple critical-line zero ordinates γ satisfy*

$$\max_{0 \leq j \leq m} \sup_{s \in K} |\zeta^{(j)}(s + i\gamma) - Z_\omega^{(j)}(s)| < \varepsilon.$$

More precisely, every weak limit ν of the empirical subprobability measures of simple critical-line prime phases satisfies $c_{\mathrm{MT}}\mu \leq \nu \leq \mu$ on Ω.

Proof. For nonnegative continuous finite-cylinder functions, the lower integral inequality follows from Theorem 13.1 by finite step approximations on Haar-continuity rectangles. Uniform approximation extends it to every nonnegative function in $C(\Omega)$. The upper inequality follows because the corresponding all-zero phase measures converge to μ and dominate the simple-critical submeasures. This proves the assertion about weak limits.

Normal convergence of the Euler products and their derivatives makes the map $z \mapsto (Z_z^{(j)}|_K)_{j \leq m}$ continuous in the product topology. The displayed approximation condition defines an open neighborhood of ω, hence has positive Haar measure. Proposition 20.2 supplies the claimed lower proportion. □

The target class in Corollary 20.3 consists of unimodular Euler-product twists in the half-plane of absolute convergence. This is distinct from approximation of arbitrary holomorphic targets in a Voronin universality strip.

Theorem 20.4 (Critical-line zeros with two edge companions)**.** *Put $B_\delta = \{z : D_E(z) < \delta\}$ and let $M_\delta(T)$ count simple critical-line zeros with $T < \gamma \leq 2T$ and $(p^{-i\gamma})_p \in B_\delta$. For every fixed sufficiently small $\delta > 0$,*

$$c_{\mathrm{MT}}\mu(B_\delta) \leq \liminf_{T \to \infty} \frac{M_\delta(T)}{N(T, 2T)} \leq \limsup_{T \to \infty} \frac{M_\delta(T)}{N(T, 2T)} \leq \mu(B_\delta). \tag{151}$$

Every counted ordinate has companions w_E, w_A satisfying

$$\zeta'(w_E) = 0, \quad \zeta(w_A) = 1,$$

$$0 < E_* - \operatorname{Re} w_E \leq C\delta, \qquad 0 < A_* - \operatorname{Re} w_A \leq C\Psi(\delta), \qquad |\operatorname{Im} w_E - \gamma| + |\operatorname{Im} w_A - \gamma| \leq C\sqrt{\delta}.$$

Both companion roots are simple. Moreover, for some $c, C > 0$,

$$\exp\left(-\frac{C\delta^{-\beta_E}}{\log(1/\delta)}\right) \leq \mu(B_\delta) \leq \exp\left(-\frac{c\delta^{-\beta_E}}{\log(1/\delta)}\right). \tag{152}$$

Proof. The continuous energy D_E is a uniformly convergent positive series. Its distribution has no atoms: condition on every coordinate except one, whose nonconstant cosine energy has an atomless distribution. Thus B_δ is a continuity set, and Proposition 20.2 proves (151). For $z_p = p^{-i\gamma}$ one has $Z_z(s) = \zeta(s + i\gamma)$. Lemma 20.1 gives the companions. Equality at either boundary would force the exact frozen configuration, impossible for an ordinary prime flow: the phases at 3 and 5 force $\gamma = 0$, while the phase at 2 then has the wrong sign.

For completeness we give the volume estimate. Write $x = K\delta^{-\beta_E}$, with a sufficiently large fixed K. The tail energy beyond x is at most $4\sum_{p>x}(\log p)p^{-E_*} \ll x^{1-E_*}$. For $p \le x$ restrict the angular distance to the frozen phase by

$$|\theta_p| \le c_0(p/x)^{E_*/2}(\log x/\log p)^{1/2}.$$

For small fixed c_0 this box has total energy less than δ. Its negative logarithmic measure is $O(x/\log x)$, since

$$\sum_{p\le x}\log(x/p) = \int_2^x \frac{\pi(u)}{u}\,du = O(x/\log x), \qquad \pi(x) = O(x/\log x).$$

The logarithm-of-logarithm term only reduces this upper bound. This proves the lower volume bound.

For the upper bound choose instead a sufficiently small fixed multiple x of $\delta^{-\beta_E}$ and keep primes in $[x, 2x]$. Their number is $m \asymp x/\log x$, each weight is comparable to $x^{-E_*}\log x$, and their expected total energy is a sufficiently large multiple of δ. The independent variables $1 - \cos\theta_p$ have a fixed nondegenerate distribution. Their Laplace transform at a fixed positive argument is strictly less than one. Exponential Markov therefore bounds the probability of energy below δ by e^{-cm}, proving the other inequality. □

Taking a diagonal sequence $\delta_j \downarrow 0$ gives triples with ordinates tending to infinity, with both ordinate displacements tending to zero and both real parts approaching their boundaries. Different critical-line zeros need not have different companions. The limits in (151) always hold for fixed δ; no asserted error term allows substitution of an arbitrary shrinking function $\delta(T)$.

For the growing-prime and probabilistic sections below, abbreviate $E = E_*$ and $A = A_*$. The local maps u_E, u_A always mean the unique simple roots in the fixed disks specified in Lemma 20.1.

21 A simultaneous height law from a growing set of primes

The fixed-dimensional theorem does not by itself supply an error rate for an infinite Euler product. We now let the number of controlled primes grow, keeping an explicit account of the Fourier cost. This produces an actual joint limit in the height and the edge tolerance. Write $\ell = \log T$ and $v = \log\ell$ throughout this section.

Lemma 21.1 (A dimension-independent all-zero estimate)**.** *Let $F(z) = \sum c_{\mathbf{n}} z^{\mathbf{n}}$ be a finite Fourier polynomial on a prime torus, $m = c_{\mathbf{0}}$, and suppose $|c_{\mathbf{n}}| \le m$. Put*

$$A_0 = \sum |c_{\mathbf{n}}|, \qquad A_2 = \sum |c_{\mathbf{n}}| \left(\sum_p n_p \log p\right)^2.$$

If all associated reduced rationals have numerator and denominator at most H, and $\log H \le \ell/16$, then

$$\sum_{T<\gamma\le 2T} F(F_S(\gamma)) = mN + O\left(\frac{mNB_S}{\ell} + \frac{NA_2}{\ell^2} + A_0H^C\ell^2\right).$$

Zeros are counted with multiplicity. The absolute exponent C and the implied constant are independent of S.

Proof. For a reduced rational $q > 1$ with numerator and denominator at most H, the uniform Landau formula of [18, Lemma 1] gives

$$\sum_{T<\gamma\le 2T} q^{\rho-1/2} = -\frac{T}{2\pi}\frac{\Lambda(q)}{\sqrt{q}} + O(H^C\ell^2),$$

where $\Lambda(q) = 0$ unless q is an integral prime power. Indeed, if q is not that nearest prime power, rational separation bounds the extra oscillatory term by a fixed power of H; also $\log q \gg H^{-1}$. Reflection $\beta \mapsto 1-\beta$ pairs the terms with the same ordinate. If $\lambda = \log q$, their difference from $q^{i\gamma}$ is controlled by $\cosh(\lambda(\beta-1/2)) - 1$. The uniform Selberg bound $N(1/2+u, 3T) \ll Ne^{-u\ell/4}$ and integration by parts give

$$\sum_{T<\gamma\le 2T} |\beta-1/2|^2 e^{|\lambda||\beta-1/2|} \ll N\ell^{-2}, \qquad |\lambda| \le \ell/16.$$

Thus replacing $q^{\rho-1/2}$ by $q^{i\gamma}$ costs $O(N\lambda^2/\ell^2)$. For $q < 1$ take the complex conjugate of the corresponding formula for q^{-1}. Sum over characters. The prime-power main terms have total modulus at most $2mTB_S$; this proves the assertion. □

Theorem 21.2 (Growing-prime phase boxes)**.** *Fix $\sigma > 1$ and a sufficiently small $b > 0$. There exists $c_0 > 0$ such that the following holds. Put $x = c_0 \log\log T$, $S_x = \{p : p \le x\}$, and*

$$r_p = b(p/x)^{\sigma/2}\sqrt{\frac{\log x}{\log p}}.$$

For arbitrary centers $\omega_p \in \mathbb{T}$, let B_T be the product of the angular arcs $|\arg(z_p/\omega_p)| < r_p$. Uniformly in these centers,

$$N_{B_T} = (1+o(1))\mu(B_T)N, \qquad N^s_{0,B_T} \ge (c_f - o(1))\mu(B_T)N,$$

where $c_f = 2 - R_f(1) > 0$ for any one fixed smooth test window with this property. In particular a fixed positive constant is available. Moreover

$$\exp(-C_2 x/\log x) \le \mu(B_T) \le \exp(-C_1 x/\log x).$$

The constants and c_0 may depend on the fixed parameters and test window. No optimality of the height scale is asserted.

Proof. For large x all radii are less than a fixed small constant, and $r_{\min} \ge x^{-C}$. The prime number theorem and partial summation give

$$\sum_{p\le x} \log(x/p) \ll x/\log x, \qquad \sum_{p\le x} \log(\log x/\log p) \ll x/\log x.$$

Together with $\pi(x) \asymp x/\log x$, these prove the stated Haar-volume bounds (the lower bound also follows by dropping the favorable square-root factor).

Let $\eta = x^{-2}$. Construct smooth product cutoffs K^- and K^+ supported in the box and in its relative $O(\eta)$ enlargement, respectively, such that $0 \le K^- \le \mathbf{1}_{B_T} \le K^+$ and each cutoff changes only across a relative boundary layer of width η. In particular, choose $K^- = 1$ on the box with radii

$(1-3\eta)r_p$ and $\operatorname{supp} K^- \subset \{|\arg(z_p/\omega_p)| < (1-2\eta)r_p\}$; this leaves a genuine buffer to the target boundary. For each coordinate use a rescaled fixed smooth transition. Then

$$\int K^{\pm} = (1+o(1))\mu(B_T),$$

since $\pi(x)\eta = o(1)$. Every fixed-order derivative norm of an individual factor is bounded by a fixed power of x. Fourier coefficient estimates of order four, followed by multiplication over the coordinates, imply

$$\sum |\widehat{K^{\pm}}(\mathbf{n})| + \sum |\widehat{K^{\pm}}(\mathbf{n})| \left(\sum n_p \log p\right)^2 \le e^{Cx}.$$

The constants are independent of the centers. For example, apply the one-dimensional weighted ℓ^1 bounds to one or two coordinates and the unweighted bound to the other coordinates; the factor $\pi(x)^2 \log^2 x$ is absorbed in e^{Cx}.

Truncate each coordinate at degree $K = \lceil e^{C'x} \rceil$. Using a sufficiently high fixed derivative order and then choosing C' large makes the uniform truncation error at most $e^{-C''x}$, with C'' as large as required. This is $o(\mu(B_T))$. The retained coefficients obey $|c_{\mathbf{n}}| \le \int K^{\pm}$, since the original cutoffs are nonnegative. Their rational heights satisfy

$$\log H \le K \sum_{p \le x} \log p \le e^{C_3 x}.$$

Choose c_0 so small that $C_3 c_0 < 1/2$ and all the constants in the Fourier second-moment bounds satisfy $Cc_0 < 1$. Lemma 21.1 then gives relative all-zero asymptotics for $K^{\pm}$, and hence for the box and all the thin enlargements used here: $A_2/\ell^2 = o(\mu(B_T))$, while $H^C = T^{o(1)}$ and $B_{S_x} \ll \sqrt{x}$.

For the Weil matrix choose

$$L = \ell - 2\log H - C_4 x - D\log \ell,$$

with fixed large C_4, D. Then $L/\ell \to 1$. Proposition 17.1, applied to the Fourier truncation of K^-, gives

$$\operatorname{tr}(D_{K^-} G D_{K^-}) = (1+o(1))\mu(B_T)N, \|D_{K^-} G D_{K^-}\|_{\mathrm{HS}}^2 \le (R_f(1)+o(1))\mu(B_T)N.$$

The cutoff itself, rather than its Fourier approximation, is used in the coordinate matrix. The uniform approximation error changes these estimates by $o(\mu(B_T)N)$, using the unweighted norm bound. The factors e^{Cx} in the arithmetic errors are absorbed by C_4 and D.

For clarity, the zero-side localization also remains uniform. The prime-phase map has Lipschitz constant $C_{S_x} \ll \sqrt{x \log x}$. The inner cutoff and the boundary of B_T are separated in physical height by

$$r_* \gg \frac{\eta r_{\min}}{C_{S_x}} \ge x^{-C_5}.$$

Retain every zero with $|\beta - 1/2| > B\log \ell/\ell$, and every shallow zero whose phase lies in B_T. The former have total multiplicity $O(N\ell^{-B/4})$, hence $o(\mu(B_T)N)$, because $\mu(B_T) = \ell^{-o(1)}$. For the discarded shallow zeros, repeated integration by parts in the compactly supported test window gives a coordinate bound

$$|u_\rho(k)| \ll_M \ell^{B/2+1}(1+L|\gamma-\alpha_k|)^{-M}.$$

The unit-interval zero count and Schur's test therefore give arbitrarily large fixed power savings in ℓ, by choosing a sufficiently large fixed M: here $Lr_* \ge \ell x^{-C_5}$. Their trace and squared Hilbert–Schmidt errors are $o(\mu(B_T)N)$. The outer height tail is subtracted exactly, as in the localization proof, and has trace norm $o(1)$. The rank–inertia inequality, with the all-zero count of B_T as budget, now gives the stated simple-zero lower bound. □

Write $D_E(t) = \sum_p (\log p) p^{-E} |p^{-it} - z_p^*|^2$, where $z_2^* = -1$ and $z_p^* = 1$ for odd p. The constants E, A and the local root estimates are established in Section 20.

Theorem 21.3 (Simultaneous shrinking Euler-edge triples)**.** *There are positive constants C_0, C_1, c_1 with the following property. Put $\delta_T = C_0(\log\log T)^{1-E}$ and let $M(T)$ count simple critical-line zero ordinates $\gamma \in (T, 2T]$ with $D_E(\gamma) \le \delta_T$. Then, for every sufficiently large T,*

$$N \exp\left(-C_1 \frac{\log\log T}{\log\log\log T}\right) \le M(T) \le N \exp\left(-c_1 \frac{\log\log T}{\log\log\log T}\right). \tag{153}$$

Each counted ordinate has simple companions w_E, w_A satisfying $\zeta'(w_E) = 0$ and $\zeta(w_A) = 1$, with

$$\begin{aligned} 0 < E - \operatorname{Re} w_E &\ll (\log\log T)^{1-E}, \\ 0 < A - \operatorname{Re} w_A &\ll (\log\log T)^{1-A}/\log\log\log T, \\ |\operatorname{Im} w_E - \gamma| + |\operatorname{Im} w_A - \gamma| &\ll (\log\log T)^{(1-E)/2}. \end{aligned}$$

Different ordinates are not asserted to have distinct companions. This is a simultaneous limit, not an iterated fixed-tolerance limit.

Proof. Use Theorem 21.2 with $\sigma = E$ and frozen centers. Inside its box,

$$\sum_{p \le x} (\log p) p^{-E} |z_p - z_p^*|^2 \le b^2 x^{-E} \log x \, \pi(x) \ll x^{1-E}.$$

The remaining tail is at most $4\sum_{p>x} (\log p) p^{-E} \ll x^{1-E}$. Taking C_0 large enough proves the lower count and all companion claims by the local edge lemma.

We prove the upper count, so that the asserted rarity has the same logarithmic order in this joint limit. Put $y = a\delta_T^{-1/(E-1)}$, where $a > 0$ is a sufficiently small fixed number, and restrict energy to primes in $[y, 2y]$. Choose $s = a_1 y^E / \log y$ with fixed $a_1 > 0$ and let

$$K(z) = \exp\left\{ -s \sum_{y \le p \le 2y} (\log p) p^{-E} |z_p - z_p^*|^2 \right\}.$$

If $h(u) = 2u - \log I_0(2u)$, then $m = \int K = \exp(-\sum h(s(\log p)p^{-E}))$. Each argument of h lies in a fixed positive compact interval; hence $m \le \exp(-cy/\log y)$. Choose a small enough that $s\delta_T \le (c/2) y/\log y$.

The Bessel Fourier expansion of each factor has nonnegative coefficients before rotation by z_p^*. Consequently its absolute Fourier coefficients form a probability distribution, and

$$A_0 = 1, \qquad A_2 = 2s \sum_{y \le p \le 2y} (\log p)^3 p^{-E} \ll y \log y.$$

Truncate each coordinate at degree $\lceil y^2 \rceil$. The factorial tail bound for $I_n(u)$, uniform for the bounded arguments here, makes the uniform error $\exp(-c' y^2 \log y)$. The rational height satisfies $\log H \ll y^3 = o(\ell)$. Lemma 21.1 therefore gives

$$\sum_{T < \gamma \le 2T} K(F(\gamma)) \le (1 + o(1)) mN + O(Ny \log y/\ell^2) + T^{o(1)}.$$

Multiply by $e^{s\delta_T}$ and use the pointwise exponential majorant for the event $D_E \le \delta_T$. Since $y \asymp \log\log T$, all the error terms are smaller than $N \exp(-c'' y/\log y)$. This bounds even the count of all zeros with the energy condition and proves the upper bound for $M(T)$. □

Remark 21.4. Equation (153) counts zeros selected by the stated phase-energy condition. It does not count every possible triple with comparable displacements, and it does not assert that the displayed approach rate is the fastest attainable one.

22 Bounded-density transfer and conditioned phase laws

The preceding localization theorem does not identify a unique phase distribution for simple critical-line zeros. Its two-sided domination nevertheless determines every diverging logarithmic probability scale. It also transports concentration statements under arbitrarily rare conditioning. The elementary distinction between these conclusions and convergence to a nondegenerate law is useful here.

Theorem 22.1 (Bounded-density transfer)**.** *Let μ be a probability measure on a measurable space Ω, let $0 < c < 1$, and let ν be a finite measure satisfying $c\mu \le \nu \le \mu$. Then the following statements hold uniformly over all such ν.*

(i) *For every nonnegative measurable F,*

$$c \int F \, d\mu \le \int F \, d\nu \le \int F \, d\mu.$$

In particular, whenever the integrals are positive and finite,

$$\left| \log \int F \, d\nu - \log \int F \, d\mu \right| \le |\log c|.$$

(ii) *Let $W \ge 0$ with $0 < \int W \, d\mu < \infty$, and normalize $W\mu$ and $W\nu$ to probability measures μ_W and ν_W. For a measurable event A, writing $p = \mu_W(A)$, one has*

$$\frac{cp}{1-(1-c)p} \le \nu_W(A) \le \frac{p}{c+(1-c)p}. \tag{154}$$

Both bounds are sharp. In particular, $c\mu_W(A) \le \nu_W(A) \le c^{-1}\mu_W(A)$.

(iii) *With total variation defined by $\mathrm{TV}(P,Q) = \sup_A |P(A) - Q(A)|$, one has the sharp bound*

$$\mathrm{TV}(\nu_W, \mu_W) \le \frac{1-\sqrt{c}}{1+\sqrt{c}}. \tag{155}$$

Moreover, putting $a_c = -c \log c/(1-c)$,

$$D_{\mathrm{KL}}(\nu_W \| \mu_W) \le a_c - 1 - \log a_c, \tag{156}$$

again with equality possible.

Proof. The Radon–Nikodym theorem gives $d\nu = h \, d\mu$ with $c \le h \le 1$ almost everywhere, proving the first assertion. For the second, put $a = \int_A h \, d\mu_W$ and $b = \int_{A^c} h \, d\mu_W$. Then $cp \le a \le p$, $c(1-p) \le b \le 1-p$, and $\nu_W(A) = a/(a+b)$. Monotonicity in a, b proves (154), and the endpoint-valued choices of h attain the bounds.

Write $\alpha = \int h \, d\mu_W \in [c, 1]$. The secant-line bound for the convex function $|x - \alpha|$ on $[c, 1]$ gives

$$\mathrm{TV}(\nu_W, \mu_W) = \frac{1}{2\alpha} \int |h - \alpha| \, d\mu_W \le \frac{(\alpha - c)(1-\alpha)}{\alpha(1-c)}.$$

The last expression is maximal at $\alpha = \sqrt{c}$ and gives (155). Equality holds when h only takes the values $c, 1$ and has this mean. Likewise, the secant-line bound for $x \log x$ yields

$$D_{\mathrm{KL}}(\nu_W \| \mu_W) \le -\log \alpha + \frac{c(1-\alpha)\log c}{\alpha(1-c)}.$$

This is maximal at $\alpha = a_c$, where its value is $a_c - 1 - \log a_c$. An endpoint-valued density with mean a_c attains equality. □

Corollary 22.2 (Logarithmic laws and concentration)**.** *Suppose $a_r \to \infty$. Under the hypotheses of Theorem 22.1, any limit*

$$\frac{1}{a_r} \log \int e^{V_r}\, d\mu$$

is unchanged when μ is replaced by ν. If $-\log \mu(A_r) \sim I a_r$ for $I > 0$, then $-\log \nu(A_r) \sim I a_r$ uniformly over admissible ν. More generally any large-deviation principle for measurable maps $X_r : \Omega \to E$ is preserved, after normalizing ν if needed.

The same conclusions hold conditionally, with μ_{W_r} and ν_{W_r} in place of μ and ν. In particular, if $X_r - x_r \to 0$ in μ_{W_r}-probability for deterministic x_r, then the convergence holds in ν_{W_r}-probability uniformly over admissible ν.

Proof. The logarithms in the first statements differ by at most a constant. After normalization or conditioning, the comparison factors are between c and c^{-1}. Division by a_r removes these constants in the open- and closed-set bounds defining a large-deviation principle. For concentration, apply the upper bound to each deviation event. □

Remark 22.3 (What the transfer does not determine)*.* Bounded-density comparison does not preserve a nondegenerate weak limit. For example, let $\Omega = [0,1]^2$ with Lebesgue measure, condition on $0 < x < \varepsilon$, and set $h = c$ for $y < 1/2$ and $h = 1$ for $y \geq 1/2$. The conditional y-law remains uniform under μ, but has two different densities under $h\mu$ for every $\varepsilon > 0$. Thus concentration at a deterministic profile transfers; a Gaussian fluctuation law requires additional information about the density.

For the application, let $\Omega = \prod_p \mathbb{T}$ with Haar measure μ and define

$$\eta_T = \frac{1}{N(T, 2T)} \sum_{\substack{T < \gamma \leq 2T \\ \zeta(1/2+i\gamma)=0,\ \zeta'(1/2+i\gamma)\neq 0}} \delta_{(p^{-i\gamma})_p}.$$

The fixed-box case of Theorem 18.1 gives $c_{\mathrm{MT}}\mu \leq \nu \leq \mu$ for every weak limit ν of η_T. To justify this on the full product, first use disjoint finite boxes to approximate nonnegative cylinder functions from below. The all-zero phase measures converge to Haar by the same box theorem. Cylinder functions are uniformly dense in $C(\Omega)$, so both inequalities pass to every weak limit on the compact product torus. The preceding theorem therefore applies with $c = c_{\mathrm{MT}}$. The sharp total-variation and relative-entropy bounds are, respectively, $0.0988640259\ldots$ and $0.0196336622\ldots$ nats. These are bounds for weak limits, including their conditional laws. They are not bounds for the finite atomic measures η_T in total variation.

Corollary 22.4 (Iterated rare-event limits)**.** *Let B_ε be Haar-continuity sets with positive measure, and suppose $-\log \mu(B_\varepsilon) \sim J(\varepsilon) \to \infty$. Put*

$$d_-(\varepsilon) = \liminf_{T\to\infty} \eta_T(B_\varepsilon), \qquad d_+(\varepsilon) = \limsup_{T\to\infty} \eta_T(B_\varepsilon).$$

Then

$$-\log d_-(\varepsilon) \sim -\log d_+(\varepsilon) \sim J(\varepsilon).$$

If $A_\varepsilon \cap B_\varepsilon$ are also Haar-continuity sets and $\mu(A_\varepsilon \mid B_\varepsilon) \to 0$, then

$$\lim_{\varepsilon\downarrow 0} \limsup_{T\to\infty} \frac{\eta_T(A_\varepsilon \cap B_\varepsilon)}{\eta_T(B_\varepsilon)} = 0.$$

Proof. Compactness of Ω gives subsequential weak limits of η_T. Their density bounds make every Haar-continuity set a continuity set for each such limit. Hence $c_{\mathrm{MT}}\mu(B_\varepsilon) \le d_-(\varepsilon) \le d_+(\varepsilon) \le \mu(B_\varepsilon)$, proving the first assertion. Along any weakly convergent subsequence the conditional ratio tends to $\nu(A_\varepsilon \cap B_\varepsilon)/\nu(B_\varepsilon)$, whose denominator is positive. Apply Theorem 22.1(ii) and then let $\varepsilon \downarrow 0$. □

All these statements take $T \to \infty$ before $\varepsilon \downarrow 0$. They do not supply estimates for a prescribed shrinking function $\varepsilon(T)$.

Finally, the Euler-product map $\omega \mapsto Z_\omega$ is a continuous injection from the compact space Ω into the space of holomorphic functions on $\operatorname{Re} s > 1$ with the compact-open topology. Injectivity follows by uniqueness of the absolutely convergent Dirichlet coefficients. It is consequently a homeomorphism onto its compact image. Pushing the measure inequalities forward gives the same bounded-density transfer for the random Euler-product law and for all compact-derivative observables. Robust root configurations are included by Rouché's theorem. The target class remains the unimodular Euler-product family in the half-plane of absolute convergence.

23 Exact phase entropy and a conditional law for the two Euler edges

Write $V(\theta) = 2(1 - \cos\theta)$ and introduce the real analytic function

$$h(u) = -\log\left(\frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-uV(\theta)}\,d\theta\right) = 2u - \log I_0(2u), \qquad u \in \mathbb{R}.$$

For $\sigma > 1$ set

$$H_\sigma = \int_0^\infty h(y^{-\sigma})\,dy, \qquad C_\sigma = \frac{H_\sigma}{\sigma} = \int_0^\infty y^{-\sigma} h'(y^{-\sigma})\,dy. \tag{157}$$

The equality follows by integration by parts. These constants are finite and strictly positive. Indeed, $h(u) = 2u + O(u^2)$ as $u \to 0$, whereas $h(u) = \frac{1}{2}\log(4\pi u) + O(u^{-1})$ as $u \to +\infty$. The method below is the classical exponential-tilting method for small deviations of weighted independent variables; the one-variable asymptotic is also a specialization of Aurzada [23], using $p_n \sim n \log n$. We include the proof to fix the integral constant and the tilt used in the joint-edge results; no new general small-deviation theorem is claimed.

Theorem 23.1 (Prime-weight small-ball specialization)**.** *Let $a > 0$, $\sigma > 1$, $\kappa \in \mathbb{R}$, and let independent uniform angles θ_p be indexed by the primes. Suppose $w_p > 0$ and $w_p \sim a p^{-\sigma}(\log p)^\kappa$ and put $D_w = \sum_p w_p V(\theta_p)$. With $\beta = (\sigma - 1)^{-1}$,*

$$-\log \mu(D_w \le \varepsilon) \sim K_{\sigma,\kappa,a}\, \varepsilon^{-\beta}[\log(1/\varepsilon)]^{(\kappa-1)\beta-1}, \tag{158}$$

where

$$K_{\sigma,\kappa,a} = (\sigma-1)^2 C_\sigma (a C_\sigma)^\beta \beta^{(\kappa-1)\beta}.$$

Changing or deleting finitely many positive weights does not change this constant.

Proof. Let $F(s) = -\log \mathbb{E}e^{-sD_w} = \sum_p h(sw_p)$, and choose $x = x(s) \to \infty$ by $sax^{-\sigma}(\log x)^\kappa = 1$. The prime number theorem and partial summation give

$$F(s) \sim H_\sigma \frac{x}{\log x}, \quad sF'(s) \sim C_\sigma \frac{x}{\log x}, \quad -s^2F''(s) \asymp \frac{x}{\log x}. \tag{159}$$

Here is a justification of the improper-integral passage. On $\eta x \le p \le Mx$, the ratio sw_p converges uniformly to $(p/x)^{-\sigma}$, and the prime number theorem applies to continuous functions on $[\eta, M]$.

For $p > Mx$ use $h(u) \le 2u$ and partial summation; the resulting normalized tail is $O(M^{1-\sigma+\xi})$ for any fixed $0 < \xi < \sigma - 1$. For $p < \eta x$, use $h(u) \ll 1 + \log(1+u)$; primes below x^c, with fixed $c < 1$, contribute $o(x/\log x)$, and the remaining normalized contribution is $O(\eta^{1-\xi}(1 + |\log \eta|))$. The same argument applies to $uh'(u)$ and $-u^2h''(u)$, which are respectively $O(\min(u, 1))$ and $O(\min(u^2, 1))$ for $u > 0$. The second-derivative integral is positive. Finite initial terms are $O(\log s)$ in the first sum and bounded in the other two.

Under the tilted probability $d\mu_s = e^{-sD_w+F(s)}d\mu$, the mean and variance of D_w are $F'(s)$ and $-F''(s)$. For the upper probability bound choose $F'(s) = \varepsilon$:

$$\mu(D_w \le \varepsilon) \le e^{s\varepsilon - F(s)}.$$

For the matching logarithmic lower bound choose a nearby tilt with mean $(1 - \eta_s)\varepsilon$, where $\eta_s \to 0$ and $\eta_s\sqrt{x/\log x} \to \infty$. Chebyshev's inequality gives

$$\mu_s((1 - 2\eta_s)\varepsilon \le D_w \le \varepsilon) = 1 - o(1).$$

Changing measure on this interval loses only $2\eta_s s\varepsilon = o(x/\log x)$ in the logarithm. Thus

$$-\log \mu(D_w \le \varepsilon) \sim (\sigma - 1)C_\sigma \frac{x}{\log x}, \qquad \varepsilon \sim aC_\sigma x^{1-\sigma}(\log x)^{\kappa-1}.$$

Solving the last relation gives $x \sim (aC_\sigma)^\beta \beta^{(\kappa-1)\beta}\varepsilon^{-\beta}[\log(1/\varepsilon)]^{(\kappa-1)\beta}$, which proves the stated constant. □

We now use $E = E_*$, $A = A_*$, $\beta = (E-1)^{-1}$, $\tau = (A-1)/(E-1)$, and $\Psi(\delta) = \delta^\tau/\log(1/\delta)$. The frozen phases are $z_2^* = -1$ and $z_p^* = 1$ for odd primes. Write $z_p = z_p^* e^{i\theta_p}$ and retain the definitions

$$D_E = \sum_p (\log p)p^{-E}V(\theta_p), \qquad D_A = \sum_p p^{-A}V(\theta_p), \qquad B_\delta = \{D_E \le \delta\}.$$

Define two explicit constants

$$\mathcal{K}_E = (E-1)^2 C_E^{E/(E-1)}, \qquad J_{E,A} = \int_0^\infty y^{-A}h'(y^{-E})\,dy, \qquad \mathcal{J}_{E,A} = (E-1)C_E^{-\tau}J_{E,A}. \tag{160}$$

The integral defining $J_{E,A}$ converges since $1 < A < E$.

Theorem 23.2 (Conditional phase-energy profile)**.** *As $\delta \downarrow 0$,*

$$-\log \mu(B_\delta) \sim \mathcal{K}_E \frac{\delta^{-\beta}}{\log(1/\delta)}. \tag{161}$$

Under the conditional law $\mu(\,\cdot \mid B_\delta)$,

$$\frac{D_E}{\delta} \longrightarrow 1, \qquad \frac{D_A}{\Psi(\delta)} \longrightarrow \mathcal{J}_{E,A} \quad \textit{in probability.} \tag{162}$$

For each fixed finite set of primes F,

$$\frac{1}{\delta}\sum_{p\in F}(\log p)p^{-E}V(\theta_p) \longrightarrow 0 \quad \textit{in conditional probability.} \tag{163}$$

The same conclusions hold, uniformly, for every finite measure ν satisfying $c\mu \le \nu \le \mu$, with any fixed $c > 0$.

Proof. The entropy formula is Theorem 23.1 with $(\sigma,\kappa,a)=(E,1,1)$. It also proves the first assertion in (162): for each $\eta>0$, the ratio $\mu(B_{(1-\eta)\delta})/\mu(B_\delta)$ tends to zero exponentially on the scale $\delta^{-\beta}/\log(1/\delta)$.

Choose a tilted law with $\mathbb{E}_s D_E=(1-\eta_s)\delta$, $\eta_s\to 0$, $\eta_s\sqrt{m}\to\infty$, where

$$s=x^E/\log x,\qquad m=x/\log x,\qquad \delta\sim C_E x^{1-E}.$$

Let $b_x=x^{1-A}/\log x$. For each real t in a fixed neighborhood of zero, prime-number-theorem summation gives

$$\frac{1}{m}\log \mathbb{E}_s \exp(tx^A D_A)\longrightarrow \Lambda(t):=\int_0^\infty \{h(y^{-E})-h(y^{-E}-ty^{-A})\}\,dy. \tag{164}$$

This integral is well defined also for $t>0$. At zero in y, y^{-E} dominates ty^{-A}, while at infinity both arguments are small and the integrand is $O(y^{-A})$. Its first two derivatives with respect to t are locally integrable uniformly near $t=0$: their behaviors near zero are bounded by constants times y^{E-A} and y^{2E-2A}, and at infinity by y^{-A} and y^{-2A}. The same bounds justify the prime-sum limits. Consequently $\Lambda'(0)=J_{E,A}$, and exponential Markov inequalities for positive and negative t imply, for every $\eta>0$ and some $c_\eta>0$,

$$\mu_s\big(|D_A/b_x-J_{E,A}|>\eta\big)\le e^{-c_\eta m}$$

for all sufficiently large x.

The change of measure used in the preceding proof implies that any event with μ_s-probability at most e^{-cm} has $\mu(\,\cdot\mid B_\delta)$-probability tending to zero: its conditional probability is at most $e^{o(m)}\mu_s(\text{event})$. Since

$$b_x\sim (E-1)C_E^{-\tau}\Psi(\delta),$$

this proves the second energy limit. For a fixed finite F put $D_F=\sum_{p\in F}(\log p)p^{-E}V(\theta_p)$. For fixed $0<q<1$,

$$\log \mathbb{E}_s e^{qsD_F}=\sum_{p\in F}\{h(sw_p)-h((1-q)sw_p)\}=O_F(1).$$

Thus $\mu_s(D_F>\eta\delta)\le\exp(-q\eta s\delta+O_F(1))$, and the same change of measure proves (163).

Finally, for every measurable event C,

$$\nu(C\mid B_\delta)\le c^{-1}\mu(C\mid B_\delta).$$

This transfers every vanishing exceptional probability, uniformly over the specified class of measures. It transfers concentration, not arbitrary nondegenerate conditional distribution laws. □

Theorem 23.3 (The phase profile of a rare Euler-edge configuration)**.** *Put* $x_\delta=(C_E/\delta)^{1/(E-1)}$. *For* $y>0$ *define the probability measure*

$$dq_y(\theta)=\exp\{-y^{-E}V(\theta)+h(y^{-E})\}\frac{d\theta}{2\pi}.$$

For every continuous g *with compact support in* $(0,\infty)\times\mathbb{T}$, *under* $\mu(\,\cdot\mid B_\delta)$ *one has*

$$\frac{\log x_\delta}{x_\delta}\sum_p g(p/x_\delta,\theta_p)\longrightarrow\int_0^\infty\int_{\mathbb{T}} g(y,\theta)\,dq_y(\theta)\,dy \tag{165}$$

in probability. Any finite family of such observables converges jointly. Both statements transfer uniformly to $c\mu\le\nu\le\mu$.

Moreover, q_y is the unique entropy-minimizing measurable family of probability measures ρ_y on the circle subject to

$$\int_0^\infty y^{-E} \int V \, d\rho_y \, dy \le C_E.$$

The minimum of $\int_0^\infty D(\rho_y \| d\theta/(2\pi))\, dy$ is $(E-1)C_E$, with uniqueness up to a set of y of measure zero.

Proof. Use the undershooting tilt from the preceding proof. On the compact interval of y supporting g, its angular marginal at the prime p converges uniformly to $q_{p/x}$. The prime number theorem gives the limit of the expected sum. There are $O(m)$ relevant independent terms, each bounded, so the exponential Markov bound (or the elementary bounded-variable exponential moment estimate) bounds any fixed deviation of their normalized sum by $e^{-c_g m}$. The previous change-of-measure argument transfers this to the conditioned law. Since $x/x_\delta \to 1$, uniform continuity of g replaces x by x_δ. A finite union bound proves joint convergence.

For the variational statement, the relative-entropy identity is

$$\int_0^\infty D(\rho_y \| d\theta/(2\pi))\, dy = \int_0^\infty D(\rho_y \| q_y)\, dy + H_E - \int_0^\infty y^{-E} \int V \, d\rho_y \, dy.$$

It holds first when the integrals are finite, by substitution of the displayed density, and then with the usual extended-integral interpretation since the energy and H_E are finite. The constraint and nonnegativity of relative entropy give the lower bound $H_E - C_E = (E-1)C_E$. The family q_y attains it by (157); equality forces the constraint to be saturated and $D(\rho_y \| q_y) = 0$ for almost every y. □

Theorem 23.4 (Typical shape of the two edge companions)**.** *Let $u_E(z)$ and $u_A(z)$ be the local simple roots defined by $Q_z(E+u_E)=0$ and $Z_z(A+u_A)=1$. Put $a_E = Q'_{z^*}(E) < 0$ and $a_A = Z'_{z^*}(A) < 0$. Under $\mu(\cdot \mid B_\delta)$, and uniformly under every $\nu(\cdot \mid B_\delta)$ with $c\mu \le \nu \le \mu$,*

$$\frac{-\operatorname{Re} u_E}{\delta} \longrightarrow \frac{1}{2|a_E|}, \qquad \frac{-\operatorname{Re} u_A}{\Psi(\delta)} \longrightarrow \frac{\mathcal{J}_{E,A}}{2|a_A|}, \qquad \frac{|\operatorname{Im} u_E| + |\operatorname{Im} u_A|}{\sqrt{\delta}} \longrightarrow 0 \tag{166}$$

in probability.

Proof. We give the additional estimates that replace the earlier worst-case inequalities by exact conditional constants. Fix a small disk about E, of radius r with $E - 2r > 1$. For its C^1 norm, Cauchy–Schwarz and a split at a fixed prime bound P give

$$\|Q_z - Q_{z^*}\|_{C^1}^2 \le C_P D_{E,\le P} + \epsilon_P D_E, \qquad \epsilon_P \longrightarrow 0.$$

For the derivative, for example, the tail constant is controlled by

$$\sum_{p>P} (\log p)^3 p^{-E+2r}.$$

Theorem 23.2, first with δ tending to zero and then with P tending to infinity, yields

$$\|Q_z - Q_{z^*}\|_{C^1}^2 = o_{\mathrm{prob}}(\delta). \tag{167}$$

For odd primes, expansion into prime powers gives

$$\operatorname{Re}\left\{ \frac{(\log p) z_p p^{-E}}{1 - z_p p^{-E}} - \frac{(\log p) p^{-E}}{1 - p^{-E}} \right\} = -\tfrac{1}{2}(\log p) p^{-E} V(\theta_p) + O((\log p) p^{-2E} V(\theta_p)).$$

Here $1-\cos(k\theta) \le k^2(1-\cos\theta)$ bounds the remainder. The prime 2 is included in the finite-prime error. Splitting the remainder at P and using (163) shows

$$\operatorname{Re} Q_z(E) = -D_E/2 + o_{\mathrm{prob}}(\delta).$$

The quantitative implicit-function expansion $u_E = -Q_z(E)/a_E + O(\|Q_z - Q_{z^*}\|_{C^1}^2)$ proves the first limit and the improved imaginary bound in (166).

For the other edge, define the normally convergent logarithm $L_z = \log Z_z(A) - \log Z_{z^*}(A)$, where $Z_{z^*}(A) = 1$. The same prime-power calculation gives

$$\operatorname{Re} L_z = -D_A/2 + O(D_{A,\le P} + P^{-A} D_A).$$

For fixed P, $D_{A,\le P} \ll_P D_E \le \delta = o(\Psi(\delta))$. Choose a fixed disk radius $r > 0$ with $2A - E - 2r > 1$ and $A - r > 1$. The strict inequality $2A - E > 1$ gives $|L_z|^2 \ll D_E$, and also a C^1 bound $\|Z_z - Z_{z^*}\|_{C^1}^2 \ll D_E$ on this disk. In fact the same fixed-prime/tail splitting improves this to $o_{\mathrm{prob}}(\delta)$: the tail coefficient for the derivative is bounded by a constant times $\sum_{p>P} (\log p) p^{E-2A+2r}$, which tends to zero. It follows also that $u_A = o_{\mathrm{prob}}(\sqrt{\delta})$. Furthermore,

$$\operatorname{Re}(Z_z(A) - 1) = -D_A/2 + o_{\mathrm{prob}}(\Psi(\delta)), \qquad u_A = -\frac{Z_z(A) - 1}{a_A} + O(\delta).$$

The second energy limit proves the asserted constant. Uniform transfer to ν follows from the conditional measure inequality. □

Corollary 23.5 (Exact logarithmic rarity along simple critical zeros)**.** *The fixed-prime phase-domination theorem proved above implies that every weak limit ν of the simple-critical-zero phase measures obeys $c_{\mathrm{MT}}\mu \le \nu \le \mu$. Let $\underline{d}(\delta)$ and $\overline{d}(\delta)$ be the lower and upper densities, relative to all zeros, of simple critical-line ordinates whose phase lies in B_δ. Then both satisfy*

$$-\log \underline{d}(\delta) \sim -\log \overline{d}(\delta) \sim \mathcal{K}_E \frac{\delta^{-\beta}}{\log(1/\delta)}.$$

Moreover, for every fixed tolerance $\eta > 0$, the upper relative proportion of exceptions to the two real-part limits in (166), among those selected simple zeros, tends to zero as $\delta \downarrow 0$, with the height limit taken first. The analogous assertion holds for $|\operatorname{Im} u_E| + |\operatorname{Im} u_A| > \eta\sqrt{\delta}$.

Proof. The density sandwich $c_{\mathrm{MT}}\mu(B_\delta) \le \underline{d}(\delta) \le \overline{d}(\delta) \le \mu(B_\delta)$ gives the logarithmic asymptotics. For the relative assertions use compactness of the phase measures and their weak-limit domination. Equivalently, bound the closed exceptional set with deviation at least η inside B_δ by its Haar measure and divide by $c_{\mathrm{MT}}\mu(B_\delta)$. The local root maps are continuous on this compact set, so the upper bound follows from Portmanteau. The conditional concentration theorem makes this bound tend to zero. No rate connecting δ to the height is asserted. □

24 A joint lower-tail rate and three active-constraint regimes

The two energies need not be tied to the typical relation found above. One can instead prescribe both tolerances and determine their joint logarithmic cost. The inequality $E < A + 1$, which follows already from $E < 2.82$ and $A > 1.93$, is useful here.

Theorem 24.1 (Joint energy rate function)**.** *Suppose $1 < A < E < A + 1$. Under product Haar measure let*

$$D_E = \sum_p (\log p) p^{-E} V(\theta_p), \qquad D_A = \sum_p p^{-A} V(\theta_p),$$

and put

$$U_x = x^{E-1} D_E, \qquad W_x = x^{A-1} (\log x) D_A, \qquad m_x = x/\log x.$$

For every $e, a > 0$,

$$-\frac{1}{m_x} \log \mu(U_x \le e,\ W_x \le a) \longrightarrow \mathcal{I}(e,a) := \max_{s,t \ge 0} \{\mathcal{H}(s,t) - se - ta\}, \tag{168}$$

where

$$\mathcal{H}(s,t) = \int_0^\infty h(sy^{-E} + ty^{-A})\, dy.$$

The maximum has a unique nonzero optimizer (s_*, t_*). Write $J_{E,A} = \int_0^\infty y^{-A} h'(y^{-E}) dy$, $J_{A,E} = \int_0^\infty y^{-E} h'(y^{-A}) dy$, and $\tau = (A-1)/(E-1)$. Its three regimes are as follows:

1. *If $a \ge J_{E,A}(e/C_E)^\tau$, then*

$$t_* = 0, \quad s_* = (C_E/e)^{E/(E-1)}, \quad \mathcal{I}(e,a) = (E-1) C_E^{E/(E-1)} e^{-1/(E-1)}.$$

2. *If $e \ge J_{A,E}(a/C_A)^{1/\tau}$, then*

$$s_* = 0, \quad t_* = (C_A/a)^{A/(A-1)}, \quad \mathcal{I}(e,a) = (A-1) C_A^{A/(A-1)} a^{-1/(A-1)}.$$

3. *Otherwise $s_*, t_* > 0$ and they are uniquely determined by*

$$\partial_s \mathcal{H}(s_*, t_*) = e, \qquad \partial_t \mathcal{H}(s_*, t_*) = a.$$

The first two parameter regions are disjoint. In the third region the joint rate is strictly larger than each separate one-constraint rate. The same logarithmic limit holds with μ replaced by any finite measure ν satisfying $c\mu \le \nu \le \mu$.

Proof. For fixed $s, t \ge 0$, the prime number theorem gives

$$-\frac{1}{m_x} \log \mathbb{E} \exp\{-m_x(sU_x + tW_x)\} \longrightarrow \mathcal{H}(s,t). \tag{169}$$

Indeed, the argument of h at the prime p is $s(p/x)^{-E}(\log p/\log x) + t(p/x)^{-A}$. The small and large prime tails are controlled exactly as in the one-variable Laplace calculation. Exponential Markov immediately gives the lower bound for the limiting negative logarithm in (168).

For existence and uniqueness of the saddle, note that h is strictly concave on $\mathbb{R}$, and $h(u+v) \le h(u) + h(v)$ for $u, v \ge 0$. Thus $\mathcal{H}$ is strictly concave away from the origin and

$$\mathcal{H}(s,t) \le H_E s^{1/E} + H_A t^{1/A}.$$

The objective is consequently coercive on the quadrant. It increases from zero along either axis sufficiently near the origin, so its unique maximum is nonzero. The gradient exists at every nonzero point of the quadrant, including the axes: at $s = 0, t > 0$, the only potentially problematic integrand is $y^{-E} h'(ty^{-A}) = O(y^{A-E})$ near zero, which is integrable by $E < A + 1$. The other cases are easier.

To prove the probability lower bound, tilt the independent prime phases by the parameter $((1+\eta)s_*, (1+\eta)t_*)$, with $\eta > 0$. Under this law the means of U_x, W_x tend to $\nabla\mathcal{H}((1+\eta)s_*, (1+\eta)t_*)$, and both limits are strictly below e, a. Both normalized energies converge in probability to these means. To see this without assuming a global variance bound, first restrict to $\epsilon x \le p \le Mx$; there each summand is $O(1/m_x)$ and the sum has variance $O(1/m_x)$. The normalized expectations of the omitted tails are uniformly small as $\epsilon \downarrow 0$, $M \uparrow \infty$, by the integrability of the two gradient integrands. Markov's inequality controls these nonnegative tails.

On a small rectangle about the limiting tilted means, contained in $\{U_x < e, W_x < a\}$, the change-of-measure formula now supplies the matching logarithmic lower probability bound. Let the rectangle width tend to zero and then let $\eta \downarrow 0$. The complementary-slackness identities

$$s_*\partial_s\mathcal{H}(s_*, t_*) = s_*e, \qquad t_*\partial_t\mathcal{H}(s_*, t_*) = t_*a$$

show that the resulting exponent is exactly $\mathcal{H}(s_*, t_*) - s_*e - t_*a$.

On the $t = 0$ axis, differentiation of $\mathcal{H}(s, 0) = H_E s^{1/E}$ gives $s_* = (C_E/e)^{E/(E-1)}$, and the unused directional derivative is $J_{E,A}s_*^{(1-A)/E} - a$. Its nonpositivity is exactly the first displayed regime. The same calculation on the other axis gives the second regime. Strict concavity excludes overlap of the two regimes and implies that in the remaining case the maximizer lies in the interior. There the optimizer differs from either axis optimizer, proving the strict increase of the joint cost. Finally, comparison by the fixed factors c and 1 changes a logarithm by at most $|\log c| = o(m_x)$. □

Corollary 24.2 (Jointly constrained phase profile)**.** *Condition on $U_x \le e, W_x \le a$, and let (s_*, t_*) be the optimizer above. For every continuous compactly supported $g : (0, \infty) \times \mathbb{T} \to \mathbb{R}$,*

$$\frac{1}{m_x}\sum_p g(p/x, \theta_p) \longrightarrow \int_0^\infty \int g(y, \theta)e^{-(s_*y^{-E}+t_*y^{-A})V(\theta)+h(s_*y^{-E}+t_*y^{-A})}\frac{d\theta}{2\pi}\,dy$$

in conditional probability. This also holds uniformly for $c\mu \le \nu \le \mu$.

Proof. Under the tilt at the exact optimizer, the normalized compact-band sum has the stated limiting mean and exponentially small fixed deviations, by independence and boundedness of its $O(m_x)$ terms. On the conditioning event the Radon–Nikodym derivative returning to Haar measure is at most $\exp(m_x(s_*e + t_*a) - F_x(s_*, t_*))$. The joint probability formula just proved equals this exponential up to $\exp(o(m_x))$. Thus exponential concentration survives conditioning. The inequality $\nu(C \mid B) \le c^{-1}\mu(C \mid B)$ completes the transfer. □

This theorem identifies when demanding additional 1-point phase coherence has no leading exponential cost, when it determines the entire cost, and when it changes the entropy-minimizing prime-phase profile. It concerns the two phase energies; it does not by itself give a local central limit theorem for the companion roots.

Corollary 24.3 (A single normalized crossover function)**.** *Put $\beta = 1/(E-1)$ and $\Phi(q) = \mathcal{I}(1, q)$. Then*

$$\mathcal{I}(e, a) = e^{-\beta}\Phi(a/e^\tau).$$

In the normalized variable $q = a/e^\tau$, the three regimes are separated by two fixed constants

$$q_- = C_A J_{A,E}^{-\tau} < q_+ = J_{E,A}C_E^{-\tau}.$$

The A-energy condition alone is active for $q \le q_-$, both conditions are active for $q_- < q < q_+$, and the E-energy condition alone is active for $q \ge q_+$.

Proof. Changing variable in the integral and then taking the dual gives

$$\mathcal{H}(\lambda^E s, \lambda^A t) = \lambda \mathcal{H}(s,t), \qquad \mathcal{I}(\lambda^{1-E} e, \lambda^{1-A} a) = \lambda \mathcal{I}(e,a).$$

Take $\lambda = e^{1/(E-1)}$ and use the two regime conditions in Theorem 24.1. Their disjointness implies the strict order of the two constants. □

25 Asymmetric transitions between the two edge constraints

The boundary between the second and third regimes in Theorem 24.1 has an explicit nonquadratic local law. The relevant exponent depends on the difference of the two classical zeta boundaries.

Theorem 25.1 (Nonquadratic joint-cost transition)**.** *Suppose* $1 < A < E < A+1$ *and* $d = E - A \in (1/2, 1)$*. Fix* $a > 0$*, and set*

$$t_0 = (C_A/a)^{A/(A-1)}, \qquad e_0 = J_{A,E} t_0^{(1-E)/A}, \qquad \mathcal{I}_A(a) = (A-1) C_A^{A/(A-1)} a^{-1/(A-1)}.$$

Define

$$K_d = \frac{\pi}{2\sin(\pi(1/d - 1))}, \qquad L_d(a) = (1-d)\left(\frac{d}{K_d}\right)^{d/(1-d)} t_0^{1/(1-d)}.$$

Then $\mathcal{I}(e,a) = \mathcal{I}_A(a)$ *for* $e \geq e_0$*, whereas*

$$\mathcal{I}(e_0 - \epsilon, a) - \mathcal{I}_A(a) \sim L_d(a)\, \epsilon^{1/(1-d)} \qquad (\epsilon \downarrow 0). \tag{170}$$

The corresponding multiplier satisfies

$$s_* \sim t_0^{1/(1-d)} \left(\frac{d\epsilon}{K_d}\right)^{d/(1-d)}, \qquad t_* = t_0 + O(s_*). \tag{171}$$

For the classical zeta boundaries the onset exponent is

$$\frac{1}{1 - (E_* - A_*)} = 7.86858\ldots.$$

Thus the additional leading exponential cost of imposing critical edge coherence begins with this fractional power, rather than a quadratic law.

Proof. We first prove the uniform expansion, for t in a compact subinterval of $(0, \infty)$,

$$\mathcal{H}(s,t) = \mathcal{H}(0,t) + s\partial_s \mathcal{H}(0,t) - K_d (s/t)^{1/d} + o(s^{1/d}), \qquad s \downarrow 0. \tag{172}$$

Put $v = ty^{-A}$, $w = sy^{-E}$, and $R(v,w) = h(v+w) - h(v) - wh'(v)$. Since h is concave, $R \leq 0$. For $v \geq 1$, the bounds $h'(v) \ll v^{-1}$ and $|h''(v)| \ll v^{-2}$ give

$$|R(v,w)| \ll \min(w/v, (w/v)^2).$$

For $v < 1$, boundedness of h'' on $[0,\infty)$ gives an integrated contribution $O(s^2)$, uniformly for the specified t.

In the remaining integral substitute $y = (s/t)^{1/d} z$. Then $w/v = z^{-d}$ and $v \to \infty$ for each fixed $z > 0$. Using $h(v) = \frac{1}{2}\log(4\pi v) + O(v^{-1})$ and $h'(v) = 1/(2v) + O(v^{-2})$, its rescaled integrand tends to

$$\frac{1}{2}\{\log(1 + z^{-d}) - z^{-d}\}.$$

It is dominated by a constant times $\min(z^{-d}, z^{-2d})$, which is integrable precisely under the restrictions $d < 1$ and $2d > 1$. Dominated convergence is uniform for t in compact positive intervals. Integration by parts followed by $u = z^d$ yields

$$\int_0^\infty \{\log(1+z^{-d}) - z^{-d}\}\, dz = -d\int_0^\infty \frac{z^{-d}}{1+z^d}\, dz$$
$$= -\int_0^\infty \frac{u^{1/d-2}}{1+u}\, du = -\frac{\pi}{\sin(\pi(1/d-1))}.$$

Because $s^2 = o(s^{1/d})$, this proves (172) with the asserted constant.

Next maximize over t. At $(0, t_0)$ the derivatives $\mathcal{H}_{tt}$ and $\mathcal{H}_{st}$ are finite, with $\mathcal{H}_{tt} < 0$. For example the mixed-derivative integrand near $y = 0$ is $O(y^{A-E}) = O(y^{-d})$, which is integrable. These two derivatives are continuous as $s \downarrow 0$, by the bound $|h''(u)| \ll (1+u)^{-2}$. It follows by a one-sided monotonicity argument that the maximizing solution of $\mathcal{H}_t(s, t_s) = a$ has $t_s = t_0 + O(s)$. Indeed, $\mathcal{H}_t(s, t_0) - a = O(s)$ and $\mathcal{H}_{tt}$ stays bounded away from zero in a small one-sided neighborhood. No extension of $\mathcal{H}$ to negative s is used.

The change in the objective caused by replacing t_s by t_0 is $O(s^2)$: the first variation of $\mathcal{H}(0,t) - at$ vanishes at t_0, and the mixed linear term changes by $O(s|t_s - t_0|)$. Hence, with $r = 1/d$,

$$\max_{t \ge 0}\{\mathcal{H}(s,t) - at\} = \mathcal{I}_A(a) + e_0 s - Bs^r + o(s^r), \qquad B = K_d t_0^{-r}. \tag{173}$$

Uniqueness and coercivity of the joint optimizer imply that as $e \uparrow e_0$ from below the optimizer tends to $(0, t_0)$. The remaining maximization is therefore locally that of $\epsilon s - Bs^r + o(s^r)$. To avoid differentiating a remainder, bound it between $\epsilon s - (B+\eta)s^r$ and $\epsilon s - (B-\eta)s^r$, maximize these explicit functions, and then let $\eta \downarrow 0$. Their common limiting value is

$$(r-1)r^{-r/(r-1)} B^{-1/(r-1)} \epsilon^{r/(r-1)}.$$

Since $r/(r-1) = 1/(1-d)$, this coefficient equals $L_d(a)$. The same rescaling, and uniqueness of the maximizer of $u - Bu^r$, gives $s_* \sim (\epsilon/(rB))^{1/(r-1)}$, which is (171). The constant branch $e \ge e_0$ is the second regime of Theorem 24.1. □

The source of the fractional onset is a divergent second derivative $\mathcal{H}_{ss}(0,t)$: its integrand behaves like y^{-2d} near zero. The first derivative stays finite because $d < 1$. The resulting nonquadratic law is compatible with the finite mixed derivative needed to optimize t; replacing the singular second derivative by a quadratic approximation would give an incorrect transition exponent.

Theorem 25.2 (Quadratic onset at the critical-edge-dominated boundary)**.** *Suppose $1 < A < E$. Fix $e > 0$ and put*

$$s_0 = (C_E/e)^{E/(E-1)}, \qquad a_0 = J_{E,A} s_0^{(1-A)/E}, \qquad \mathcal{I}_E(e) = (E-1)C_E^{E/(E-1)} e^{-1/(E-1)}.$$

Define the positive Gram matrix

$$M_{bc}(s_0) = \int_0^\infty y^{-b-c}[-h''(s_0 y^{-E})]\, dy, \qquad b, c \in \{E, A\},$$

and its positive Schur complement

$$\kappa_e = M_{AA}(s_0) - \frac{M_{EA}(s_0)^2}{M_{EE}(s_0)} > 0.$$

Then $\mathcal{I}(e,a)=\mathcal{I}_E(e)$ for $a\ge a_0$, and

$$\mathcal{I}(e,a_0-\epsilon)-\mathcal{I}_E(e)\sim\frac{\epsilon^2}{2\kappa_e},\qquad t_*\sim\frac{\epsilon}{\kappa_e}\qquad(\epsilon\downarrow 0).\tag{174}$$

In particular, the two transition boundaries have different leading orders for the zeta-edge parameters: quadratic at this boundary, and order 7.86858… at the boundary of Theorem 25.1.

When $2A-E>1$, as holds for the classical zeta boundaries, the curvature constant can also be written

$$\kappa_e=\frac{s_0^{(1-2A)/E}}{E}\left\{(2A-E-1)J_{E,\,2A-E}-\frac{(A-1)^2J_{E,A}^2}{(E-1)C_E}\right\},\tag{175}$$

where $J_{E,b}=\int_0^\infty y^{-b}h'(y^{-E})\,dy$.

Proof. Every entry of M is finite. Near zero its integrand is $O(y^{2E-b-c})$, and at infinity it is $O(y^{-b-c})$. The exponents are integrable for $b,c\in\{E,A\}$ and $1<A<E$. Since $-h''(u)$ is the strictly positive variance of V under a nondegenerate angular tilt, the matrix M is the Gram matrix of the linearly independent functions y^{-E} and y^{-A} in a positively weighted L^2 space. Thus it is positive definite and $\kappa_e>0$.

All second derivatives of $\mathcal{H}$ are continuous in a one-sided neighborhood of $(s_0,0)$. Maximizing first in s gives a unique solution $s_t=s_0+O(t)$ of $\mathcal{H}_s(s_t,t)=e$, and ordinary second-order expansion gives

$$\max_{s\ge0}\{\mathcal{H}(s,t)-se\}=\mathcal{I}_E(e)+a_0t-\tfrac12\kappa_et^2+o(t^2).$$

Indeed its second derivative is $\mathcal{H}_{tt}-\mathcal{H}_{ts}^2/\mathcal{H}_{ss}=-\kappa_e$. As in the preceding proof, uniqueness and coercivity localize the joint maximizer near the axis. Maximization of $\epsilon t-\kappa_et^2/2+o(t^2)$, using upper and lower quadratic bounds, proves (174).

Finally, scaling the integrals and differentiating the axis formulas give

$$M_{EE}(s)=\frac{(E-1)C_E}{E}s^{(1-2E)/E},\qquad M_{EA}(s)=\frac{(A-1)J_{E,A}}{E}s^{(1-E-A)/E}.$$

If $2A-E>1$, integration by parts also gives

$$M_{AA}(s)=\frac{2A-E-1}{E}J_{E,\,2A-E}s^{(1-2A)/E}.$$

Substitution yields (175). □

26 Zero-free half-planes and the first height of an extreme point

This section concerns the ordinary prime flow and phase patterns that agree with a Dirichlet character outside finitely many primes. It does not assert a first-height obstruction for arbitrary Euler-product twists. For a primitive character χ of conductor q, including the character of conductor 1, and $1/2\le\vartheta<1$, write

$$\mathrm{H}_\vartheta(\chi):\qquad L(\rho,\chi)=0,\quad 0<\operatorname{Re}\rho<1\quad\Longrightarrow\quad\operatorname{Re}\rho\le\vartheta.\tag{176}$$

The numerical specialization $\vartheta=7/8$ invokes [17, Theorem 1.1], which asserts this hypothesis for every Dirichlet character. The transfer proved below remains valid under its explicit hypothesis independently of that input.

Lemma 26.1 (A smooth prime block)**.** *Fix a nonnegative nonzero $\phi \in C_c^\infty((1,2))$ and put*

$$\widehat{\phi}(s) = \int_0^\infty \phi(u) u^{s-1}\, du, \qquad I_\phi = \widehat{\phi}(1) > 0.$$

Under $\mathrm{H}_\vartheta(\chi)$*, uniformly for $X \geq 2$ and real t,*

$$S_\chi(X,t) := \sum_p (\log p)\phi(p/X)\chi(p)p^{-it} = \mathbf{1}_{\chi=1} X^{1-it}\widehat{\phi}(1-it) + O_\phi\Big(X^\vartheta \log(q(|t|+3))\Big). \tag{177}$$

Here $\chi = 1$ denotes the character of conductor 1.

Proof. Mellin inversion gives

$$\sum_n \Lambda(n)\chi(n)n^{-it}\phi(n/X) = \frac{1}{2\pi i}\int_{(2)} -\frac{L'}{L}(s+it,\chi)X^s\widehat{\phi}(s)\, ds.$$

Moving the contour to $\operatorname{Re} s = -1/2$ gives the smooth explicit formula

$$\mathbf{1}_{\chi=1}X^{1-it}\widehat{\phi}(1-it) - \sum_\rho X^{\rho-it}\widehat{\phi}(\rho-it) + O_\phi\Big(1 + X^{-1/2}\log(q(|t|+3))\Big). \tag{178}$$

The sum runs over the nontrivial zeros with multiplicity. A possible trivial zero at 0 contributes the bounded term. On the new line the functional equation gives $L'/L(-1/2+iu,\chi) = O(\log(q(|u|+3)))$; rapid vertical decay of $\widehat{\phi}$ bounds the remaining integral. Horizontal contours avoiding zeros justify the shift; see the classical explicit formula and zero counts in [6, Chapters 10 and 12].

For every fixed N, uniformly for $0 \leq \beta \leq 1$, $|\widehat{\phi}(\beta+iv)| \leq C_{\phi,N}(1+|v|)^{-N}$. Together with the unit-interval zero count $\#\{\rho : u \leq \operatorname{Im}\rho \leq u+1\} \ll \log(q(|u|+3))$, this makes the zero sum absolutely convergent and gives

$$\sum_\rho |X^{\rho-it}\widehat{\phi}(\rho-it)| \ll_\phi X^\vartheta \log(q(|t|+3)).$$

The prime-power terms of exponent at least two are

$$\ll \sum_{2\leq k\leq \log(2X)/\log 2}\ \sum_{p\leq (2X)^{1/k}} \log p \ll X^{1/2} + X^{1/3}\log X \ll X^{1/2}.$$

They are absorbed since $\vartheta \geq 1/2$. This proves (177), without a factor $|t|$ or an extra $\log X$. □

Fix a finite set S and unit complex numbers b_p, for $p \nmid q$, satisfying $b_p = \chi(p)$ outside S. For $w_p \asymp p^{-\sigma}(\log p)^\kappa$, $\sigma > 1$, define

$$\mathcal{C}_{b,w}(t;u) = \sum_{p\nmid q} w_p(1 - \operatorname{Re}(ub_p p^{-it})), \qquad |u| = 1. \tag{179}$$

For later use define the inverse energy scale

$$X_{\sigma,\kappa}(\varepsilon) = \varepsilon^{-1/(\sigma-1)}[\log(1/\varepsilon)]^{(\kappa-1)/(\sigma-1)}.$$

The optional common phase includes the simultaneous rotation that occurs for relative prime frequencies.

Theorem 26.2 (A deterministic coherence barrier)**.** *Assume* $\mathrm{H}_\vartheta(\chi)$ *and put* $d=(\sigma-1)/(1-\vartheta)$. *There are* $c>0$ *and* T_0 *such that, for all* $|t|\ge T_0$ *and* $|u|=1$,

$$\mathcal{C}_{b,w}(t;u)\ge c\big(\log(|t|+3)\big)^{-d}\big(\log\log(|t|+3)\big)^{\kappa-1}. \tag{180}$$

Consequently, for sufficiently small ε, *a hit* $\mathcal{C}_{b,w}(t;u)\le C_1\varepsilon$ *with* $|t|\ge T_0$ *requires*

$$\log(|t|+3)\ge cX_{\sigma,\kappa}(\varepsilon)^{1-\vartheta}. \tag{181}$$

Constants may depend on the fixed character, exceptional set and weights.

Proof. Put $L=\log(q(|t|+3))$ and $X=KL^{1/(1-\vartheta)}$, where K is a large fixed constant. For large $|t|$, the block primes lie outside S and are coprime to q. The prime number theorem gives

$$M(X):=\sum_p(\log p)\phi(p/X)=I_\phi X+o(X).$$

The principal main term of (177) has modulus at most $I_\phi X/8$ for large $|t|$. Its error is at most $CK^{-(1-\vartheta)}X$, also at most $I_\phi X/8$ after choosing K. Hence $M(X)-\mathrm{Re}(uS_\chi(X,t))\ge I_\phi X/2$, uniformly in u. On this block,

$$w_p\ge cX^{-\sigma}(\log X)^{\kappa-1}\log p.$$

Nonnegativity and boundedness of ϕ imply $\mathcal{C}_{b,w}(t;u)\ge cX^{1-\sigma}(\log X)^{\kappa-1}$. Substitution proves (180).

For the last assertion take $X=c_2X_{\sigma,\kappa}(\varepsilon)$ with $c_2>0$ small. The energy assumption bounds $M(X)-\mathrm{Re}(uS_\chi(X,t))$ by $I_\phi X/4$. Thus $|S_\chi(X,t)|\gg X$, and (177), after absorbing its principal term, forces $X^{1-\vartheta}\ll\log(q(|t|+3))$. □

Theorem 26.3 (A local inverse statement)**.** *No hypothesis* H_ϑ *is assumed here. Fix* $\eta>0$ *and* $c_1>0$. *There are* $C_1,C_2>0$ *such that, for sufficiently large* X *and* $|t|$, *if*

$$|S_\chi(X,t)|\ge c_1X,\qquad X^{1-b}>C_1\log(q(|t|+3)),\qquad \tfrac12\le b<1,$$

then $L(s,\chi)$ *has a nontrivial zero* $\rho=\beta+i\gamma$ *with*

$$\beta>b,\qquad |\gamma-t|\le C_2\big(\log(q(|t|+3))\big)^{\eta}. \tag{182}$$

The constants are uniform in b.

Proof. Put $L=\log(q(|t|+3))$. Formula (178) is unconditional. Its principal term is $o(X)$ as $|t|\to\infty$. The condition $L\ll X^{1-b}$ with $b\ge1/2$ makes the remaining integral $O(1)$, while the prime powers contribute $O(X^{1/2})$. The absolute zero sum is consequently at least $c_1X/2$.

Choose an integer $N>1+1/\eta$ and put $H=C_2L^\eta$. If there is no zero as in (182), zeros with $|\gamma-t|\le H$ contribute at most C_NX^bL; the others contribute at most

$$C_NX\sum_{|\gamma-t|>H}(1+|\gamma-t|)^{-N}\ll_N XLH^{1-N}.$$

The unit-interval count proves the last estimate; its extra $\log H$ is absorbed by L. This term is less than $c_1X/8$ after choosing C_2 and the lower threshold, while $X^{1-b}>C_1L$ makes the near term less than $c_1X/8$ after choosing C_1. This is a contradiction. □

26.1 Consequences at the two ordinary-zeta edges

Recall $z_2^* = -1$, $z_p^* = 1$ for odd p. The phase-defect estimates already proved above give

$$\zeta'(E_* - \delta + it) = 0 \implies \sum_p (\log p) p^{-E_*} |p^{-it} - z_p^*|^2 \ll \delta,$$

and

$$\zeta(A_* - \varepsilon + it) = 1 \implies \sum_p p^{-A_*} |p^{-it} - z_p^*|^2 \ll \varepsilon.$$

Apply Theorem 26.2 with $\chi = 1$, $b_p = \overline{z_p^*}$ and $u = 1$. Under $\mathrm{H}_\vartheta(\zeta)$ this yields

$$E_* - \operatorname{Re}\rho \geq c\big(\log(|\operatorname{Im}\rho| + 3)\big)^{-(E_*-1)/(1-\vartheta)}, \qquad \zeta'(\rho) = 0, \tag{183}$$

$$A_* - \operatorname{Re} s \geq \frac{c\big(\log(|\operatorname{Im} s| + 3)\big)^{-(A_*-1)/(1-\vartheta)}}{\log\log(|\operatorname{Im} s| + 3)}, \qquad \zeta(s) = 1. \tag{184}$$

These statements apply to sufficiently small positive edge deficits. Such points have unbounded ordinates: the exact extremal pattern cannot occur on a bounded segment of the ordinary prime flow, and its continuous defect has a positive minimum there.

Let $T_E(\delta)$ and $T_A(\varepsilon)$ denote the first positive ordinates in the respective edge strips. The deterministic recurrence and local root-transfer upper bounds in Section 8 combine with these lower bounds to give, with $\beta_E = (E_* - 1)^{-1}$ and $\beta_A = (A_* - 1)^{-1}$,

$$\exp(c\delta^{-\beta_E(1-\vartheta)}) \leq T_E(\delta) \leq \exp(C\delta^{-\beta_E}), \tag{185}$$

$$\exp\Big(c[\varepsilon\log(1/\varepsilon)]^{-\beta_A(1-\vartheta)}\Big) \leq T_A(\varepsilon) \leq \exp\Big(C[\varepsilon\log(1/\varepsilon)]^{-\beta_A}\Big). \tag{186}$$

For $\vartheta = 7/8$ the lower-height powers are

$$\frac{1}{8(E_* - 1)} = 0.0689459643\ldots, \qquad \frac{1}{8(A_* - 1)} = 0.1329643401\ldots.$$

The boundary powers are $8(E_* - 1) = 14.5041121620\ldots$ and $8(A_* - 1) = 7.5208134699\ldots$.

The local inverse theorem also gives an unconditional alternative. If an ordinary-zeta critical point of deficit δ occurs at a height satisfying

$$\delta^{-(1-b)/(E_*-1)} > C\log(|t| + 3),$$

then a zeta zero lies to the right of b within $O((\log(|t| + 3))^\eta)$ of that same ordinate, for any fixed $\eta > 0$. For a 1-point replace the left side by $[\varepsilon\log(1/\varepsilon)]^{-(1-b)/(A_*-1)}$.

These first-height restrictions for arithmetic phase patterns neither identify an exact first-height exponent nor improve the deterministic uniform covering bound. In particular, a zero-free half-plane supplies no new small-divisor bound for $\sum_p k_p \log p = \log(a/b)$ beyond those used in the covering argument of Theorem 2.2.

27 Gaussian inverse principles and unconditional edge repulsion

This section improves the vertical localization in the inverse coherence theorem and removes any fixed-half-plane hypothesis from a weaker first-height obstruction. The analytic input is the classical explicit formula; the additional ingredient is to exploit positive lognormal smoothing on the prime side. Throughout, χ is a fixed primitive Dirichlet character of conductor q, including $q = 1$. Write

$$L_t = \log(q(|t| + 3)), \qquad M_t = \log L_t.$$

All assertions involving t are for sufficiently large $|t|$.

27.1 A Gaussian prime block

For $X \geq 2$ put

$$G_\chi(X,t) = \sum_p (\log p) \exp\left(-\frac{\log^2(p/X)}{2}\right) \chi(p) p^{-it}.$$

The series converges absolutely. Mellin inversion, followed by a contour shift to $\operatorname{Re} s = -1/2$, gives

$$G_\chi(X,t) = \mathbf{1}_{\chi=1}\sqrt{2\pi}\, X^{1-it} e^{(1-it)^2/2} - \sqrt{2\pi} \sum_\rho X^{\rho-it} e^{(\rho-it)^2/2} + O\Big(X^{1/2} + 1 + X^{-1/2} L_t\Big). \tag{187}$$

The zero sum includes multiplicities and is absolutely convergent. Indeed, the Mellin transform of $e^{-\log^2 u/2}$ is $\sqrt{2\pi} e^{s^2/2}$, and

$$|e^{(\beta+i(\gamma-t))^2/2}| = e^{\beta^2/2 - (\gamma-t)^2/2}.$$

The unit-interval zero count is $O(\log(q(|\gamma|+3)))$. The functional equation bounds the new contour by $O(X^{-1/2} L_t)$. A possible trivial zero at zero gives $O(1)$. Finally, the prime-power counting function of exponents at least two is $O(y^{1/2})$; integration by parts against the lognormal weight bounds their contribution by $O(X^{1/2})$. Gaussian decay on horizontal contours justifies the shift. Thus every term in (187) is uniform in real t.

Theorem 27.1 (Gaussian localization of a forced zero)**.** *Fix $c_0 > 0$. There are constants $C_0, C_1 > 0$ such that, if*

$$|G_\chi(X,t)| \geq c_0 X, \qquad X^{1-b} > C_0 L_t, \qquad \tfrac{1}{2} \leq b < 1,$$

then $L(s,\chi)$ has a nontrivial zero $\beta + i\gamma$ with

$$\beta > b, \qquad |\gamma - t| \leq C_1 \sqrt{\log L_t}.$$

The constants are uniform in b.

Proof. The pole term in (187) is $o(X)$ as $|t| \to \infty$. The threshold implies $X^{-1/2} L_t \ll X^{1/2-b} \leq 1$, and the prime powers are $o(X)$. Consequently the absolute zero sum is at least a fixed multiple of X. Suppose the asserted zero does not exist, and set $H = C_1\sqrt{\log L_t}$. Zeros with $|\gamma - t| \leq H$ have $\beta \leq b$, so their contribution is $O(X^b L_t)$. For the remaining zeros use $X^\beta \leq X$ and split the Gaussian:

$$\sum_{|\gamma-t|>H} X^\beta e^{-(\gamma-t)^2/2} \ll X L_t e^{-H^2/4}.$$

Here the residual sum with weight $e^{-(\gamma-t)^2/4}$ is $O(L_t)$ by the unit-interval zero count. Choosing C_0 and C_1 large makes both contributions smaller than the required multiple of X, a contradiction. □

Let $w_p \asymp p^{-\sigma}(\log p)^\kappa$, where $\sigma > 1$ and $\kappa \in \mathbb{R}$. Let b_p be unit complex numbers agreeing with $\chi(p)$ outside a fixed finite set, for $p \nmid q$, and set

$$\mathcal{C}(t;u) = \sum_{p \nmid q} w_p (1 - \operatorname{Re}(u b_p p^{-it})), \qquad |u| = 1.$$

As in the preceding phase-defect applications, the common phase u may depend on t. Define

$$X_{\sigma,\kappa}(\varepsilon) = \varepsilon^{-1/(\sigma-1)} [\log(1/\varepsilon)]^{(\kappa-1)/(\sigma-1)}.$$

Completing a square in $\log p$ shows that

$$(\log p)e^{-\log^2(p/X)/2} \le CX^{\sigma}(\log X)^{1-\kappa}w_p \qquad (X \ge 2). \tag{188}$$

For completeness, after division by w_p the exponential factor is $X^{\sigma}e^{\sigma^2/2}e^{-(\log p-\log X-\sigma)^2/2}$, multiplied by $(\log p)^{1-\kappa}$. Its supremum is $O((\log X)^{1-\kappa})$ after the Gaussian is included; the portion $\log p < \frac{1}{2}\log X$ is exponentially smaller than every power of $\log X$. Also, the prime number theorem gives $G_1(X,0) \sim \sqrt{2\pi e}\,X$. Fixed exceptional primes contribute $o(X)$. Therefore $\mathcal{C}(t;u) \le \varepsilon$ implies $|G_\chi(X,t)| \gg X$ at $X = cX_{\sigma,\kappa}(\varepsilon)$ with a sufficiently small fixed $c > 0$. This proves the following quantitative version.

Corollary 27.2 (A nearby rightward zero from early coherence)**.** *There are $C_0, C_1 > 0$ such that a hit $\mathcal{C}(t;u) \le \varepsilon$ forces a zero with*

$$\beta > b, \qquad |\gamma - t| \le C_1\sqrt{\log L_t},$$

whenever $X_{\sigma,\kappa}(\varepsilon)^{1-b} > C_0L_t$ and $1/2 \le b < 1$. In particular, if $X = cX_{\sigma,\kappa}(\varepsilon)$ and $\log X \ge 2\log(2C_0L_t)$, one may take

$$b = 1 - \frac{\log(2C_0L_t)}{\log X}.$$

27.2 Adaptive smoothing and joint localization

The preceding window still grows, although very slowly. Allowing the Gaussian width to increase gives a complementary conclusion: exceptionally small phase energy forces a zero close both to the 1-line and to the same ordinate. For $v \ge 1$ put

$$k_v(y) = \frac{1}{\sqrt{2\pi v}}\exp\left(-\frac{v}{2} - \frac{\log^2 y}{2v}\right), \qquad \mathcal{H}_\chi(v,t) = \sum_p (\log p)k_v(p)\chi(p)p^{-it}.$$

The Mellin transform is exactly

$$\int_0^\infty k_v(y)y^{s-1}\,dy = e^{v(s^2-1)/2}. \tag{189}$$

The prime number theorem implies

$$\mathcal{H}_1(v,0) = 1 + o(1) \qquad (v \to \infty). \tag{190}$$

One way to see this uniformly is to integrate against $\vartheta(y) = y + o(y)$. The continuous comparison measure has $\log y$ normally distributed with mean v and variance v, so its mass on every bounded interval tends to zero.

The same contour shift as before gives, uniformly for $v \ge 1$,

$$\mathcal{H}_\chi(v,t) = \mathbf{1}_{\chi=1}e^{-vt^2/2-ivt} - \sum_\rho e^{v((\rho-it)^2-1)/2} + O\Big(L_te^{-3v/8}\Big). \tag{191}$$

On $\operatorname{Re} s = -1/2$, the Gaussian has modulus $e^{-3v/8}e^{-v(\operatorname{Im} s)^2/2}$, giving the stated contour bound. The possible zero at zero contributes at most $e^{-v/2}$. The prime-power contribution is $O(e^{-3v/8})$: integrate the bound $\sum_{p^j \le y,\, j \ge 2}\log p \ll y^{1/2}$ against $k_v(y)$ and complete a square. These observations also establish the uniformity required when v grows with t.

Theorem 27.3 (A quadratic zero constraint)**.** *Fix σ, κ, the weights, the character and the finite exceptional set. There are constants $C, c > 0$ with the following property. Suppose*

$$\mathcal{C}(t; u) \le \varepsilon, \qquad R = \log(1/\varepsilon) \ge C \log L_t.$$

Then a nontrivial zero $\rho = \beta + i\gamma$ satisfies

$$\beta^2 - (\gamma - t)^2 > 1 - c\frac{\log L_t}{R}. \tag{192}$$

Whenever the right-hand side is positive, this implies

$$\beta > \sqrt{1 - c\frac{\log L_t}{R}}, \qquad |\gamma - t| < \sqrt{c\frac{\log L_t}{R}}.$$

In particular, along any sequence of such hits for which $R/\log L_t \to \infty$, the forced zeros satisfy $\beta \to 1$ and $\gamma - t \to 0$.

Proof. Completing the square once more gives the pointwise estimate

$$\frac{(\log p)k_v(p)}{w_p} \le C v^{1/2-\kappa} e^{v(\sigma^2-1)/2}. \tag{193}$$

Indeed, apart from a constant, the left side equals

$$v^{-1/2} e^{v(\sigma^2-1)/2} (\log p)^{1-\kappa} e^{-(\log p - \sigma v)^2/(2v)}.$$

Its remaining supremum is $O(v^{1-\kappa})$ for $v \ge 1$; near σv this is immediate, and the two Gaussian tails absorb all polynomial factors. Fixed exceptional primes and primes dividing q have total weight $O(e^{-v/2})$.

Take $v = R/(\sigma^2 - 1)$. Equations (190) and (193) imply

$$\mathrm{Re}(u\mathcal{H}_\chi(v, t)) \ge 1 - o(1) - Ce^{-R/2} R^{1/2-\kappa},$$

and hence $|\mathcal{H}_\chi(v, t)| \ge 1/2$. By increasing the constant in $R \ge C \log L_t$, the error in (191) is less than $1/8$; its pole term is also less than $1/8$. Thus

$$\sum_\rho e^{-v(1-\beta^2+(\gamma-t)^2)/2} \ge \tfrac{1}{4}. \tag{194}$$

Since $0 < \beta < 1$, all quantities $1 - \beta^2$ are nonnegative. If every zero satisfied $v(1 - \beta^2 + (\gamma - t)^2) > 4 \log(KL_t)$, splitting the exponential into two equal factors would bound the left side by

$$\frac{1}{KL_t} \sum_\rho e^{-v(\gamma-t)^2/4} \ll \frac{1}{K}.$$

The last bound is uniform for $v \ge 1$ by the unit-interval zero count. Choosing K large contradicts (194). Substituting $v = R/(\sigma^2 - 1)$ and absorbing the fixed constant inside $\log(KL_t)$ proves (192). □

27.3 An unconditional first-height obstruction

The classical Vinogradov–Korobov zero-free region states that, for some $a > 0$, every sufficiently high zeta zero satisfies

$$1 - \beta \geq \frac{a}{(\log(|\gamma| + 3))^{2/3}(\log\log(|\gamma| + 3))^{1/3}}. \tag{195}$$

For example, this form appears in Ford's zero-free-region paper [25]; explicit refinements do not affect the argument below.

Corollary 27.4 (Unconditional arithmetic coherence repulsion)**.** *Take $\chi = 1$ in the definition of $\mathcal{C}$. Uniformly for $|u| = 1$, there is $C > 0$ such that*

$$\mathcal{C}(t; u) \geq \exp\Big[-C(\log(|t| + 3))^{2/3}(\log\log(|t| + 3))^{4/3}\Big]. \tag{196}$$

Consequently a hit $\mathcal{C}(t; u) \leq \varepsilon$, with $|t|$ sufficiently large and ε sufficiently small, requires

$$\log(|t| + 3) \geq c\frac{[\log(1/\varepsilon)]^{3/2}}{[\log\log(1/\varepsilon)]^2}. \tag{197}$$

Proof. Put $L = \log(|t| + 3)$ and choose

$$b = 1 - \frac{a_1}{L^{2/3}(\log L)^{1/3}}$$

with $a_1 > 0$ smaller than the constant in (195). Every zero in the window $|\gamma - t| \leq C_1\sqrt{\log L}$ has $\beta \leq b$. Corollary 27.2 therefore requires

$$(1 - b)\log X_{\sigma,\kappa}(\varepsilon) \leq \log(C_0 L)$$

for any sufficiently small coherence hit. Since

$$\log X_{\sigma,\kappa}(\varepsilon) = \frac{\log(1/\varepsilon) + (\kappa - 1)\log\log(1/\varepsilon)}{\sigma - 1},$$

this gives (196). Inverting $R \leq CL^{2/3}(\log L)^{4/3}$ gives $L \geq cR^{3/2}/(\log R)^2$, proving (197). □

Apply this corollary to the frozen ordinary-zeta phase pattern $z_2^* = -1$, $z_p^* = 1$ for odd p, using the phase-defect necessity at the two Euler edges. For $\zeta'(E_* - \delta + it) = 0$ use $\sigma = E_*$, $\kappa = 1$ and $\mathcal{C}(t; 1) \ll \delta$; for $\zeta(A_* - \varepsilon + it) = 1$ use $\sigma = A_*$, $\kappa = 0$ and $\mathcal{C}(t; 1) \ll \varepsilon$. Define $T_E(\delta)$ as the infimum of positive ordinates of critical points with $0 < E_* - \operatorname{Re} s \leq \delta$, and define $T_A(\varepsilon)$ analogously for 1-points with $0 < A_* - \operatorname{Re} s \leq \varepsilon$. Existence for every positive tolerance follows from Theorem 21.3. These first positive edge ordinates satisfy, unconditionally,

$$\begin{aligned}\log T_E(\delta) &\geq c\frac{[\log(1/\delta)]^{3/2}}{[\log\log(1/\delta)]^2},\\ \log T_A(\varepsilon) &\geq c\frac{[\log(1/\varepsilon)]^{3/2}}{[\log\log(1/\varepsilon)]^2}.\end{aligned} \tag{198}$$

As before, sufficiently small edge deficits cannot occur in a fixed bounded interval because the exact frozen phase pattern is absent from the ordinary prime flow. These inequalities are weaker than the consequences of a fixed zero-free half-plane, but have no such hypothesis. The inverse statements do not establish a converse equivalence with the Riemann hypothesis: an off-line zero alone does not supply the almost complete prime-phase coherence assumed here.

Comparison with other inverse theorems. Gaussian zero detection is an established method, not a new principle claimed here. In particular, Dong, Wang, Wang and Zhang's preprint *Large zeta sums and zeros of the Riemann zeta function* [26] gives a zero-forcing theorem from large ordinary partial sums $\sum_{n\le x} n^{it}$, adapting the character-sum method of Granville and Soundararajan. The present hypotheses concern near-complete weighted prime coherence and therefore differ from their hypotheses. The conclusions above are direct coherence and Euler-edge applications with explicit vertical windows; no claim of a general improvement over that partial-sum theorem is intended.

28 Exact spectral surgery and a sharp fourth-moment certificate

This section concerns finite Hermitian matrices. Its conclusions are unconditional linear algebra. The application to zeta in Corollary 28.4 explicitly assumes two additional trace asymptotics. The numerical constant 16/21 also appears in the public conditional certificates of Jesse/SolidSF [22]; no priority claim for that constant is made here. Our purpose is to give a self-contained reduction, an exact spectral feasibility criterion, and a matching block extremal.

Write $n_+(H)$ for the number of strictly positive eigenvalues of a Hermitian matrix, with multiplicity.

Proposition 28.1 (Exact spectral surgery)**.** *Let G be Hermitian and let $\lambda_1 \ge \cdots \ge \lambda_r > 0$ be its positive eigenvalues. For integers $s, b \ge 0$ and $\tau \ge 0$, a matrix P satisfying*

$$P \succeq 0, \qquad \operatorname{rank} P \le s, \qquad \operatorname{Tr} P \le \tau, \qquad n_+(G-P) \le b$$

exists if and only if

$$(r-b)_+ \le s, \qquad \sum_{j=b+1}^{r} \lambda_j \le \tau,$$

where an empty sum is zero.

Proof. Put $Q = G - P$. Since $G \preceq P + Q_+$ and $\operatorname{rank} Q_+ \le b$, the min–max principle gives

$$\lambda_{b+j}(G) \le \lambda_j(P) \qquad (1 \le j \le r-b).$$

This proves necessity. Conversely, with orthonormal eigenvectors u_j for the positive spectrum, set

$$P = \sum_{j=b+1}^{r} \lambda_j u_j u_j^*.$$

This has exactly the required rank and trace and leaves at most b positive eigenvalues. □

The criterion makes no positivity assumption on G or Q. If the additional condition $n_-(G-P) \le b$ is imposed, the further necessary and sufficient condition is $n_-(G) \le b$: necessity follows from $G \succeq G - P$, and the displayed construction leaves the negative spectrum unchanged.

Lemma 28.2 (Spectral certificates)**.** *Suppose*

$$G = P + Q, \quad P \succeq 0, \quad \operatorname{rank} P \le s, \quad \operatorname{Tr} P \le s, \quad n_+(Q) \le b, \quad s + 2b \le N. \tag{199}$$

Let $A, B, C \ge 0$ and let q be a real function on $\mathbb{R}$ satisfying

$$q(x) \le 0 \quad (x \le 0), \qquad q(x) \le \min\{A + Bx, C\} \quad (x > 0).$$

Then

$$\operatorname{Tr} q(G) \le (A+B)s + Cb.$$

In particular, if $A = 1$ *and* $C = 2B$*, then*

$$s \ge \operatorname{Tr} q(G) - BN.$$

Proof. The nonpositive spectrum contributes at most zero. Bound the largest b positive eigenvalues by C each. Proposition 28.1 bounds the number of remaining positive eigenvalues by s and their sum by s. Their contribution is therefore at most $(A+B)s$. The case of fewer than b positive eigenvalues has an empty tail. The final assertion follows from $2b \le N - s$. □

Theorem 28.3 (A fourth-moment ratio)**.** *Under* (199)*, define*

$$L = 4 \operatorname{Tr} G - \operatorname{Tr} G^2 - 2N, \qquad D = \operatorname{Tr}\left[(G(2I - G))^2\right].$$

If $D > 0$*, then*

$$s \ge \frac{(L_+)^2}{D}. \tag{200}$$

If $D = 0$*, the hypotheses imply* $L \le 0$*.*

Proof. For $\theta > 0$ put

$$q_\theta(x) = \frac{8\theta x - (4 + 2\theta)x^2 + 4x^3 - x^4}{\theta^2}.$$

Every term in its numerator is nonpositive when $x \le 0$. Moreover the exact identities

$$1 + \frac{4x}{\theta} - q_\theta(x) = \frac{(x^2 - 2x + \theta)^2}{\theta^2},$$
$$\frac{8}{\theta} - q_\theta(x) = \frac{(x-2)^2(x^2 + 2\theta)}{\theta^2}$$

give the hypotheses of Lemma 28.2 with $A = 1$, $B = 4/\theta$, and $C = 8/\theta$. Consequently

$$s \ge \operatorname{Tr} q_\theta(G) - \frac{4N}{\theta} = \frac{2L}{\theta} - \frac{D}{\theta^2}.$$

For $L, D > 0$, take $\theta = D/L$. When $L \le 0$, use $s \ge 0$. If $D = 0$ and $L > 0$, letting $\theta \downarrow 0$ contradicts the displayed inequality. □

Corollary 28.4 (Conditional moment transfer)**.** *Consider a sequence of decompositions satisfying* (199)*, with* $N \to \infty$*, and suppose*

$$\frac{1}{N}(\operatorname{Tr} G, \operatorname{Tr} G^2, \operatorname{Tr} G^3, \operatorname{Tr} G^4) \longrightarrow \left(1, \frac{4}{3}, 2, \frac{13}{4}\right).$$

Then $\liminf s/N \ge 16/21$*. The same conclusion holds if each count constraint has an additive* $o(N)$ *error.*

Proof. We have $L/N \to 2/3$ and $D/N \to 7/12$, and apply Theorem 28.3. For the version with errors, use the fixed parameter $\theta = 7/8$ in its proof; every error then remains $o(N)$. □

At $\theta = 7/8$, the certificate is

$$q(x) = \frac{448x - 368x^2 + 256x^3 - 64x^4}{49}, \qquad B = \frac{32}{7}.$$

The four prescribed moments give $N^{-1} \operatorname{Tr} q(G) \to 16/3$, and $16/3 - 32/7 = 16/21$. More generally, if the normalized moments tend to $(1, m_2, m_3, m_4)$ with $m_4 - 4m_3 + 4m_2 > 0$, then the same argument gives

$$\liminf \frac{s}{N} \geq \frac{((2 - m_2)_+)^2}{m_4 - 4m_3 + 4m_2}.$$

Proposition 28.5 (Sharp block extremal)**.** *The constant* $16/21$ *cannot be increased using only the four moments in Corollary 28.4, the constraints* (199)*, and the additional requirements that* $d = N$*, that* P *be a sum of* s *unit rank-one operators, and that* Q *be a sum of positive multiplicity-two blocks.*

Proof. Take $N = d = 42$ and $a = 1/(2\sqrt{2})$. Let P have sixteen eigenvalues $1 - a$, sixteen eigenvalues $1 + a$, and ten zero eigenvalues. Let Q equal twice an orthogonal projection of rank five supported in $\ker P$. Then $s = 32$, $b = 5$, and $G = P + Q$ has five zero eigenvalues. Directly,

$$(\operatorname{Tr} G, \operatorname{Tr} G^2, \operatorname{Tr} G^3, \operatorname{Tr} G^4) = \left(42, 56, 84, \frac{273}{2}\right).$$

Thus $L = 28$, $D = 49/2$, and $(L_+)^2/D = 32 = 16N/21$.

For each pair of eigenvectors e_+, e_- of P, set

$$v_\pm = \sqrt{\frac{1 + a}{2}}\, e_+ \pm \sqrt{\frac{1 - a}{2}}\, e_-.$$

Both vectors have norm one, and the sum of their rank-one operators is $\operatorname{diag}(1 + a, 1 - a)$. Sixteen such pairs realize P as the required sum. The five summands $2ee^*$ realize Q. Orthogonal sums of the construction give arbitrarily large exact examples. □

Theorem 28.6 (A surface of sharp moment data)**.** *Let* $1 \leq R < 2$*, set* $u = 2 - R$*, and suppose* $u^2 \leq D_0 \leq u$*. Among asymptotic matrix sequences satisfying* (199)*,* $d = N$*, and normalized moment limits*

$$(m_1, m_2, m_3, m_4) = (1, R, 3R - 2, 8R - 8 + D_0),$$

the optimal lower bound for $\liminf s/N$ *is exactly* u^2/D_0*. Sharpness persists when* P *is required to be a sum of unit rank-one operators and* Q *a sum of positive multiplicity-two blocks.*

Proof. Theorem 28.3 gives the lower bound. Put $s_0 = u^2/D_0$ and $\theta = D_0/u$, so $0 < s_0 \leq 1$ and $0 < \theta \leq 1$. Consider the spectral probability measure with masses $(1 - s_0)/2$ at each of $0, 2$ and masses $s_0/2$ at each of $1 - \sqrt{1 - \theta}, 1 + \sqrt{1 - \theta}$. Assign the latter two masses to P and the mass at 2 to Q. Its first moment is one, and its next three moments are

$$2 - s_0\theta = R, \qquad 4 - 3s_0\theta = 3R - 2, \qquad 8 - 8s_0\theta + s_0\theta^2 = 8R - 8 + D_0.$$

For rational s_0, choose dimensions making all multiplicities integral. For general s_0, choose even integers s_N with $s_N/N \to s_0$ and $N - s_N$ even, and keep θ fixed in the same construction. Then $\operatorname{rank} P = \operatorname{Tr} P = s_N$, and Q has rank $(N - s_N)/2$. The two-vector construction in Proposition 28.5 supplies the required unit-vector realization, and the moments converge to the prescribed values. □

These extremals concern the matrix count model, not realizability by actual zeta ordinates or by a band-limited evaluation frame. Such geometry is additional information. In the Weil-compression application, the first two moments are available unconditionally at the critical bandwidth. The third and fourth moments assumed in Corollary 28.4 are the unproved analytic input; this section supplies no unconditional improvement to a zeta zero proportion.

29 Spectral stability and limitations of moment amplification

Continue with the matrix count model (199). The results here refine the preceding certificates and remain finite-dimensional statements; they assert no new trace asymptotics for the zeta compression.

Proposition 29.1 (A dimension obstruction)**.** *For every Hermitian matrix G of dimension $d \le N/2$, (199) is feasible with $s = 0$. If, in addition, $G \succeq 0$, $\operatorname{tr} G = N$, and N is even, then G is a sum of $N/2$ operators of the form $2vv^*$ with $\|v\| = 1$.*

Proof. For the first assertion take $P = 0$, $Q = G$, and $b = n_+(G)$. For the second, set $n = N/2$. The nonnegative eigenvalues of $G/2$, padded with zeros to length n, have sum n and majorize $(1, \dots, 1)$. The finite Schur–Horn theorem gives a positive semidefinite n-by-n Gram matrix with this spectrum and diagonal $(1, \dots, 1)$. Factoring it gives n unit vectors whose frame operator is unitarily equivalent to $G/2$. □

The unit vectors in this construction need not be evaluation vectors of a band-limited function space. Such geometry is extra information. For a sampling grid of spacing $2\pi/L$ in $[T, 2T]$, the ratio of dimension to total zero count tends to η when $L/\log T \to \eta$. The elementary fourth-moment diagonal argument for prime products of length e^{2L} has the safe range $\eta < 1/2$. Proposition 29.1 explains why this direct shortening of bandwidth cannot, by itself, produce a positive simple-zero proportion in the count model.

Proposition 29.2 (A third-moment obstruction)**.** *For $1 \le R \le 2$, there are asymptotic examples satisfying (199), with $d = N$, such that*

$$\frac{1}{N}(\operatorname{tr} G, \operatorname{tr} G^2, \operatorname{tr} G^3) \longrightarrow (1, R, 3R - 2), \qquad \frac{s}{N} \longrightarrow 2 - R.$$

Here P can be a sum of unit rank-one operators and Q a sum of multiplicity-two unit rank-one operators.

Proof. Put $u = 2 - R$. Assign spectral masses u to 1 and $(1 - u)/2$ to each of $0, 2$. Put the 1-eigenspace in P and the 2-eigenspace in Q. All constraints are equalities. The moments are as stated, and the fourth normalized moment is $7R - 6$. Rational approximation of the multiplicities treats general u. □

In particular the three moments $(1, 4/3, 2)$ alone cannot improve $2/3$ within this model. There is also an elementary certificate obstruction: a nonconstant real polynomial which is nonpositive on $(-\infty, 0]$ and bounded above on $[0, \infty)$ has even degree and negative leading coefficient. Thus a certificate in this class of degree at most three already has degree at most two.

Theorem 29.3 (An exact stability remainder)**.** *Assume (199). Set*

$$L_0 = 4 \operatorname{tr} G - \operatorname{tr} G^2 - 2N, \qquad D_0 = \operatorname{tr}\Big(\big(G(2I - G)\big)^2 \Big).$$

For $0 < \theta < 1$ define

$$\mathcal{S}_\theta = \{0, 2, 1 - \sqrt{1 - \theta}, 1 + \sqrt{1 - \theta}\}, \qquad c_\theta = \min\left\{ \frac{2}{\theta}, \frac{1 - \theta}{\theta^2} \right\}.$$

Writing all eigenvalues of G with multiplicity as λ_j, one has

$$s \ge \frac{2L_0}{\theta} - \frac{D_0}{\theta^2} + c_\theta \sum_{j=1}^{d} \operatorname{dist}(\lambda_j, \mathcal{S}_\theta)^2. \tag{201}$$

If $L_0, D_0 > 0$ and $D_0/L_0 \in (0, 1)$, the first two terms at $\theta = D_0/L_0$ equal L_0^2/D_0.

Proof. Let $\lambda_1 \geq \cdots \geq \lambda_r > 0$ be the positive spectrum, put $k = \min(b, r)$, $n = r - k$, and $S = \sum_{j=k+1}^{r} \lambda_j$. The min–max principle applied to $G \preceq P + Q_+$ gives $n \leq s$ and $S \leq s$. Set

$$q_\theta(x) = \frac{8\theta x - (4 + 2\theta)x^2 + 4x^3 - x^4}{\theta^2}.$$

The identities

$$\frac{8}{\theta} - q_\theta(x) = \frac{(x-2)^2(x^2 + 2\theta)}{\theta^2},$$
$$1 + \frac{4x}{\theta} - q_\theta(x) = \frac{(x^2 - 2x + \theta)^2}{\theta^2}$$

give the exact decomposition

$$s - \frac{2L_0}{\theta} + \frac{D_0}{\theta^2} = \frac{4}{\theta}(N - s - 2b) + (s - n) + \frac{4}{\theta}(s - S) + \frac{8}{\theta}(b - k) + \sum_{j=1}^{k} \frac{(\lambda_j - 2)^2(\lambda_j^2 + 2\theta)}{\theta^2} + \sum_{j=k+1}^{r} \frac{(\lambda_j^2 - 2\lambda_j + \theta)^2}{\theta^2} - \sum_{\lambda_j < 0} q_\theta(\lambda_j). \tag{202}$$

Every term on the right is nonnegative. The first spectral sum dominates $(2/\theta)\sum_{j \leq k} |\lambda_j - 2|^2$. For the second put $a_\pm = 1 \pm \sqrt{1 - \theta}$. At least one of $|x - a_-|, |x - a_+|$ is at least $\sqrt{1 - \theta}$, so

$$(x^2 - 2x + \theta)^2 \geq (1 - \theta)\min\{|x - a_-|, |x - a_+|\}^2.$$

Finally, for $y \geq 0$,

$$-q_\theta(-y) = \frac{8\theta y + (4 + 2\theta)y^2 + 4y^3 + y^4}{\theta^2} \geq \frac{4 + 2\theta}{\theta^2} y^2.$$

Comparison with the distance to $\mathcal{S}_\theta$ proves (201). □

Corollary 29.4 (A geometric strengthening of the conditional target)**.** *Suppose a sequence satisfying* (199) *has normalized first four moments tending to* $(1, 4/3, 2, 13/4)$*. If*

$$\liminf \frac{1}{N} \sum_{j=1}^{d} \operatorname{dist}\left(\lambda_j, \left\{0, 2, 1 - \frac{1}{2\sqrt{2}}, 1 + \frac{1}{2\sqrt{2}}\right\}\right)^2 \geq \varepsilon,$$

then

$$\liminf \frac{s}{N} \geq \frac{16}{21} + \frac{8}{49}\varepsilon.$$

Proof. Use $\theta = 7/8$ in Theorem 29.3. □

The numerical value 16/21 is already present in public conditional certificates [22]; the additional statement here is the explicit spectral-distance remainder. Neither the four-moment hypothesis nor a positive spectral-distance lower bound is proved here for the zeta compression. The corollary isolates the extra information that could make evaluation geometry stronger than the unrestricted matrix model.

Proposition 29.5 (Exact nonattainment for sinc evaluation vectors)**.** *Let v_x be unit vectors in a Hilbert space, indexed by real x, with*

$$\langle v_x, v_y \rangle = \frac{\sin \pi(x - y)}{\pi(x - y)}.$$

The kernel takes its continuous value 1 *on the diagonal. Let* $x_1, \dots, x_s, y_1, \dots, y_b$ *be distinct real numbers, with* $s, b > 0$*, and put*

$$P = \sum_{j=1}^{s} v_{x_j} v_{x_j}^*, \qquad Q = 2 \sum_{j=1}^{b} v_{y_j} v_{y_j}^*, \qquad G = P + Q, \qquad N = s + 2b.$$

For every $0 < \theta < 1$*, the defect on the left of* (202) *is strictly positive.*

Proof. Work on the finite-dimensional span of the displayed vectors. If the defect were zero, its nonnegative decomposition would force exactly b top positive eigenvalues equal to 2, all other positive eigenvalues in $\{a_-, a_+\}$, and a positive tail of size s, where $a_\pm = 1 \pm \sqrt{1-\theta} < 2$. Thus $G \preceq 2I$ and, since $0 \preceq Q \preceq G$, $Q \preceq 2I$. But $\operatorname{rank} Q \le b$ and $\operatorname{tr} Q = 2b$, so Q is twice an orthogonal projection of rank b. Positivity of $2I - G$ then shows that P vanishes on the range of Q. Consequently every v_{x_j} is orthogonal to v_{y_1}, and hence $x_j - y_1$ is a nonzero integer. The distinct x_j therefore differ from one another by nonzero integers, so the v_{x_j} are mutually orthogonal. Thus P is a projection. As P and Q have orthogonal ranges, the positive tail spectrum of G is identically 1, contradicting $a_\pm \neq 1$. □

This finite nonattainment statement does not supply a uniform positive defect per zero as the dimension grows. It also concerns real sinc evaluation vectors and a positive simple-plus-double decomposition; off-line Weil pairs need not satisfy these hypotheses. Converting nonattainment into a quantitative asymptotic gap is a separate problem.

30 Remaining analytic questions

The fixed-phase theorem and its uniform extensions have different quantifiers. Theorem 19.1 treats a fixed finite prime set, whereas Theorem 21.3 uses a growing set at its stated tolerance. The conditional profiles of Sections 22–25 take the height limit first. No argument above permits exchanging these orders of limits at an arbitrary prescribed tolerance.

Further analytic questions are whether one can remove the extra power of $\log\log T$ from fixed-prime resolution, obtain the exact entropy constant in a joint height-and-tolerance limit, improve the edge approach rate, or determine the limiting phase density of all simple critical zeros. A nondegenerate joint fluctuation law for the companion roots is not supplied by bounded-density comparison.

Higher-moment amplification has two distinct missing inputs: the required third and fourth trace asymptotics at a usable bandwidth, and quantitative information on the geometry of the evaluation vectors. The exact matrix extremals show what the unrestricted count model allows; finite sinc nonattainment does not furnish an asymptotic spectral gap. Similarly, a zero-free half-plane bounds smooth prime sums but does not automatically improve the rational frequency separation used in deterministic covering.

Part III
Rare-event laws and the recovery of hidden spectra

The preceding prime-phase analysis leads to an inverse question: which features of a weighted independent-site model survive in its rare events? The common local probability law and the individual spectral locations are different unknowns. A normalized continuum rate can recover the first while erasing the second; a centered fluctuation law can recover the information lost at the continuum scale. This part proves that separation, calibrates unknown power weights, and then examines what changes when the local variables are correlated vectors.

All identification assertions use exact functions or exact probability laws, with the normalizations and model classes stated in each result. An open patch of a rate function is infinite information. A channel count is not a sample-complexity bound, and local derivative data are not supplied automatically by noisy probability measurements. The hard-conditioning statements below concern the specified independent product model. The final sinc construction is an abstract sampling obstruction and does not construct a configuration of zeta zeros.

Relation to established methods. Scalar weighted small deviations are treated by Aurzada [23], and Mellin methods for small-deviation expansions by Fill and Torcaso [27]. Nonvanishing Mellin transforms as an inverse and cancellation principle are central to Jacobsen, Mikosch, Rosiński, and Samorodnitsky [28]. Spectral uniqueness for Gamma convolutions is classical [29]; second-chaos cumulants and limits are studied in [30, 31]. Joint constraints and condensation are treated in [32]. General Υ-transforms, matrix mixing, and homogeneous-measure decompositions are established subjects [34, 35, 36]; dominant-atom signed Lévy expansions and random-integral inversion are likewise classical [37, 38]. The results below concern particular observation designs and the information they retain. The use of these classical mechanisms is not itself a novelty claim.

31 From a regularly varying spectrum to joint rare events

This section supplies the probabilistic meaning of the continuum transform. Let $0 < r_1 \le r_2 \le \cdots \to \infty$, with repetitions permitted, and put

$$N(x) = \#\{n : r_n \le x\}.$$

Assume that N is regularly varying with positive index ρ: $N(cx)/N(x) \to c^\rho$ for every $c > 0$. Let V_n be independent copies of a nonnegative, nonconstant random variable V with essential infimum zero and $\mathbb{E}V^2 < \infty$. Throughout this section suppose

$$h(u) = -\log \mathbb{E}e^{-uV}, \qquad h'(u) \sim \frac{\alpha}{u}, \qquad -h''(u) \sim \frac{\alpha}{u^2} \quad (u \to \infty), \qquad \alpha > 0. \tag{203}$$

In particular $h(u) = O(1+\log(1+u))$, and $g(u) := -h''(u) > 0$. The function h is twice continuously differentiable at zero and smooth for $u > 0$. The assumptions hold, for example, when V has a density $cv^{\alpha-1}(1 + O(v^\eta))$ near zero, with $c, \eta > 0$, and finite second moment. They include both bounded phase potentials and Gamma laws.

Fix $E > A > \rho$ and define the almost surely finite sums

$$D_E = \sum_n r_n^{-E} V_n, \qquad D_A = \sum_n r_n^{-A} V_n, \qquad X_x = \frac{x^E D_E}{N(x)}, \quad Y_x = \frac{x^A D_A}{N(x)}.$$

The continuum Laplace exponent and its lower-tail dual are

$$H(s,t) = \rho \int_0^\infty y^{\rho-1} h(sy^{-E} + ty^{-A})\, dy, \tag{204}$$

$$I(e,a) = \sup_{s,t\geq 0} \{H(s,t) - se - ta\}, \qquad e,a > 0. \tag{205}$$

Theorem 31.1 (Joint lower-tail limit)**.** *Under these assumptions, for every $e, a > 0$,*

$$\lim_{x\to\infty} -\frac{1}{N(x)} \log \mathbb{P}(X_x \leq e,\ Y_x \leq a) = I(e,a).$$

The optimizing pair in (205) exists, is unique, and is nonzero. Furthermore,

$$H(s,t) = \inf_{e,a>0} \{I(e,a) + se + ta\}, \qquad s,t > 0. \tag{206}$$

Proof. Regular variation gives vague convergence

$$\frac{1}{N(x)} \sum_n \delta_{r_n/x} \ \longrightarrow\ \rho y^{\rho-1}\, dy \quad \text{on } (0,\infty).$$

This convergence extends to the integrand defining H. Here are the endpoint estimates needed for that extension. Fix $0 < \delta < \min(\rho, A-\rho)$. Potter bounds control the normalized number of points below xy by $Cy^{\rho-\delta}$ for $y \leq 1$ and by $Cy^{\rho+\delta}$ for $y \geq 1$, apart from finitely many original points. Near zero the integrand is $O(1 + |\log y|)$; near infinity it is $O(y^{-A})$. Integration by parts therefore makes both discarded tails uniformly small. Each exceptional fixed point contributes $O(\log x)/N(x) = o(1)$. Consequently

$$\frac{1}{N(x)} \sum_n h\big(s(x/r_n)^E + t(x/r_n)^A\big) \longrightarrow H(s,t). \tag{207}$$

The same argument applies locally uniformly for (s,t) in the positive quadrant, and to first and second derivatives there. For example, at $y \downarrow 0$ the ss-derivative is bounded by $Cy^{\rho-1}$ when $s > 0$; the other derivatives are no worse. At infinity use boundedness of the first two derivatives of h and $A > \rho$.

The function H is strictly concave on the nonnegative quadrant: apply the strict concavity of h pointwise, observing that two distinct parameter pairs give different arguments except possibly at one value of y. In the positive quadrant its negative directional second derivative is the integral of

$$g(sy^{-E} + ty^{-A})(uy^{-E} + vy^{-A})^2,$$

which is positive because $E \neq A$. Since h is increasing and concave with $h(0) = 0$, it is subadditive. Scaling yields

$$H(s,t) \leq H(s,0) + H(0,t) = C_E s^{\rho/E} + C_A t^{\rho/A}$$

for positive constants C_E, C_A. Thus the objective in (205) is coercive and has a unique maximizer. The maximizer is not the origin, since $s^{\rho/E}$ dominates s as $s \downarrow 0$.

Chernoff's inequality and (207) give the asserted lower bound on the negative logarithm. For the reverse bound first consider positive tilt parameters (s,t). Change the product law by the density proportional to

$$\exp\{-sx^E D_E - tx^A D_A\}.$$

Under this law (X_x, Y_x) converges in probability to $(H_s(s,t), H_t(s,t))$: expectations converge by the derivative version of (207), and both variances are $O(1/N(x))$. On a small rectangle around this limit, the inverse density has a uniform exponential lower bound. This proves the matching probability lower bound at all gradients of positive parameters.

Boundary optimizers require one endpoint observation. At $(s,0)$ with $s>0$, both first derivatives are finite. At $(0,t)$ with $t>0$, $H_s(0,t)$ is finite precisely when $E-A<\rho$, because its integrand near zero is asymptotic to $(\rho\alpha/t)y^{\rho-1-(E-A)}$. If this derivative is infinite, $(0,t)$ cannot maximize the objective for finite e. If it is finite, the tilted first moments converge; concentration follows by restricting to $\eta \le r_n/x \le R$, where the variance is $O(1/N(x))$, and making the omitted nonnegative tails small in expectation. This avoids any assumption of a finite boundary Hessian.

At any maximizing pair (s_*, t_*), multiply both coordinates by $1+\varepsilon$. Each finite tilted first moment strictly decreases. The active constraints then have a strict margin below their target; the inactive constraints already have a weak margin by the one-sided optimality inequalities and also strictly decrease. Apply the preceding tilted lower bound on a sufficiently small rectangle. Let its width tend to zero and then let $\varepsilon \downarrow 0$. Terms belonging to inactive constraints have zero multiplier, so the limit of the change-of-measure cost is exactly $I(e,a)$.

Finally, the definition of I gives one inequality in (206). For its reverse, take $(e,a) = \nabla H(s,t)$, whose coordinates are positive and finite. Concavity makes (s,t) the optimizer for this pair, giving equality. □

Corollary 31.2 (Universality and its lost information)**.** *For fixed ρ, E, A and microscopic law V, all spectra with a regularly varying counting function of index ρ have the same normalized joint rate I. In particular, when $\rho = 1$, the positive integers and the primes have identical continuum rates after normalization by their respective counting functions.*

The normalization is essential: $N(x) \sim x$ for integers and $N(x) \sim x/\log x$ for primes. The corollary says that the continuum rate forgets the individual locations of the spectrum, not that the unnormalized probabilities are equal. The later fluctuation theorem recovers information that this leading-order limit discards.

32 Microscopic recovery from a joint rate

Fix $E > A > \rho$ and use the assumptions and notation of Section 31. The single-coordinate continuum rate retains only one number from the law of V. The joint rate, by contrast, determines that law completely.

Theorem 32.1 (Two-energy identifiability)**.** *For fixed ρ, E, A, knowledge of $I(e,a)$ for all $e, a > 0$ uniquely determines the distribution of V. More explicitly, recover H by (206), and put*

$$K(t) = -H_{st}(1,t), \qquad g(u) = -h''(u), \qquad (\mathcal{M}f)(z) = \int_0^\infty u^{z-1} f(u)\, du.$$

For

$$a(z) = 1 + \frac{A-\rho}{E} - \frac{A}{E}z, \qquad \omega(z) = 1 + \frac{A-\rho}{E} + \frac{E-A}{E}z,$$

one has

$$(\mathcal{M}K)(z) = \frac{\rho}{E}\frac{\Gamma(a(z))\Gamma(z)}{\Gamma(\omega(z))}(\mathcal{M}g)(\omega(z)), \qquad \operatorname{Re} z = 1. \tag{208}$$

The multiplier on the right is nonzero. Hence this identity gives the Mellin transform of g on the whole vertical line $\operatorname{Re} w = 2 - \rho/E$*, and uniquely determines g. The remaining recovery is*

$$h(u) = \int_0^\infty \min(u, v) g(v)\, dv, \qquad \mathbb{E}e^{-uV} = e^{-h(u)}. \tag{209}$$

Proof. The hypotheses imply $0 < g(u) \le C(1+u)^{-2}$, since g is continuous at zero and has the specified asymptotic at infinity. Differentiating H in the positive quadrant gives

$$K(t) = \rho \int_0^\infty y^{\rho - E - A - 1} g(y^{-E} + t y^{-A})\, dy.$$

For real $z = 1$, Tonelli's theorem permits exchanging integrals. For complex z on this line the same nonnegative integral proves absolute convergence and hence Fubini's theorem applies. Substitute $u = ty^{-A}$ and then $v = y^{-E}$. This gives

$$(\mathcal{M}K)(z) = \frac{\rho}{E} \int_0^\infty \int_0^\infty v^{a(z)-1} u^{z-1} g(v+u)\, du\, dv.$$

On the specified line, $\operatorname{Re} a(z) = 1 - \rho/E > 0$ and $\operatorname{Re} \omega(z) = 2 - \rho/E \in (1, 2)$. Changing to $r = u + v$ and $q = u/(u+v)$ evaluates the inner beta integral and proves (208). Its absolute convergence also follows directly from $g(u) = O(1)$ near zero and $g(u) = O(u^{-2})$ at infinity.

The Gamma function has neither zeros nor poles in the right half-plane. The multiplier is therefore finite and nonzero. As $\operatorname{Im} z$ ranges over $\mathbb{R}$, so does $\operatorname{Im} \omega(z)$ because $E \neq A$. Mellin uniqueness here is simply Fourier uniqueness applied to the integrable function $v \mapsto e^{cv} g(e^v)$, where $c = 2 - \rho/E$. Thus g is determined almost everywhere and, by continuity, everywhere. Finally $h'(u) \to 0$ and $h(0) = 0$. Integrating $-h'' = g$ twice gives (209); both integrals converge by the preceding bounds. Uniqueness of the Laplace transform of a probability measure finishes the proof. □

The assertion is uniqueness from exact limiting data. It is not a claim that finitely many noisy rare-event measurements give stable recovery. The explicit multiplier will quantify that distinction below.

Proposition 32.2 (No finite list of marginal rates is sufficient)**.** *Fix any finite list $\sigma_1, \dots, \sigma_m > \rho$. There exist distinct laws satisfying (203) whose scalar continuum transforms*

$$H_{\sigma_j}(s) = \rho \int_0^\infty y^{\rho - 1} h(s y^{-\sigma_j})\, dy$$

agree for every $s \ge 0$ and every j. Their scalar lower-tail rates therefore agree as well. The examples can have the same value of α in (203).

Proof. Choose $m + 2$ distinct positive scales $c_1, \dots, c_{m+2}$. The $(m+1) \times (m+2)$ real matrix with rows

$$(1, \dots, 1), \qquad (c_1^{\rho/\sigma_j}, \dots, c_{m+2}^{\rho/\sigma_j}), \quad 1 \le j \le m,$$

has a nonzero null vector b. Choose $\eta > 0$ so small that $\alpha_i^{\pm} = 1 \pm \eta b_i > 0$. Let $V^{\pm}$ be the sum of independent Gamma variables of shapes $\alpha_i^{\pm}$ and scales c_i. Their Laplace exponents are

$$h_{\pm}(u) = \sum_i \alpha_i^{\pm} \log(1 + c_i u).$$

These laws have finite second moments, essential infimum zero, and the same asymptotic coefficient $\alpha = \sum_i \alpha_i^{\pm} = m + 2$. Direct scaling and one integration by parts give

$$H_{\sigma_j,\pm}(s) = \frac{\pi}{\sin(\pi\rho/\sigma_j)} s^{\rho/\sigma_j} \sum_i \alpha_i^{\pm} c_i^{\rho/\sigma_j}.$$

The null-vector equations make these expressions identical. The laws are nevertheless distinct: equality of h'_+ and h'_- would imply $\sum_i b_i c_i/(1 + c_i u) = 0$ for all $u > 0$. Distinct poles of this rational function force every $b_i = 0$, a contradiction. □

32.1 Exact data on one active patch suffice

Lemma 32.3 (Analyticity in the positive quadrant)**.** *Under* (203)*, H is real analytic on $(0,\infty)^2$. Its Hessian is negative definite there.*

Proof. Negative definiteness was proved in Theorem 31.1. For analyticity let $L(u) = \mathbb{E}e^{-uV}$ and note that $M = \sup_{u>0} uh'(u) < \infty$. For $u > 0$ and $|z - u| \le \delta u$ with $0 < \delta < 1$, integration along the segment from u to z yields

$$\left|\frac{L(z)}{L(u)} - 1\right| \le \delta u \, \frac{L((1-\delta)u)}{L(u)} h'((1-\delta)u) \le \frac{\delta M}{(1-\delta)^{M+1}}.$$

Choose δ so that the last expression is less than $1/2$. Thus L is nonzero on these relative disks, and the branch of $-\log L$ agreeing with h on the positive axis is holomorphic there. Fix $s_0, t_0 > 0$. For sufficiently small complex relative perturbations of s_0, t_0, the number $sy^{-E} + ty^{-A}$ lies in such a disk about $s_0 y^{-E} + t_0 y^{-A}$, uniformly in y. Near $y = 0$ the complex integrand is bounded by $C(1 + |\log y|)$. Near infinity it is $O(y^{-A})$: use $|L(z) - 1| \le |z|\mathbb{E}V$ for $\operatorname{Re} z \ge 0$ and the logarithm near 1. These bounds are integrable against $y^{\rho-1}dy$. Holomorphic parameter integration proves the assertion. □

Theorem 32.4 (Local active-patch identifiability)**.** *Let I_1, I_2 arise from two microscopic laws satisfying the stated assumptions, with the same known ρ, E, A. Suppose they agree on a nonempty open set $U \subset (0,\infty)^2$ on which both optimizing multipliers are positive for both laws. Then the two microscopic laws coincide.*

Proof. For a positive optimizer, the inverse function theorem applied to ∇H gives a locally real analytic optimizer as a function of (e, a). On this region

$$(s,t) = -\nabla I(e,a), \qquad \nabla^2 I(e,a) = -[\nabla^2 H(s,t)]^{-1}, \qquad H(s,t) = I(e,a) + se + ta.$$

Consequently agreement of the two rates on U gives agreement of their H functions on a nonempty open set in the positive quadrant. The real analytic identity theorem on that connected quadrant extends the agreement everywhere. Theorem 32.1 applies. □

The active-patch condition is essential. In a region where only one constraint is active, the rate is independent of the other coordinate and supplies only marginal information. Thus the information gain comes from observing the genuinely joint part of the rate surface.

33 Exact smoothing and a regularized inverse

The injectivity theorem concerns exact data. Its inverse is strongly ill-conditioned, and the degree of ill-conditioning can be computed explicitly. This section concerns the linear map from the tilted variance $g = -h''$ to the mixed derivative $K(t) = -H_{st}(1,t)$; it does not assert a minimax lower bound over probability distributions.

Put

$$r = \frac{\rho}{E}, \qquad b = \frac{A}{E}, \qquad q = 1 - b, \qquad c = 2 - r,$$

so that $0 < r < b < 1$. Define logarithmic coordinates

$$f(v) = e^{cv} g(e^v), \qquad k(v) = e^v K(e^v), \qquad d(v) = qk(qv).$$

We use the Fourier convention $\widehat{f}(\xi) = \int_{\mathbb{R}} e^{i\xi v} f(v)\, dv$. The bounds $g(u) = O(1)$ at zero and $g(u) = O(u^{-2})$ at infinity imply $f \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$. The Mellin identity gives

$$\widehat{d}(\xi) = m(\xi)\widehat{f}(\xi), \qquad m(\xi) = rB\Big(1 - r - i\frac{b}{q}\xi, 1 + i\frac{\xi}{q}\Big). \tag{210}$$

In particular $d \in L^2(\mathbb{R})$.

Proposition 33.1 (Exact smoothing scale)**.** *The multiplier in* (210) *never vanishes, and*

$$|m(\xi)| \sim r\sqrt{2\pi}\, b^{1/2-r} q^{r-1} |\xi|^{-1/2} \exp\Big(-\frac{\pi A}{E - A}|\xi|\Big) \qquad (|\xi| \longrightarrow \infty). \tag{211}$$

Consequently the inverse of this linear map is unbounded on its L^2 range.

Proof. The two arguments of the beta function have positive real parts, as does their sum. The gamma function has no zeros and has no poles at these arguments. Thus m does not vanish. Apply

$$|\Gamma(a + it)| \sim \sqrt{2\pi}\, |t|^{a-1/2} e^{-\pi|t|/2}$$

to the two gamma functions in the numerator and the one in the denominator. The imaginary parts are $-b\xi/q$, ξ/q, and ξ; their exponential factors give $\exp(-\pi b|\xi|/q)$. The powers of $|\xi|$ combine to $|\xi|^{-1/2}$, and the remaining factors give the displayed constant. For unboundedness, take real L^2 functions of norm one whose Fourier transforms are supported in $[n, n+1] \cup [-n-1, -n]$. Their images have norm at most $\sup_{|\xi| \geq n} |m(\xi)|$, which tends to zero. These perturbations are ambient L^2 functions; they need not be tilted variances of probability laws. □

Proposition 33.2 (Spectral cutoff bound)**.** *Let $\ell > 0$ and suppose $\|f\|_{H^\ell} \leq M$. If measured data satisfy $\|d_\delta - d\|_2 \leq \delta$, define*

$$\widehat{f}_{\delta,R}(\xi) = \mathbf{1}_{\{|\xi| \leq R\}} \frac{\widehat{d}_\delta(\xi)}{m(\xi)}.$$

For $R \geq 1$,

$$\|f_{\delta,R} - f\|_2 \leq C\delta(1 + R)^{1/2} \exp\Big(\frac{\pi A}{E - A} R\Big) + M(1 + R^2)^{-\ell/2}, \tag{212}$$

where C depends only on E, A, ρ. For any fixed $0 < \eta < 1$, the choice

$$R = \frac{\eta(E - A)}{\pi A} \log(1/\delta)$$

gives, for sufficiently small δ and fixed $M > 0$, $\|f_{\delta,R} - f\|_2 = O((\log(1/\delta))^{-\ell})$.

Proof. Continuity, nonvanishing, and (211) imply

$$\sup_{|\xi|\le R} |m(\xi)|^{-1} \le C(1+R)^{1/2} e^{\pi AR/(E-A)}.$$

Plancherel's theorem bounds the error on $[-R,R]$ by this supremum times δ. The Sobolev bound gives the second term outside that interval. For the stated choice of R, the first term is $O(\delta^{1-\eta}\sqrt{\log(1/\delta)})$ and the second is $O((\log(1/\delta))^{-\ell})$. □

Remark 33.3. If the error is instead specified as $\|k_\delta - k\|_2 \le \delta$, then $\|d_\delta - d\|_2 \le \sqrt{q}\,\delta$, and the same conclusion holds. The noise assumption is on a mixed derivative of the recovered free energy, after passage to logarithmic coordinates. A noise bound on the original rate surface does not automatically imply this assumption: convex duality and numerical differentiation require a separate error analysis. The factor $\pi A/(E-A)$ tends to infinity when the two exponents coalesce. At $E = A$ the transform ceases to be injective.

34 Microscopic fluctuations retain the discrete spectrum

The following statement requires no counting law for the weights. It isolates information that a leading large-deviation integral can erase. Let $V \ge 0$ have finite second moment and write

$$h(u) = -\log \mathbb{E}e^{-uV}, \qquad g(u) = -h''(u) = \mathrm{Var}_u(V), \qquad u \ge 0,$$

where $\mathbb{P}_u$ denotes the one-coordinate exponential tilt. Assume, for a fixed $\alpha > 0$,

$$uh'(u) \longrightarrow \alpha, \qquad u^2 g(u) \longrightarrow \alpha \quad (u \to \infty). \tag{213}$$

In particular $M := \sup_{u\ge 0} u^2 g(u) < \infty$; finiteness near zero follows from $\mathbb{E}V^2 < \infty$.

Theorem 34.1 (Square-summable ratio fluctuation theorem)**.** *Let V_n be independent copies of V, and let $b_n > 0$ satisfy $\sum_n b_n < \infty$. For a real sequence $a = (a_n) \in \ell^2$ set*

$$D = \sum_n b_n V_n, \qquad X = \sum_n a_n b_n V_n, \qquad d\mathbb{P}_t = \frac{e^{-tD}}{\mathbb{E}e^{-tD}}\, d\mathbb{P}.$$

Both series are absolutely convergent almost surely. Put $m_t = \mathbb{E}_t X = \sum_n a_n b_n h'(tb_n)$. Then, as $t \to \infty$,

$$t(X - m_t) \implies Y_a := \sum_{n\ge 1} a_n (G_n - \alpha), \qquad G_n \overset{\text{iid}}{\sim} \mathrm{Gamma}(\alpha, 1). \tag{214}$$

The series defining Y_a converges in L^2 and almost surely. The variances also converge:

$$t^2 \,\mathrm{Var}_t(X) \longrightarrow \alpha \sum_n a_n^2.$$

The characteristic function and the cumulants of the limiting law are

$$\mathbb{E}e^{izY_a} = \prod_n e^{-i\alpha z a_n}(1 - iza_n)^{-\alpha}, \tag{215}$$

$$\kappa_k(Y_a) = \alpha(k-1)! \sum_n a_n^k, \qquad k \ge 2. \tag{216}$$

In particular the limit is non-Gaussian whenever $a \ne 0$.

Proof. Cauchy–Schwarz gives $\sum_n |a_n|b_n \le \|a\|_2\|b\|_2 < \infty$. Together with $\mathbb{E}V < \infty$, this proves absolute convergence of both original series. Exponential tilting preserves independence, and coordinate n has parameter $u = tb_n$. For $z \ge 0$, the Laplace transform of uV under $\mathbb{P}_u$ is

$$\mathbb{E}_u e^{-zuV} = \exp\{h(u) - h((1+z)u)\} \longrightarrow (1+z)^{-\alpha},$$

because $h((1+z)u) - h(u) = \int_1^{1+z} uh'(uv)\,dv \to \alpha\log(1+z)$. Also $\mathbb{E}_u(uV) = uh'(u) \to \alpha$ and $\mathrm{Var}_u(uV) = u^2 g(u) \to \alpha$. Thus each finite vector of the centered coordinates converges to independent copies of $G - \alpha$.

For every R and every $t > 0$, independence gives the uniform tail bound

$$\mathbb{E}_t \left| \sum_{n>R} a_n\{tb_nV_n - tb_nh'(tb_n)\} \right|^2 \le M \sum_{n>R} a_n^2. \tag{217}$$

Finite-dimensional convergence followed by $R \to \infty$ proves (214). Dominated convergence in $\sum_n a_n^2(tb_n)^2 g(tb_n)$ proves variance convergence. The limiting independent centered series has summable variances, which also proves its almost-sure convergence.

The logarithm of each factor in (215) is $O_z(a_n^2)$ for all sufficiently large n, so the product converges. Since an ℓ^2 sequence is bounded, the logarithm is analytic in a fixed neighborhood of zero and can be expanded absolutely there. This gives (216). For $a \ne 0$ its fourth cumulant is $6\alpha\sum_n a_n^4 > 0$, excluding a Gaussian law. □

Corollary 34.2 (The law determines the nonzero coefficient spectrum)**.** *For fixed known $\alpha > 0$, the law of Y_a determines the multiset of nonzero real coefficients a_n, including multiplicities.*

Proof. The finite measure $\lambda_a = \sum_n a_n^2 \delta_{a_n}$ has compact support. All its moments are determined by the law, since

$$\int x^j\, d\lambda_a(x) = \frac{\kappa_{j+2}(Y_a)}{\alpha(j+1)!}, \qquad j \ge 0.$$

Compactly supported finite measures are determined by their moments: on a compact interval containing the supports of two candidates, polynomials are uniformly dense in continuous functions. For each nonzero x, its multiplicity is $\lambda_a(\{x\})/x^2$. Zero coefficients are invisible and are expressly excluded. □

Remark 34.3 (Scope of the inversion)*.* The recovery corollary is exact identifiability, not a quantitative or numerically stable recovery algorithm. It reconstructs a multiset; the ordering of the coefficients and the underlying weights b_n cannot be recovered without additional structure. Neither theorem asserts that conditioning on $D \le \varepsilon$ may replace the canonical tilt. Such a replacement needs a separate local limit argument. Nor can a nondegenerate limiting law be transferred merely from bounds $c\mu \le \nu \le \mu$.

35 Identifying the law and the spectrum together

The preceding results can be combined into an exact inverse theorem for two different layers of a model. All parameters and normalizations in the following statement are part of the observed model; in particular, the two power exponents are known.

Theorem 35.1 (Two-scale identifiability)**.** *Fix $E > A > \rho > 0$ with $2(E - A) > \rho$. For each of two candidate models, let its positive spectrum tend to infinity with a counting function regularly varying of index ρ, and let its common microscopic law satisfy (203) and have finite second moment. Use the normalization of Theorem 31.1. Suppose the models have the following two pieces of identical data:*

(i) their joint lower-tail rate surfaces agree on a nonempty open patch where both constraints are active for both models;

(ii) under the tilt proportional to e^{-tD_A}, their limiting centered fluctuation laws of $t(D_E - \mathbb{E}_t D_E)$ agree.

Then the microscopic laws coincide and the spectral location multisets coincide, including multiplicities.

Proof. The first datum and Theorem 32.4 identify the microscopic law, and in particular the coefficient α. Let $d = E - A$. Regular variation and $2d > \rho$ imply $\sum_n r_n^{-2d} < \infty$. Apply Theorem 34.1 with $b_n = r_n^{-A}$ and $a_n = r_n^{-d}$. The second datum and Corollary 34.2 identify the positive coefficient multiset (r_n^{-d}). The injective map $a \mapsto a^{-1/d}$ recovers the original spectral locations. □

The two data sources are mathematically distinct. The first is a limiting logarithmic probability surface, normalized by the number of modes. The second is an order-one distributional limit under a specified tilted law. Neither is replaced here by a finite sample or by a bounded comparison between probability measures. The theorem is a uniqueness statement for exact asymptotic data; a statistical estimation theorem would require additional assumptions and error analysis.

36 A two-scale distinction between integers and primes

The two inverse questions concern different objects. The continuum rate identifies the common law of the independent variables, whereas a non-Gaussian fluctuation law can identify the individual spectral locations. The following example separates these statements explicitly.

Take independent unit exponential variables V_n, so $h(u) = \log(1+u)$, $g(u) = (1+u)^{-2}$, and $\alpha = 1$. For a spectrum $\mathcal{R}$, put

$$D_2^{\mathcal{R}} = \sum_{r\in\mathcal{R}} r^{-2} V_r, \qquad D_3^{\mathcal{R}} = \sum_{r\in\mathcal{R}} r^{-3} V_r.$$

Consider either $\mathcal{R} = \mathbb{N}$ or $\mathcal{R} = \mathbb{P}$, the set of primes. The counting functions are respectively asymptotic to x and $x/\log x$. Both have index $\rho = 1$, so their normalized joint lower-tail rate is the same function:

$$H(s,t) = \int_0^\infty \log(1 + sy^{-3} + ty^{-2})\, dy, \qquad I(e,a) = \sup_{s,t\ge 0}\{H(s,t) - se - ta\}.$$

The inverse theorem recovers the unit exponential microscopic law from this common surface. It cannot recover whether the spectrum consists of primes or all positive integers from this surface alone.

Now impose the canonical tilt proportional to $e^{-tD_2^{\mathcal{R}}}$. Under it the variables remain independent and V_r has exponential rate $1 + t/r^2$. Writing $m_{\mathcal{R}}(t)$ for the tilted mean of $D_3^{\mathcal{R}}$, the fluctuation theorem gives

$$t(D_3^{\mathcal{R}} - m_{\mathcal{R}}(t)) \implies Z_{\mathcal{R}} := \sum_{r\in\mathcal{R}} \frac{G_r - 1}{r}, \qquad G_r \text{ independent unit exponentials.} \tag{218}$$

The series converges in L^2 and almost surely. Its characteristic function and cumulants are

$$\mathbb{E}e^{iuZ_{\mathcal{R}}} = \prod_{r\in\mathcal{R}} e^{-iu/r}(1 - iu/r)^{-1}, \qquad \kappa_k(Z_{\mathcal{R}}) = (k-1)! \sum_{r\in\mathcal{R}} r^{-k}, \quad k \ge 2.$$

For $\mathcal{R} = \mathbb{N}$, the classical Weierstrass product for the Gamma function [33] gives

$$\mathbb{E}e^{iuZ_{\mathbb{N}}} = e^{-i\gamma u}\Gamma(1 - iu).$$

Thus $Z_{\mathbb{N}}$ is a standard Gumbel random variable centered at its mean γ. For primes one instead obtains

$$\kappa_k(Z_{\mathbb{P}}) = (k-1)!P(k), \qquad P(z) = \sum_p p^{-z} \quad (\operatorname{Re} z > 1).$$

Already the variances differ: $\zeta(2)$ for the integer spectrum and $P(2)$ for the prime spectrum. All cumulants together determine the entire reciprocal spectrum, including multiplicities, by the compact moment argument in the fluctuation section.

Remark 36.1 (What the example does and does not establish). The Gamma product, the Gumbel representation, and uniqueness of the associated spectral measure are classical ingredients. Their role here is to exhibit exactly which information is discarded by the continuum rare-event limit and recovered by a second asymptotic scale. Equation (218) is a statement under an explicit probability tilt. It is not, by itself, a limit law along the zeros of the Riemann zeta function, nor a prime-detection algorithm from finitely many noisy observations.

36.1 An exactly computable transform check

For this exponential microscopic law,

$$(\mathcal{M}g)(w) = \Gamma(w)\Gamma(2-w), \qquad 0 < \operatorname{Re} w < 2.$$

The mixed Mellin data therefore equal

$$(\mathcal{M}K)(z) = \frac{\rho}{E}B(a(z), z)\Gamma(\omega(z))\Gamma(2-\omega(z)),$$

with $a(z)$ and $\omega(z)$ defined in the inverse theorem. In particular

$$\int_0^\infty K(t)\,dt = \frac{\pi\rho/E}{\sin(\pi\rho/E)}.$$

This identity checks both the density factor ρ and the Mellin normalization without a numerical inverse problem.

37 Analytic and probabilistic foundations

We extend the continuum foundation to any finite number of distinct scalar power channels and record the inversion facts needed for blind calibration. The two-power and fluctuation proofs have already been given above and will be used directly. Throughout this section $V \geq 0$ is nondegenerate, $\mathbb{E}V^2 < \infty$, and

$$h(u) = -\log \mathbb{E}e^{-uV}, \qquad uh'(u) \longrightarrow \alpha > 0, \qquad u^2 g(u) \longrightarrow \alpha, \qquad g = -h''. \tag{219}$$

Differentiation of the Laplace transform shows that $h'(u)$ is the mean and $g(u) > 0$ is the variance of V under the tilt proportional to e^{-uV}. In particular

$$h(u) = O(\log(2+u)), \quad h(u) \leq u\mathbb{E}V, \quad h'(u) \leq \frac{C}{1+u}, \quad 0 < g(u) \leq \frac{C}{(1+u)^2}. \tag{220}$$

Here the first bound is meant globally and follows by integrating h'; continuity at zero follows from the finite second moment.

Lemma 37.1 (Continuum transform and the fully active rate)**.** *Let $r_n > 0$ tend to infinity and suppose $N(x) = \#\{n : r_n \le x\}$ is regularly varying of index $\rho > 0$: $N(xy)/N(x) \to y^\rho$ for every $y > 0$. Fix distinct powers $\sigma_1, \dots, \sigma_m > \rho$ and define*

$$D_j = \sum_n r_n^{-\sigma_j} V_n, \qquad Z_{x,j} = \frac{x^{\sigma_j} D_j}{N(x)}, \qquad H(s) = \rho \int_0^\infty y^{\rho-1} h\left(\sum_j s_j y^{-\sigma_j}\right) dy,$$

where the V_n are independent copies of V. The series D_j are finite almost surely. For s in the positive orthant,

$$-\frac{1}{N(x)} \log \mathbb{E} e^{-N(x) s \cdot Z_x} = \frac{1}{N(x)} \sum_n h\left(\sum_j s_j (x/r_n)^{\sigma_j}\right) \longrightarrow H(s).$$

The convergence also holds for derivatives through order two locally in that orthant. The function H is real analytic and strictly concave. At every $z = \nabla H(s)$ with $s > 0$,

$$-\frac{1}{N(x)} \log \mathbb{P}\{Z_{x,j} \le z_j \text{ for all } j\} \longrightarrow I(z) := H(s) - s \cdot z = \sup_{v \ge 0} \{H(v) - v \cdot z\}. \tag{221}$$

Near each such point, $-\nabla I(z) = s$ and $\nabla^2 I(z) = -[\nabla^2 H(s)]^{-1}$.

Proof. Regular variation gives $N(x) = O(x^{\rho+\eta})$ for every fixed $\eta > 0$, so summation over dyadic intervals proves $\sum_n r_n^{-\sigma_j} < \infty$. This and $\mathbb{E}V < \infty$ prove almost-sure summability. The measures $N(x)^{-1} \sum_n \delta_{r_n/x}$ converge on compact subintervals of $(0, \infty)$ to $\rho y^{\rho-1} dy$. We detail the required tail control so that compact convergence suffices here. For every sufficiently small $\eta > 0$, the regular-variation ratio bound gives, whenever x and xy exceed a fixed constant,

$$\frac{N(xy)}{N(x)} \le C_\eta \begin{cases} y^{\rho-\eta}, & 0 < y \le 1, \\ y^{\rho+\eta}, & y \ge 1. \end{cases}$$

This standard bound follows by iterating the defining ratio limit on a fixed geometric mesh and using monotonicity between mesh points. Near $y = 0$ the summand is $O(1 + |\log y|)$, by (220); near infinity it is $O(y^{-\min_j \sigma_j})$. Summing over geometric annuli and choosing $\eta < \min(\rho, \min_j \sigma_j - \rho)$ gives uniformly vanishing tails. The finitely many terms with bounded r_n contribute at most $O(\log x / N(x)) = o(1)$.

For the first two derivatives the relevant summands are $y^{-\sigma_i} h'(u)$ and $y^{-\sigma_i - \sigma_j} g(u)$, with $u = \sum_k s_k y^{-\sigma_k}$. On compact positive parameter sets, both are bounded near zero, since each $s_k y^{-\sigma_k} \le u$; near infinity they are bounded by the respective integrable powers. The same argument gives locally uniform derivative convergence. Moreover $-\nabla^2 H$ is the Gram matrix of the functions $y^{-\sigma_j}$ in the positive measure $\rho y^{\rho-1} g(u) dy$. Distinct powers are linearly independent, so the matrix is positive definite.

For analyticity write $L(u) = \mathbb{E} e^{-uV}$ and $M_1 = \sup_{u>0} u h'(u) < \infty$. If $|w - u| \le \delta u$, $u > 0$, and $0 < \delta < 1$, then

$$\left| \frac{L(w)}{L(u)} - 1 \right| \le \delta u \, \frac{L((1-\delta)u)}{L(u)} h'((1-\delta)u) \le \frac{\delta M_1}{(1-\delta)^{M_1+1}}.$$

The first inequality follows by integrating L' along the segment; the second follows by integrating $h'(v) \le M_1/v$. Choose δ so the final bound is less than $1/2$. Thus L has a holomorphic logarithm in every such relative disk. A sufficiently small complex relative perturbation of all s_j keeps u in the corresponding disk, uniformly in y. The resulting logarithm is bounded by $C(1 + |\log y|)$ near

zero and $Cy^{-\min \sigma_j}$ near infinity. For the latter bound use $|L(w) - 1| \le |w|\mathbb{E}V$ when $\operatorname{Re} w \ge 0$. Holomorphic parameter integration proves real analyticity of H.

Write H_x for the normalized logarithmic Laplace transform. The exponential tilt with density $\exp\{-N(x)s \cdot Z_x + N(x)H_x(s)\}$ has mean $\nabla H_x(s)$ and covariance $-\nabla^2 H_x(s)/N(x)$ for Z_x. The established derivative convergence therefore gives concentration at $\nabla H(s)$. The Chernoff bound proves the lower bound on the negative logarithm in (221). For its reverse, use the tilt with parameter $(1+\varepsilon)s$. Every coordinate of $\nabla H((1+\varepsilon)s)$ is strictly less than the corresponding coordinate of z, because all entries of $\nabla^2 H$ are negative. A small box around this new mean lies inside the lower orthant with corner z and has tilted probability tending to one. Undo the tilt on that box; then let its radius tend to zero and $\varepsilon \downarrow 0$. This gives the reverse bound. Strict concavity makes s the unique optimizer, and the inverse function theorem gives the stated derivative identities. □

Lemma 37.2 (Two-power inversion from an active patch)**.** *For known $E > A > \rho > 0$, the values of the joint rate I on any nonempty open fully active patch uniquely determine the function h and hence the distribution of V. More explicitly, once H is recovered, put $K(t) = -H_{st}(1,t)$ and $(\mathcal{M}f)(z) = \int_0^\infty u^{z-1} f(u)du$. For $\operatorname{Re} z = 1$, set*

$$a(z) = 1 + \frac{A-\rho}{E} - \frac{A}{E}z, \qquad w(z) = 1 + \frac{A-\rho}{E} + \frac{E-A}{E}z.$$

Then

$$(\mathcal{M}K)(z) = \frac{\rho}{E}\frac{\Gamma(a(z))\Gamma(z)}{\Gamma(w(z))}(\mathcal{M}g)(w(z)), \qquad h(u) = \int_0^\infty \min(u,v)g(v)dv. \tag{222}$$

The same inversion determines a positive constant multiple of a rescaled h if such a function is used in place of h.

Proof. The local-patch assertion is Theorem 32.4, and the displayed Mellin identity and reconstruction are exactly Theorem 32.1, with $w = \omega$. Those proofs establish absolute convergence, the nonvanishing multiplier, and Mellin uniqueness. All inversion steps are linear in h or g, and their bounds persist under positive multiplication and positive rescaling of the argument. This proves the final assertion without assuming that the resulting function is itself a probability Laplace exponent. □

Lemma 37.3 (Canonical Gamma-series fluctuation limit)**.** *Let $b_n > 0$ with $\sum_n b_n < \infty$, and let $a = (a_n) \in \ell^2$ be real. For independent copies V_n of V, put $D = \sum_n b_n V_n$, $X = \sum_n a_n b_n V_n$, and let $\mathbb{P}_t$ have density $e^{-tD}/\mathbb{E}e^{-tD}$. Both original series converge absolutely almost surely, and*

$$t(X - \mathbb{E}_t X) \implies Y_a = \sum_n a_n(G_n - \alpha), \qquad G_n \overset{\text{iid}}{\sim} \operatorname{Gamma}(\alpha, 1). \tag{223}$$

The limiting series converges almost surely and in L^2, its variance is $\alpha \sum_n a_n^2$, and its cumulants are

$$\kappa_k(Y_a) = \alpha(k-1)! \sum_n a_n^k, \qquad k \ge 2. \tag{224}$$

For known α, its law determines the multiset of nonzero coefficients a_n, including multiplicities.

Proof. The convergence, summability, variance, and cumulant assertions are Theorem 34.1. Recovery of the nonzero coefficient multiset, including multiplicity, is Corollary 34.2. Those results allow arbitrary real square-summable coefficients, as required here. □

For $b_n = r_n^{-A}$ and $a_n = r_n^{-(E-A)}$, regular variation, $A > \rho$, and $2(E-A) > \rho$ give precisely the summability required in Lemma 37.3. This identifies the continuum inversion and the microscopic fluctuation statements as two distinct, fully justified pieces of data.

38 Blind calibration and the exact coordinate ambiguity

The power exponents need not be supplied to the inverse problem. Their ratios to the counting exponent can be recovered from the local curvature of the joint rate. This is a finite-dimensional consequence of weighted Euler homogeneity, with a sign structure that rules out singular calibration matrices.

Let $m \geq 2$, let $\sigma_1, \dots, \sigma_m > \rho > 0$ be distinct, and set $p_j = \sigma_j/\rho$. For the one-coordinate Laplace exponent $h(u) = -\log \mathbb{E}e^{-uV}$, assume $V \geq 0$ is nondegenerate, $\mathbb{E}V^2 < \infty$, and, for some $\alpha > 0$,

$$uh'(u) \longrightarrow \alpha, \qquad -u^2 h''(u) \longrightarrow \alpha.$$

Put

$$H(s) = \rho \int_0^\infty y^{\rho-1} h\left(\sum_{j=1}^m s_j y^{-\sigma_j}\right) dy = \int_0^\infty h\left(\sum_{j=1}^m s_j x^{-p_j}\right) dx, \qquad s \in (0,\infty)^m,$$

and $I(z) = \sup_{s \geq 0}\{H(s) - s \cdot z\}$. The second representation follows from $x = y^\rho$. The Hessian of H is negative definite: with $g = -h'' > 0$, its negative is the Gram matrix of the linearly independent functions x^{-p_j} in the measure $g(\sum_j s_j x^{-p_j})dx$. Every entry of $-\nabla^2 H$ is positive.

Theorem 38.1 (Calibration from a single second-order jet)**.** *At any point z where the optimizing vector is strictly positive, define, using only the first and second derivatives of the observed rate,*

$$s = -\nabla I(z), \qquad M = [\nabla^2 I(z)]^{-1}, \qquad B = \operatorname{diag}(z) - M \operatorname{diag}(s).$$

Then B is invertible, B^{-1} is entrywise nonnegative, and

$$(p_1, \dots, p_m)^{\mathsf{T}} = B^{-1} z. \tag{225}$$

Thus the first and second derivatives of I at a single fully active point determine every ratio σ_j/ρ, independently of the unknown microscopic distribution.

Proof. The change of variable $x = \lambda v$ gives $H(\lambda^{p_1} s_1, \dots, \lambda^{p_m} s_m) = \lambda H(s)$. Euler differentiation gives $\sum_j p_j s_j H_j = H$. Differentiating once more, for every i, yields

$$p_i H_i + \sum_j p_j s_j H_{ij} = H_i.$$

At the saddle point, $\nabla H = z$ and $-\nabla^2 H = M$, so these identities say $Bp = z$. The off-diagonal entries of B are negative. Let $P = \operatorname{diag}(p)$. The matrix BP has positive row sums z_i, nonpositive off-diagonal entries, and therefore positive diagonal entries satisfying strict row diagonal dominance. Writing $BP = D - C$ with D positive diagonal and $C \geq 0$ having zero diagonal, the row sums of $D^{-1}C$ are strictly less than one. Hence

$$(BP)^{-1} = \sum_{k \geq 0} (D^{-1}C)^k D^{-1} \geq 0.$$

This proves invertibility and inverse nonnegativity, and then the formula. The identities connecting the derivatives of I and H follow from the inverse function theorem and strict concavity. □

The matrix argument is the classical nonsingular M-matrix criterion; its role here is to prove that calibration cannot fail at an exceptional fully active point; see also [39]. The theorem concerns exact derivatives. It does not replace derivative estimation from noisy data. There is also a useful exact

residual certificate. For any trial vector $\widehat{p}$, put $r = B\widehat{p} - z$. Since $P^{-1}B^{-1}\operatorname{diag}(z)$ is nonnegative and has every row sum equal to one,

$$\max_j \left| \frac{\widehat{p}_j - p_j}{p_j} \right| \le \max_i \frac{|r_i|}{z_i}. \tag{226}$$

The right side is computable from the exact local data without knowing p.

Theorem 38.2 (Blind two-scale identification modulo a power change)**.** *Consider two models with unknown parameters $E > A > \rho > 0$, positive spectra $r_n \to \infty$ whose counting functions are regularly varying of index ρ, and microscopic laws satisfying the preceding assumptions. Write $N(x) = \#\{n : r_n \le x\}$, $D_E = \sum_n r_n^{-E} V_n$, and $D_A = \sum_n r_n^{-A} V_n$. For each model use $N(x)$ as the large-deviation speed and $(x^E D_E/N(x), x^A D_A/N(x))$ as the normalized threshold variables, and assume $2(E-A) > \rho$. The data below are only the resulting limiting functions and laws, not the full pointwise function N. Suppose that*

1. *the normalized joint rate surfaces agree on a nonempty open patch on which both constraints are active;*
2. *under the tilt proportional to e^{-tD_A}, the limiting laws of $t(D_E - \mathbb{E}_t D_E)$ agree.*

Then the microscopic laws coincide, the ratios $p = E/\rho$ and $q = A/\rho$ coincide, and the multisets

$$\{r_n^\rho : n \ge 1\}$$

coincide, including multiplicities. Equivalently, the only remaining ambiguity is

$$(\rho, E, A, r_n) \longmapsto (c\rho, cE, cA, r_n^{1/c}), \qquad c > 0, \tag{227}$$

up to a permutation of the spectrum. Every transformation in (227) does preserve both observations.

Proof. Theorem 38.1 identifies p, q from any one point of the common patch. Replacing y by y^ρ reduces the continuum transform to the same known exponents p, q and counting exponent one. The active patch recovers a patch of H by Legendre duality. Lemma 37.2 then identifies h and the law of V, including α.

By Lemma 37.3, the microscopic fluctuation limit is

$$Y = \sum_n a_n (G_n - \alpha), \qquad a_n = r_n^{-(E-A)}, \quad G_n \overset{\text{iid}}{\sim} \operatorname{Gamma}(\alpha, 1).$$

Here $\sum_n a_n^2 < \infty$ by regular variation and the assumed inequality. Its cumulants determine $\sum_n a_n^2 \delta_{a_n}$, a compactly supported finite measure, because $\kappa_{k+2}(Y)/[\alpha(k+1)!] = \sum_n a_n^{k+2}$. Moment uniqueness on a compact interval determines every positive a_n and its multiplicity. Hence the data determine

$$r_n^\rho = a_n^{-1/(p-q)}.$$

Writing $c = \rho_2/\rho_1$ gives precisely (227). Conversely, this transformation leaves every coefficient r_n^{-E} and r_n^{-A} unchanged, and sends $N(x)$ to $N(x^c)$, which is regularly varying with index $c\rho$. It also leaves p, q, h and the normalized continuum transform unchanged. Both observations are therefore preserved. □

One calibrated exponent, or the value of ρ, removes the ambiguity. Alternatively, one identified spectral location $r_* \neq 1$ together with its recovered canonical location $R_* = r_*^\rho$ determines $\rho = \log R_* / \log r_*$, and therefore every absolute exponent and spectral location. Knowing a location equal to one supplies no such calibration. Since the power change leaves all weighted sums unchanged pathwise, further observations formed from the same transformed powers cannot distinguish the equivalent models.

Proposition 38.3 (Finite-jet non-identification of the microscopic law)**.** *Fix distinct powers $p_1, \dots, p_m > 1$, finitely many positive parameter points, and any finite derivative order $k \geq 2$. There exist two distinct admissible microscopic laws having the same α such that their continuum transforms have identical jets through order k at all those points. At the corresponding common fully active threshold points, the rate functions also have identical jets through order k. On any fixed admissible spectrum, their limiting centered Gamma-series fluctuation laws also agree, since those limits depend on the microscopic law only through α.*

Proof. Choose positive distinct scales $c_1, \dots, c_N$, where N is larger than the total number of prescribed real jet entries plus one. Let

$$h_\pm(u) = \sum_{\ell=1}^{N} (1 \pm \eta b_\ell) \log(1 + c_\ell u).$$

Every prescribed derivative of the corresponding H is linear in the coefficients $1 \pm \eta b_\ell$. Impose equality of all of them, and impose $\sum_\ell b_\ell = 0$. This homogeneous real linear system has a nonzero solution b. Choose $\eta > 0$ small enough that all shapes $1 \pm \eta b_\ell$ are positive. Then $h_\pm$ are the Laplace exponents of sums of independent Gamma variables, with the same asymptotic coefficient $\alpha = N$, and their specified jets agree. The laws differ: equality of their derivatives would give the rational identity $\sum_\ell b_\ell c_\ell / (1 + c_\ell u) = 0$; its distinct poles force all $b_\ell = 0$.

Matching gradients give the same threshold points. Strict concavity makes the specified positive parameter points their unique optimizers. The derivatives of an analytic Legendre transform through order k are determined by the corresponding jet through order k whenever the Hessian is nonsingular. This follows inductively by differentiating $\nabla H(s(z)) = z$, and proves the last assertion. □

The hierarchy is therefore strict: local curvature calibrates all exponent ratios, but no finite list of finite-order derivatives determines the microscopic law. An exact open active patch supplies the missing infinite information; the fluctuation law subsequently recovers the discrete canonical spectrum.

Remark 38.4 (Why the speed normalization is substantive)*.* If a one-coordinate law is infinitely divisible, duplicate every spectral location k times and replace that law by its kth convolution root. The complete joint law of every finite collection of weighted sums is unchanged, since the k variables at each location sum to the original one-coordinate law. For example, $\mathrm{Gamma}(\alpha, 1)$ becomes k independent copies of $\mathrm{Gamma}(\alpha/k, 1)$ at each site. The raw observations cannot distinguish these descriptions. The counting function, however, is multiplied by k, the normalized threshold variables are divided by k, and the continuum transform is divided by k. Thus the normalization defining the limiting data in Theorem 38.2 is substantive, rather than an innocuous omission. Without it, microscopic multiplicity and convolution divisibility create an additional ambiguity.

Proposition 38.5 (The exact shape–multiplicity ambiguity)**.** *Let $a_n > 0$, $\sum_n a_n^2 < \infty$, and let $Y = \sum_n a_n (G_n - \alpha)$ with unknown $\alpha > 0$. Its law determines the set of distinct coefficient values x and the positive numbers $w_x = \alpha m_x$, where m_x is the multiplicity of x. There is a largest number*

$\alpha_* > 0$ such that w_x/α_* is an integer for every x. All representations of the given law within this class have exactly the shapes and multiplicities

$$\alpha = \frac{\alpha_*}{k}, \qquad m_x = k\frac{w_x}{\alpha_*}, \qquad k = 1, 2, \dots.$$

In particular, imposing the normalization $\gcd_x m_x = 1$ identifies both α and the coefficient multiset. A simple spectrum automatically has this normalization.

Proof. Without knowing α, the cumulants determine all moments of the compactly supported finite measure $\lambda = \alpha \sum_n a_n^2 \delta_{a_n}$:

$$\int x^j \, d\lambda(x) = \frac{\kappa_{j+2}(Y)}{(j+1)!}, \qquad j \geq 0.$$

Moment uniqueness recovers λ, and hence $w_x = \lambda(\{x\})/x^2$ for every nonzero atom. In any one representation put $g = \gcd_x m_x$ and $\alpha_* = \alpha g$. The gcd of an infinite family of positive integers is the gcd of a finite subfamily, so Bézout's identity applies to the integers m_x/g. If $c > 0$ satisfies $w_x/c \in \mathbb{N}$ for all x, then $(\alpha_*/c)(m_x/g)$ is an integer for every x. A finite Bézout combination proves $\alpha_*/c \in \mathbb{N}$. Thus $c = \alpha_*/k$, and the stated multiplicities follow. Conversely, splitting a Gamma variable into k independent Gamma variables of shape $1/k$ times the original shape preserves the centered series law and gives every listed representation. □

Consequently a microscopic fluctuation law can itself calibrate the counting granularity, provided primitive multiplicities are part of the model. For instance, suppose the measured continuum transform is $\widetilde{H} = cH$ for an unknown constant $c > 0$, with the Laplace coordinates fixed. Its derivative calibration still recovers the power ratios, and Mellin inversion recovers $\widetilde{h} = ch$. The coefficient $\widetilde{\alpha} = \lim_{u\to\infty} u\widetilde{h}'(u)$ is therefore known. The primitive microscopic law gives α, and $c = \widetilde{\alpha}/\alpha$ identifies the missing amplitude. The qualification about fixed Laplace coordinates matters: multiplying a rate by a constant while keeping its threshold coordinates fixed also rescales the dual Laplace coordinates.

Corollary 38.6 (Calibration of an unknown rate amplitude)**.** *Suppose that the observed rate is $J = cI$ on a nonempty open fully active patch, with unknown $c > 0$ and with threshold coordinates fixed. Suppose also that the microscopic fluctuation law is observed as in Theorem 38.2, and restrict the model class to spectra with primitive multiplicities. These observations identify c, h, all exponent ratios, and the canonical spectrum $\{r_n^\rho\}$.*

Proof. Multiplying I by c sends s to cs and M to M/c, so the calibration matrix B is unchanged. The exponent ratios are therefore recovered directly from J. The dual transform of J is

$$H_J(s) = cH(s/c) = \int_0^\infty \psi\left(\sum_j s_j x^{-p_j}\right) dx, \qquad \psi(u) = ch(u/c).$$

Lemma 37.2 recovers ψ; the inversion is linear in its second derivative and does not require $e^{-\psi}$ to be a probability Laplace transform. Its coefficient is $\widetilde{\alpha} = \lim_{u\to\infty} u\psi'(u) = c\alpha$. Proposition 38.5 identifies α from the primitive microscopic law, so that $c = \widetilde{\alpha}/\alpha$ and $h(u) = \psi(cu)/c$. Spectral recovery then proceeds as before. □

39 A finite-observation obstruction for microscopic spectra

The inverse theorem based on the full fluctuation law cannot in general be replaced by a fixed finite list of cumulants. The following elementary construction makes this distinction explicit while leaving the entire tail of the microscopic spectrum unchanged. Its algebraic ingredients are the classical power-sum identities; we do not claim a new general theory of truncated moments.

Proposition 39.1 (A flow invisible to finitely many cumulants)**.** *Fix an integer $K \geq 2$, $\alpha > 0$, and distinct positive numbers $x_1 > \cdots > x_K > 0$. There is a real-analytic curve $x(\tau) = (x_1(\tau), \ldots, x_K(\tau))$, defined for sufficiently small $|\tau|$, through $x(0) = x$, with all coordinates positive and distinct, such that*

$$\sum_{i=1}^{K} x_i(\tau)^r = \sum_{i=1}^{K} x_i^r, \qquad 2 \leq r \leq K,$$
$$\sum_{i=1}^{K} x_i(\tau)^{K+1} = \sum_{i=1}^{K} x_i^{K+1} + (K+1)\tau.$$

Moreover $x_1'(0) > 0$. Append any fixed positive square-summable tail (c_n) with $\sup_n c_n < x_K$. The centered Gamma series

$$Y_\tau = \sum_{i=1}^{K} x_i(\tau)(G_i - \alpha) + \sum_{n \geq 1} c_n(\widetilde{G}_n - \alpha), \qquad G_i, \widetilde{G}_n \overset{\text{iid}}{\sim} \text{Gamma}(\alpha, 1),$$

have exactly the same cumulants through order K, although their largest coefficients vary. Their next cumulant satisfies

$$\kappa_{K+1}(Y_\tau) - \kappa_{K+1}(Y_0) = \alpha(K+1)!\tau.$$

Any asymptotic counting law depending only on the common coefficient-sequence tail is unchanged. This assertion does not concern the probability tails of Y_τ: its right-tail exponential rate changes when its largest coefficient changes.

Proof. For $P_\tau(z) = \prod_{j=1}^{K}(z - x_j(\tau))$, solve the analytic ODE

$$x_i'(\tau) = \frac{1}{x_i(\tau) P_\tau'(x_i(\tau))}.$$

Its vector field is analytic on the open set of distinct nonzero coordinates, so the asserted local solution exists and remains ordered and positive. Lagrange interpolation gives

$$\sum_{i=1}^{K} \frac{x_i^m}{P'(x_i)} = 0 \quad (0 \leq m \leq K-2), \qquad \sum_{i=1}^{K} \frac{x_i^{K-1}}{P'(x_i)} = 1.$$

Therefore the derivative of the rth power sum is zero for $2 \leq r \leq K$, and equals $K+1$ for $r = K+1$. Since $x_1 > \cdots > x_K > 0$, every factor of $P'(x_1)$ is positive, and hence $x_1'(0) > 0$. Positivity of the separation from the appended tail persists after shortening the parameter interval.

The series converge in L^2, and their cumulants are $\alpha(r-1)!$ times the corresponding power sums for $r \geq 2$. This proves all assertions. In fact, using the same Gamma variables for each parameter gives the quantitative continuity estimate

$$W_2(\mathcal{L}(Y_\tau), \mathcal{L}(Y_0)) \leq \sqrt{\alpha}\, \|x(\tau) - x(0)\|_2 = O(|\tau|).$$

□

Remark 39.2 (An exact polynomial evolution). The construction can equivalently be written

$$\partial_\tau P_\tau(z) = \frac{P_\tau(z)/P_\tau(0) - 1}{z}.$$

The apparent singularity at $z = 0$ is removable. Partial fractions give

$$\sum_i \frac{1}{x_i P'(x_i)} = -\frac{1}{P(0)}.$$

This supplies a direct finite algebraic description of the invisible deformation.

Remark 39.3 (Scope). The proposition proves noninjectivity for arbitrary fixed finite cumulant data within the positive square-summable Gamma-series class. It does not contradict finite-cumulant convergence criteria when the target spectrum has fixed finite rank, nor reconstruction under a specified finite-dimensional model. Both are substantial classical subjects. It also does not say that every prescribed moment vector has such a deformation: boundary moment vectors can be rigid.

40 The same arithmetic fluctuations under a hard energy constraint

For this section let $V(\theta) = 2(1 - \cos\theta)$ with θ uniform on $[-\pi, \pi]$, and let h and $g = -h''$ be as above. The law of V has density $[\pi\sqrt{v(4-v)}]^{-1}$ on $(0, 4)$, and the one-site parameter in (213) is $\alpha = 1/2$. Fix $A > 1$ and write

$$D_A = \sum_p p^{-A} V_p, \qquad F(t) = \sum_p h(tp^{-A}), \qquad \delta_t = F'(t) = \mathbb{E}_t D_A.$$

Theorem 40.1 (Hard conditioning and an independent boundary slack)**.** *For every real $a = (a_p)_p \in \ell^2$ set*

$$X_a = \sum_p a_p p^{-A} V_p, \qquad m_t = \mathbb{E}_t X_a.$$

Under the original product law conditioned on $D_A \le \delta_t$,

$$\big(t(X_a - m_t),\ t(\delta_t - D_A)\big) \implies (Y_a, R), \tag{228}$$

where $Y_a = \sum_p a_p(G_p - 1/2)$, $G_p \overset{\text{iid}}{\sim} \mathrm{Gamma}(1/2, 1)$, and $R \sim \mathrm{Exp}(1)$ is independent of all the G_p. Consequently, for $E - A > 1/2$ and $D_E = \sum_p (\log p) p^{-E} V_p$, one may take $a_p = (\log p) p^{A-E}$. The centering is the exact tilted mean; replacing it by a leading prime-number-theorem equivalent is not justified at this fluctuation scale.

Proof. We supply the local argument that is missing from a purely logarithmic Gibbs-conditioning calculation. Put $x = t^{1/A}$ and $m = x/\log x$. Under $\mathbb{P}_t$ let

$$Z_t = t(D_A - \delta_t), \qquad \sigma_t^2 = \mathrm{Var}_t(Z_t) = \sum_p (tp^{-A})^2 g(tp^{-A}).$$

Prime-number-theorem summation gives

$$\sigma_t^2 \sim v_A m, \qquad v_A = \int_0^\infty y^{-2A} g(y^{-A})\, dy \in (0, \infty). \tag{229}$$

Indeed the integrand is bounded near zero and $O(y^{-2A})$ at infinity. The same assertion, with half the constant, holds if the primes are restricted to either the odd or the even indices in the increasing sequence of primes. Their counting functions differ from half the prime counting function by at most one. Deleting any fixed finite set does not change these asymptotics.

Here is a uniform local central limit theorem for these sums. For $u > 0$ let $W_u = uV$ under its one-coordinate tilt. Direct integration against the displayed arcsine density gives

$$\mathbb{E}_u|W_u - \mathbb{E}_u W_u|^3 \le C\min(u^3, 1).$$

For $u \ge 1$, the estimate follows by splitting $0 < v < 4$ at $v = 2$: the first part is bounded by a Gamma integral after $w = uv$, and the second is exponentially small despite its integrable endpoint singularity. For $u \le 1$, use $0 \le W_u \le 4u$. The sum of these third moments is $O(m)$, so the normalized sums satisfy the central limit theorem by the Lyapunov condition.

For primes in $[x, 2x]$, the parameter u lies in the compact interval $[2^{-A}, 1]$. Uniformly on this interval, the characteristic function $\psi_u(\xi) = \mathbb{E}_u e^{i\xi W_u}$ satisfies

$$|\psi_u(\xi)| \le e^{-c\xi^2} \quad (|\xi| \le \eta), \qquad \sup_{\eta \le |\xi| \le B} |\psi_u(\xi)| < 1, \qquad |\psi_u(\xi)| \le C|\xi|^{-1/2} \quad (|\xi| \ge 1).$$

The first bound follows from the uniformly positive variances, the second from the continuous nonlattice densities and compactness, and the last by splitting the arcsine integral at its endpoints and one integration by parts on the intervening interval. There are $\asymp m$ such primes in each of the two index classes. Multiplication of their characteristic functions, followed by Fourier inversion, upgrades the central limit theorem to

$$\sup_{z \in \mathbb{R}} \left| \sigma_t f_t(\sigma_t z) - (2\pi)^{-1/2} e^{-z^2/2} \right| \longrightarrow 0, \tag{230}$$

where f_t is the density of Z_t. For completeness, the Fourier integral splits into $|\xi| \le \eta$, $\eta < |\xi| \le B$, and $|\xi| > B$. The first normalized integrand is dominated by $e^{-c'v^2}$; the second contributes $O(\sigma_t \rho^{cm})$ with $\rho < 1$; choosing $B > C^2$ makes the integral of $(C|\xi|^{-1/2})^{cm}$ on the last region exponentially small. This proves uniform Fourier-integral convergence. The same proof applies after deleting finitely many primes and to either index class. In particular all these reservoir densities have supremum $O(m^{-1/2})$.

Changing measure gives

$$\mathbb{P}(D_A \le \delta_t) = e^{t\delta_t - F(t)} I_t, \qquad I_t = \int_{-\infty}^{0} e^z f_t(z)\, dz \sim (2\pi)^{-1/2} \sigma_t^{-1}.$$

Let I be a fixed finite set of primes, and center its tilted energy as $Z_I = \sum_{p \in I} (tp^{-A} V_p - tp^{-A} h'(tp^{-A}))$. Let $f_{I^c,t}$ be the density of the independently centered complementary energy. The joint conditional law of the coordinates in I and the slack $r = t(\delta_t - D_A)$ has density, relative to $d\mathbb{P}_{t,I}\, dr$, equal to

$$\mathbf{1}_{r \ge 0} \frac{e^{-r} f_{I^c,t}(-Z_I - r)}{I_t}. \tag{231}$$

By (230), the density ratio tends to e^{-r} uniformly for bounded Z_I, r, and is bounded by Ce^{-r} everywhere. Since the finite coordinates $tp^{-A}V_p$ converge to independent Gamma$(1/2, 1)$ variables under $\mathbb{P}_t$, this proves their joint conditional convergence with an independent exponential slack.

It remains to pass from finitely many coordinates to X_a. Whenever a set of observed primes leaves one whole index class unobserved, integrating (231) in r (the formula also holds for an infinite

observed subseries) bounds its conditional marginal density relative to its tilted marginal by a fixed constant. This follows from the reservoir supremum $O(m^{-1/2})$ and $I_t \asymp m^{-1/2}$; convolution with any extra coordinates preserves that supremum bound. Split the primes exceeding a fixed cutoff Q into their two index classes. Each part has the other class as a reservoir. Therefore independence under the tilted law and the bound $\sup_u u^2 g(u) < \infty$ give

$$\limsup_{t\to\infty} \mathbb{E}\left[\left|\sum_{p>Q} a_p\{tp^{-A}V_p - tp^{-A}h'(tp^{-A})\}\right|^2 \;\middle|\; D_A \le \delta_t\right] \le C\sum_{p>Q} a_p^2.$$

The right side tends to zero. The finite-coordinate joint limit therefore proves (228). □

Remark 40.2 (A limitation for zeta-zero sampling). The hard-conditioning theorem concerns product Haar measure. A comparison $c\mu \le \nu \le \mu$ transfers concentration at a deterministic limit but does not determine the distribution of Y_a. For example, weighting by a bounded nonconstant function of the fluctuation changes its limiting law while preserving such a comparison. Thus this theorem alone gives no Gamma-series law for the full set of simple zeta zeros.

41 Activation universality and the spatial response profile

We retain the assumptions and normalization of Section 31. Write $d = E - A$. The change from one active constraint to two is governed by d/ρ, rather than by the complete microscopic law. Its spatial mechanism changes at $d = \rho/2$.

Fix $a > 0$. Let $t_0 > 0$ be the unique solution of $H_t(0, t_0) = a$, and put

$$I_A(a) = H(0, t_0) - t_0 a.$$

If $d < \rho$, define $e_0 = H_s(0, t_0) < \infty$. Then $I(e, a) = I_A(a)$ for $e \ge e_0$: the E-constraint is inactive. For $e < e_0$ sufficiently close to e_0, both multipliers are positive.

Theorem 41.1 (Three activation regimes)**.** *Suppose $0 < d < \rho$ and let $\varepsilon \downarrow 0$.*

1. *If $d < \rho/2$, set*

$$M_{bc} = \rho\int_0^\infty y^{\rho-1-b-c} g(t_0 y^{-A})\,dy \quad (b, c \in \{E, A\}), \qquad \kappa = M_{EE} - \frac{M_{EA}^2}{M_{AA}} > 0.$$

Then

$$I(e_0 - \varepsilon, a) - I_A(a) \sim \frac{\varepsilon^2}{2\kappa}.$$

2. *If $d = \rho/2$, then*

$$I(e_0 - \varepsilon, a) - I_A(a) \sim \frac{t_0^2}{4\alpha}\frac{\varepsilon^2}{\log(1/\varepsilon)}.$$

3. *If $\rho/2 < d < \rho$, put*

$$q = \frac{\rho}{d} \in (1, 2), \qquad K = \frac{\alpha\pi}{\sin(\pi(q-1))}.$$

Then

$$I(e_0 - \varepsilon, a) - I_A(a) \sim (q-1)q^{-q/(q-1)}K^{-1/(q-1)}t_0^{q/(q-1)}\varepsilon^{q/(q-1)}. \tag{232}$$

If $d \ge \rho$, then $H_s(0, t) = \infty$ for every $t > 0$; there is no A-only optimizing face at finite (e, a).

Proof. The last assertion follows from $y^{\rho-1-E}h'(ty^{-A}) \sim (\alpha/t)y^{\rho-1-d}$ at zero. Assume henceforth $d<\rho$. The mixed derivative $H_{st}(0,t)$ and $H_{tt}(0,t)$ are finite, continuous from the positive quadrant, and $H_{tt}(0,t)<0$. The implicit function theorem applied to $H_t(s,t_s)=a$ therefore gives $t_s=t_0+O(s)$ for $s\downarrow 0$. For clarity, the theorem can be applied after extending H_t across $s=0$ by a C^1 extension; equivalently, its usual one-sided proof uses the mean value theorem and $H_{tt}(0,t_0)\neq 0$. Define globally

$$F(s)=\sup_{t\ge 0}\{H(s,t)-at\},\qquad D(s)=F(0)+e_0 s-F(s).$$

Locally, $F(s)=H(s,t_s)-at_s$. Strict concavity shows that the maximizing pair for $e\uparrow e_0$ converges to $(0,t_0)$, and consequently

$$I(e_0-\varepsilon,a)-I_A(a)=\sup_{s\ge 0}\{\varepsilon s-D(s)\} \tag{233}$$

with the maximizer tending to zero.

When $2d<\rho$, the full boundary Hessian is finite. Differentiation of $H_t(s,t_s)=a$ gives $D(s)=\kappa s^2/2+o(s^2)$. The strict Cauchy–Schwarz inequality gives $\kappa>0$, since y^{-E} and y^{-A} are not proportional. This proves the first case.

For the other two cases it suffices first to keep $t=t_0$ fixed: profiling t changes $H(s,t)-at$ by $O(s^2)$. Indeed its first t-derivative at t_0 is $O(s)$, its second derivative remains bounded, and $t_s-t_0=O(s)$. Put $y_s=(s/t)^{1/d}$. The bounds

$$0<h'(u)\le\frac{C}{1+u},\qquad 0<g(u)\le\frac{C}{(1+u)^2} \tag{234}$$

follow from (203) and continuity on compact intervals. In the case $1<q<2$, substitution $y=y_s z$ and dominated convergence give

$$\begin{aligned}&H(s,t)-H(0,t)-sH_s(0,t)\\&\quad\sim\rho\alpha(s/t)^q\int_0^\infty z^{\rho-1}\{\log(1+z^{-d})-z^{-d}\}\,dz=-K(s/t)^q.\end{aligned}$$

The domination is $Cz^{\rho-1}\min(z^{-d},z^{-2d})$, integrable at both endpoints precisely for $d<\rho<2d$. The integral is evaluated by setting $v=z^{-d}$ and integrating by parts:

$$q\int_0^\infty(v-\log(1+v))v^{-q-1}\,dv=\int_0^\infty\frac{v^{1-q}}{1+v}\,dv=\frac{\pi}{\sin(\pi(q-1))}.$$

Thus $D(s)\sim Kt_0^{-q}s^q$.

At $2d=\rho$, split the integral defining $-H_{ss}(s,t)$ at y_s and at a small fixed number. On the middle range its integrand is asymptotic to $(\rho\alpha/t^2)y^{-1}$, uniformly away from the two ends. The lower range contributes $O(1)$ and the upper range also contributes $O(1)$ after the fixed splitting point is chosen. More explicitly, (234) bounds the lower range by $Cs^{-2}y_s^\rho=O(1)$; the middle logarithmic coefficient follows by making ty^{-A} large and $sy^{-E}/(ty^{-A})$ small. It follows that

$$-H_{ss}(s,t)\sim\frac{2\alpha}{t^2}\log(1/s),\qquad D(s)\sim\frac{\alpha}{t_0^2}s^2\log(1/s).$$

Integrating the second derivative twice is justified by its logarithmic bound. Finally optimize the three displayed asymptotics in (233). Quadratic and power functions give the first and third constants. In the critical case the maximizing scale is $s\sim t_0^2\varepsilon/(2\alpha\log(1/\varepsilon))$, giving the stated constant. These optimizations can alternatively be obtained by upper and lower multiplicative bounds on D; no differentiation of an unspecified remainder is needed. □

The preceding exponent has a spatial explanation. The following statement concerns a deterministic measure of the change in tilted mean energy; it is not a claim that a randomly chosen coordinate has the stated distribution. For fixed $t > 0$ and $0 < d < \rho$, define the positive measure

$$dR_{s,t}(y) = \rho y^{\rho-1-E}\{h'(ty^{-A}) - h'(sy^{-E} + ty^{-A})\}\, dy.$$

Its mass is the reduction of the E-energy mean when the multiplier s is activated at fixed t.

Theorem 41.2 (Universal activation profile)**.** *Suppose $\rho/2 < d < \rho$, and normalize $R_{s,t}$ to mass one. Under the change of variable $W = (y/y_s)^d$, where $y_s = (s/t)^{1/d}$, these measures converge weakly as $s \downarrow 0$ to the beta-prime law*

$$\frac{\sin(\pi(q-1))}{\pi}\,\frac{w^{q-2}}{1+w}\,dw, \qquad w > 0, \quad q = \rho/d. \tag{235}$$

At $d = \rho/2$, the image of the normalized measure under

$$y \longmapsto \frac{\log(y/y_s)}{\log(1/y_s)}$$

converges weakly on $\mathbb{R}$ to the uniform law on $[0,1]$. For $d < \rho/2$, there is instead an unscaled limit: $s^{-1}dR_{s,t}(y)$ converges in total mass to $\rho y^{\rho-1-2E} g(ty^{-A})\, dy$.

Proof. For fixed $z > 0$, after $y = y_s z$, the density of $R_{s,t}$ divided by $(\rho\alpha/t)y_s^{\rho-d}$ tends to

$$\frac{z^{\rho-d-1}}{1+z^d}.$$

The bounds (234) dominate it by $Cz^{\rho-d-1}\min(1, z^{-d})$. Hence convergence holds in L^1 when $d < \rho < 2d$. Its integral is $\pi/[d\sin(\pi(q-1))]$; setting $w = z^d$ proves (235). At $2d = \rho$, the density on $y_s \ll y \ll 1$ is asymptotic to $(\rho\alpha s/t^2)dy/y$. Each fixed closed subinterval of the logarithmic coordinate inside $(0,1)$ therefore receives asymptotically its length times the total mass. The two endpoint ranges are controlled by (234): below y_s and above a fixed positive constant their masses are $O(s)$, while the full mass is asymptotic to $(2\alpha/t^2)s\log(1/s)$. Letting the closed subintervals increase to $(0,1)$ proves the critical assertion. If $2d < \rho$, ordinary dominated convergence applied to the difference quotient proves the last assertion. □

The fractional response profile also describes the leading response along the constrained path $t = t_s$. Define the signed measure

$$d\widetilde{R}_s(y) = \rho y^{\rho-1-E}\{h'(t_0 y^{-A}) - h'(sy^{-E} + t_s y^{-A})\}\, dy.$$

Its difference from R_{s,t_0} has total variation $O(s)$, because $|t_s - t_0| = O(s)$ and H_{st} is locally bounded. This is negligible compared with s^{q-1} in the fractional regime, or with $s\log(1/s)$ in the critical case. The signed measure therefore has the same normalized leading limit as the positive response measure above.

There is a further local interpretation. Let Q_u be the tilt of the law of V with density $e^{-uV+h(u)}$. At a fixed rescaled location $y = y_s z$, put $u_0 = ty^{-A}$. Under $Q_{u_0(1+z^{-d})}$, the random variable $u_0 V$ tends to a Gamma law with shape α and rate $1 + z^{-d}$. Indeed, for $\lambda \geq 0$ its Laplace transform tends to

$$\exp\{h(u_0(1+z^{-d})) - h(u_0(1+z^{-d}+\lambda))\} \longrightarrow \left(\frac{1+z^{-d}}{1+z^{-d}+\lambda}\right)^{\alpha},$$

by integrating $h'(u) \sim \alpha/u$. Thus the singular regime has a universal moving response profile, the critical regime spreads across logarithmic scales, and the regular regime retains a finite quadratic information metric. These conclusions use the continuum limit first and activation second; they do not assert a uniform two-parameter limit for an arbitrary regularly varying spectrum.

42 One-prime and collective tails of the arithmetic fluctuation

Let $1/2 < d < 1$ and define

$$a_p = (\log p)p^{-d}, \qquad Y_d = \sum_p a_p(G_p - 1/2), \qquad G_p \overset{\text{iid}}{\sim} \text{Gamma}(1/2, 1).$$

This is the hard-conditioned fluctuation in Theorem 40.1 when $d = E - A$. Write

$$K_d = \frac{\pi}{2\sin(\pi(1/d-1))}, \qquad B_d = (1-d)^2\left(\frac{d}{K_d}\right)^{d/(1-d)}, \qquad a_* = \max_p a_p.$$

Theorem 42.1 (Asymmetric arithmetic tails)**.** *As $z \to \infty$,*

$$-\log \mathbb{P}(Y_d \le -z) \sim B_d \frac{z^{1/(1-d)}}{\log z}, \tag{236}$$

whereas

$$-\log \mathbb{P}(Y_d \ge z) \sim z/a_*. \tag{237}$$

If a_* is attained at a unique prime p_*, the sharper right-tail equivalent is

$$\mathbb{P}(Y_d \ge z) \sim \sqrt{\frac{a_*}{\pi}} e^{-1/2} \prod_{p \ne p_*} \left[e^{-a_p/(2a_*)}(1 - a_p/a_*)^{-1/2}\right] z^{-1/2} e^{-z/a_*}.$$

For $d = 2.813014020252898 - 1.940101683743625$, the unique maximizing prime is $p_* = 3$, and the exponent in (236) is approximately 7.86858435. The numerical values only illustrate the exact formulas.

Proof. For $s \ge 0$, independence gives the everywhere-finite transform

$$L(s) = \log \mathbb{E}e^{-sY_d} = \frac{1}{2}\sum_p \{sa_p - \log(1 + sa_p)\}.$$

Let $x = x(s)$ be the large solution of $s(\log x)x^{-d} = 1$. The prime number theorem and the change of variable $p = xy$ yield

$$L(s) \sim \frac{x}{\log x}\frac{1}{2}\int_0^\infty \{y^{-d} - \log(1 + y^{-d})\}\, dy = K_d \frac{x}{\log x}.$$

The improper passage is valid because the limit superior, as $x \to \infty$ with fixed η, R, of the normalized small-prime tail is $O(\eta^{1-d})$ for $p < \eta x$, and that of the large-prime tail is $O(R^{1-2d})$ for $p > Rx$. These bounds follow respectively from $u - \log(1 + u) \le u$ and $u - \log(1 + u) \le u^2/2$, followed by partial summation. To evaluate the integral substitute $u = y^{-d}$ and integrate by parts; the result is $\frac{1}{2}\int_0^\infty u^{1-1/d}/(1 + u)\, du = K_d$. Since $x \sim d^{-1/d}s^{1/d}(\log s)^{1/d}$, this gives

$$L(s) \sim K_d d^{1-1/d} s^{1/d} (\log s)^{1/d-1}. \tag{238}$$

For clarity we spell out the exponential-Tauberian step. Convexity and regular variation in (238) imply $L'(s) \sim L(s)/(ds)$. This also follows by direct differentiation of the prime sum. Moreover

$$0 < L''(s) = \frac{1}{2}\sum_p \frac{a_p^2}{(1 + sa_p)^2} \le L'(s)/s.$$

For $Z = -Y_d$, the exponential tilt by $e^{sZ-L(s)}$ has mean $L'(s)$ and variance $L''(s)$. At the saddle $L'(s) = z$, the relative variance tends to zero. Exponential Markov gives the upper probability bound. For the reverse bound, choose a saddle with $L'(s) = (1+\eta)z$ and restrict Z to $[z, (1+2\eta)z]$. Its tilted probability tends to one by the variance estimate. Changing measure and then letting $\eta \downarrow 0$ proves $-\log \mathbb{P}(Z \geq z) \sim \sup_{s \geq 0}\{sz - L(s)\}$. Legendre optimization of (238) gives exactly the constant B_d in (236).

The moment generating function of Y_d is finite for $0 \leq q < 1/a_*$, yielding $\liminf z^{-1}[-\log \mathbb{P}(Y_d \geq z)] \geq 1/a_*$. Keeping one maximizing Gamma variable and restricting the independent remaining sum to a fixed interval of positive probability gives the reverse inequality. If the maximum is unique, write $Y_d = a_*(G_{p_*} - 1/2) + R$. Since $a_p \to 0$, the second largest coefficient is strictly smaller than a_*. Thus R has a finite moment generating function in a neighborhood of $1/a_*$. Integrating the Gamma tail equivalent $\mathbb{P}(G \geq u) \sim \pi^{-1/2} u^{-1/2} e^{-u}$ against R gives the stated constant; the extra exponential moment controls the complement of bounded R/z. Finally, $u \mapsto (\log u)u^{-d}$ increases up to $u = e^{1/d}$ and decreases thereafter. At the displayed value of d, $3 < e^{1/d} < 5$, and direct comparison of a_3 and a_5 selects $p_* = 3$. □

The left-tail exponent is the same singular exponent that occurs when the second constraint activates at the A-only edge. Here it has a different, microscopic interpretation: downward fluctuation requires coordinated suppression of infinitely many centered Gamma terms, whereas the far upward fluctuation is set by a single coefficient. The use of exponential tilting and exponential Tauberian reasoning is classical; the assertion identifies their concrete arithmetic consequences in this prime-indexed limit law.

43 Correlated local laws: exact aliases and one-channel restoration

The preceding sections use a common scalar local law. We now change the model explicitly: let X be a nonnegative random vector, set $h(u) = -\log \mathbb{E}e^{-u\cdot X}$, and identify a nonnegative linear functional with its coefficient vector. Channels ℓ_i with powers $F_i > \rho > 0$ have response

$$H(s) = \rho \int_0^\infty y^{\rho-1} h\left(\sum_i s_i y^{-F_i} \ell_i\right) dy. \tag{239}$$

The coordinates of X need not be independent. Dilation can now be compensated by convolution, so even the entire joint response can lose the local law. A repeated component or a positive mixed component at a second power can remove this ambiguity under the respective hypotheses below. These classes must be kept distinct: a gated law with an atom at zero does not satisfy the positive Gamma-shape asymptotic used in the scalar calibration theorem.

Fix $d \geq 2$, $\rho > 0$, and $E_1, \ldots, E_d > \rho$. For a probability measure μ on $\mathbb{R}_+^d$ write

$$h_\mu(v) = -\log \int e^{-v\cdot x}\, \mu(dx), \qquad \delta_c v = (c^{E_1} v_1, \ldots, c^{E_d} v_d).$$

We use the following class, which includes arbitrary continuous distributions away from the origin as well as finite discrete examples:

$$\mathcal{C}_d = \left\{ p_0\delta_0 + (1-p_0)\nu : 0 < p_0 < 1, \quad \operatorname{supp}\nu \subset [a,b]^d \text{ for some } 0 < a < b < \infty \right\}. \tag{240}$$

The constants p_0, a, b need not be known and can differ between laws. For $\mu \in \mathcal{C}_d$ its homogeneous response is

$$\mathcal{H}_\mu(s) = \rho \int_0^\infty y^{\rho-1} h_\mu(s_1 y^{-E_1}, \ldots, s_d y^{-E_d})\, dy. \tag{241}$$

This integral is finite: $h_\mu \le -\log p_0$ and $h_\mu(v) \le \mathbb{E}_\mu X \cdot v$. It is concave and differentiable on the positive orthant, and

$$\mathcal{H}_\mu(\delta_c s) = c^\rho \mathcal{H}_\mu(s).$$

The term "response" here refers to this limiting log-Laplace object. It does not assert that $\exp(-\mathcal{H}_\mu)$ is a Laplace transform of an infinitely divisible law for every μ.

Theorem 43.1 (Dilation-convolution aliases)**.** *Let $X^{(1)}, \dots, X^{(n)}$ be independent copies with law $\mu \in \mathcal{C}_d$, and let $c_1, \dots, c_n > 0$. Define*

$$Y_j = \sum_{i=1}^{n} c_i^{E_j} X_j^{(i)}, \qquad 1 \le j \le d.$$

Then the law μ_c of Y belongs to $\mathcal{C}_d$, and

$$\mathcal{H}_{\mu_c}(s) = \left(\sum_{i=1}^{n} c_i^\rho \right) \mathcal{H}_\mu(s). \tag{242}$$

Consequently the full function $\mathcal{H}_\mu$ does not determine μ in $\mathcal{C}_d$.

Proof. Independence gives $h_{\mu_c}(v) = \sum_i h_\mu(c_i^{E_1} v_1, \dots, c_i^{E_d} v_d)$. In the ith integral in (241), substitute $y = c_i r$. This proves (242). The atom at zero has mass p_0^n, and every nonzero outcome has all coordinates positive and bounded away from zero, uniformly on the support. Thus $\mu_c \in \mathcal{C}_d$. Taking $n = 2$ and $c_1 = c_2 = 2^{-1/\rho}$ preserves $\mathcal{H}$ but changes $\mathbb{E}Y_j$ by the factor $2^{1-E_j/\rho} < 1$. □

Proposition 43.2 (Aliases survive every finite calibration)**.** *Fix any integer $M \ge 1$. There exist distinct $\mu, \widetilde{\mu} \in \mathcal{C}_d$ with the same homogeneous response and the same mixed moments of total degree at most M. They may also be required to have identical masses at zero and identical coordinatewise upper support bounds. In addition, any prescribed finite collection of one-component responses*

$$\mathcal{R}_{j,F}(s) = \rho \int_0^\infty y^{\rho-1} h_\mu(0, \dots, s y^{-F}, \dots, 0)\, dy, \qquad F > \rho,$$

can be made identical between the two laws, for all $s > 0$.

Proof. Let B be Bernoulli with a transcendental success probability in $(0,1)$, and start with $X = (B, \dots, B)$. Every cumulant $\kappa_m(B)$ is a nonzero polynomial in its success probability with integer coefficients: its coefficient of the linear term is one. Hence every $\kappa_m(B)$ is nonzero. For the dilation-convolution law above, the joint cumulant of multi-order $k = (k_1, \dots, k_d) \ne 0$ is

$$\kappa_k(Y) = \kappa_{|k|}(B) \sum_i c_i^{k \cdot E}. \tag{243}$$

Thus equality of power sums for the finite set

$$P = \{\rho\} \cup \{k \cdot E : 1 \le |k| \le M\}$$

is enough. To preserve the additional scalar responses, also include $E_j \rho / F$ for every specified pair (j, F): direct substitution gives $\mathcal{R}_{j,F}^{\mu_c} = (\sum_i c_i^{E_j \rho / F}) \mathcal{R}_{j,F}^{\mu}$. Remove repeated elements of P, and put $q = |P|$.

Choose $q+1$ distinct positive numbers c_i. The Jacobian of $c \mapsto (\sum_i c_i^p)_{p \in P}$ has rank q. Indeed, after harmless nonzero row factors, its entries are c_i^{p-1}; distinct real powers form a Chebyshev system on $(0, \infty)$. For completeness, after $c_i = e^{u_i}$ this assertion follows by induction from Rolle's theorem:

divide a nonzero linear combination of exponentials by its smallest exponential, differentiate, and count its zeros. The implicit function theorem therefore gives a nonconstant smooth curve of scale vectors along which all these power sums are constant.

Choose an integer m with $mE_1 > \max P$. The augmented Jacobian obtained by adding $\sum_i c_i^{mE_1}$ has full rank $q+1$, by the same argument. Consequently this extra power sum has nonzero derivative along the local one-dimensional level set. Two sufficiently close points on that level set give different mth cumulants of Y_1, by (243), and thus distinct probability laws. All required lower moments match because moments are polynomials in cumulants. The number $q+1$ of Bernoulli summands is fixed, so the atom at zero matches. The coordinatewise maxima are $\sum_i c_i^{E_j}$ and also match, since $M \geq 1$. □

Remark 43.3 (A concrete alias). For $d = 2$, $\rho = 1$, $(E_1, E_2) = (3, 2)$, and any nondegenerate Bernoulli B, the laws

$$(B, B) \quad \text{and} \quad ((B_1 + B_2)/8, (B_1 + B_2)/4)$$

have identical complete two-component responses. A new channel observing the first component with exponent $C = 4$ distinguishes this particular pair even through its marginal response: the second response is $2^{1/4}$ times the first. The preceding proposition shows why finitely many separate marginal measurements do not suffice in general.

43.1 One redundant channel restores the entire correlated law

Choose $C > \rho$ with $C \neq E_1$. We observe the first component once more, with this different exponent, and define

$$\mathcal{K}_\mu(s_1, \dots, s_d, u) = \rho \int_0^\infty y^{\rho-1} h_\mu(s_1 y^{-E_1} + u y^{-C}, s_2 y^{-E_2}, \dots, s_d y^{-E_d})\, dy. \tag{244}$$

The new channel measures no new local random variable. Its information comes from applying a different spatial weight to the same component.

Theorem 43.4 (Identification by a repeated component)**.** *For $d \geq 2$, the map $\mu \mapsto \mathcal{K}_\mu$ is injective on $\mathcal{C}_d$. In particular, d coordinate channels fail to identify arbitrary correlated local laws even after any finite moment calibration, whereas $d+1$ channels in (244) identify them. This is a threshold for this specified observation architecture, not a claim about all possible measurement schemes.*

Proof. Let

$$f_\mu(v) = -\partial_{v_1}^2 h_\mu(v) = \mathrm{Var}_{\mu,v}(X_1),$$

where μ, v denotes exponential tilting by $e^{-v \cdot X}$. The support and atom assumptions imply

$$0 \leq f_\mu(v) \leq \frac{b^2}{p_0} \exp\{-a(v_1 + \cdots + v_d)\}, \qquad v \in \mathbb{R}_+^d. \tag{245}$$

In particular its d-fold Mellin transform exists whenever all real parts of its arguments are positive.

Set $s_d = 1$ and define

$$G(s_1, u, s_2, \dots, s_{d-1}) = -\partial_{s_1} \partial_u \mathcal{K}_\mu(s_1, s_2, \dots, s_{d-1}, 1, u).$$

Differentiation under the integral is justified by (245) at zero and by $E_1 + C > \rho$ at infinity. For complex $z, w, z_2, \dots, z_{d-1}$ put

$$a_1 = z + w, \qquad a_j = z_j \quad (2 \leq j < d),$$
$$a_d = \frac{E_1 + C - \rho - E_1 z - C w - \sum_{j=2}^{d-1} E_j z_j}{E_d}.$$

Whenever $\operatorname{Re} z, \operatorname{Re} w, \operatorname{Re} z_j, \operatorname{Re} a_d > 0$, its Mellin transform satisfies

$$\mathcal{M}G(z, w, z_2, \dots, z_{d-1}) = \frac{\rho}{E_d}\,\mathrm{B}(z, w)\mathcal{M}f_\mu(a_1, \dots, a_d). \tag{246}$$

To verify this, substitute $v_1 = s_1 y^{-E_1}$ and $v_0 = u y^{-C}$ in the first two Mellin integrals, and $v_j = s_j y^{-E_j}$ in the others. The first two integrals combine by

$$\int_0^\infty \int_0^\infty v_1^{z-1} v_0^{w-1} F(v_1 + v_0)\, dv_1\, dv_0 = \mathrm{B}(z, w) \int_0^\infty r^{z+w-1} F(r)\, dr.$$

The remaining substitution $v_d = y^{-E_d}$ gives the factor $1/E_d$ and the exponent a_d in (246). For real parameters Tonelli's theorem applies. For complex parameters absolute convergence follows from the same calculation with real parts.

Choose small positive real $z_0, w_0, z_{j,0}$ so that $a_{d,0} > 0$. The affine map from $(z, w, z_2, \dots, z_{d-1})$ to $(a_1, \dots, a_d)$ has nonzero determinant, up to sign,

$$\frac{E_1 - C}{E_d}.$$

Therefore all imaginary parts of $(a_1, \dots, a_d)$ are obtained while their real parts stay fixed at a positive vector. The beta factor has no zeros in the domain in question. Equation (246) determines the Mellin transform of f_μ on an entire vertical d-plane. Uniqueness of the Fourier transform, after logarithmic coordinates, determines f_μ. The bound (245) supplies the required integrability. Continuity makes the identification pointwise.

It remains to recover the integration constants lost by differentiating. As $r \to \infty$, uniformly for fixed $v' \in \mathbb{R}_+^{d-1}$,

$$h_\mu(r, v') \longrightarrow c_\mu = -\log p_0, \qquad \partial_1 h_\mu(r, v') \longrightarrow 0.$$

Consequently

$$h_\mu(v_1, v') = c_\mu - \int_{v_1}^\infty (r - v_1) f_\mu(r, v')\, dr, \qquad c_\mu = \int_0^\infty r\, f_\mu(r, 0, \dots, 0)\, dr. \tag{247}$$

The second equality uses $h_\mu(0) = 0$; the boundary values of f_μ are obtained by continuity and dominated convergence from the positive orthant. Hence h_μ, then its Laplace transform e^{-h_μ}, and finally μ are determined. □

Corollary 43.5 (The same distinction for dual rate functions)**.** *Define, for either response $F = \mathcal{H}_\mu$ or $\mathcal{K}_\mu$,*

$$I_F(e) = \sup_{q \geq 0} \{F(q) - q \cdot e\}.$$

Then I_F determines F on the positive orthant by

$$F(q) = \inf_{e \geq 0} \{I_F(e) + q \cdot e\}.$$

Thus the nonidentifiability and restoration conclusions above hold unchanged when the exact complete dual rate function is the observable.

Proof. The inequality $F(q) \leq I_F(e) + q \cdot e$ is the definition of I_F. For equality choose $e = \nabla F(q)$, which is coordinatewise nonnegative. Concavity implies that q maximizes $F(v) - v \cdot e$, proving the formula. The statement is about exact functions; it does not identify them from a finite noisy sample, nor by itself establish an array large-deviation principle. □

Remark 43.6 (Conditioning and stability). The reconstruction is explicit but not uniformly stable. In the simplest case $d = 2$, along $z = z_0 + iR$, $w = w_0 - iR$, the beta multiplier satisfies

$$|\mathrm{B}(z, w)| \asymp |R|^{z_0+w_0-1} e^{-\pi|R|}.$$

Recovering the corresponding Mellin modes therefore requires exponential amplification. This multiplier calculation alone is not a minimax lower bound over the nonlinear probability-law class. Such a claim would require separate admissible perturbations and a statistical noise model.

43.2 A scale window gives a different, local reconstruction

The information loss is also removed by retaining one finite radial window. For $c > 1$ define

$$A_{\mu,c}(s) = \rho \int_1^c y^{\rho-1} h_\mu(\delta_{1/y} s)\, dy, \qquad D = \sum_{j=1}^d E_j s_j \partial_{s_j}.$$

The following inversion only needs a nonnegative law with finite means; it does not need the atom or support-gap assumptions.

Proposition 43.7 (One-window inversion)**.** *Let $E_* := \min_j E_j > \rho$. Then*

$$g(s) := \frac{1}{\rho}(D - \rho) A_{\mu,c}(s) = h_\mu(s) - c^\rho h_\mu(\delta_{1/c} s), \tag{248}$$

$$h_\mu(s) = \sum_{n=0}^\infty c^{n\rho} g(\delta_{c^{-n}} s). \tag{249}$$

The series converges absolutely and locally uniformly. In the norm $\|f\|_* = \sup_{s \in \mathbb{R}^d_+ \setminus \{0\}} |f(s)|/(s_1 + \cdots + s_d)$, its linear inverse has norm at most $(1 - c^{\rho - E_*})^{-1}$. Thus recovering h from the differentiated window data loses no further exponential factor.

Proof. Since $Dh(\delta_{1/y} s) = -y \partial_y h(\delta_{1/y} s)$, integration by parts proves (248). Iterate this identity; the remainder after n terms is $c^{n\rho} h(\delta_{c^{-n}} s)$, bounded by $c^{n(\rho - E_*)} (\max_j \mathbb{E} X_j) \sum_j s_j$. It tends to zero. The dilation operator $Tf(s) = c^\rho f(\delta_{1/c} s)$ has norm at most $c^{\rho - E_*} < 1$, so the Neumann series gives the stated bound. □

Prior-art boundary. Homogeneous dilation, convolution homomorphisms, and inversion of random-integral transforms are established themes in probability; Jurek's work on random integral mappings [38] is directly relevant. The beta-integral identity and Mellin uniqueness used above are also classical. The prospective contribution here is the exact information comparison for these particular rare-event response maps: finite calibrations do not eliminate genuine correlated aliases, while one repeated component restores the entire joint local law. No priority claim for this combination should be made without a more complete literature review. In particular, the response $\mathcal{H}_\mu$ for a general bounded non-infinitely-divisible local law must not be identified automatically with an operator-stable Laplace exponent.

44 Weighted-series realization of the correlated responses

The response functions above are realized by genuine rare events without assuming independence between the coordinates at one site. The following theorem also permits repeated local coordinates.

Theorem 44.1 (Joint lower tails with arbitrary local correlations)**.** *Let $k \geq 1$, $\rho > 0$, and $F_i > \rho$ for $1 \leq i \leq k$. Let $Y^{(n)}$ be independent identically distributed vectors with law $p_0\delta_0 + (1-p_0)\nu$, where $0 < p_0 < 1$ and $\operatorname{supp}\nu \subset [a,b]^k$ for some $0 < a < b$. Coordinates of each vector may be arbitrarily dependent, including identical. Put*

$$D_i = \sum_{n\geq 1} n^{-F_i/\rho} Y_i^{(n)}, \qquad U_{x,i} = x^{F_i-\rho} D_i,$$

and

$$H(q) = \rho \int_0^\infty y^{\rho-1} h(q_1 y^{-F_1}, \ldots, q_k y^{-F_k})\, dy, \quad h(v) = -\log \mathbb{E} e^{-v\cdot Y}.$$

Then, for every vector $e \in (0,\infty)^k$,

$$\lim_{x\to\infty} -x^{-\rho} \log \mathbb{P}\{U_{x,i} \leq e_i \text{ for every } i\} = I(e) := \sup_{q\in[0,\infty)^k} \{H(q) - q\cdot e\}. \tag{250}$$

In particular, the $d+1$-channel response in (244) is obtained by taking $Y = (X_1, \ldots, X_d, X_1)$ and $F = (E_1, \ldots, E_d, C)$.

Proof. All D_i are almost surely finite, by finite expectation and $F_i/\rho > 1$. Set $M = x^\rho$. The exact normalized negative logarithm of the Laplace transform is

$$H_x(q) := -M^{-1} \log \mathbb{E} e^{-Mq\cdot U_x} = M^{-1} \sum_{n\geq 1} h(q_1 (n/M)^{-F_1/\rho}, \ldots, q_k (n/M)^{-F_k/\rho}).$$

For each fixed $q \neq 0$, this Riemann sum converges to $H(q)$. Indeed, $h \leq -\log p_0$ controls the origin, and $h(v) \leq b\sum_i v_i$ supplies an integrable bound at infinity. The same Riemann-sum argument gives convergence of its first and second partial derivatives at any fixed $q \in [0,\infty)^k \setminus \{0\}$, including one-sided derivatives at zero coordinates. Near the origin of the Riemann-sum variable, the tilted first moments and covariances have exponential bounds of the form

$$|\mathbb{E}_v Y_i| + |\operatorname{Cov}_v(Y_i, Y_j)| \leq K \exp\{-a\textstyle\sum_\ell v_\ell\};$$

at infinity, the factors $t^{-F_i/\rho}$ and $t^{-(F_i+F_j)/\rho}$ are integrable. Splitting into a compact interval and these two tails justifies the derivative limits.

Under the tilted probability measure with density proportional to $e^{-Mq\cdot U_x}$, one has

$$\mathbb{E}_q U_x = \nabla H_x(q) \longrightarrow \nabla H(q), \qquad \operatorname{Cov}_q(U_x) = -M^{-1}\nabla^2 H_x(q) = O(M^{-1}).$$

Thus $U_x \to \nabla H(q)$ in probability under every such fixed tilt.

For $e > 0$, the supremum defining $I(e)$ is attained. To see coercivity, write $c_0 = -\log p_0$ and use

$$h(v) \leq \min(c_0, b\textstyle\sum_i v_i) \leq \sum_i \min(c_0, bv_i).$$

Integration gives $H(q) \leq \sum_i K_i q_i^{\rho/F_i}$, whereas $q\cdot e$ grows linearly. The maximizing vector q_* is nonzero: on every coordinate axis, $H(te_i) = t^{\rho/F_i} H(e_i)$, so for small positive t the objective exceeds its value zero at the origin. Concavity and the optimality conditions give

$$m_i := \partial_i H(q_*) \leq e_i, \qquad q_{*,i}(m_i - e_i) = 0.$$

Every m_i is strictly positive. No sign assumption on off-diagonal covariances is used.

For $c > 1$, put $q_{c,i} = c^{F_i} q_{*,i}$. Homogeneity yields

$$H(q_c) = c^{\rho} H(q_*), \qquad \partial_i H(q_c) = c^{\rho - F_i} m_i < e_i.$$

Thus a sufficiently small fixed neighborhood of $\nabla H(q_c)$ is contained in the lower orthant $\{u \le e\}$ and has tilted probability tending to one. Changing measure on that neighborhood gives

$$\limsup_{x \to \infty} -M^{-1} \log \mathbb{P}\{U_x \le e\} \le H(q_c) - q_c \cdot \nabla H(q_c) = c^{\rho} I(e),$$

where the neighborhood radius is sent to zero first. Letting $c \downarrow 1$ proves the upper bound on the logarithmic cost. Chernoff's inequality, valid for every $q \ge 0$, gives the reverse bound. This proves (250). □

Remark 44.2. The anisotropic dilation in this proof is useful when coordinates are negatively correlated: increasing all ordinary tilt coordinates need not reduce every tilted mean. The homogeneity identity gives a simultaneous strict decrease by the explicit factors $c^{\rho - F_i}$ instead.

Proposition 44.3 (Active lower tails for arbitrary bounded gated vectors)**.** *In the preceding weighted-series construction, let Y be any bounded nonnegative vector with $\mathbb{P}(Y = 0) > 0$. At each strictly positive tilt q, write $e = \nabla H(q)$. After deleting identically zero coordinates, $e > 0$ and the limit in (250) holds at this e. In particular, this applies to bounded vectors observed through nonnegative linear channels and a shared Bernoulli gate.*

Proof. The log-Laplace Riemann sums have the same endpoint bounds as before. For derivatives at $q > 0$, put $S = \sum_i q_i y^{-F_i} Y_i$. Each $y^{-F_i} Y_i \le S/q_i$, and the tilted denominator is at least $\mathbb{P}(Y = 0)$. Since $S^j e^{-S}$ is bounded for $j = 1, 2$, the first and second derivative integrands are bounded multiples of $y^{\rho - 1}$ at zero. At infinity boundedness of Y gives the same integrable power bounds as before. Consequently the derivative Riemann sums converge and the tilted variances vanish. Concavity makes q an optimizer at $e = \nabla H(q)$. Under $q_{c,i} = c^{F_i} q_i$, every positive mean is multiplied by $c^{\rho - F_i} < 1$. The change-of-measure proof just given then applies. No sign condition on correlations or support gap is needed at a strictly positive tilt. □

44.1 Power-law behavior at the origin in a positive cone

The origin atom in Theorem 43.4 can be replaced by a power law near zero. The following hypothesis permits arbitrary dependence between radial size and direction.

Definition 44.4. A bounded nonnegative random vector X belongs to $\mathcal{P}_d$ if there are a random variable $R \in (0, 1]$, constants $a, b, c_-, c_+ > 0$, and $\kappa > 0$ such that almost surely

$$aR \le X_j \le bR \quad (1 \le j \le d), \qquad c_- r^{\kappa} \le \mathbb{P}\{R \le r\} \le c_+ r^{\kappa} \quad (0 < r \le 1).$$

The constants and exponent may depend on the law. No independence of R and the direction of X is required.

Proposition 44.5 (Restoration for power-law radial behavior)**.** *The injectivity conclusion of Theorem 43.4 holds with $\mathcal{C}_d$ replaced by $\mathcal{P}_d$. In addition, the exact alias construction of Theorem 43.1 remains within $\mathcal{P}_d$. Distinct aliases can preserve any prescribed finite collection of mixed moments and one-component responses as in Proposition 43.2.*

Proof. Write $T = \sum_j v_j$. The assumptions and integration by parts for the distribution function of R give, for each integer $m \ge 0$,

$$\mathbb{E}e^{-v\cdot X} \ge c(1+T)^{-\kappa}, \qquad \mathbb{E}[R^m e^{-v\cdot X}] \le C_m(1+T)^{-\kappa-m}.$$

For the lower bound restrict to $R \le (1+T)^{-1}$. For the upper bound use $v\cdot X \ge aTR$; the estimate follows by splitting into intervals of length $(1+T)^{-1}$ and summing an exponentially decaying series, or by Stieltjes integration by parts. Thus every tilted moment of total order m is $O((1+T)^{-m})$. In particular,

$$f(v) := -h_{11}(v) = O((1+T)^{-2}), \quad h_j(v) = O((1+T)^{-1}), \quad |\partial_j f(v)| = O((1+T)^{-3}). \tag{251}$$

Also $h(v) = O(1+\log(1+T))$ as $T \to \infty$, so both response integrals remain finite. The d-fold Mellin transform of f exists on the nonempty tube

$$\operatorname{Re} a_j > 0, \qquad \sum_j \operatorname{Re} a_j < 2.$$

The identity (246) is still valid whenever its transformed parameters lie in this tube. Such a vertical plane is attainable: take all the Mellin variables small and positive except the variable attached to the larger of E_1 and C, and choose that variable just below

$$\frac{E_1 + C - \rho}{\max(E_1, C)} < 2.$$

The remaining transformed parameter a_d can be made positive and arbitrarily small, while the sum of all a_j stays below two. The same invertible affine map and Mellin uniqueness therefore identify f.

To check the integration constants without an atom at zero, suppose two laws give the same f. Their logarithmic Laplace transforms differ by $A(v')v_1 + B(v')$. The derivative bound in (251) forces $A(v') = 0$. Furthermore, for each fixed $v' \ge 0$,

$$|h(v_1, v') - h(v_1, 0)| \le C\|v'\|_1/(1+v_1) \longrightarrow 0.$$

It follows that $B(v') = B(0) = 0$, since both transforms vanish at the origin. Thus the complete law is identified. If an explicit reconstruction is desired, the same bounds justify

$$h(v_1, 0) = \int_0^\infty \min(v_1, r) f(r, 0)\, dr,$$
$$h(v_1, v') - h(v_1, 0) = -\int_{v_1}^\infty (r - v_1)[f(r, v') - f(r, 0)]\, dr.$$

The second integrand is integrable by the third estimate in (251).

Finally, a finite dilation-convolution is comparable in every coordinate to a positive weighted sum of independent copies of R. Its lower-tail distribution is bounded above and below by constants times $r^{n\kappa}$: use the inclusions that a sum at most r requires each summand at most r, and is guaranteed when each summand is at most r/n, with the fixed positive coefficients absorbed into constants. After a fixed rescaling this proves membership in $\mathcal{P}_d$.

For the finite-calibration assertion, replace the Bernoulli seed in Proposition 43.2 by a common Beta$(\alpha, 1)$ variable, with a transcendental $\alpha > 0$. Every cumulant is a rational function of α, since its mth moment is $\alpha/(\alpha+m)$. That rational function is nonzero: as $\alpha \downarrow 0$, its leading term is α/m. Consequently all cumulants are nonzero at transcendental α. The identical power-sum and implicit-function argument now supplies the required distinct laws in $\mathcal{P}_d$. □

Remark 44.6. At fully active maximizing tilts, the lower-tail proof of Theorem 44.1 also applies to $\mathcal{P}_k$. For clarity, the necessary replacements are $h = O(1 + \log(1 + T))$, integrable at the lower endpoint of the response integral, and the tilted moment estimates above. Near zero in the Riemann-sum variable, at a strictly positive tilt, the first and second derivative integrands are bounded because the same positive tilt contains all coordinates; hence that proof immediately covers fully active maximizing tilts. Boundary maximizing tilts require separate integrability checks when some exponents differ and are not asserted here without them.

45 One additional mixed probe identifies a vector law

For unrestricted nonnegative vector laws, repeating one coordinate is insufficient to extend scalar two-weight identification to a vector. We instead add one positive linear combination of all coordinates at a different power. The construction is an explicit anisotropic polar decomposition, within the classical theory of Lévy mixing and homogeneous measures [34, 35, 36]. The candidate contribution is the sharp observation design and its applicability after one common random gate, not a new Lévy representation theorem.

45.1 An explicit signed-measure inverse

Put $C_k = [0,\infty)^k \setminus \{0\}$ and use the ℓ^1 norm. Fix $\rho > 0$, $p, q > \rho$, $p \neq q$, and $v \in (0,\infty)^k$. Let ν be a signed Radon measure on C_k such that

$$\int_{C_k} (\|x\|^{\rho/p} + \|x\|^{\rho/q})|\nu|(dx) < \infty. \tag{252}$$

Define

$$h_\nu(a) = \int_{C_k} (1 - e^{-a\cdot x})\nu(dx), \tag{253}$$

$$\mathcal{H}_\nu(a,t) = \int_0^\infty \rho y^{\rho-1} h_\nu(y^{-p}a + t y^{-q} v)\, dy, \qquad a > 0,\ t > 0. \tag{254}$$

For $\nu \geq 0$, h_ν is the Laplace exponent of a drift-free nonnegative infinitely divisible vector. There are k coordinate probes at power p and one shared probe $v \cdot X$ at power q.

Theorem 45.1 (Explicit mixed-probe recovery)**.** *Exact values of $\mathcal{H}_\nu$ on any nonempty open subset of $(0,\infty)^{k+1}$ determine ν uniquely. Let η be the unique signed Radon measure on C_{k+1}, with $\int (1 \wedge \|b\|)|\eta|(db) < \infty$, satisfying*

$$\mathcal{H}_\nu(a,t) = \int (1 - e^{-a\cdot z - tw})\eta(dz, dw). \tag{255}$$

It is supported on $z \neq 0, w > 0$. Set

$$R(z,w) = \left(\frac{w}{v\cdot z}\right)^{1/(p-q)}, \qquad X(z,w) = R(z,w)^p z. \tag{256}$$

For each continuous f compactly supported in C_k,

$$\int f\, d\nu = \frac{1}{2^\rho - 1} \int_{1 \leq R(z,w) \leq 2} f(X(z,w))\eta(dz, dw). \tag{257}$$

Proof. For $r > \rho$ and $c \geq 0$, substitution and integration by parts give

$$\int_0^\infty \rho y^{\rho-1}(1 - e^{-cy^{-r}})\,dy = \Gamma(1-\rho/r)c^{\rho/r}. \tag{258}$$

Together with $1 - e^{-(A+B)} \leq (1 - e^{-A}) + (1 - e^{-B})$, this proves the absolute integrability needed below. Push $\rho y^{\rho-1}dy\,\nu(dx)$ forward by

$$F(y, x) = (y^{-p}x,\ y^{-q}v \cdot x).$$

Its total variation is bounded by the same pushforward of $\rho y^{\rho-1}dy\,|\nu|(dx)$. Equation (258) and the comparability of $1 - e^{-u}$ and $1 \wedge u$ show that this image η has the asserted integrability. Absolute Fubini proves (255).

The signed representation is unique by finite-measure Laplace uniqueness. Indeed, the sum of all first partial derivatives, evaluated at $(a, t) = (1, \ldots, 1) + \lambda$, is the Laplace transform at $\lambda \geq 0$ of the finite signed measure $\|b\|e^{-\|b\|}\eta(db)$. Its weight is positive on C_{k+1}. Finite Laplace uniqueness itself follows by setting $u_i = e^{-b_i}$ and using density of polynomials on $[0, 1]^{k+1}$. The integral is holomorphic when all parameters have positive real parts, so real-analytic continuation makes any real open patch sufficient.

Finally $w/(v \cdot z) = y^{p-q}$. Thus F is a Borel isomorphism from $(0, \infty) \times C_k$ onto $\{(z, w) : z \neq 0, w > 0\}$, with inverse (256). Pull back η and restrict y to $[1, 2]$. Its mass in the scale variable is $2^\rho - 1$, which gives (257). □

Remark 45.2. The data are a full continuum function, not $k + 1$ numbers. The number of probes counts aggregate energies whose joint limiting Laplace exponent is supplied. Analytic continuation and Laplace inversion are part of recovery; the explicit scale formula alone does not establish numerical stability.

Corollary 45.3 (Unequal base powers)**.** *Identification still holds with coordinate powers $p_i > \rho$ and a shared power $q > \max_i p_i$ or $\rho < q < \min_i p_i$, assuming*

$$\int \left(\sum_i x_i^{\rho/p_i} + (v \cdot x)^{\rho/q}\right) |\nu|(dx) < \infty.$$

The inverse scale is the unique positive solution of $w = \sum_i v_i z_i y^{p_i - q}$; then $x_i = y^{p_i} z_i$.

Proof. Use $z_i = y^{-p_i}x_i$ in the preceding proof. The right side of the scale equation is strictly monotone, onto $(0, \infty)$, since every nonzero summand has the same exponent sign. □

Corollary 45.4 (Unknown counting exponent)**.** *For nonzero $\nu \geq 0$, knowledge of p, q, v and $\mathcal{H}_\nu$ also determines ρ through*

$$\mathcal{H}_\nu(\lambda^p a, \lambda^q t) = \lambda^\rho \mathcal{H}_\nu(a, t).$$

Proof. Substitute $y = \lambda u$. The function is positive, so its homogeneity degree is unique. □

45.2 A sharp observation threshold

For nonnegative linear forms ℓ_i, powers $p_i > \rho$, and m probes, set

$$\mathcal{K}_\nu(s) = \int_0^\infty \rho y^{\rho-1} \int_{C_k} \left(1 - \exp\left\{-\sum_{i=1}^m s_i y^{-p_i}\ell_i(x)\right\}\right) \nu(dx)\,dy. \tag{259}$$

Theorem 45.5 (At most k linear probes never suffice)**.** *For $m \le k$, $\nu \mapsto \mathcal{K}_\nu$ is not injective even on finite positive measures with finite support in $(0,\infty)^k$. The two measures may have the same total mass, as small as desired. Thus the associated distinct compound-Poisson laws may have the same prescribed atom at zero, arbitrarily close to one. The minimum number for uniform identification of the stated drift-free infinitely divisible class is therefore exactly $k+1$.*

Proof. Let M be the $m \times k$ matrix of the forms. If $\operatorname{rank} M < k$, choose $x_0 > 0$ and nonzero $u \in \ker M$. Small $\epsilon \neq 0$ keeps $x_0 + \epsilon u > 0$, and $\lambda\delta_{x_0}$ and $\lambda\delta_{x_0+\epsilon u}$ are indistinguishable.

Otherwise $m = k$ and M is invertible. For c close to one put $x_c = M^{-1}\operatorname{diag}(c^{p_i})Mx_0 > 0$. Substitution $y = cu$ shows $\mathcal{K}_{\delta_{x_c}} = c^\rho \mathcal{K}_{\delta_{x_0}}$. Choose $c_- < 1 < c_+$ close to one and $a \in (0,1)$ with $ac_-^\rho + (1-a)c_+^\rho = 1$. The distinct measures

$$\nu_1 = \lambda\delta_{x_0}, \qquad \nu_2 = \lambda\{a\delta_{x_{c_-}} + (1-a)\delta_{x_{c_+}}\}$$

have the same observations and mass λ. Their compound-Poisson laws have atom $e^{-\lambda}$ at zero because all jumps are positive. The upper bound is Theorem 45.1. □

Proposition 45.6 (The extra probe must see every hidden coordinate)**.** *The two-dimensional design*

$$\int_0^\infty \rho y^{\rho-1} h(sy^{-E} + ty^{-A}, uy^{-B})\, dy$$

does not identify the joint law, even for independent coordinates and distinct $E, A, B > \rho$.

Proof. Independence gives $h(a,b) = h_V(a) + h_W(b)$. The second coordinate contributes only $u^{\rho/B}\int_0^\infty \rho y^{\rho-1} h_W(y^{-B})\, dy$. For a compound-Poisson W with jump measure ξ, this coefficient is $\Gamma(1-\rho/B)\int x^{\rho/B}\xi(dx)$. Choose two distinct ξ of the same mass and fractional moment, one atom versus a mixture of atoms on either side, and hold V fixed. □

At $p = q$ even the positive shared probe fails, since

$$\mathcal{H}_\nu(a,t) = \Gamma(1-\rho/p)\int ((a+tv)\cdot x)^{\rho/p}\nu(dx).$$

The additional power breaks the otherwise invisible dilation.

45.3 A common Bernoulli gate gives universal recovery

Corollary 45.7 (Arbitrary vector laws after gating)**.** *Let μ be any law on $[0,\infty)^k$ satisfying $\mathbb{E}\|X\|^{\rho/p} + \mathbb{E}\|X\|^{\rho/q} < \infty$; every bounded nonnegative vector is allowed. Let B be independent of X, with known $\mathbb{P}(B=0) = p_0 \in (1/2, 1)$ and $\mathbb{P}(B=1) = 1-p_0$. The mixed-probe continuum function of BX uniquely determines the original joint law μ. The same Bernoulli variable must multiply all coordinates of each local copy; different copies use independent gates.*

Proof. Put $r = (1-p_0)/p_0 < 1$. The classical logarithm series gives the signed measure, with its atom at zero discarded,

$$\nu = \sum_{n\ge1} \frac{(-1)^{n+1} r^n}{n} \mu^{*n} \quad \text{on } C_k, \tag{260}$$

and

$$-\log(p_0 + (1-p_0)\mathbb{E}e^{-a\cdot X}) = \int (1 - e^{-a\cdot x})\nu(dx).$$

For $0 < \alpha < 1$, subadditivity yields

$$\int \|x\|^\alpha |\nu|(dx) \le \sum_{n\ge 1} \frac{r^n}{n} \mathbb{E}\|X_1 + \cdots + X_n\|^\alpha \le \frac{r}{1-r} \mathbb{E}\|X\|^\alpha.$$

Apply Theorem 45.1; the recovered h_ν gives $\mathbb{E}e^{-a\cdot X} = (e^{-h_\nu(a)} - p_0)/(1-p_0)$, which determines μ. □

The representation (260) is Cuppens' classical quasi-infinite-divisibility construction, reproduced in [37, Theorem 3.2]. Gating is an additional intervention, not an inference from the ungated data of an arbitrary law.

Theorem 45.8 (Sharpness within the bounded gated class)**.** *For any fixed known $p_0 \in (1/2, 1)$, no collection of at most k nonnegative linear power probes with exponents above ρ identifies every bounded nonnegative vector law from the continuum functions of its copies gated with probability p_0.*

Proof. Write M for the probe matrix. Its kernel gives distinct positive vectors with identical probe values if $\operatorname{rank} M < k$. Gating their point masses proves that case.

Otherwise M is invertible. Choose $x_0 > 0$ and write $T_c = M^{-1} \operatorname{diag}(c^{p_i}) M$ for c close to one. Fix $b \in (\sqrt{p_0}, 1)$, and let Z_1, Z_2 be independent, each equal to zero with probability b and to x_0 otherwise. For sufficiently small $\epsilon > 0$ put

$$d_\pm = (1 \pm \epsilon)^{1/\rho}, \qquad Y_1 = Z_1 + Z_2, \qquad Y_2 = T_{d_+} Z_1 + T_{d_-} Z_2.$$

Both vectors are bounded and nonnegative and have atom $b^2 > p_0$ at zero. Thus each has the form $p_0\delta_0 + (1-p_0)\mu_i$ with a bounded probability law μ_i. Independence makes logarithmic Laplace transforms additive, and substitution $y = cu$ multiplies the contribution of $T_c Z$ by c^ρ. Since $d_+^\rho + d_-^\rho = 2$, the complete observed functions of Y_1 and Y_2 agree.

The laws differ: for any nonzero probe row ℓ_j the ratio of its mean under Y_2 to that under Y_1 is

$$\frac{d_+^{p_j} + d_-^{p_j}}{2} = \frac{(1+\epsilon)^{p_j/\rho} + (1-\epsilon)^{p_j/\rho}}{2} > 1$$

by strict convexity. Hence $\mu_1 \ne \mu_2$. Combined with Corollary 45.7, this proves the exact threshold. □

Proposition 45.9 (Exact placement of the extra exponent)**.** *Fix $p_0 \in (1/2, 1)$ and let $\mathcal{P}_b$ be the class of all compactly supported probability laws on $[0,\infty)^k$, allowing mass at the origin. For $\mu \in \mathcal{P}_b$, write*

$$P_\mu = p_0\delta_0 + (1-p_0)\mu, \qquad h_\mu(a) = -\log \int e^{-a\cdot x} P_\mu(dx).$$

Fix $p_1, \dots, p_k, q > \rho > 0$ and $v_i > 0$, and define

$$H_\mu(a,t) = \int_0^\infty \rho y^{\rho-1} h_\mu\big((a_i y^{-p_i})_{i=1}^k + t y^{-q} v\big)\, dy.$$

If $q \in [\min_i p_i, \max_i p_i]$, the map $\mu \mapsto H_\mu$ is not injective on $\mathcal{P}_b$. Consequently, together with the inversion theorem, universal identification in this fixed-gate class holds exactly when $q < \min_i p_i$ or $q > \max_i p_i$.

Proof. Suppose first that $\min_i p_i < q < \max_i p_i$. Choose distinct adjacent positive integers n, m and put $c = (n/m)^{1/\rho}$. The coefficients $v_i(c^{p_i} - c^q)$ have both signs. We can therefore choose $x \in (0,\infty)^k$ such that

$$\sum_i v_i(c^{p_i} - c^q)x_i = 0.$$

Set $x_i' = c^{p_i}x_i$; then $v \cdot x' = c^q v \cdot x$. Choose $\theta \in (0,1)$ small enough that $(1-\theta)^{\max(n,m)} > p_0$, and let

$$V = xB_n, \qquad V' = x'B_m, \qquad B_n \sim \mathrm{Bin}(n,\theta), \quad B_m \sim \mathrm{Bin}(m,\theta).$$

Their laws P, P' are distinct, since their masses at zero are $(1-\theta)^n$ and $(1-\theta)^m$. Both exceed p_0, so

$$\mu = \frac{P - p_0\delta_0}{1-p_0}, \qquad \mu' = \frac{P' - p_0\delta_0}{1-p_0}$$

are distinct elements of $\mathcal{P}_b$. Writing $g(u) = -\log(1 - \theta + \theta e^{-u})$, their Laplace exponents are $h_\mu(a) = ng(a \cdot x)$ and $h_{\mu'}(a) = mg(a \cdot x')$. The substitution $y = cu$ gives, for all $a \in [0,\infty)^k$ and $t \geq 0$,

$$\begin{aligned} H_{\mu'}(a,t) &= m\int_0^\infty \rho y^{\rho-1} g\left(\sum_i a_i c^{p_i} x_i y^{-p_i} + tc^q (v\cdot x) y^{-q}\right) dy \\ &= mc^\rho \int_0^\infty \rho u^{\rho-1} g\left(\sum_i a_i x_i u^{-p_i} + t(v\cdot x)u^{-q}\right) du = H_\mu(a,t), \end{aligned}$$

because $mc^\rho = n$. The integrals converge: g is bounded at infinity and is $O(u)$ at zero, while every probe exponent exceeds ρ.

If q equals an endpoint of the exponent interval, choose an index j with $p_j = q$, take $x = e_j$, and put $x' = c^q e_j$. The same construction and calculation apply. This also covers the case $p_1 = \cdots = p_k = q$. □

Remark 45.10. The allowance of an atom at zero in μ is essential to this particular counterexample: the fixed gate p_0 is a known lower contribution to the observed zero atom, not an assertion that the observed zero atom equals p_0. If one instead requires $\mu(\{0\}) = 0$, the above de-gating argument does not prove the stated lower bound for that smaller class.

45.4 Local rate data and a singular power collision

Example 45.11 (An interior exponent can add no identifying power). Take $\rho = 1$, coordinate exponents $(2, 4)$, and mixed exponent 3 with $v = (1, 1)$. For independent binomial counts of common success probability θ, the two vectors

$$(2,1)\,\mathrm{Bin}(2,\theta) \quad \text{and} \quad (8,16)\,\mathrm{Bin}(1,\theta)$$

have identical three-channel responses. Indeed, $c = 2$ scales the coordinates by $2^2, 2^4$, while $24 = 2^3 \cdot 3$ scales their sum; the factor $c^\rho = 2$ compensates for the number of summands. At $\theta = 1/100$, both zero atoms exceed the same known gate probability $p_0 = 3/4$, so de-gating gives two different bounded input laws with identical observed responses. Replacing exponent 3 by 5 gives a universally identifying design in this gated class.

Corollary 45.12 (An active rate patch suffices)**.** *Assume* $\nu \geq 0$, (252), *and* $\mathrm{span}(\mathrm{supp}\,\nu) = \mathbb{R}^k$. *Define* $\mathcal{I}(e) = \sup_{\lambda \geq 0}\{\mathcal{H}_\nu(\lambda) - \lambda \cdot e\}$. *Any nonempty exact open patch of* $\mathcal{I}$ *on which the maximizer exists and has positive coordinates determines* ν.

Proof. The negative Hessian of $\mathcal{H}_\nu$ has quadratic form $\int (b\cdot c)^2 e^{-\lambda\cdot b}\eta(db)$. If this vanishes for $c=(c_0,c_1)$, then $y^{-p}c_0\cdot x + c_1 y^{-q} v\cdot x = 0$ for almost every (y,x). The distinct powers and $v\cdot x>0$ force $c_1=0$ and then $c_0=0$ by the span assumption. The Hessian is therefore negative definite. The inverse function theorem and strict concavity give

$$\lambda = -\nabla \mathcal{I}(e), \qquad \mathcal{H}_\nu(\lambda) = \mathcal{I}(e) - e\cdot \nabla \mathcal{I}(e)$$

on an open image set. Apply Theorem 45.1. □

Although $\log R = (\log w - \log(v\cdot z))/(p-q)$ explicitly exhibits one conditioning factor, much stronger degeneracies occur.

Proposition 45.13 (Arbitrarily high-order coalescence)**.** *Fix $p>\rho$, $v>0$, and $\theta\in(0,\infty)^k$. For each integer $m\ge 0$ there are distinct finite positive measures $\nu_\pm$ of equal mass, supported on finitely many points of the ray $\{r\theta : r>0\}$, such that*

$$\partial_q^j \mathcal{H}_{\nu_+}(a,t;q)\Big|_{q=p} = \partial_q^j \mathcal{H}_{\nu_-}(a,t;q)\Big|_{q=p}, \qquad 0\le j\le m, \quad a>0,\ t>0.$$

Their difference is $O(|q-p|^{m+1})$ uniformly on compact positive parameter sets, although every fixed $q\neq p$ distinguishes the two measures.

Proof. Set $\alpha=\rho/p$. Choose $m+3$ distinct positive radii r_i in a compact interval and a nonzero real vector (c_i) satisfying

$$\sum_i c_i = 0, \qquad \sum_i c_i r_i^{\alpha}(\log r_i)^j = 0 \quad (0\le j\le m).$$

There are more unknowns than equations. Perturb positive weights by $\pm\epsilon c_i$ to obtain the two positive measures. Substitution $y=r^{1/p}z$ gives

$$\mathcal{H}_{\delta_{r\theta}}(a,t;q) = r^{\alpha}\int_0^\infty \rho z^{\rho-1}\left(1-e^{-z^{-p}a\cdot\theta - t r^{1-q/p} z^{-q} v\cdot\theta}\right)dz.$$

Its jth q-derivative at p is r^α times a polynomial of degree at most j in $\log r$. The coefficient functions do not depend on r, so the annihilating moments prove the claim. Differentiation and Taylor remainders are uniform on compact sets: as $z\downarrow 0$, $e^{-cz^{-p}}$ dominates every derivative; as $z\to\infty$, the bound is $Cz^{\rho-1-q_*}(1+|\log z|^{m+1})$ for some $q_*>\rho$. □

These are uniqueness results for exact continuum information. They provide neither a finite-sample statistical guarantee nor a priority claim for the classical transform machinery.

46 A periodic sampling obstruction to a spectral improvement

The following construction explains why exact finite-dimensional nonattainment of a moment certificate need not produce an asymptotic improvement, even for genuine band-limited evaluation vectors with uniform spacing and exceptionally regular weighted counting density. It concerns an abstract real sampling configuration, not the actual zeros of the zeta function.

Theorem 46.1 (A regular sinc configuration saturating the quartic target)**.** *Let $v_x(t)=e^{2\pi i x t}$ in $L^2([-1/2,1/2])$, so that $\langle v_x, v_y\rangle = \operatorname{sinc}(x-y)$, where $\operatorname{sinc}(u)=\sin(\pi u)/(\pi u)$ with its continuous value at zero. Put*

$$c=\frac{1}{2\sqrt{2}}, \qquad d=\frac{2}{\pi}\arccos c\in(0,1), \qquad \theta=\frac{7}{8}.$$

For each integer $k \ge 1$, take the simple locations

$$\{2n, 2n+d : 0 \le n < 16k\}$$

and the double locations

$$\{2n : 16k \le n < 21k\}.$$

Define

$$G_k = \sum_{0 \le n < 16k} (v_{2n} v_{2n}^* + v_{2n+d} v_{2n+d}^*) + 2 \sum_{16k \le n < 21k} v_{2n} v_{2n}^*, \qquad N_k = 42k.$$

Then the number of simple locations is $s_k = 32k$, the number of double locations is $b_k = 5k$, and

$$\frac{1}{N_k}(\operatorname{tr} G_k, \operatorname{tr} G_k^2, \operatorname{tr} G_k^3, \operatorname{tr} G_k^4) = \left(1, \frac{4}{3}, 2, \frac{13}{4}\right) + O\left(\frac{\log(k+2)}{k}\right), \qquad \frac{s_k}{N_k} = \frac{16}{21}. \tag{261}$$

The minimum distance between distinct locations is at least d. If $\mathcal{N}_k(I)$ counts these locations with their indicated multiplicities, then every interval $I \subset [0, N_k)$ satisfies

$$|\mathcal{N}_k(I) - |I|| \le 4. \tag{262}$$

For

$$L_k = 4 \operatorname{tr} G_k - \operatorname{tr} G_k^2 - 2N_k, \qquad D_k = \operatorname{tr}\big(G_k(2I - G_k)\big)^2,$$

the nonnegative quartic certificate defect obeys

$$0 \le s_k - \frac{2L_k}{\theta} + \frac{D_k}{\theta^2} = O(\log(k+2)). \tag{263}$$

Consequently separation and the bounded counting discrepancy (262) alone cannot force a positive normalized improvement over $16/21$ from this quartic certificate.

Proof. We first record an elementary two-coset identity. The doubly infinite matrix

$$C = (\operatorname{sinc}(2(i-j)+d))_{i,j \in \mathbb{Z}}$$

defines an operator on $\ell^2(\mathbb{Z})$ satisfying

$$CC^* = C^*C = \cos^2(\pi d/2) I = c^2 I. \tag{264}$$

Indeed, the families (v_{2n}) and (v_{2n+d}) are individually orthonormal. Identify functions on $[-1/2, 1/2]$ with two-component functions on $[-1/2, 0]$ by $f(t) \mapsto (f(t), f(t+1/2))$. The respective closed spans have fiber directions $(1,1)/\sqrt{2}$ and $e^{2\pi i d t}(1, e^{\pi i d})/\sqrt{2}$. Their fiberwise inner product has constant modulus $|(1 + e^{\pi i d})/2| = c$. Multiplication by its phase is unitary, which proves (264) in the orthonormal bases indexed by n.

Let C_m be the compression of C to $0 \le i, j < m$. If $\sigma_1, \dots, \sigma_m$ are its singular values, then $0 \le \sigma_j \le c$ and

$$\begin{aligned} \sum_{j=1}^{m} (c^2 - \sigma_j^2) &= \sum_{0 \le i < m} \sum_{j \notin [0, m-1]} |\operatorname{sinc}(2(i-j)+d)|^2 \\ &= O(\log(m+2)). \end{aligned} \tag{265}$$

For the last estimate use $|\operatorname{sinc}(2r+d)| \ll_d (1+|r|)^{-1}$; the sum of the exterior tails for row i is $O((i+1)^{-1} + (m-i)^{-1})$.

The Gram matrix A_m of the $2m$ simple vectors has block form

$$A_m = \begin{pmatrix} I & C_m \\ C_m^* & I \end{pmatrix}$$

and eigenvalues $1 \pm \sigma_j$. For each fixed integer $1 \le r \le 4$, the expression $(1+\sigma)^r + (1-\sigma)^r$ is a polynomial in σ^2. It follows from (265) that

$$\operatorname{tr} A_m^r = m\big((1+c)^r + (1-c)^r\big) + O(\log(m+2)). \tag{266}$$

The assertion for $r = 1$ is exact.

The nonzero spectrum of G_k is the spectrum of the weighted Gram matrix K_k of the vectors with factors 1 at simple locations and $\sqrt{2}$ at double locations. With $m = 16k$ and $b = 5k$, write

$$K_k = B_k + E_k, \qquad B_k = \operatorname{diag}(A_m, 2I_b).$$

Here E_k has only off-diagonal blocks between the simple and double groups. The inner products between base vectors v_{2i} and double vectors v_{2j} vanish. The other inner products give

$$\|E_k\|_{\mathrm{HS}}^2 \ll_d \sum_{0 \le i < m} \sum_{m \le j < m+b} (1 + j - i)^{-2} = O(\log(k+2)).$$

All operator norms are bounded independently of k: the base vectors v_{2n} are orthonormal and carry weights at most 2, while the offset vectors v_{2n+d} are another orthonormal family carrying weight 1. Thus $\|G_k\| = \|K_k\| \le 3$; also $\|B_k\| \le 2$ and $\|E_k\| \le 5$. For $1 \le r \le 4$, expand the trace of $(B_k + E_k)^r$ into words. A word with exactly one E_k has zero trace, because B_k is block diagonal and E_k is block off-diagonal. Every word with at least two E_k has absolute trace bounded by a constant times $\|E_k\|_{\mathrm{HS}}^2$, by the Hilbert–Schmidt trace inequality and the uniform operator norm bounds. Therefore

$$\operatorname{tr} K_k^r = m\big((1+c)^r + (1-c)^r\big) + b2^r + O(\log(k+2)).$$

Substituting $m = 16k$, $b = 5k$, $c^2 = 1/8$, and $N_k = 42k$ gives (261). More explicitly the simple-part normalized second, third, and fourth moments are $6/7$, $22/21$, and $113/84$; the double-part contributions are $10/21$, $20/21$, and $40/21$, respectively.

Every cell $[2n, 2n+2)$ contains total weight exactly 2. Removing the at most two partial cells at the ends of an interval proves (262). Within a simple cell the two locations have distance d; all other consecutive gaps are at least $2 - d > d$. This proves the spacing assertion.

Finally, we verify nonnegativity without invoking any zeta input. Write $G_k = P + Q$, where P is the simple part and Q the double part. In the decreasing positive spectrum, remove the top b_k eigenvalues. Rank interlacing with Q, which has rank b_k, shows that the remaining number is at most s_k and their sum is at most $\operatorname{tr} P = s_k$. For

$$q_\theta(x) = \frac{8\theta x - (4+2\theta)x^2 + 4x^3 - x^4}{\theta^2},$$

the identities

$$\frac{8}{\theta} - q_\theta(x) = \frac{(x-2)^2(x^2+2\theta)}{\theta^2}, \qquad 1 + \frac{4x}{\theta} - q_\theta(x) = \frac{(x^2 - 2x + \theta)^2}{\theta^2}$$

give $\operatorname{tr} q_\theta(G_k) \le 8b_k/\theta + s_k + 4s_k/\theta = s_k + 4N_k/\theta$. The difference on the right is precisely the defect in (263). The moment limits above give $L_k = (2/3)N_k + O(\log(k+2))$ and $D_k = (7/12)N_k + O(\log(k+2))$. At $\theta = 7/8$ their certificate main term is $(16/21)N_k = s_k$, proving (263). □

The same construction works for any $0 < \theta < 1$ by taking $d = (2/\pi)\arccos\sqrt{1-\theta}$ and any fixed positive proportion of simple cells. It produces the two simple spectral values $1 \pm \sqrt{1-\theta}$ together with the double spectral value 2, with only logarithmic finite-section and interface errors. The obstruction is phase separation in the type of sampling cell, despite uniformly regular multiplicity-weighted counting. Additional information about the distribution of multiplicities, beyond ordinary spacing or density bounds, is therefore necessary to exclude this mechanism for zeta zeros.

47 Boundaries of the rare-event reconstruction results

The results distinguish complete continuum responses, local curvature, and entire microscopic fluctuation laws. A homogeneous response can retain angular dependence while losing scale; an extra channel can separate the dilation orbits. Scalar local curvature determines all weight ratios, but finitely many jets do not determine an unrestricted local law. A full Gamma-series fluctuation law determines its nonzero coefficients, whereas finitely many cumulants can miss even the largest.

The noisy inverse problem remains separate from these uniqueness statements. Proposition 33.2 starts from an error bound on mixed-derivative data. Controlling the preceding duality and differentiation operations under a specified observation model is an additional problem. Likewise, the mixed-probe inverse requires quantitative control of the reconstruction of a signed measure from its Laplace response before the exact channel threshold yields a statistical guarantee. Analytic continuation from a patch is used only for uniqueness.

Unknown scalar exponents are handled by Theorem 38.1, and the remaining coordinate and normalization ambiguities are classified in Theorem 38.2 and Proposition 38.5. Genuinely correlated vector laws lack the entrywise covariance positivity used in that scalar argument; blind calibration for those classes requires additional hypotheses. The repeated-component inverse also has no unrestricted extension, as Proposition 45.6 demonstrates.

At $2(E-A) = \rho$, slowly varying counting factors can decide whether the discrete coefficient ratios belong to ℓ^2. A uniform classification there is not furnished by the continuum activation calculation. The hard-conditioning result uses the arcsine density and a large prime-indexed independent reservoir; its extension needs an appropriate local limit estimate, not merely a logarithmic probability asymptotic. Finally, transferring a nondegenerate fluctuation law to sampling by simple zeta zeros requires more than a bounded comparison with product Haar measure. The sinc obstruction shows why ordinary spacing and multiplicity-weighted counting bounds alone do not close that gap.

Part IV
Controlled observation and blind calibration

48 From observation design to hidden structure

The inverse problems of the preceding part motivate a change in the observation law itself. A continuously varied Bernoulli gate first recovers an ordered Laplace profile; arithmetic structure in that profile then separates the original coordinate powers from the powers generated by mixed moments. A positive interior baseline extends this mechanism to arbitrary nonnegative summable laws, without assuming moments of the original distribution. The finite-window normalization also permits an unknown observation gain and a common unknown baseline to be recovered.

The reconstruction rests on three operations that remain distinct. Compact moment inversion recovers a distribution of profile values. Monotonicity restores their order in the observation parameter. Finally, positivity of mixed moments identifies the original exponents as the additively indecomposable nonzero atoms of an exponent measure. The observation is a full function, not a single scalar value. Inactive coordinates and uncalibrated coordinate amplitudes retain the explicit ambiguities described below.

After the exact reconstruction results, the finite-data analysis proceeds from noisy profile values to normalized gate derivatives and then to ordinary gate values. These guarantees require quantitative visibility conditions and recover dimension and powers; they do not provide uniform recovery of the full law. Separated flat scales give a complementary interior probe: its actual values determine every finite multivariate measure, while all derivatives at its endpoint can fail to distinguish smooth positive densities. Radial windows and adaptive linear queries then give two different sharp information bounds.

The logarithmic and Stieltjes-transform relation and compact moment uniqueness are classical [48, 46, 41]. Known-power arithmetic encoding has antecedents in [44, 43]; the distinction here is the blind recovery of exponents and cardinality, followed by the explicitly stated observation modifications and obstructions. Restricted-direction determination is related to sharp Cramér–Wold theory [40], but that theory observes projection laws rather than one Laplace value at each point of a nonlinear curve. The later quantitative arguments use classical Prony, approximation, convolution, and information-width methods. Their hypotheses and precise recovery targets are recorded with each result.

49 Inverting a continuously varied Bernoulli gate

Let $X = (X_1, \dots, X_d)$ be an almost surely finite nonnegative random vector, and write

$$L_X(s) = \mathbb{E}\exp(-s \cdot X), \qquad s \in [0,\infty)^d.$$

Fix $\rho > 0$, powers $p_i > \rho$, and $a_i > 0$. Put $q_i = p_i/\rho$ and $\alpha_i = 1/q_i$. A shared Bernoulli gate with known success probability $\theta \in (0,1)$ replaces X by $B_\theta X$, where B_θ is independent of X. Its continuum response is

$$G_a(\theta) = \int_0^\infty \rho y^{\rho-1} \left[-\log\left(1-\theta+\theta L_X(a_1 y^{-p_1}, \dots, a_d y^{-p_d})\right)\right] dy. \tag{267}$$

Only the gate probability is varied here; the amplitudes and powers remain fixed. The substitution $r = y^{\rho}$ gives

$$f_a(r) = 1 - L_X(a_1 r^{-q_1}, \dots, a_d r^{-q_d}), \qquad G_a(\theta) = \int_0^{\infty} -\log(1 - \theta f_a(r))\, dr. \tag{268}$$

The usefulness of the gate is that its logarithm supplies every power of f_a, rather than only one homogeneous average.

Lemma 49.1 (Exact integrability condition)**.** *The following are equivalent:*

(i) $G_a(\theta) < \infty$ *for some* $\theta \in (0,1)$*;*

(ii) $G_a(\theta) < \infty$ *for every* $\theta \in (0,1)$*;*

(iii) $\mathbb{E}X_i^{\alpha_i} < \infty$ *for every* i*.*

More precisely,

$$\max_i \{\Gamma(1-\alpha_i) a_i^{\alpha_i} \mathbb{E}X_i^{\alpha_i}\} \le \int_0^{\infty} f_a(r)\, dr \le \sum_i \Gamma(1-\alpha_i) a_i^{\alpha_i} \mathbb{E}X_i^{\alpha_i}. \tag{269}$$

Proof. For nonnegative z_i,

$$\max_i (1 - e^{-z_i}) \le 1 - e^{-\sum_i z_i} \le \sum_i (1 - e^{-z_i}).$$

Tonelli's theorem and the change of variable $t = cr^{-q}$ give

$$\int_0^{\infty} (1 - e^{-cr^{-q}})\, dr = \Gamma(1 - 1/q) c^{1/q}, \qquad q > 1,$$

including $c = 0$. This proves (269). Finally, for $0 \le u \le 1$ and $0 < \theta < 1$,

$$\theta u \le -\log(1 - \theta u) \le \frac{\theta}{1-\theta} u.$$

□

Theorem 49.2 (Gate inversion on one anisotropic ray)**.** *Assume the equivalent conditions of Lemma 49.1. The function* G_a *on any nonempty open subinterval of* $(0,1)$ *determines* $f_a(r)$ *for every* $r > 0$*. It therefore determines the Laplace transform of* X *on the entire curve*

$$\{(a_1 r^{-q_1}, \dots, a_d r^{-q_d}) : r > 0\}.$$

The same conclusion holds if G_a *is known on a set of distinct points having an accumulation point in* $[0,1)$*.*

An explicit inversion is as follows. Define the finite measure

$$\nu_a(A) = \int_0^{\infty} f_a(r) \mathbf{1}_{\{f_a(r) \in A\}}\, dr, \qquad A \subset [0,1] \text{ Borel.} \tag{270}$$

Its moments are

$$\int_0^1 u^n\, \nu_a(du) = \frac{G_a^{(n+1)}(0)}{n!}, \qquad n \ge 0. \tag{271}$$

If X *is not almost surely zero, put* $m = 1 - \mathbb{P}(X = 0)$ *and*

$$R_a(t) = \int_{[t,1]} u^{-1}\, \nu_a(du), \qquad 0 < t < m. \tag{272}$$

Then R_a *is the ordinary inverse of the strictly decreasing continuous function* $f_a : (0,\infty) \to (0,m)$*. In particular,* $L_X(a) = 1 - t_a$*, where* t_a *is the unique solution of* $R_a(t_a) = 1$*.*

Proof. For $|\theta| < 1$, the series

$$-\log(1-\theta f_a(r)) = \sum_{n\geq 1} \frac{\theta^n f_a(r)^n}{n}$$

converges after integration, locally uniformly in θ, because $0 \leq f_a \leq 1$ and $\int f_a < \infty$. Thus G_a is holomorphic in the unit disk and (271) follows. The identity theorem proves the assertion concerning restricted gate data. Compactly supported finite measures are determined by their moments: equality on polynomials implies equality on continuous functions by uniform polynomial approximation. Consequently the data determine ν_a.

Suppose $\mathbb{P}(X \neq 0) > 0$. As r increases, every nonzero realization of X strictly decreases $\sum_i a_i X_i r^{-q_i}$; hence f_a is strictly decreasing. Dominated convergence gives continuity and the endpoint limits $f_a(0+) = m$, $f_a(\infty) = 0$. The measure ν_a has no atoms, since the preimage of a point under f_a is at most a singleton. For $0 < t < m$, cancellation of the factor $f_a(r)$ gives

$$R_a(t) = \int_0^\infty \mathbf{1}_{\{f_a(r)\geq t\}}\, dr = f_a^{-1}(t).$$

This proves every assertion. If $X = 0$ almost surely then $G_a = 0$; conversely, vanishing G_a implies $f_a = 0$ almost everywhere and hence everywhere. □

Corollary 49.3 (Recovery of an arbitrary correlated law)**.** *For fixed known p_i, ρ, the responses $G_a(\theta)$ for $a \in (0,\infty)^d$ and θ in any nonempty open interval determine the law of every nonnegative X satisfying $\mathbb{E}X_i^{\rho/p_i} < \infty$. No independence, infinite divisibility, support gap, or atom at zero is required.*

Proof. Theorem 49.2 recovers $L_X(a)$ for every positive a. Uniqueness of the multivariate Laplace transform concludes the proof. □

The moment inversion can be centered at a gate value that is actually observed. This avoids making analytic continuation to zero part of the formal inverse, although recovering all derivatives remains an exact-data operation.

Proposition 49.4 (Inversion from the full jet at one gate value)**.** *Fix $\theta_0 \in (0,1)$ and let*

$$v_a(r) = \frac{f_a(r)}{1-\theta_0 f_a(r)}, \qquad \nu_{a,\theta_0}(A) = \int_0^\infty v_a(r)\mathbf{1}_{\{v_a(r)\in A\}}\, dr.$$

This is a finite measure supported in $[0,(1-\theta_0)^{-1}]$, and

$$\int v^n\, \nu_{a,\theta_0}(dv) = \frac{G_a^{(n+1)}(\theta_0)}{n!}, \qquad n \geq 0. \tag{273}$$

The tail $\int_{[t,\infty)} v^{-1}\nu_{a,\theta_0}(dv)$ recovers $v_a^{-1}(t)$, and $f_a = v_a/(1+\theta_0 v_a)$. Thus the full derivative sequence at one gate value determines the same ray data.

Proof. Differentiate under the integral, using $v_a \leq f_a/(1-\theta_0)$ and a neighborhood of θ_0 contained in $(0,1)$. Then $G_a^{(n+1)}(\theta_0) = n!\int v_a(r)^{n+1}\, dr$. The rest repeats the compact moment and monotone inversion arguments. □

Remark 49.5 (What is classical in the inverse). The compact moment uniqueness and the relation between logarithmic transforms and Stieltjes transforms are classical. For example,

$$\frac{1}{z}G'_a(1/z) = \int \frac{\nu_a(du)}{z-u}, \qquad |z| > 1.$$

This is the Stieltjes transform of the finite measure ν_a; compare the Shannon-transform viewpoint in [48, 46] and the compact moment problem in [41]. The additional step here is that the mixing variable is a strictly monotone Laplace profile. Consequently its pushforward measure determines its order in r, and not merely its unordered value distribution. Without monotonicity, arbitrary measure-preserving rearrangements of f_a leave G_a unchanged.

49.1 No finite set of gate values is universally sufficient

The continuum gate parameter cannot be replaced by any fixed finite list on the full class of bounded scalar laws. Here "a profile" includes all positive amplitudes, not just finitely many measured values.

Theorem 49.6 (Finite-gate obstruction with moment calibration). *Fix $0 < b < B < \infty$, $p > \rho > 0$, finitely many gate values $\theta_j \in (0,1)$, and nonnegative integers k_j. Also fix an integer $Q \geq 0$. There are distinct finitely supported scalar probability laws μ_+ and μ_- supported in $[b, B]$ with the same first Q ordinary moments such that, for every $s > 0$,*

$$\partial_\theta^\ell G_s^{\mu_+}(\theta)\Big|_{\theta=\theta_j} = \partial_\theta^\ell G_s^{\mu_-}(\theta)\Big|_{\theta=\theta_j}, \qquad 0 \leq \ell \leq k_j. \tag{274}$$

Indeed, the common data have a positive-dimensional local family of distinct laws. Thus two gate values, or any fixed finite number of gate values and finite jets, do not identify arbitrary bounded scalar laws.

Proof. Write $\alpha = \rho/p$. Scaling in (267) gives

$$G_s^\mu(\theta) = s^\alpha C_\mu(\theta), \qquad C_\mu(\theta) = \alpha \int_0^\infty -\log\{1-\theta+\theta L_\mu(u)\}\, u^{-\alpha-1}\, du.$$

Consequently all the equalities in (274) reduce to $M = \sum_j (k_j+1)$ real scalar constraints.

Choose $N = M + Q + 2$ distinct points $x_1, \ldots, x_N \in [b, B]$. In the open probability simplex consider weights $w = (w_i)$ with $\sum_i w_i x_i^q$ fixed at their uniform-weight values for $1 \leq q \leq Q$. Together with $\sum_i w_i = 1$, these constraints have rank $Q+1$ by the Vandermonde determinant. They define an open subset U of an affine space of dimension $N - Q - 1 = M + 1$.

The map

$$F : U \longrightarrow \mathbb{R}^M, \qquad w \longmapsto \left(C^{(\ell)}_{\sum_i w_i \delta_{x_i}}(\theta_j) \right)_{j,\, 0 \leq \ell \leq k_j}$$

is smooth. To justify differentiation, write any tangent perturbation as $\sum_i h_i = 0$. Then $\sum_i h_i e^{-ux_i} = O(u)$ at zero and $O(e^{-bu})$ at infinity, whereas the gated denominator is bounded below by $1 - \theta_j > 0$. The same bounds apply after finitely many gate or weight derivatives.

Let r be the maximum rank of DF on U and choose a point where that rank is attained. Some r by r minor remains nonzero in a neighborhood, and maximality makes the rank constant there. The constant-rank theorem then gives local fibers of dimension $M + 1 - r \geq 1$. Any two distinct weights in such a fiber give the desired laws. Their finite supports are the same, and all their weights can be kept strictly positive. □

Remark 49.7 (Scope of the obstruction). Embedding these scalar laws on a coordinate axis gives the same failure for any fixed vector dimension and any prescribed coordinate powers, provided the class allows laws supported on that axis. The theorem does not assert that finitely many gate values fail in every more restrictive model, nor that joint observations at additional distinct powers cannot restore information. It concerns the stated observation architecture and the unrestricted bounded-law class.

49.2 A scalar exponent can be calibrated from a gate curve

Proposition 49.8 (Blind scalar calibration under a finite mean). *Let $d = 1$, $a = 1$, and suppose $0 < \mathbb{E}X < \infty$. The function $G_1(\theta)$ on any open gate interval determines both the unknown ratio $q = p/\rho > 1$ and the law of X. In terms of the recovered profile f,*

$$q = -\lim_{r\to\infty} \frac{\log f(r)}{\log r}, \qquad \mathbb{E}X = \lim_{r\to\infty} r^q f(r), \qquad L_X(s) = 1 - f(s^{-1/q}). \tag{275}$$

Proof. The gate inverse reconstructs f without using the exponent. The finite mean assumption gives $1 - L_X(s) \sim s\mathbb{E}X$ as $s \downarrow 0$ and hence $f(r) \sim (\mathbb{E}X)r^{-q}$. This proves the formulas. □

Remark 49.9 (Why the mean assumption matters). Without a tail restriction the exponent and the law can be confounded. If $q' > q > 1$, put $\beta = q/q' \in (0, 1)$ and let X' be the positive β-stable random variable with Laplace transform $L_{X'}(s) = e^{-s^\beta}$. The models $(X \equiv 1, q)$ and (X', q') have the identical profile $1 - e^{-r^{-q}}$ and hence the identical gate response. Both satisfy the fractional-moment condition needed above, while X' has infinite mean. The gate by itself therefore determines a Laplace profile before it determines the exponent used to parameterize that profile.

50 The gate curve as a scalar rare-event observable

We give a direct probabilistic realization of the continuum data. This also specifies exactly what information is meant by a measured gate curve. Let $X_n = (X_{1,n}, \dots, X_{d,n})$, $n \geq 1$, be independent copies of a bounded nonnegative vector X with $\mathbb{P}(X \neq 0) > 0$. Fix $a_i > 0$, $q_i > 1$, and $\theta \in (0, 1)$, and let $B_{\theta,n}$ be independent Bernoulli variables with success probability θ, independent of the vectors. For positive integers N define the single scalar statistic

$$Z_N = \frac{1}{N} \sum_{n\geq 1} B_{\theta,n} \sum_{i=1}^{d} a_i (N/n)^{q_i} X_{i,n}. \tag{276}$$

For each fixed N this sum is bounded by a finite deterministic constant, because X is bounded and each $q_i > 1$. Set

$$\begin{aligned} T(r) &= \sum_i a_i r^{-q_i} X_i, \\ h_\theta(\lambda, r) &= -\log\{1 - \theta + \theta \mathbb{E} e^{-\lambda T(r)}\}, \\ F_\theta(\lambda) &= \int_0^\infty h_\theta(\lambda, r)\, dr, \qquad \lambda > 0. \end{aligned} \tag{277}$$

The function $F_\theta(1)$ is exactly the gate response $G_a(\theta)$ under the normalization $r = y^\rho$, $q_i = p_i/\rho$.

Theorem 50.1 (Scalar lower-tail principle and dual recovery)**.** *Under the preceding assumptions, for every $\lambda > 0$,*

$$-\frac{1}{N}\log \mathbb{E}e^{-N\lambda Z_N} \longrightarrow F_\theta(\lambda). \tag{278}$$

The function F_θ is twice continuously differentiable and strictly concave on $(0,\infty)$, with

$$F'_\theta(0+) = \infty, \qquad F'_\theta(\infty) = 0.$$

Consequently, for each $e > 0$ there is a unique $\lambda_e > 0$ satisfying $F'_\theta(\lambda_e) = e$, and

$$I_\theta(e) = \sup_{\lambda>0}\{F_\theta(\lambda) - \lambda e\} = F_\theta(\lambda_e) - \lambda_e e \tag{279}$$

is finite and continuous on $(0,\infty)$. For every $e > 0$,

$$\lim_{N\to\infty}\frac{1}{N}\log \mathbb{P}(Z_N \le e) = -I_\theta(e). \tag{280}$$

The full lower-tail rate function determines the gate response by

$$G_a(\theta) = F_\theta(1) = \inf_{e>0}\{I_\theta(e) + e\}. \tag{281}$$

Thus a family of exact scalar lower-tail rate functions, indexed by the known gate probability, supplies the data used in the inverse theorems.

Proof. Independence and convergence of the defining sums give

$$F_{\theta,N}(\lambda) := -\frac{1}{N}\log \mathbb{E}e^{-N\lambda Z_N} = \frac{1}{N}\sum_{n\ge 1} h_\theta(\lambda, n/N). \tag{282}$$

Write $q_{\min} = \min_i q_i > 1$. Uniformly for λ in a compact subinterval of $(0,\infty)$, the integrand is bounded at $0 < r \le 1$ by $-\log(1-\theta)$ and is $O(r^{-q_{\min}})$ for $r \ge 1$. It is continuous in $r > 0$. Splitting the integral and the Riemann sum into $(0,\delta)$, $[\delta, R]$, and (R,∞) proves (278): the first part is $O(\delta + N^{-1})$, the last is $O(R^{1-q_{\min}})$, and the middle part is an ordinary Riemann sum.

At each r, tilt the law of $B_\theta T(r)$ by the density proportional to $e^{-\lambda B_\theta T(r)}$. Its mean and variance satisfy

$$\partial_\lambda h_\theta(\lambda, r) = \mathbb{E}_{\lambda,r}[B_\theta T(r)], \qquad -\partial_\lambda^2 h_\theta(\lambda, r) = \mathrm{Var}_{\lambda,r}(B_\theta T(r)). \tag{283}$$

For $j = 1, 2$, the tilted jth raw moment is bounded by

$$\frac{\theta}{1-\theta}\mathbb{E}[T(r)^j e^{-\lambda T(r)}].$$

On $0 < r \le 1$ this is bounded uniformly for λ in compact positive intervals, using $\sup_{t\ge 0} t^j e^{-\lambda t} < \infty$. On $r \ge 1$, boundedness of X makes it $O(r^{-jq_{\min}})$. These bounds justify differentiating the continuum integral and show, by the same three-part Riemann-sum argument, that

$$F^{(j)}_{\theta,N}(\lambda) \longrightarrow F^{(j)}_\theta(\lambda), \qquad j = 1, 2. \tag{284}$$

The variance in (283) is strictly positive for every r: the Bernoulli variable supplies a zero atom, while $\mathbb{P}(T(r) > 0) > 0$. Hence $F''_\theta(\lambda) < 0$.

As $\lambda \downarrow 0$, the tilted mean increases to $\theta\mathbb{E}T(r)$, whose integral over $0 < r < 1$ is infinite because at least one $\mathbb{E}X_i$ is positive and $q_i > 1$. Monotone convergence gives $F'_\theta(0+) = \infty$. As $\lambda \to \infty$, the

tilted mean tends pointwise to zero, since the denominator stays above $1-\theta$. It is bounded by its value at any fixed positive tilt, an integrable function. Dominated convergence gives $F'_\theta(\infty)=0$. These facts establish the unique maximizer in (279). Continuity follows from the strict monotonicity and continuity of F'_θ and its inverse. Also $F_\theta(0+)=0$ by dominated convergence, so no missing endpoint term is involved in the supremum.

For the upper bound in (280), Markov's inequality gives, for every $\lambda>0$,

$$\mathbb{P}(Z_N\le e)\le \exp\{N\lambda e - NF_{\theta,N}(\lambda)\}.$$

Taking the limit and optimizing proves the required upper bound. For the lower bound, fix $0<\delta<e$ and let λ_δ be the unique tilt with $F'_\theta(\lambda_\delta)=e-\delta$. Define the tilted law by

$$\frac{d\mathbb{Q}_{N,\delta}}{d\mathbb{P}} = \exp\{-N\lambda_\delta Z_N + NF_{\theta,N}(\lambda_\delta)\}.$$

Equations (284) and elementary differentiation of the log-Laplace transform show that

$$\mathbb{E}_{\mathbb{Q}_{N,\delta}} Z_N \longrightarrow e-\delta, \qquad \mathrm{Var}_{\mathbb{Q}_{N,\delta}}\, Z_N = -N^{-1}F''_{\theta,N}(\lambda_\delta) = O(N^{-1}).$$

Therefore, for every $0<\varepsilon<\min\{\delta, e-\delta\}$, the event $A_N=\{|Z_N-(e-\delta)|<\varepsilon\}$ has tilted probability tending to one and is contained in $\{Z_N\le e\}$. Changing measure on A_N yields

$$\mathbb{P}(Z_N\le e)\ge \exp\{-NF_{\theta,N}(\lambda_\delta)+N\lambda_\delta(e-\delta-\varepsilon)\}\mathbb{Q}_{N,\delta}(A_N).$$

Take the logarithmic limit, then let $\varepsilon\downarrow 0$ and $\delta\downarrow 0$. The result is $-I_\theta(e)$ by continuity, which proves the lower bound.

Finally, the definition of I_θ gives $I_\theta(e)+\lambda e\ge F_\theta(\lambda)$ for every $e>0$, with equality at $e=F'_\theta(\lambda)$. Taking $\lambda=1$ proves (281). □

51 A sharp arithmetic criterion for one aggregate

We now keep the positive coefficient vector $a=(a_1,\dots,a_d)$ fixed. The observed statistic is a single aggregate with coordinate-dependent spatial powers. Its gate profile is

$$G_{\mu,a,p}(\theta)=\int_0^\infty \rho y^{\rho-1}\left\{-\log\left[1-\theta+\theta L_\mu(a_1y^{-p_1},\dots,a_dy^{-p_d})\right]\right\}\,dy, \qquad 0<\theta<1, \tag{285}$$

where $p_i>\rho$ and $L_\mu(s)=\int e^{-s\cdot x}\mu(dx)$. The gate inversion established above says that this profile determines

$$F_\mu(t)=L_\mu(a_1t^{p_1},\dots,a_dt^{p_d}), \qquad t>0. \tag{286}$$

It is therefore meaningful to ask when this one-dimensional restriction determines a genuinely multivariate law. The positive coefficient-encoding argument below is a finite-dimensional instance of the prescribed-index method in [43, Lemmas 3.2–3.3]; it is reproduced to make the sharp criterion and its converse self-contained.

Theorem 51.1 (Exact resonance criterion)**.** *Fix $a\in(0,\infty)^d$, $p\in(\rho,\infty)^d$, and a nonempty open interval $J\subset(0,1)$. The following are equivalent.*

1. *The numbers $p_1,\dots,p_d$ are linearly independent over $\mathbb{Q}$.*
2. *The map $\mu\mapsto G_{\mu,a,p}|_J$ is injective on the class of compactly supported probability measures on $[0,\infty)^d$.*

3. The map $\mu \mapsto F_\mu$ is injective on that class.

If rational independence fails, then, in every prescribed nonempty open box $B \subset (0,\infty)^d$ and for every integer $M \geq 0$, there are distinct probability measures μ_+ and μ_- with smooth compactly supported densities in B such that their gate profiles agree for all $0 < \theta < 1$ and

$$\int x^n \, \mu_+(dx) = \int x^n \, \mu_-(dx) \qquad (n \in \mathbb{N}_0^d, \ |n| \leq M). \tag{287}$$

Proof. For a compactly supported law, the multivariate Laplace transform is entire and, for $t > 0$,

$$F_\mu(t) = \sum_{n \in \mathbb{N}_0^d} \frac{(-1)^{|n|} a^n}{n!} \left(\int x^n \mu(dx) \right) t^{p \cdot n}. \tag{288}$$

This series converges absolutely. Since $p_i > 0$, the set of indices with $p \cdot n \leq R$ is finite for every R. Under rational independence, all the exponents $p \cdot n$ are distinct. We can therefore list them as $0 = \lambda_0 < \lambda_1 < \cdots \to \infty$. Writing $F_\mu(t) = \sum_{k \geq 0} c_k t^{\lambda_k}$, each coefficient is recovered recursively by

$$c_k = \lim_{t \downarrow 0} t^{-\lambda_k} \left[F_\mu(t) - \sum_{j<k} c_j t^{\lambda_j} \right]. \tag{289}$$

To justify this limit without any separation estimate on adjacent exponents, choose an integer N with $N \min_i p_i > \lambda_k$. The Taylor remainder containing terms of total degree at least N is $O(t^{N \min_i p_i})$ for $0 < t \leq 1$; only finitely many other terms remain. Formula (289) follows. Thus F_μ determines all mixed moments. If two compactly supported laws have the same moments, put both supports in one compact box and apply polynomial approximation on that box to obtain equality of the laws. This proves the positive direction. The previously established gate inverse and analyticity in θ make the observations on J equivalent to the full curve (286).

For the converse, rational dependence and positivity of the p_i give distinct nonzero multi-indices $\alpha, \beta \in \mathbb{N}_0^d$ such that $p \cdot \alpha = p \cdot \beta$. Define the nonzero polynomial

$$P(s) = a^{-\alpha} s^\alpha - a^{-\beta} s^\beta. \tag{290}$$

It vanishes on $s = (a_1 t^{p_1}, \ldots, a_d t^{p_d})$. Choose a nonzero nonnegative $\varphi \in C_c^\infty(B)$ and set

$$Q(s) = s_1^{M+1} P(s), \qquad g = Q(\partial)\varphi. \tag{291}$$

Repeated integration by parts gives

$$\int e^{-s \cdot x} g(x) \, dx = Q(s) \int e^{-s \cdot x} \varphi(x) \, dx. \tag{292}$$

The last Laplace transform is strictly positive for real positive s, and Q is a nonzero polynomial. Consequently $g \not\equiv 0$. Every monomial of Q has degree greater than M, so integration by parts also gives $\int x^n g(x) \, dx = 0$ for $|n| \leq M$, including zero total mass. Take a smooth compactly supported probability density f_0 in B which is strictly positive on $\operatorname{supp} \varphi$. Since $\operatorname{supp} g \subset \operatorname{supp} \varphi$, the functions $f_0 \pm \varepsilon g$ are nonnegative probability densities for all sufficiently small $\varepsilon > 0$. Their difference is nonzero, all the moments in (287) agree, and (292) shows equality of their Laplace transforms on the entire curve. Their gate profiles therefore agree as well. □

Corollary 51.2 (Moment-determinate laws)**.** *The injectivity conclusion for rationally independent powers also holds on the class of nonnegative laws having all mixed moments finite and determined by those moments. Compact support and a finite joint exponential moment are sufficient, but are not necessary, hypotheses.*

Proof. For a fixed integer N, Taylor's theorem on the nonnegative real axis gives

$$\left| e^{-z} - \sum_{k=0}^{N} \frac{(-z)^k}{k!} \right| \le \frac{z^{N+1}}{(N+1)!}, \qquad z \ge 0.$$

With $z = \sum_i a_i t^{p_i} X_i$ and $0 < t \le 1$, taking expectations shows that the terms with $|n| \le N$ in (288) have a remainder bounded by

$$\frac{\mathbb{E}(\sum_i a_i X_i)^{N+1}}{(N+1)!} t^{(N+1)\min_i p_i}.$$

This finite-order expansion justifies (289) at every prescribed exponent, without requiring convergence of the infinite moment series. Thus equal gate profiles imply equal mixed moments, and the assumed moment determinacy identifies the law. The gate integrability follows already from finite first moments and $p_i > \rho$. □

Remark 51.3 (Observation architecture)*.* The aggregate in (285) assigns the distinct weights y^{-p_i} to the different coordinates before summing. It is not a channel of the form $y^{-p}\ell(X)$ with one common exponent and one fixed linear functional ℓ. Moreover the data contain an exact continuum of gate probabilities. Thus the theorem neither recovers a multivariate law from a single scalar number nor contradicts lower bounds for finitely many fixed linear channels with a fixed gate.

Corollary 51.4 (Finite collections of resonant designs)**.** *Fix finitely many designs $(a^{(j)}, p^{(j)})$, $1 \le j \le m$, each with positive coefficients and powers greater than ρ. Their combined full gate profiles identify every compactly supported nonnegative d-dimensional law if and only if at least one vector $p^{(j)}$ has rationally independent coordinates. If all the designs are resonant, the indistinguishable laws can have smooth densities in any prescribed positive box and matching mixed moments through any prescribed finite total degree.*

Proof. Sufficiency follows from Theorem 51.1. If every design is resonant, construct its polynomial P_j as in (290) and use $Q(s) = s_1^{M+1} \prod_{j=1}^{m} P_j(s)$ in (291). The product is nonzero and vanishes on every observed curve. The same density perturbation proves all the assertions. □

Example 51.5 (An irrational design and a resonant design)*.* For $\rho = 1$ and $d = 2$, the exponent vector $(2, 2\sqrt{2})$ is rationally independent, so one full gate profile with any fixed $a_1, a_2 > 0$ determines every compactly supported joint law. For $(2, 4)$ the polynomial $s_1^2/a_1^2 - s_2/a_2$ generates invisible smooth perturbations. Merely changing the coefficients a finite number of times does not remove this obstruction. Pairwise irrational ratios are not enough in dimension three: $(2, 2\sqrt{2}, 2 + 2\sqrt{2})$ has a rational relation.

51.1 Instability near a resonance

Exact injectivity has no uniform quantitative meaning as a design approaches a rational relation. The next estimate makes this precise without asserting a minimax rate for a fixed design.

Proposition 51.6 (Deterioration near resonant powers)**.** *Fix $a \in (0,\infty)^d$, $R > 0$, $\eta > 0$, and $0 < \theta_* < 1$. For every law μ supported in $[0,R]^d$ and every $p, q \in [\rho+\eta,\infty)^d$,*

$$\sup_{0\le\theta\le\theta_*} |G_{\mu,a,p}(\theta) - G_{\mu,a,q}(\theta)| \le \sum_{i=1}^{d} C_i |p_i - q_i|, \quad C_i = \frac{\theta_*}{1-\theta_*}\left(\frac{1}{e\rho} + \frac{\rho a_i R}{\eta^2}\right). \tag{293}$$

In particular, for every resonant $p^0 \in (\rho,\infty)^d$ there are fixed distinct smooth compactly supported laws $\mu_\pm$ and a sequence of rationally independent vectors $p^{(n)} \to p^0$ such that their gate data differ by $O(\|p^{(n)} - p^0\|_1)$ on $[0,\theta_*]$, although their Wasserstein distance is a fixed strictly positive number.

Proof. Write $A = \sum_i a_i X_i y^{-p_i}$ and $u = 1 - \mathbb{E}e^{-A}$. Differentiation gives

$$|\partial_{p_i} u| \le |\log y|\, \mathbb{E}[a_i X_i y^{-p_i} e^{-A}].$$

For $0 < y \le 1$, the expectation is at most $\mathbb{E}[Ae^{-A}] \le e^{-1}$. For $y \ge 1$, it is at most $a_i R y^{-p_i}$. Also $\theta/(1-\theta u) \le \theta_*/(1-\theta_*)$. The two needed integrals are

$$\int_0^1 \rho y^{\rho-1} |\log y|\, dy = \rho^{-1}, \qquad \int_1^\infty \rho y^{\rho-1-p_i} \log y\, dy = \frac{\rho}{(p_i-\rho)^2}.$$

These bounds justify differentiation under the integral and give (293) by integrating along the segment joining p and q. Choose $\mu_\pm$ from Theorem 51.1 for p^0. Their gate data agree at p^0, so apply (293) twice. Rationally independent vectors are dense, being the complement of countably many proper hyperplanes. Finally $W_1(\mu_+,\mu_-) > 0$ because the two compactly supported probability measures are distinct. □

Thus no inverse modulus $W_1 \le \omega(\|\Delta G\|)$ with $\omega(\varepsilon) \to 0$ can hold uniformly over rationally independent designs approaching a resonance. In particular, a Hölder estimate cannot have a uniform constant there. The rational independence criterion is an exact identifiability result, not a claim of numerical robustness.

51.2 Why a tail hypothesis is essential

Proposition 51.7 (Stable-law aliasing)**.** *The compact-support assumption in Theorem 51.1 cannot be replaced by the fractional-moment conditions needed to make the gate profile finite. For any $p_1 > p_2 > \rho$ and any $a_1, a_2 > 0$, there are distinct nonnegative two-dimensional laws, one bounded and one unbounded, with identical full gate profiles for the design (a,p). This remains true when p_1/p_2 is irrational.*

Proof. Put $\alpha = p_2/p_1 \in (0,1)$, and let S be the positive α-stable random variable normalized by $\mathbb{E}e^{-sS} = e^{-s^\alpha}$. For completeness, its existence follows from the subordinator Lévy measure $\alpha x^{-1-\alpha}\, dx/\Gamma(1-\alpha)$, whose Laplace exponent is s^α. Set $c = a_2/a_1^\alpha$ and

$$X = (c^{1/\alpha} S, 0), \qquad Y = (0,1).$$

Then for every $t > 0$,

$$\mathbb{E}e^{-a_1 t^{p_1} X_1 - a_2 t^{p_2} X_2} = e^{-c(a_1 t^{p_1})^\alpha} = e^{-a_2 t^{p_2}} = \mathbb{E}e^{-a_1 t^{p_1} Y_1 - a_2 t^{p_2} Y_2}.$$

Hence the gate profiles coincide. Both profiles are finite since their common curve satisfies $1 - e^{-a_2 y^{-p_2}} = O(y^{-p_2})$ at infinity and $p_2 > \rho$. They also satisfy the natural coordinatewise fractional-moment conditions: $\mathbb{E}S^r < \infty$ for $0 < r < \alpha$, whereas $\rho/p_1 < \alpha$. The moment assertion follows directly from

$$\mathbb{E}S^r = \frac{r}{\Gamma(1-r)} \int_0^\infty (1 - e^{-s^\alpha}) s^{-r-1}\, ds, \qquad 0 < r < \alpha.$$

Adding the same deterministic vector in $(0,\infty)^2$ to X and Y gives an example supported strictly inside the positive orthant, with the same conclusion. □

Corollary 51.2 allows tails well beyond compact support. The stable example isolates a different failure mechanism: a fractional tail power can imitate another coordinate's spatial power when the required higher moments do not exist.

51.3 An explicit pair and its near-resonant measurements

The following pair gives an elementary concrete instance of the preceding phenomenon. On $[1,2]^2$, put $t_i = x_i - 1$,

$$\phi(t) = t^3(1-t)^3, \qquad g(x) = \phi''(t_1)\phi(t_2) - \phi(t_1)\phi'(t_2), \qquad f_\pm(x) = 1 \pm 32g(x).$$

Extend these densities by zero outside the box. They are probability densities: $\int g = 0$, and $\|\phi\|_\infty \le 1/64$, $\|\phi'\|_\infty \le 3/16$, $\|\phi''\|_\infty \le 3/8$ imply $|32g| \le 9/32 < 1$. Their Laplace-transform difference is exactly

$$L_+(s_1,s_2) - L_-(s_1,s_2) = 64(s_1^2 - s_2)\Phi(s_1)\Phi(s_2), \qquad \Phi(s) = \int_1^2 e^{-sx}\phi(x-1)\,dx.$$

Integration by parts justifies this identity, since ϕ and its first derivative vanish at both endpoints. Thus at $\rho = 1$, $a = (1,1)$, and $p = (2,4)$, their complete gate profiles agree exactly. They nevertheless satisfy the fixed separation

$$W_1(\mu_+,\mu_-) \ge |\mathbb{E}_+X_2 - \mathbb{E}_-X_2| = \frac{64}{19600} > 0,$$

because $x \mapsto x_2$ is 1-Lipschitz and $\int_0^1 \phi = 1/140$.

For the irrational designs $p = (2, 4+\delta)$ with $\delta = \sqrt{2}\,10^{-k}$, exact uniqueness holds, but these fixed laws become harder to distinguish as k grows. The table gives deterministic quadrature values; it is an illustration, not an input to the proof or a statistical experiment.

δ	$\lvert G_+(1/2) - G_-(1/2)\rvert$
$\sqrt{2}\,10^{-1}$	1.559669×10^{-5}
$\sqrt{2}\,10^{-2}$	1.631627×10^{-6}
$\sqrt{2}\,10^{-3}$	1.639186×10^{-7}
$\sqrt{2}\,10^{-4}$	1.639946×10^{-8}

The computation integrates in $z = \log y$ and uses the stable formula $-2e^z \operatorname{arctanh}(\theta D/(1-\theta+\theta L_0))$, where L_0 is the Laplace transform of the uniform law on the box and $D = 32(s_1^2 - s_2)\Phi(s_1)\Phi(s_2)$. An 80-node Gauss–Legendre rule evaluates Φ; adaptive quadrature uses $-6 \le z \le 24$. The reported quadrature error estimates are below 10^{-12}. All equality and separation assertions above follow from the exact formulas, independently of numerical error estimates.

52 Blind recovery of dimension, powers, and the correlated law

The arithmetic criterion has a stronger consequence when the coordinate powers are not known in advance. Nonnegativity makes the pattern of nonzero mixed moments hereditary under taking

sub-multi-indices. This allows the primitive spatial powers to be read from the asymptotic exponents themselves.

For a subset $S \subset (0,\infty)$ define its set of *additively indecomposable elements* by

$$\mathcal{A}(S) = \{s \in S : s \neq u + v \text{ for every } u, v \in S\}. \tag{294}$$

No closure of S under addition is assumed. That qualification matters because a joint moment can vanish when the corresponding coordinates are never simultaneously positive.

Lemma 52.1 (Primitive exponents from positive moments)**.** *Let $q_1, \dots, q_d > 0$ be linearly independent over $\mathbb{Q}$. Let $X \in [0,\infty)^d$ have finite mixed moments of every order and $\mathbb{P}(X_i > 0) > 0$ for each i. Define*

$$S_X(q) = \{q \cdot n : n \in \mathbb{N}_0^d \setminus \{0\}, \mathbb{E}X^n > 0\}. \tag{295}$$

Then

$$\mathcal{A}(S_X(q)) = \{q_1, \dots, q_d\}. \tag{296}$$

The same conclusion holds for a finite or countably infinite index set and a nonzero finite positive measure in place of the probability law, provided all finite mixed moments exist and every coordinate has positive first moment. In that form all multi-indices have finite support, and linear independence means independence of every finite subfamily.

Proof. The first moments are strictly positive, so $q_i \in S_X(q)$. If $q_i = q\cdot m + q\cdot n$ for nonzero $m, n \in \mathbb{N}_0^d$, rational independence implies $e_i = m + n$, which is impossible. Thus every q_i is indecomposable.

Conversely, suppose $s = q \cdot n \in S_X(q)$ and $|n| \geq 2$. Choose i with $n_i > 0$. Since $X^n \geq 0$ and $\mathbb{E}X^n > 0$, the event $\{X^n > 0\}$ has positive probability. On this event both X_i and X^{n-e_i} are positive. Their expectations are therefore strictly positive, so both q_i and $q \cdot (n - e_i)$ belong to $S_X(q)$. They sum to s, which is consequently decomposable. The argument involves only the finitely many coordinates in the support of the relevant multi-index; it therefore proves the countable and finite-positive-measure versions as well, with expectations replaced by integrals. □

Theorem 52.2 (Fully blind identification from one gate curve)**.** *Consider the class of models (d, q, μ) satisfying:*

(i) d is an unknown finite positive integer;

(ii) $q = (q_1, \dots, q_d) \in (1,\infty)^d$ has coordinates linearly independent over $\mathbb{Q}$;

(iii) μ is a probability law on $[0,\infty)^d$ with all mixed moments finite, determined by those moments;

(iv) every coordinate is nontrivial: $\mu\{x : x_i > 0\} > 0$.

For $X \sim \mu$, observe only

$$G(\theta) = \int_0^\infty -\log\left[1 - \theta + \theta\mathbb{E}\exp\left(-\sum_{i=1}^d X_i r^{-q_i}\right)\right] dr \tag{297}$$

on a nonempty open interval of $\theta \in (0,1)$, with the displayed normalization of dr known. Then G determines d, the unordered set $\{q_1, \dots, q_d\}$, and the joint law μ, up to the same permutation of powers and coordinates. In particular, the result holds for every bounded nonnegative vector with nontrivial coordinates.

More explicitly, the gate inverse recovers

$$F(t) = \mathbb{E}\exp\left(-\sum_{i=1}^{d} X_i t^{q_i}\right), \qquad t > 0. \tag{298}$$

Let $S(F)$ be the positive support of its generalized asymptotic expansion at zero. Then

$$\{q_1, \dots, q_d\} = \mathcal{A}(S(F)), \qquad d = |\mathcal{A}(S(F))|. \tag{299}$$

If $c_{q\cdot n}$ denotes the coefficient at exponent $q \cdot n$, with coefficient zero when that exponent is absent, then

$$\mathbb{E}X^n = (-1)^{|n|} n!\, c_{q\cdot n}, \qquad n \in \mathbb{N}_0^d. \tag{300}$$

Proof. Finite first moments imply the fractional-moment condition in Lemma 49.1. The gate inversion applies directly to (297); it does not require the values of d or q. It determines the strictly decreasing function $f(r) = 1 - F(1/r)$, hence determines F.

We make the asymptotic-support assertion precise. For every integer $N \geq 0$, Taylor's theorem and nonnegativity give, for $0 < t \leq 1$,

$$F(t) = \sum_{|n|\leq N} \frac{(-1)^{|n|}}{n!} \mathbb{E}X^n\, t^{q\cdot n} + O\left(t^{(N+1)q_{\min}}\right), \qquad q_{\min} = \min_i q_i. \tag{301}$$

Indeed, the absolute remainder is at most

$$\frac{\mathbb{E}(\sum_i X_i)^{N+1}}{(N+1)!} t^{(N+1)q_{\min}}.$$

The set $\{q \cdot n : n \in \mathbb{N}_0^d\}$ is locally finite, because all q_i are strictly positive. Rational independence makes all these exponents distinct. Consequently the nonzero asymptotic coefficients of $F - 1$ have precisely the support $S_X(q)$ in (295); no coefficients can cancel at a common exponent.

This support and its coefficients are intrinsic, even though q is initially unknown. To see this, compare two candidate models satisfying the hypotheses. The union of their two locally finite exponent sets is again locally finite. If their coefficient arrays differ, there is a smallest exponent λ where they differ. Choose the Taylor order in each model so that its remainder in (301) has order strictly greater than λ. Subtract the finite expansions and divide by t^λ. As $t \downarrow 0$, the difference tends to the nonzero difference of the two coefficients. This contradicts equality of their functions F. Thus both models have the same asymptotic support and the same coefficients on it.

Lemma 52.1 now identifies the power set from that support and gives (299). Relabel the coordinates so that the powers are in increasing order. Rational independence identifies each multi-index n from its exponent $q \cdot n$, and (300) recovers every mixed moment. The moment-determinacy hypothesis concludes the proof. □

Example 52.3 (Mutually exclusive coordinates). Suppose $d = 2$, $q_1/q_2 \notin \mathbb{Q}$, and $X = (U, 0)$ with probability $1/2$ and $X = (0, V)$ with probability $1/2$, where U, V are strictly positive bounded variables. Every mixed moment involving both coordinates vanishes. The observed support is

$$S(F) = \{nq_1 : n \geq 1\} \cup \{nq_2 : n \geq 1\},$$

which is not closed under addition. Nevertheless its only indecomposable elements are q_1 and q_2. The missing mixed coefficients recover the mutual exclusion, so the theorem does not require a positive joint density or simultaneous positivity.

Remark 52.4 (The remaining gauges and necessary qualifications). An identically zero coordinate can always be appended without changing the data; the nontrivial-coordinate condition removes this unavoidable dimension ambiguity. If unknown positive amplitudes a_i are inserted in (297), the identified variables are $a_i X_i$; their amplitudes and marginal scales cannot be separated without an external calibration. In the original variables $q_i = p_i/\rho$, the curve identifies these ratios and not the common absolute scale of $p_1, \dots, p_d, \rho$.

The normalization of the continuum measure is also material. If the unknown factor $C > 0$ multiplies dr, substitution $r' = Cr$ gives

$$C \int_0^\infty -\log\left[1 - \theta + \theta \mathbb{E} e^{-\sum_i X_i r^{-q_i}}\right] dr = \int_0^\infty -\log\left[1 - \theta + \theta \mathbb{E} e^{-\sum_i (C^{q_i} X_i)(r')^{-q_i}}\right] dr'.$$

Thus an unknown intensity cannot be separated from the corresponding anisotropic rescaling of the law. Finally, the theorem uses the exact full gate curve and arbitrarily deep asymptotics; it is not a finite sample procedure or a stable dimension-selection algorithm.

Proposition 52.5 (Dimension is not uniformly stable). *Fix an admissible bounded model (d, q, X) from Theorem 52.2 and $0 < \theta_* < 1$. Choose $q_* > 1$ such that $q_1, \dots, q_d, q_*$ are rationally independent. For $\varepsilon > 0$, append the deterministic coordinate $X_{d+1} = \varepsilon$ with power q_*. The enlarged model is admissible, has dimension $d + 1$, and its gate curve G_ε satisfies*

$$\sup_{0 \le \theta \le \theta_*} |G_\varepsilon(\theta) - G(\theta)| \le \frac{\theta_*}{1 - \theta_*} \Gamma(1 - 1/q_*) \varepsilon^{1/q_*}. \tag{302}$$

Therefore latent dimension has no uniform continuity in the full gate data on such intervals, even when the powers are fixed and no resonance is approached.

Proof. Writing $L(r) = \mathbb{E} \exp(-\sum_i X_i r^{-q_i})$, the new Laplace profile is $L(r) e^{-\varepsilon r^{-q_*}}$. Its difference from $L(r)$ is at most $1 - e^{-\varepsilon r^{-q_*}}$. The derivative in $u \in [0, 1]$ of $-\log(1 - \theta u)$ is at most $\theta_*/(1 - \theta_*)$. Integrating and applying the scalar identity in the proof of Lemma 49.1 proves (302). Boundedness implies moment determinacy, and all the coordinates of the enlarged model are nontrivial. □

53 Blind recovery with accumulating exponent spectra

The finite-dimensional reconstruction admits a countable-dimensional extension. In particular, an accumulation of weighting exponents does not destroy exact identification. The appropriate intermediate object is a finite signed measure on the exponent axis, rather than a generalized series that can be ordered by successive smallest exponents.

We use the signed-exponent-measure encoding already employed by Lenzi [43, Lemmas 3.2–3.3] for prescribed rationally independent indices and the recovery of fractional power-sum profiles. The further step here identifies the unknown indices themselves: they are precisely the additively indecomposable nonzero atoms of that measure. Positivity of the latent coordinates is essential to this step.

Let J be a nonempty finite or countably infinite index set. Write $\ell^1_+(J)$ for the nonnegative summable families indexed by J, equipped with its norm Borel σ-algebra. Consider a family $q = (q_j)_{j \in J}$ and an $\ell^1_+(J)$-valued random variable $X = (X_j)_{j \in J}$ satisfying the following conditions:

(i) there are finite constants $1 < q_- \le q_+ < \infty$ such that $q_- \le q_j \le q_+$ for every j;

(ii) every finite rational relation between the q_j is trivial;

(iii) $\mathbb{P}(X_j > 0) > 0$ for every j;

(iv) for some finite M, $\sum_{j\in J} X_j \le M$ almost surely.

The index set, exponents, constants in these assumptions, and joint law are all allowed to be unknown. Known positive amplitudes can be absorbed into X_j; if the amplitudes are themselves unknown, the theorem recovers these scaled coordinates. No independence between the coordinates of X is assumed.

Define the normalized continuum gate profile

$$\mathcal{G}_{q,X}(\theta) = \int_0^\infty -\log\left(1-\theta+\theta\mathbb{E}\exp\left\{-\sum_{j\in J} r^{-q_j}X_j\right\}\right)dr, \qquad 0<\theta<1. \tag{303}$$

The exponents in (303) are the normalized ratios $q_j = p_j/\rho$ when the original radial measure is $\rho y^{\rho-1}\,dy$ and $r = y^\rho$.

Theorem 53.1 (Blind countable identification)**.** *Within the class just specified, the exact function $\mathcal{G}_{q,X}$ on any nonempty open subinterval of $(0,1)$ determines the index cardinality, the exponent set $\{q_j : j \in J\}$, and the joint law of X, up to the simultaneous relabeling of coordinates and exponents.*

More explicitly, if two such models (J, q, X) and $(\widetilde{J}, \widetilde{q}, \widetilde{X})$ have the same profile on such an interval, then there is a bijection $\pi : J \to \widetilde{J}$ such that

$$q_j = \widetilde{q}_{\pi(j)} \quad (j \in J),$$
$$(X_j)_{j\in J} \overset{d}{=} (\widetilde{X}_{\pi(j)})_{j\in J} \quad \text{as random elements of } \ell^1_+(J).$$

The exponents may have finite accumulation points. The theorem is an identification statement for exact function data and does not assert a finite-sample reconstruction rate.

Proof. We give the reconstruction in four steps.

Step 1: recover the monotone radial profile. Put

$$f(r) = 1 - \mathbb{E}\exp\left\{-\sum_{j\in J} r^{-q_j}X_j\right\}.$$

For each $r > 0$ the series in the exponent is finite almost surely, since

$$\sum_j r^{-q_j}X_j \le M\max\{r^{-q_-}, r^{-q_+}\}.$$

The same estimate gives continuity in r on compact subintervals of $(0,\infty)$. In addition,

$$0 \le f(r) \le 1 \quad (0<r\le 1), \qquad 0 \le f(r) \le Mr^{-q_-} \quad (r \ge 1). \tag{304}$$

Thus $\int_0^\infty f(r)\,dr \le 1 + M/(q_- - 1) < \infty$. For $r_1 < r_2$ the series at r_1 is strictly larger than that at r_2 on the event $\sum_j X_j > 0$, which has positive probability. Therefore f is strictly decreasing. By dominated convergence its endpoint limits are $m = \mathbb{P}(\sum_j X_j > 0)$ at zero and zero at infinity.

The hypotheses used in the proof of Theorem 49.2 are precisely continuity, strict monotonicity, and integrability of f, all just verified. That proof does not use finite dimension. Applying its weighted pushforward and tail inverse recovers f, and therefore

$$g(t) = 1 - f(1/t) = \mathbb{E}\exp\left\{-\sum_j t^{q_j}X_j\right\}, \qquad t > 0. \tag{305}$$

Step 2: recover a finite signed measure of exponents. Let $\mathcal{I}_J$ be the countable set of nonnegative integer multiindices $n = (n_j)_{j\in J}$ with finite support. Set

$$|n| = \sum_j n_j, \qquad n! = \prod_j n_j!, \qquad X^n = \prod_j X_j^{n_j}, \qquad n \cdot q = \sum_j n_j q_j,$$

using the empty-product convention. Define

$$c_n = \frac{(-1)^{|n|}}{n!}\mathbb{E}X^n, \qquad \eta = \sum_{n\in\mathcal{I}_J} c_n \delta_{n\cdot q}. \tag{306}$$

Tonelli's theorem and increasing finite subsets of J give

$$\sum_{n\in\mathcal{I}_J} |c_n| = \mathbb{E}\exp\left(\sum_j X_j\right) \le e^M. \tag{307}$$

Consequently (306) defines a finite signed Borel measure on $[0,\infty)$. For $s > 0$, absolute convergence allows expansion of the exponential in (305), yielding

$$g(e^{-s}) = \sum_{n\in\mathcal{I}_J} c_n e^{-s(n\cdot q)} = \int_{[0,\infty)} e^{-s\lambda}\,\eta(d\lambda). \tag{308}$$

Thus g determines η by uniqueness of the Laplace transform of finite signed measures. For completeness, if a finite signed measure ζ has zero Laplace transform on $(0,\infty)$, push $e^{-\lambda}\zeta(d\lambda)$ forward by $u = e^{-\lambda}$ to a measure on $[0,1]$. Every nonnegative integer moment of the resulting measure is zero. Polynomial density makes that measure zero, and then restriction to $[0,R]$ and multiplication by e^{λ} show $\zeta = 0$. This proof is valid in the presence of accumulating atoms.

Step 3: identify the unknown coordinates by indecomposability. Let

$$A = \{\lambda > 0 : \eta(\{\lambda\}) \neq 0\}, \qquad Q = A \setminus (A + A), \qquad A + A = \{a + b : a, b \in A\}. \tag{309}$$

Here A is the set of nonzero atoms, not the closed topological support of η. Rational independence makes the map $n \mapsto n \cdot q$ injective. Since X is nonnegative, there can be no cancellation within an atom, and

$$A = \{n \cdot q : n \in \mathcal{I}_J \setminus \{0\},\ \mathbb{E}X^n > 0\}. \tag{310}$$

The countable form of Lemma 52.1 now applies directly to (310). It proves the intrinsic formula

$$Q = \{q_j : j \in J\}. \tag{311}$$

In particular no discreteness or local finiteness of A is needed. The argument uses hereditary positivity under taking submulti-indices, not closure of A under addition.

Step 4: recover the joint law. The recovered set Q determines the unknown cardinality. Choose any enumeration of Q to label the coordinates. Rational independence and (306) give, for every finite-support multiindex,

$$\mathbb{E}X^n = (-1)^{|n|} n!\,\eta\left(\left\{\sum_j n_j q_j\right\}\right). \tag{312}$$

This formula also recovers zero moments. Each finite subvector is bounded, so its mixed moments determine its law. When comparing two models with different unknown bounds, use the larger of

the two bounds for this moment-determinacy argument. Agreement of every finite marginal then determines the law on the coordinate σ-algebra.

Finally, on $\ell^1(J)$ the coordinate σ-algebra equals the norm Borel σ-algebra. One inclusion follows from continuity of the coordinate maps. For the reverse inclusion, for each fixed $x \in \ell^1(J)$ the function $z \mapsto \|z - x\|_1$ is the increasing limit of measurable finite sums of coordinate distances. Norm balls are therefore coordinate measurable; separability of $\ell^1(J)$ gives a countable base of such balls. Thus the recovered law is the law as an ℓ^1-valued random element, and (311) establishes the asserted simultaneous relabeling. □

Remark 53.2 (What is being recovered). The theorem identifies a model satisfying the structural assumptions; it does not infer those assumptions from a finite experiment. A coordinate that is identically zero is invisible and is excluded. Likewise, the normalized ratios q_j do not separate an unknown counting exponent ρ from the physical powers p_j. The data consist of a full gate-response function, so this result does not contradict a lower bound proved for finitely many fixed-gate channels.

54 Correlated calibration: a local obstruction and its repair

Let $W = (W_1, \dots, W_m)$ be a bounded nonnegative random vector, assume $\mathbb{P}(W = 0) > 0$ and $\mathbb{P}(W_j > 0) > 0$ for every j, and let $p_j > 1$. Write

$$h(u) = -\log \mathbb{E}e^{-u\cdot W}, \qquad H(s) = \int_0^\infty h(s_1 x^{-p_1}, \dots, s_m x^{-p_m})\, dx.$$

The counting exponent has been normalized to one; thus p_j denotes the physical exponent divided by the counting exponent. The function H is finite, increasing, concave, and real analytic on the positive orthant. It satisfies

$$\sum_{j=1}^m p_j s_j H_j(s) = H(s). \tag{313}$$

For completeness, analyticity does not require independence or a positive covariance assumption. If s is positive and $|v_j| \le \delta s_j$, the perturbation of the Laplace transform at $(s_j x^{-p_j})_j$ is bounded by $\mathbb{E}[e^{-T}(e^{\delta T} - 1)]$, where $T = \sum_j s_j x^{-p_j} W_j$. This bound tends to zero uniformly in x as $\delta \downarrow 0$, while the unperturbed transform is bounded below by $\mathbb{P}(W = 0)$. A fixed relative complex neighborhood therefore has a zero-free Laplace transform and a consistent logarithm. Its logarithm is uniformly bounded for $x \le 1$, and is $O(x^{-\min_j p_j})$ for $x \ge 1$. Holomorphic integration proves the assertion. Homogeneity follows by a change of variables.

At a point s, put $z = \nabla H(s)$, $M = -\nabla^2 H(s)$, and

$$B = \operatorname{diag}(z) - M \operatorname{diag}(s).$$

Differentiation of (313) gives $Bp = z$. For common-scalar or nonnegative infinitely divisible models, all entries of M are nonnegative, and the usual nonsingular M-matrix argument proves that B is invertible. Correlation can remove this sign property. The following is an exact failure of identification, rather than merely a failure of that proof.

Theorem 54.1 (A full local curvature jet can conceal the exponents)**.** *Fix any gate probability $\pi_0 \in (1/2, 1)$. There exist $P > 1$, $\epsilon_0 > 0$, and a smooth family of bounded random vectors $W^{(\epsilon)} \in [0, \infty)^2$, $|\epsilon| < \epsilon_0$, such that:*

1. $\mathbb{P}(W^{(\epsilon)} = 0) = \pi_0$ *for every* ϵ*; conditionally on being nonzero, each law has two strictly positive, linearly independent support points;*

2. *the associated exponents are* $(p_1, p_2) = (P + \epsilon, P - \epsilon)$*;*

3. *the values, gradients, and Hessians of all associated transforms* H_ϵ *at* $s = (1, 1)$ *are identical.*

All these Hessians are negative definite. Consequently the local Legendre rates $I_\epsilon(z) = \sup_{s \geq 0}\{H_\epsilon(s) - s \cdot z\}$ *have the same full second-order jet at a common fully active point. In particular, even a known shared Bernoulli gate with arbitrarily large zero probability does not make one-point curvature calibration valid for all correlated laws.*

Proof. Set $c = (1 - \pi_0)/\pi_0 > 0$. For $0 < a < 1$, define

$$Q(t) = \frac{ce^{-t}}{1 + ce^{-t}}, \quad J_1(a) = \int_0^\infty t^{-a} Q(t)\, dt, \quad J_2(a) = \int_0^\infty t^{1-a} Q(t)\, dt.$$

As $a \downarrow 0$,

$$J_1(a) \longrightarrow \log(1 + c), \qquad J_2(a) \longrightarrow \int_0^c \frac{\log(1 + u)}{u}\, du.$$

The latter limit is strictly larger than the former: their difference vanishes at $c = 0$ and has derivative $\log(1 + c)/c - 1/(1 + c) > 0$. Choose a small enough that $J_2(a) > J_1(a)$, put $P = 1/a$, and set $b_0 = \sqrt{J_1(a)/J_2(a)} \in (0, 1)$.

Consider laws depending on (L, θ, b) near $(1, 0, b_0)$. Give the origin probability π_0, and give the points

$$x_+ = \frac{L}{2}(1 + b, 1 - b), \qquad x_- = \frac{L}{2}(1 - b, 1 + b)$$

probabilities $(1 - \pi_0)(1 + \theta)/2$ and $(1 - \pi_0)(1 - \theta)/2$, respectively. At equal exponents (P, P) and $s = (1, 1)$, direct differentiation and the substitution $t = Lx^{-P}$ give

$$\begin{aligned} H &= L^a J_1, \\ D := H_1 - H_2 &= aL^a \theta b J_1, \\ R := H_{11} + H_{22} - 2H_{12} &= -aL^a b^2 \{J_2 - \theta^2 J_3\}, \qquad J_3 = \int_0^\infty t^{1-a} Q(t)^2\, dt. \end{aligned}$$

For the first identity one integrates by parts in $a \int_0^\infty t^{-a-1} \log[(1 + c)/(1 + ce^{-t})]dt$. The Jacobian of (H, D, R) with respect to (L, θ, b) at $(1, 0, b_0)$ is nonsingular: its three successive diagonal entries are aJ_1, ab_0J_1, and $-2ab_0J_2$.

At this base point, $z_1 = z_2 = z = aJ_1/2$ and $H_{11} - H_{12} = R/2 = -z$. Thus $B(1, -1)^\mathsf{T} = 0$. In addition to the true Euler identities with weight vector (P, P), the entire base second-order jet therefore satisfies the Euler identities with every weight vector $(P + \epsilon, P - \epsilon)$.

Apply the implicit function theorem to the three observations (H, D, R) as the exponents vary from (P, P) to $(P + \epsilon, P - \epsilon)$. It provides smooth functions $L(\epsilon), \theta(\epsilon), b(\epsilon)$ keeping these observations fixed. At any positive exponent pair, the three Euler equations

$$p_1H_1 + p_2H_2 = H, \quad p_1H_{11} + p_2H_{12} = (1 - p_1)H_1, \quad p_1H_{12} + p_2H_{22} = (1 - p_2)H_2$$

together with D and R uniquely determine all remaining entries of the second-order jet. Indeed the first equation and D determine the gradient; the final two equations and R determine the three Hessian entries. The coefficient matrix in the latter step has nonzero determinant of absolute value $(p_1 + p_2)^2$. Since the base jet also satisfies these same equations for every ϵ, all jets agree exactly.

The three support points $0, x_+, x_-$ are affinely independent and have positive probabilities. Every tilted covariance matrix is therefore positive definite, and integration preserves strict negative definiteness of the Hessian of H_ϵ. The common positive saddle $s = (1, 1)$ has a common gradient z, and Legendre inversion gives $I_\epsilon(z) = H_\epsilon(1, 1) - 2z$, $\nabla I_\epsilon(z) = -(1, 1)$, and $\nabla^2 I_\epsilon(z) = [-\nabla^2 H_\epsilon(1, 1)]^{-1}$. This proves the rate assertion. □

For example, $c = 1/2$ and $P = 10$ give numerically $J_1 \simeq 0.4271863541$, $J_2 \simeq 0.4280990318$, and $b_0 \simeq 0.9989334657$. These numbers illustrate the construction; existence in the theorem follows from the preceding analytic argument. Taking two distinct small positive values of ϵ yields distinct, ordered exponent pairs, so the phenomenon is not merely an interchange of coordinate labels.

Theorem 54.2 (Generic multipoint calibration and eventual finite jets)**.** *Under the bounded-vector assumptions at the beginning of this section:*

1. *the exponent vector p is uniquely determined by H on any nonempty open subset of the positive orthant;*
2. *in every such subset, a generic m-tuple of points $s^{(1)}, \dots, s^{(m)}$ determines p from only the values and first derivatives of H, by the linear system*

$$\sum_j s_j^{(i)} H_j(s^{(i)}) p_j = H(s^{(i)}), \qquad 1 \le i \le m; \tag{314}$$

3. *at every fixed positive point, there is some finite derivative order, possibly depending on the law and the point, whose jet uniquely determines p by differentiated Euler identities.*

Here "generic" means outside the zero set of a nonzero real-analytic determinant; in particular, the exceptional set has empty interior and Lebesgue measure zero. No uniform bound on the derivative order in the last assertion is claimed.

Proof. The transform extends continuously to the nonnegative orthant and has coordinate restrictions

$$H(te_j) = C_j t^{1/p_j}, \qquad C_j = \int_0^\infty -\log \mathbb{E} e^{-x^{-p_j} W_j}\, dx > 0. \tag{315}$$

Continuity follows by dominated convergence, using the atom at zero near $x = 0$ and boundedness of W near $x = \infty$. Analytic continuation of a known open patch and (315) already prove the first assertion.

For the constructive statement, put $A(s) = (s_1 H_1(s), \dots, s_m H_m(s))$. These vectors span $\mathbb{R}^m$ as s ranges over any nonempty open set U. Otherwise some nonzero q satisfies $\sum_j q_j s_j H_j(s) = 0$ on U, hence throughout the positive orthant by analyticity. Integration along the flow gives $H(e^{tq_1} s_1, \dots, e^{tq_m} s_m) = H(s)$ for every real t. Passing to a coordinate axis and using (315) forces $e^{tq_j/p_j} = 1$ for every j, hence $q = 0$, a contradiction. Thus there is an invertible matrix with m rows $A(s^{(i)})$ in U. Its determinant is analytic and not identically zero on the connected positive orthant product, which proves the generic claim.

For the final statement, differentiate Euler's identity. For each multi-index β one obtains the linear equation

$$\sum_j \left(s_j \partial^{\beta + e_j} H + \beta_j \partial^\beta H \right) p_j = \partial^\beta H. \tag{316}$$

If the rows of these equations did not span $\mathbb{R}^m$ at the chosen point, a nonzero q would annihilate all of them. Every Taylor coefficient of the analytic function $\sum_j q_j s_j H_j(s)$ would then vanish there.

The same dilation-flow argument would imply $q = 0$. The rows therefore span a finite-dimensional space of full dimension, so some finite collection of them already has full rank. They involve a finite jet and recover p uniquely. □

When an active rate I is observed instead, one sets $s = -\nabla I(z)$ and $H(s) = I(z) + s \cdot z$. Thus the multipoint system can be written directly as

$$\sum_j s_j^{(i)} z_j^{(i)} p_j = I(z^{(i)}) + s^{(i)} \cdot z^{(i)}.$$

The argument distinguishes exact identifiability from numerical conditioning: the error amplification of the finite system is controlled by its smallest singular value, which need not be bounded away from zero uniformly over the model class. Differentiated Euler identities and symmetry-based rank tests are classical tools; the specific contribution here is the explicit gated correlated family with coincident curvature data, together with a calibration repair that does not assume signs for cross-covariances.

55 Blind recovery by an interior power probe

Moving a Laplace probe from the boundary of the positive orthant to its interior changes the relevant moment problem. The moments to be decoded are then exponentially tilted moments, which exist and determine the tilted law without any moment assumption on the original law. For known powers, rationally independent moment encoding is the method of [43, Lemmas 3.2–3.3]; the contribution considered here is its combination with blind recovery of the powers, the dimension, a common unknown baseline, and an unknown observation gain.

Let I be an unknown nonempty finite or countably infinite index set. Write $\ell^1_+(I)$ for the nonnegative summable sequences, and let $X = (X_i)_{i\in I}$ have a Borel probability law P on this space. Set $S(x) = \sum_i x_i$. The probe parameters are an unknown baseline $a > 0$, an unknown gain $C > 0$, and unknown positive powers q_i satisfying

$$\inf_{i\in I} q_i > 0, \qquad \{q_i : i \in I\} \text{ is linearly independent over } \mathbb{Q}, \qquad P(X_i > 0) > 0 \quad (i \in I). \tag{317}$$

Linear independence in the countable case means independence of every finite subfamily. No upper bound on the powers is imposed. Define

$$F(t) = \mathbb{E}_P \exp\left[-aS(X) - \sum_{i\in I} t^{q_i} X_i\right], \qquad 0 \le t \le 1, \tag{318}$$

$$\mathcal{G}(\theta) = C \int_0^1 -\log[1 - \theta + \theta F(t)]\, dt, \qquad 0 < \theta < 1. \tag{319}$$

The parameter t has the displayed known window and Lebesgue clock; the gain C is not assumed known. Since $S(X) < \infty$ almost surely, these quantities exist for every probability law under consideration. In particular, no positive ordinary moment need exist.

Theorem 55.1 (Blind dimension, powers, baseline, gain, and law)**.** *Knowledge of $\mathcal{G}$ on any nonempty open interval contained in $(0, 1)$ determines C, a, the cardinality of I, the unordered set $\{q_i : i \in I\}$, and the joint law P, up to the same permutation of the powers and coordinates, throughout the class* (317)–(319).

More precisely, the observation first determines the curve F and then the finite measure

$$\nu(dx) = e^{-aS(x)}P(dx). \tag{320}$$

The baseline is the unique positive solution of

$$\int e^{bS(x)}\nu(dx) = 1, \tag{321}$$

and $P(dx) = e^{aS(x)}\nu(dx)$.

Proof. Step 1: gate inversion and the gain. Put $f(t) = 1 - F(t)$. Dominated convergence, first for the summable series and then for the bounded exponential, shows that F is continuous on $[0,1]$. It is strictly decreasing: if $0 \le t < u \le 1$, then $\sum_i (u^{q_i} - t^{q_i})X_i > 0$ on the event $\{S(X) > 0\}$, which has positive probability. Also

$$0 < f(0) \le f(t) \le f(1) < 1.$$

The first strict inequality follows from $a > 0$ and $P(S(X) > 0) > 0$; the last follows from $S(X) < \infty$ almost surely. The function $\mathcal{G}$ extends holomorphically to $|\theta| < 1$, so the prescribed data determine all its derivatives at zero. Define the finite positive measure on $[0,1]$

$$\kappa(A) = C\int_0^1 f(t)\mathbf{1}_A(f(t))\,dt.$$

For every $n \ge 0$,

$$\int u^n \kappa(du) = \frac{\mathcal{G}^{(n+1)}(0)}{n!}. \tag{322}$$

Compact moment uniqueness determines κ. Consequently it determines

$$\sigma(du) = \mathbf{1}_{(0,1]}(u)u^{-1}\kappa(du), \qquad C = \sigma([0,1]).$$

The probability measure σ/C is the distribution of $f(U)$ for U uniform on $[0,1]$. Its quantile function is the continuous strictly increasing function f, including the endpoints by limits. Thus C and the whole curve F are recovered.

Step 2: a finite signed measure of exponents. For a finitely supported multi-index $n \in \mathbb{N}_0^{(I)}$, put $|n| = \sum_i n_i$, $n! = \prod_i n_i!$, $n \cdot q = \sum_i n_i q_i$, and

$$m_n = \int x^n \nu(dx).$$

All these moments are finite, because $x^n e^{-aS(x)} \le S(x)^{|n|}e^{-aS(x)}$ is bounded. Let $q_- = \inf_i q_i > 0$ and choose $0 < \tau < 1$ so small that $\tau^{q_-} < a$. The signed atomic measure

$$\eta_\tau = \sum_{n\in\mathbb{N}_0^{(I)}} \frac{(-1)^{|n|}\tau^{n\cdot q}m_n}{n!}\,\delta_{n\cdot q} \tag{323}$$

has finite total variation. Indeed, the countable multinomial theorem and monotone convergence give

$$\begin{aligned}\sum_n \frac{\tau^{n\cdot q}m_n}{n!} &= \int \exp\left(\sum_i \tau^{q_i}x_i\right)\nu(dx)\\ &\le \mathbb{E}_P e^{-(a-\tau^{q_-})S(X)} \le 1.\end{aligned} \tag{324}$$

The absolutely convergent expansion therefore yields

$$F(\tau e^{-s}) = \int_{[0,\infty)} e^{-s\lambda}\eta_\tau(d\lambda), \qquad s \geq 0. \tag{325}$$

Uniqueness of the Laplace transform of finite signed measures recovers η_τ. One elementary proof pushes the measure forward under $\lambda \mapsto e^{-\lambda}$ and uses compact moment uniqueness at the nonnegative integer values of s.

Since a and q_- are unknown, the choice of τ is not asserted to be a data-effective stopping rule. For the identifiability claim, compare two admissible models with the same data and choose a common τ small enough for both. Step 1 gives the same F, and (325) then gives the same signed measure η_τ. This comparison is sufficient for all subsequent steps.

Step 3: the primitive powers. Use the set of actual nonzero positive atoms

$$A_\tau = \{\lambda > 0 : \eta_\tau(\{\lambda\}) \neq 0\},$$

not its topological closure. Rational independence prevents collisions between distinct multi-indices in (323), so

$$A_\tau = \{n \cdot q : n \neq 0,\ m_n > 0\}.$$

The finite-positive-measure version of Lemma 52.1, applied to ν, shows that the additively indecomposable elements of A_τ are exactly the powers q_i. Indeed $m_{e_i} > 0$, and a positive mixed moment remains positive upon deleting a coordinate factor. The lemma uses actual atoms rather than topological support, so it remains valid when the powers and their finite sums accumulate. It identifies whether the dimension is finite or countably infinite as well as its cardinality. Once the powers have been matched, all moments are recovered from

$$m_n = (-1)^{|n|} n! \tau^{-n\cdot q} \eta_\tau(\{n \cdot q\}). \tag{326}$$

In particular the zeroth moment is $\nu(\ell^1_+) = F(0) > 0$.

Step 4: the tilted law, its normalization, and the baseline. For every finite $J \subset I$ and $0 < b \leq a$,

$$\int \exp\left(b \sum_{i\in J} x_i\right) \nu(dx) \leq 1.$$

Each finite-coordinate marginal of ν thus has a moment generating function in a neighborhood of zero and is determined by its moments. The finite-coordinate marginals determine the measure on the countable product, and therefore on $\ell^1_+(I)$: its Borel sigma field equals the trace of the coordinate sigma field. Thus the recovered moments determine ν.

Finally $H(b) = \int e^{bS(x)}\nu(dx)$ is strictly increasing wherever finite, because $\nu(S > 0) > 0$. It satisfies $H(a) = 1$, whereas $H(0) = F(0) < 1$. It can therefore equal one for only one positive b. This identifies a, and then (320) identifies P. □

Remark 55.2 (What is and is not assumed). The original law can have infinite mean, no finite positive ordinary moment, or moment-indeterminate finite moments. In finite dimension, every nonnegative finite-valued vector satisfies the summability condition. In countable dimension the condition is $S(X) < \infty$ almost surely, with no deterministic bound and no integrability condition on S. The lower bound on the powers is a hypothesis of the countable theorem. An identically zero coordinate remains unobservable and has explicitly been excluded. Unknown coefficients multiplying $t^{q_i}X_i$ introduce the usual coordinate scale ambiguity; those coefficients are normalized to one here.

This is an exact-data identification theorem. An open interval of the gate curve contains infinitely much data; neither finite-sample recovery nor a stable procedure for resolving infinitely many powers is asserted. The construction of the finite signed exponent measure is an adaptation of the known-power method in [43, Lemmas 3.2–3.3]; exponential tilting and compact moment uniqueness are classical. The extra blind parameters and the precise ambiguity below are the candidate contribution, without a claim that its priority has been exhaustively established.

Example 55.3 (How the interior probe breaks a stable-law alias). Let $0 < \beta < 1$ and let Z_β have Laplace transform $\mathbb{E}e^{-sZ_\beta} = e^{-s^\beta}$. At zero baseline, the deterministic variable $X = 1$ with power q and the variable Z_β with power q/β give the identical curve e^{-t^q}. For positive baselines a and b, their interior curves instead are

$$e^{-a-t^q}, \qquad e^{-(b+t^{q/\beta})^\beta}.$$

Their first nonconstant exponents at $t = 0$ are respectively q and q/β, with nonzero coefficients. Thus they cannot agree, even if their baselines are independently unknown. The gate and gain inverse ensures that an unknown positive observation gain cannot conceal this difference either. This explicitly removes the heavy-tail alias rather than imposing a moment condition that excludes it.

55.1 The exact ambiguity for an unknown vector baseline

For finite d, replace the scalar a in (318) by an unknown vector $b = (b_1, \dots, b_d) \in (0, \infty)^d$:

$$F(t) = \int \exp\left[-b \cdot x - \sum_{i=1}^{d} t^{q_i} x_i\right] P(dx).$$

The preceding proof still identifies the gain, the powers, and the finite measure $\nu(dx) = e^{-b\cdot x} P(dx)$.

Proposition 55.4 (Complete vector-baseline fiber)**.** *After powers and coordinates have been matched, all models producing this same data are exactly*

$$\mathcal{B}_\nu = \left\{ c \in (0, \infty)^d : \int e^{c\cdot x} \nu(dx) = 1 \right\}, \qquad P_c(dx) = e^{c\cdot x} \nu(dx). \tag{327}$$

If ν has compact support and every coordinate is nontrivial, $\mathcal{B}_\nu$ is locally a smooth hypersurface of dimension $d-1$ near every one of its points. Restricting the baseline to a known positive ray $c = av$, $v_i > 0$, leaves at most one admissible a, and exactly one whenever a model on that ray is assumed to exist.

Proof. The equality in (327) is precisely what makes P_c a probability measure. Substitution shows that its probe is $\int e^{-\sum_i t^{q_i} x_i} \nu(dx)$, independent of c. Conversely, recovery of ν in the preceding proof shows that every admissible model has this form. In the compactly supported case the function $H(c) = \int e^{c\cdot x} \nu(dx)$ is smooth, and every partial derivative $\partial_i H = \int x_i e^{c\cdot x} \nu(dx)$ is strictly positive. The implicit function theorem gives the hypersurface statement. Along the prescribed ray, $a \mapsto H(av)$ is strictly increasing wherever finite, proving uniqueness there. □

55.2 A sharp fixed-power criterion

The positive baseline removes tail assumptions, but it does not remove arithmetic resonance. This remains a genuine observational obstruction.

Proposition 55.5 (Necessity and sufficiency of rational independence)**.** *Fix a finite positive baseline vector b, a positive gain C, and positive powers $q_1, \dots, q_d$. The map taking an arbitrary probability law P on $[0,\infty)^d$ to its finite-window gate curve is injective if and only if the powers are linearly independent over $\mathbb{Q}$. If the powers are dependent, any prescribed positive open box contains two distinct smooth compactly supported probability densities with identical full gate curves. They can additionally be chosen to have the same ordinary mixed moments through any prescribed finite order.*

Proof. The zero curve corresponds exactly to $P = \delta_0$. For every other law, sufficiency follows from the tilted-moment part of the preceding proof, without requiring every coordinate to be nontrivial when the dimension and powers are fixed. For necessity, choose distinct $\alpha, \beta \in \mathbb{N}_0^d$ with $\alpha \cdot q = \beta \cdot q$. The nonzero polynomial

$$R(s) = (s-b)^\alpha - (s-b)^\beta$$

vanishes along $s = b + (t^{q_1}, \dots, t^{q_d})$. If moments through order N are to be preserved, put $Q(s) = s_1^{N+1} R(s)$. Choose a nonzero nonnegative smooth bump φ supported inside the prescribed box, and set $g = Q(\partial)\varphi$. Repeated integration by parts gives

$$\int e^{-s\cdot x} g(x)\, dx = Q(s) \int e^{-s\cdot x} \varphi(x)\, dx.$$

Thus the Laplace transform of g vanishes on the probe curve, and all its derivatives through order N at zero vanish. In particular $\int g = 0$, and its ordinary moments through that order vanish. The function g is nonzero: the displayed transform cannot vanish identically, since Q is nonzero and the Laplace transform of φ is strictly positive at every real positive s. Choose a smooth probability density p_0 supported in the same box and bounded below by a positive constant on the support of g. For sufficiently small $\varepsilon > 0$, the functions $p_0 \pm \varepsilon g$ are nonnegative probability densities. Their Laplace curves, and consequently their full gate curves, agree, while the densities are distinct and their prescribed moments agree. □

55.3 Universal identification does not give uniform tail stability

Proposition 55.6 (An exponential tail barrier)**.** *Even with known dimension $d = 1$, baseline $a > 0$, power $q > 0$, and gain $C > 0$, the inverse observation map is not uniformly continuous over all nonnegative probability laws in total variation or in the Wasserstein distance W_1, even if the entire gate interval including $\theta = 1$ is observed. More precisely, fix $0 < \varepsilon < 1$ and, for $R > 0$, put*

$$P_R = (1-\varepsilon)\delta_0 + \varepsilon \delta_R, \qquad Q_R = (1-\varepsilon)\delta_0 + \varepsilon \delta_{R+1}.$$

With total variation normalized to lie in $[0,1]$, one has

$$\mathrm{TV}(P_R, Q_R) = W_1(P_R, Q_R) = \varepsilon,$$

whereas

$$\sup_{0 \le \theta \le 1} |\mathcal{G}_{P_R}(\theta) - \mathcal{G}_{Q_R}(\theta)| \le \frac{C\varepsilon}{1-\varepsilon} e^{-aR}. \tag{328}$$

Proof. The total variation is immediate. Transporting the mass ε from R to $R+1$ gives $W_1 \le \varepsilon$, and the difference of means gives the opposite inequality. Their Laplace profiles differ by at most εe^{-aR} uniformly on $0 \le t \le 1$, and both profiles are at least $1-\varepsilon$. For every $0 \le \theta \le 1$, the function $u \mapsto -\log(1-\theta+\theta u)$ has derivative of absolute value at most $(1-\varepsilon)^{-1}$ on $[1-\varepsilon, 1]$. Integrating proves (328). Letting $R \to \infty$ gives the failure of every uniform inverse modulus tending to zero. □

The proposition does not rule out stability on a fixed tight class with additional quantitative restrictions. It separates the universal exact identification theorem from a uniform noisy-data recovery claim; the common atom at zero shows that this instability persists even when the endpoint $\theta = 1$ is available.

Example 55.7 (Countably many coordinates with no positive moments). Let Y have density $(1+y)^{-2}$ on $[0,\infty)$, put $T = e^Y - 1$, and set $X_i = 2^{-i}T$ for $i \geq 1$. Then $\sum_i X_i = T < \infty$ almost surely, but

$$\mathbb{E}X_i^r = \infty \qquad (i \geq 1,\ r > 0).$$

Indeed the exponential growth of $(e^y - 1)^r$ dominates the polynomial decay of the density. Choose $q_i = \sqrt{p_i}$, where p_i is the ith prime. The square roots of distinct primes are rationally independent: in each finite multiquadratic field, the automorphism reversing one square root and fixing the others isolates its coefficient in a putative rational relation. These powers have a positive lower bound and no upper bound. For every $a, C > 0$, the response is well defined and Theorem 55.1 identifies the model within its stated class. This example simultaneously uses countably many coordinates, unbounded powers, arbitrary dependence, and the absence of every positive ordinary moment.

56 An interior probe invisible to every ordinary jet

The origin is the wrong expansion point if the unknown distribution has heavy tails. At a strictly positive Laplace parameter every finite measure has an analytic transform, regardless of its ordinary moments. We first give an explicit uniqueness curve for analytic germs and then apply this elementary analytic fact to probability measures. Analytic uniqueness sequences are an established subject [49]; multidimensional Laplace uniqueness sequences are studied in [50]. The results below give a self-contained construction, without a claim of priority for the general uniqueness-set principle.

Definition 56.1 (Separated scales)**.** Let $x_j : (0,\epsilon) \to (0,1)$, $1 \leq j \leq d$, tend to zero as $t \downarrow 0$. They are separated if

$$\frac{\log(1/x_j(t))}{\log(1/x_{j-1}(t))} \longrightarrow \infty, \qquad 2 \leq j \leq d.$$

Theorem 56.2 (A leading monomial on separated scales)**.** *For separated scales and every nonzero real or complex analytic germ A at $0 \in \mathbb{R}^d$, there exist a unique $\alpha \in \mathbb{N}_0^d$ and $c \neq 0$ such that*

$$A(x_1(t),\dots,x_d(t)) = c\prod_{j=1}^{d} x_j(t)^{\alpha_j}(1 + o(1)). \tag{329}$$

Here α is the smallest index with a nonzero Taylor coefficient when indices are compared first in coordinate d, then $d-1$, and so on. In particular, the composition has no zero for all sufficiently small positive t.

Proof. For $d = 1$, factor the ordinary order of vanishing from the convergent power series. Suppose the assertion holds in dimension $d-1$. The nonempty set of powers of z_d occurring in A has a least element m. Convergence of the series gives the analytic factorization

$$A(z', z_d) = z_d^m(A_0(z') + z_d B(z', z_d)), \qquad A_0 \not\equiv 0,$$

where B is bounded on a sufficiently small closed polydisc. By induction, $A_0(x_1(t),\dots,x_{d-1}(t))$ is asymptotic to $c\prod_{j<d} x_j(t)^{\alpha_j}$. Writing $\ell_j(t) = \log(1/x_j(t))$, separation implies $\sum_{j<d}\alpha_j\ell_j(t) =$

$o(\ell_d(t))$. Consequently

$$\frac{x_d(t)}{\prod_{j<d} x_j(t)^{\alpha_j}} = \exp\left(-\ell_d(t) + \sum_{j<d} \alpha_j \ell_j(t)\right) \longrightarrow 0.$$

The term containing B is negligible. This proves the assertion with $\alpha_d = m$, including the description of its coefficient. Two distinct monomials on these scales have a ratio tending to zero or infinity, so the leading index and coefficient are unique. □

Corollary 56.3 (A universal discrete interior probe)**.** *Fix $a \in (0,\infty)^d$ and any sequence $t_n > 0$ tending to zero. For $d \geq 2$, put*

$$c(t) = (t, e^{-1/t}, e^{-1/t^2}, \dots, e^{-1/t^{d-1}}), \qquad s_n = a + c(t_n). \tag{330}$$

For $d = 1$, put $c(t) = t$. If finite Borel measures μ, ν on $[0,\infty)^d$ satisfy

$$\int e^{-\langle s_n, x\rangle}\, \mu(dx) = \int e^{-\langle s_n, x\rangle}\, \nu(dx) \qquad \text{for every } n,$$

then $\mu = \nu$. No moment, tail, support, independence, or moment-determinacy assumption is required. The same statement holds for finite signed or complex measures.

Proof. The difference of the transforms is holomorphic on $\{s \in \mathbb{C}^d : \operatorname{Re} s_j > 0 \text{ for all } j\}$. Indeed, differentiation of every order is justified on a neighborhood of a by

$$\sup_{x \geq 0} x^\beta \exp\left(-\frac{1}{2}\sum_j a_j x_j\right) < \infty,$$

with the product convention for x^β. The scales in (330) are separated. If the transform difference had a nonzero germ at a, Theorem 56.2 would make its values nonzero at all sufficiently late s_n, a contradiction. Thus the transforms agree on a neighborhood of a, and the identity theorem extends this equality to the connected positive tube. For a fully explicit final uniqueness step, fix $b \in (0,\infty)^d$. The finite signed or complex measure $e^{-\langle b,x\rangle}(\mu - \nu)(dx)$ has Fourier transform equal to the transform difference at $b + i\xi$, hence zero everywhere. Fourier uniqueness and positivity of the exponential weight give $\mu = \nu$. □

Corollary 56.4 (Exponential approximation and weak convergence)**.** *For the sample parameters s_n of Corollary 56.3, the linear span of*

$$x \longmapsto e^{-\langle s_n, x\rangle}, \qquad n \geq 1,$$

is dense in $C_0([0,\infty)^d)$ in the uniform norm and in $L^p([0,\infty)^d)$ for every $1 \leq p < \infty$. Moreover, for probability measures μ_k, μ on $[0,\infty)^d$,

$$\mu_k \Rightarrow \mu \quad \Longleftrightarrow \quad L_{\mu_k}(s_n) \longrightarrow L_\mu(s_n) \text{ for every } n.$$

In particular,

$$d_c(\mu,\nu) = \sum_{n \geq 1} 2^{-n} |L_\mu(s_n) - L_\nu(s_n)|$$

is a metric inducing the weak topology on probability measures.

Proof. If the span were not dense in C_0, Hahn–Banach and Riesz–Markov would give a nonzero finite signed or complex measure annihilating it, contrary to Corollary 56.3. For L^p, an annihilator is a function $h \in L^{p'}$. Fix $b \in (0,\infty)^d$ with $b_j < a_j$. Hölder's inequality shows that $e^{-\langle b,x\rangle}h(x)\,dx$ is a finite complex measure. Its Laplace transform vanishes at $s_n - b$, so the same corollary forces $h = 0$.

Weak convergence implies convergence of these bounded continuous test functions. Conversely, uniform density and the common total mass bound imply convergence of integrals against every function in C_0. This is vague convergence, and because the limit μ has total mass one it implies weak convergence. Finally, the summable weights and $0 \le L_\mu(s_n) \le 1$ show that convergence in d_c is equivalent to coordinatewise convergence of the probe values. □

Remark 56.5 (The anchoring step is not exclusive to flat probes). If $q_1, \dots, q_d > 0$ are linearly independent over $\mathbb{Q}$, the shifted power curve $a + (t^{q_1}, \dots, t^{q_d})$ also determines every finite measure. To see this, substitute into the convergent Taylor series at a. The exponents $\alpha \cdot q$ are distinct and form a locally finite set. A nonzero series has a least exponent and a nonzero leading coefficient. The same argument as above then applies. Thus removal of tail assumptions comes from interior anchoring. Separated flat scales provide a different, explicit design with no arithmetic independence requirement.

Proposition 56.6 (The entire endpoint jet is insufficient)**.** *For $d \ge 2$, extend $c(0) = 0$ in (330). For any analytic germ A at a, the function $A(a + c(t))$ is smooth from the right at zero and*

$$\frac{d^k}{dt^k} A(a + c(t))\bigg|_{t=0+} = \partial_1^k A(a), \qquad k \ge 0. \tag{331}$$

There exist two distinct compactly supported probability measures whose Laplace observations along the curve have identical derivatives of every order at zero, although their observations differ for every sufficiently small positive t.

Proof. Every component of c after the first is flat at zero. By writing the difference $A(a + c(t)) - A(a + (t, 0, \dots, 0))$ as the integral of the derivatives of A along the line segment between these two arguments, one sees that the difference is a smooth flat function. This proves (331).

For an explicit pair, take $a = (1, \dots, 1)$, $p = 1/(e+1)$, and

$$\mu = \delta_{(0,1,0,\dots,0)}, \qquad \nu = p\delta_0 + (1-p)\delta_{(0,2,0,\dots,0)}.$$

Their transform difference is $D(s) = e^{-s_2} - p - (1-p)e^{-2s_2}$. It satisfies $D(a) = 0$ and is independent of s_1, so all the derivatives in (331) vanish. On the other hand,

$$D(a + c(t)) = \frac{1-e}{e(e+1)} e^{-1/t} + O(e^{-2/t}),$$

which is nonzero for small positive t. The measures have disjoint supports and total variation distance one. □

Remark 56.7 (What is measured). The data in Corollary 56.3 are countably many exact real numbers with an accumulation point, not one scalar measurement. The corollary is a uniqueness result, not a noise-stable recovery algorithm. Proposition 56.6 also shows that arbitrarily accurate ordinary Taylor models can omit the information responsible for uniqueness.

56.1 Smooth laws with identical infinite jets

The atomic endpoint-jet example above admits a stronger smooth-density version with a quantified observation gap. The observations remain direct Laplace values along the same interior flat curve.

For a probability measure μ on $[0,1]^d$, write

$$L_\mu(s) = \int_{[0,1]^d} e^{-s\cdot x}\, d\mu(x).$$

Fix $a \in (0,\infty)^d$ and $d \geq 2$, and consider the particular curve

$$\Gamma_1(t) = a_1 + t, \qquad \Gamma_j(t) = a_j + \exp(-t^{-(j-1)}) \quad (2 \leq j \leq d), \qquad t > 0. \tag{332}$$

Set $\Gamma(0) = a$. Every component is C^∞ on $[0,\infty)$, although the components with index at least two are not analytic at zero.

Proposition 56.8 (Identical infinite jets with separated laws)**.** *For every positive integer M, there exist two distinct probability measures $\mu_{M,+}, \mu_{M,-}$ with smooth densities on $[0,1]^d$, bounded between $1/2$ and $3/2$, and constants $c_M, C_M > 0$, such that*

$$W_1(\mu_{M,+}, \mu_{M,-}) \geq c_M, \tag{333}$$

$$\frac{d^n}{dt^n} L_{\mu_{M,+}}(\Gamma(t))\Big|_{t=0} = \frac{d^n}{dt^n} L_{\mu_{M,-}}(\Gamma(t))\Big|_{t=0} \qquad (n \geq 0), \tag{334}$$

$$\sup_{0\leq t\leq \epsilon} \left| L_{\mu_{M,+}}(\Gamma(t)) - L_{\mu_{M,-}}(\Gamma(t)) \right| \leq C_M \exp(-M\epsilon^{-(d-1)}) \qquad (\epsilon > 0). \tag{335}$$

The measures and constants in this statement do not depend on ϵ. Their one-dimensional marginals in coordinates $1, \ldots, d-1$ are identical.

Proof. In $L^2[0,1]$, let V_M be the finite-dimensional span of

$$1, \quad e^{-a_d x}, \quad x e^{-a_d x}, \quad \ldots, \quad x^{M-1} e^{-a_d x},$$

and let $h = x - \mathrm{proj}_{V_M} x$. This is a smooth real-valued function, and $h \neq 0$: an identity $x = A + P(x)e^{-a_d x}$ on $[0,1]$ would extend to all real x, which is impossible as $x \to +\infty$. Orthogonality gives

$$\int_0^1 h = 0, \qquad \int_0^1 x^k e^{-a_d x} h(x)\, dx = 0 \quad (0 \leq k < M), \qquad \int_0^1 x h(x)\, dx = \|h\|_2^2 > 0. \tag{336}$$

Set $b = (2\|h\|_\infty)^{-1}$ and define densities $f_\pm(x_1, \ldots, x_d) = 1 \pm b h(x_d)$. They are probability densities in the stated range. The test function $x \mapsto x_d$ is 1-Lipschitz, so Kantorovich duality gives $W_1(\mu_+, \mu_-) \geq 2b\|h\|_2^2 =: c_M$.

Put $\Psi(s) = \int_0^1 e^{-sx} h(x)\, dx$ and $\Phi(s) = \int_0^1 e^{-sx}\, dx$. Then

$$L_{\mu_+}(s) - L_{\mu_-}(s) = 2b\Psi(s_d) \prod_{j<d} \Phi(s_j).$$

By (336), Ψ has a zero of order at least M at a_d. Taylor's formula, using $s \geq a_d > 0$, yields

$$|\Psi(a_d + u)| \leq \frac{u^M}{M!} \int_0^1 x^M |h(x)|\, dx \quad (u \geq 0).$$

Since $0 < \Phi(s) \leq 1$ for $s \geq 0$, this proves (335) with $C_M = 2b(M!)^{-1} \int_0^1 x^M |h(x)|\, dx$. The function $\Psi(a_d + u)$ factors as u^M times an analytic function. Substituting $u = \exp(-t^{-(d-1)})$ therefore gives a smooth function flat at zero. The other factors are smooth there, proving (334). □

Corollary 56.9 (A horizon-dependent noise obstruction)**.** *Suppose the entire function $L_\mu(\Gamma(t))$, $0 \le t \le \epsilon$, is observed with adversarial error at most δ in supremum norm. If*

$$2\delta \ge C_M \exp(-M\epsilon^{-(d-1)}), \tag{337}$$

then the worst-case W_1 reconstruction error over smooth densities $1/2 \le f \le 3/2$ is at least $c_M/2$.

Proof. The midpoint of the two observation curves lies within δ of each curve. Any single reconstruction from this common noisy observation has distance at least half the separation from one of the two laws. □

In particular, even complete knowledge of the Taylor jet at the base point need not identify a law that the full curve identifies. The constants c_M may shrink with M; no uniform lower bound in M is asserted. Also, this result varies the accessible horizon ϵ. It does not contradict continuity of the inverse on a fixed compact class when a fixed identifying curve segment is observed exactly.

57 A finite resolution theorem for the hidden dimension

The exact gate inverse and finite noisy sampling are different observation models. In this section the available data are noisy values of the *Laplace profile*

$$F(t) = \mathbb{E} \exp\left(-\sum_{i=1}^{d} X_i t^{q_i}\right), \qquad 0 < t \le 1.$$

No stability assertion for recovering these values from noisy gate data is implicit. That additional step is supplied by Sections 58–60. The result below is an application of classical approximate Prony and matrix-pencil ideas; compare [52, 53]. Its purpose is to give a complete finite resolution guarantee for the present nonlinear random-vector model, including its approximation error.

Theorem 57.1 (Finite noisy recovery of dimension and powers)**.** *Fix an integer $D \ge 1$, numbers $0 < a \le b < 2a$, $m > 0$, $S > 0$, and $0 < \Delta \le b$. Consider all nonnegative random vectors with unknown dimension $1 \le d \le D$ such that*

$$a \le q_i \le b, \qquad |q_i - q_j| \ge \Delta \ (i \ne j), \qquad \mathbb{E}X_i \ge m, \qquad \mathbb{E}\left(\sum_i X_i\right)^2 \le S. \tag{338}$$

The separation condition is vacuous when $d = 1$. In particular, no rational independence of the powers is assumed. Put

$$\zeta = \frac{\Delta}{eb}, \qquad \kappa = \frac{1}{D}\left(\frac{\zeta}{2}\right)^{2D-2}. \tag{339}$$

Choose $0 < \tau \le 1$ and suppose that the $2D$ real observations satisfy

$$t_j = \tau e^{-j/b}, \qquad |\widetilde{F}_j - F(t_j)| \le \delta, \qquad j = 0, \dots, 2D-1.$$

Define

$$\epsilon = \delta + \frac{S}{2}\tau^{2a}, \qquad E = D\epsilon, \qquad s = m\kappa\tau^b. \tag{340}$$

If $E < s/2$, then the number of eigenvalues greater than $s/2$ of

$$\widetilde{H}_D = (1 - \widetilde{F}_{j+k})_{j,k=0}^{D-1}$$

is exactly d. Thus the first $2D-1$ observations already determine the hidden dimension throughout this class.

After recovering d, form the d by d matrices

$$\widetilde{H}_0 = (1-\widetilde{F}_{j+k})_{j,k=0}^{d-1}, \qquad \widetilde{H}_1 = (1-\widetilde{F}_{j+k+1})_{j,k=0}^{d-1}.$$

The first matrix is positive definite. There is a matching of the generalized eigenvalues $\widetilde{z}_i$ of $(\widetilde{H}_1, \widetilde{H}_0)$ to $z_i = e^{-q_i/b}$ for which

$$|\widetilde{z}_i - z_i| \le \eta := \frac{2E}{s-E}. \tag{341}$$

If also $\eta < 1/(2e)$, the estimates $\widetilde{q}_i = -b\log\widetilde{z}_i$ are real and satisfy

$$\min_{\pi}\max_{i} |\widetilde{q}_i - q_{\pi(i)}| \le 2eb\eta. \tag{342}$$

Here the minimum ranges over permutations.

Proof. For $u \ge 0$ one has $0 \le u - (1-e^{-u}) \le u^2/2$. Since $t \le 1$ and $q_i \ge a$,

$$1 - F(t) = \sum_i \mu_i t^{q_i} + R(t), \qquad |R(t)| \le \frac{S}{2}t^{2a}, \qquad \mu_i = \mathbb{E}X_i. \tag{343}$$

Set $w_i = \mu_i \tau^{q_i}$ and $z_i = e^{-q_i/b}$. The principal part of the sampled sequence is therefore

$$y_j = \sum_i w_i z_i^j, \qquad w_i \ge m\tau^b.$$

The total entrywise error in $1-\widetilde{F}_j$ is at most ϵ. The nodes lie in $[e^{-1}, e^{-a/b}]$, and the mean value theorem gives $|z_i - z_j| \ge \zeta$.

We record an elementary Vandermonde bound. Let $V_d = (z_i^j)_{0\le j\le d-1,\,1\le i\le d}$. The coefficients of each Lagrange polynomial

$$\ell_i(z) = \prod_{k\ne i} \frac{z - z_k}{z_i - z_k}$$

have total absolute value at most $(2/\zeta)^{d-1}$. These coefficient vectors are the rows of V_d^{-1}. Consequently

$$\|V_d^{-1}\|_\infty \le (2/\zeta)^{d-1}, \qquad \|V_d^{-1}\|_1 \le d(2/\zeta)^{d-1},$$

and hence

$$\sigma_{\min}(V_d)^2 \ge \frac{1}{d}(\zeta/2)^{2d-2} \ge \kappa. \tag{344}$$

Adding rows cannot decrease the smallest singular value of a matrix with full column rank, so the same bound applies to $V_D = (z_i^j)_{0\le j\le D-1,\,1\le i\le d}$.

The ideal matrix $H_D = (y_{j+k})_{j,k=0}^{D-1} = V_D \operatorname{diag}(w_i) V_D^T$ is positive semidefinite of rank d, and each nonzero eigenvalue is at least s. The perturbation has operator norm at most $D\epsilon = E$. Weyl's inequality now puts exactly d eigenvalues of $\widetilde{H}_D$ above $s/2$.

For the recovered size d, the ideal matrices satisfy

$$H_0 = V_d \operatorname{diag}(w_i) V_d^T, \qquad H_1 = V_d \operatorname{diag}(w_i z_i) V_d^T.$$

Thus $H_0 \succeq sI$, their generalized eigenvalues are the z_i, and each perturbation has norm at most $d\epsilon \le E$. For a nonzero real vector v, write $r(v) = v^T H_1 v/(v^T H_0 v)$. This quotient lies in $[0,1]$. Its perturbed version obeys

$$|\widetilde{r}(v) - r(v)| \le \frac{\|\widetilde{H}_1 - H_1\| + |r(v)|\|\widetilde{H}_0 - H_0\|}{s - E} \le \frac{2E}{s - E}.$$

The generalized Courant–Fischer principle gives (341) in sorted order. When $\eta < 1/(2e)$, the true and estimated nodes are at least $1/(2e)$. The derivative of $-b\log z$ on this interval is bounded by $2eb$, which proves (342). □

Corollary 57.2 (An explicit noise threshold)**.** *Under the assumptions of Theorem 57.1, suppose $0 < \delta \le S$ and choose*

$$\tau = (\delta/S)^{1/(2a)}, \qquad \alpha = 1 - \frac{b}{2a} > 0.$$

A sufficient condition for exact dimension recovery is

$$\delta^{\alpha} < \frac{m\kappa}{4DS^{b/(2a)}}. \tag{345}$$

For all sufficiently small δ, the same $2D$ samples give

$$\min_{\pi} \max_{i} |\widetilde{q}_i - q_{\pi(i)}| \le \frac{16ebDS^{b/(2a)}}{m\kappa}\, \delta^{\,1-b/(2a)}.$$

The last assertion is an upper bound for this procedure, not a claim of minimax optimality.

Proof. For the displayed choice of τ, one has $\epsilon = 3\delta/2 \le 2\delta$, and therefore

$$\frac{E}{s} \le \frac{2DS^{b/(2a)}}{m\kappa}\delta^{\alpha}.$$

Condition (345) makes this ratio smaller than $1/2$. Then $\eta = 2E/(s-E) \le 4E/s$. As $\delta \downarrow 0$, the additional logarithm condition is eventually satisfied, and the claimed estimate follows. □

Corollary 57.3 (Positive tilting supplies the moment bound)**.** *Let P be any probability law of a finite nonnegative vector, without assuming that any positive moment of P is finite. Write $T = \sum_i X_i$. Fix a known baseline $c \ge c_0 > 0$ and suppose*

$$\nu(dx) = e^{-c\sum_i x_i} P(dx), \qquad Z = \nu(\mathbb{R}^d_+) \ge z_0 > 0.$$

Consider the normalized tilted profile

$$F_c(t) = \frac{1}{Z}\int \exp\left(-\sum_i t^{q_i} x_i\right) \nu(dx).$$

If $1 \le d \le D$, the powers satisfy the band and separation assumptions of Theorem 57.1, and each tilted coordinate mean satisfies $\int x_i\, \nu(dx)/Z \ge m$, that theorem applies with the explicit choice

$$S = \frac{4}{e^2 c_0^2 z_0}. \tag{346}$$

In particular, the finite dimension and power guarantees can hold even when every positive moment of the original law P is infinite.

If noisy unnormalized profile values have error at most δ_F, and a mass estimate satisfies $|\widetilde{Z} - Z| \le \delta_Z \le z_0/2$, then dividing those observations by $\widetilde{Z}$ gives the normalized error bound

$$\delta = \frac{2(\delta_F + \delta_Z)}{z_0}.$$

These are errors at the tilted-profile and mass stages. Section 59 connects them to errors in gate measurements.

Proof. The normalized tilted law satisfies

$$\frac{1}{Z}\mathbb{E}_P[T^2 e^{-cT}] \le \frac{1}{Z}\sup_{u\ge 0} u^2 e^{-cu} = \frac{4}{e^2 c^2 Z} \le \frac{4}{e^2 c_0^2 z_0}.$$

It is therefore in the class of Theorem 57.1. For the error claim, write the exact unnormalized profile value as A, so that $0 \le A \le Z$. Since $\widetilde{Z} \ge z_0/2$,

$$\left|\frac{\widetilde{A}}{\widetilde{Z}} - \frac{A}{Z}\right| \le \frac{\delta_F}{\widetilde{Z}} + \frac{A\delta_Z}{Z\widetilde{Z}} \le \frac{2(\delta_F + \delta_Z)}{z_0}.$$

□

Remark 57.4 (Scope and the role of the gap). The class includes arbitrarily dependent coordinates and unbounded laws with the indicated second-moment bound. Compact support with $\sum_i X_i \le M$ implies the hypothesis with $S = M^2$. The condition $b < 2a$ separates the original powers from all quadratic and higher powers in the expansion of F. It allows recovery of dimension and powers even when the full joint law is not identifiable because of a higher-order rational resonance. For example, the powers 2 and 3 lie below the first quadratic threshold 4, although $3 \cdot 2 = 2 \cdot 3$ produces a higher-order resonance. The condition is sufficient for this simple method; its failure is not an impossibility theorem. The constants are deliberately conservative. Neither the Vandermonde mechanism nor the matrix pencil is asserted to be new.

Proposition 57.5 (Two obstructions despite a positive mean floor)**.** *Fix $m > 0$. The observation in this proposition is the direct profile $F(t)$ on $0 < t \le 1$.*

1. *If no uniform tail bound is imposed, a positive lower bound on every coordinate mean does not make dimension uniformly stable, even with fixed separated rationally independent powers. More precisely, append to any fixed admissible vector an independent coordinate which equals m/p with probability p and zero otherwise, using a fixed new power $q_* > 0$. Its mean is m, whereas the profiles satisfy*
$$\sup_{0<t\le 1} |F_p(t) - F(t)| \le p \longrightarrow 0.$$

2. *If power separation is removed, dimension is not uniformly stable even for deterministic bounded vectors with the same positive mean floor and a uniform second-moment bound. Fix $q > 0$. Compare the one-coordinate model $(X, q) = (2m, q)$ with the two-coordinate model $(X_1, X_2) = (m, m)$ and powers $(q, q + u)$, where $u \downarrow 0$ and u/q is irrational. Their profiles satisfy*
$$\sup_{0<t\le 1} |F_u(t) - F_0(t)| \le \frac{mu}{eq}.$$

Proof. For the first assertion, independence gives

$$F_p(t) = F(t)(1 - p + pe^{-(m/p)t^{q_*}}).$$

Since $0 \le F(t) \le 1$, the difference is at most p. The new coordinate has second moment m^2/p, so this example does not satisfy the uniform second-moment hypothesis of Theorem 57.1.

For the second assertion, on $0 < t \le 1$ the difference between the two exponents inside the Laplace transform is at most $mu\,t^q|\log t| \le mu/(eq)$. The map $v \mapsto e^{-v}$ is 1-Lipschitz on $[0,\infty)$, which proves the profile bound. Both models have total mass $2m$ deterministically, and therefore the common second-moment bound is $S = 4m^2$. □

58 Finite gate jets control the entire ordered profile

This section provides a quantitative bridge from finitely many gate derivatives to the function encoded by the gate experiment. It is a deterministic pairwise stability statement. Derivatives are measured in the factorial normalization below; noise in an unnormalized high derivative is a different observation model.

Fix constants $0 < \beta < 1$, $0 < C_- \le C_+ < \infty$, and $L > 0$. An admissible pair (C, f) consists of $C \in [C_-, C_+]$ and a nondecreasing L-Lipschitz function $f : [0,1] \to [\beta, 1]$. Strict increase, though available in the applications, is not needed here. Define

$$G_{C,f}(\theta) = C \int_0^1 -\log(1 - \theta f(t))\, dt, \qquad J_k(C,f) = \frac{G_{C,f}^{(k)}(0)}{(k-1)!} = C \int_0^1 f(t)^k\, dt \quad (k \ge 1). \tag{347}$$

Theorem 58.1 (A finite-jet modulus for gate inversion)**.** *There is an absolute constant $K > 0$ with the following property; $K = 1024$ is an admissible choice. Let (C, f) and $(\widetilde{C}, \widetilde{f})$ be admissible, and suppose for some integer $N \ge 2$ and $\eta \ge 0$ that*

$$|J_k(C,f) - J_k(\widetilde{C}, \widetilde{f})| \le \eta \qquad (1 \le k \le N). \tag{348}$$

Set

$$D_N = 2C_+(1-\beta)^N + (2^N - 1)\eta, \tag{349}$$

$$E_N = \frac{1}{C_-}\left\{ \frac{2C_+K}{N} + 5\,6^N\eta + 2C_+(1-\beta)^N \right\}. \tag{350}$$

Then

$$|C - \widetilde{C}| \le D_N, \qquad \|f - \widetilde{f}\|_\infty \le 2E_N + \sqrt{2LE_N}. \tag{351}$$

In particular, if $0 < \eta \le 6^{-4}$ and

$$N = \left\lfloor \frac{\log(1/\eta)}{2\log 6} \right\rfloor, \tag{352}$$

then, as $\eta \downarrow 0$ with the model constants fixed,

$$|C - \widetilde{C}| + \|f - \widetilde{f}\|_\infty = O\big((\log(1/\eta))^{-1/2}\big). \tag{353}$$

No optimality of this modulus is asserted.

Lemma 58.2 (Polynomial approximation with controlled coefficients)**.** *The absolute constant K can be chosen so that for every 1-Lipschitz $h : [0,1] \to \mathbb{R}$ with $h(0) = 0$ and every integer $N \geq 2$, there is a polynomial $P(u) = \sum_{k=1}^{N} b_k u^k$ such that*

$$\|h - P\|_\infty \leq K/N, \qquad \sum_{k=1}^{N} |b_k| \leq 4\,6^N. \tag{354}$$

Proof. We include a standard Jackson-kernel construction to specify the coefficient cost needed here. The even 2π-periodic function $H(v) = h((1+\cos v)/2)$ has supremum norm at most one and Lipschitz constant at most $1/2$ with respect to circular distance. For an integer $n \geq 2$, normalize

$$\left(\frac{\sin(nv/2)}{\sin(v/2)}\right)^4$$

to a nonnegative trigonometric polynomial K_n with $(2\pi)^{-1}\int_{-\pi}^{\pi} K_n(v)\,dv = 1$. Its degree is $2(n-1)$. The estimates

$$\left|\frac{\sin(nv/2)}{\sin(v/2)}\right| \leq \min\{n, \pi/|v|\}, \qquad \left|\frac{\sin(nv/2)}{\sin(v/2)}\right| \geq 2n/\pi \quad (|v| \leq 1/n)$$

show that the unnormalized integral is at least $32n^3/\pi^4$, while its first absolute moment is at most $(1+\pi^4)n^2$. Consequently

$$\frac{1}{2\pi}\int_{-\pi}^{\pi} |v| K_n(v)\,dv \leq A/n$$

with $A = \pi^4(1+\pi^4)/32 < 300$. Convolution on the circle gives an even trigonometric polynomial $Q_n = H * K_n$ with $\|Q_n - H\|_\infty \leq A/(2n)$ and $\|Q_n\|_\infty \leq 1$.

Take $n = \lfloor N/2 \rfloor + 1$. Write $Q_n(v) = a_0 + \sum_{k=1}^{2(n-1)} a_k \cos(kv)$, where $|a_0| \leq 1$ and $|a_k| \leq 2$. Substituting $\cos(kv) = T_k(2u-1)$ produces a polynomial P_0 of degree at most N with $\|P_0 - h\|_\infty \leq A/N$. Here T_k is the Chebyshev polynomial. All its shifted roots lie in $(0,1)$, so the coefficients of $T_k(2u-1)$ alternate in sign. Thus their sum of absolute values is

$$|T_k(-3)| = T_k(3) = \tfrac{1}{2}((3+2\sqrt{2})^k + (3-2\sqrt{2})^k) \leq (3+2\sqrt{2})^k.$$

It follows that the sum of absolute values of the monomial coefficients of P_0 is at most $1 + 2\sum_{k=1}^{N}(3+2\sqrt{2})^k \leq 4\,6^N$. Finally set $P = P_0 - P_0(0)$. Its constant coefficient vanishes, its nonconstant coefficients are unchanged, and $\|P - h\|_\infty \leq 2A/N$. This proves the lemma with $K = 2A$, and hence with the convenient choice $K = 1024$. □

Proof of Theorem 58.1. Let $\nu = f_{\#}dt$ and $\widetilde{\nu} = \widetilde{f}_{\#}dt$ be probability measures on $[\beta, 1]$, and set $\sigma = C\nu$, $\widetilde{\sigma} = \widetilde{C}\widetilde{\nu}$. The observed values J_k are the nonzero-order moments of these finite measures.

First, the polynomial $p_N(u) = 1 - (1-u)^N$ has no constant coefficient and the sum of absolute values of its coefficients is $2^N - 1$. Moreover,

$$0 \leq C - \int p_N\,d\sigma = \int (1-u)^N\,d\sigma \leq C_+(1-\beta)^N.$$

The analogous estimate for $\widetilde{\sigma}$, followed by (348), proves $|C - \widetilde{C}| \leq D_N$.

Next, for any 1-Lipschitz h with $h(0) = 0$, apply Lemma 58.2 to obtain

$$\left|\int h\,d\sigma - \int h\,d\widetilde{\sigma}\right| \leq \frac{2C_+K}{N} + 4\,6^N\eta.$$

Since $\|h\|_\infty \le 1$, division by C and comparison with $\widetilde{C}$ gives

$$\left|\int h\,d\nu - \int h\,d\widetilde{\nu}\right| \le \frac{1}{C_-}\left\{\frac{2C_+K}{N} + 4\,6^N\eta + D_N\right\} \le E_N.$$

Kantorovich duality on $[0,1]$ therefore gives $W_1(\nu,\widetilde{\nu}) \le E_N$. Notice that approximation was performed before normalizing the measures. Normalizing their moments first would unnecessarily amplify the mass error.

Because f and $\widetilde{f}$ are continuous and nondecreasing, they are quantile representatives of their pushforward laws; hence

$$W_1(\nu,\widetilde{\nu}) = \int_0^1 |f(t) - \widetilde{f}(t)|\,dt. \tag{355}$$

The difference $r = f - \widetilde{f}$ is L-Lipschitz, rather than merely $2L$-Lipschitz: for $s < t$, both increments belong to $[0, L(t-s)]$. If $b = \|r\|_\infty$ is attained at t_0, then $|r(t)| \ge (b - L|t - t_0|)_+$. Integrating this lower bound over $[0,1]$ is minimized when t_0 is an endpoint. It follows that

$$W_1(\nu,\widetilde{\nu}) \ge \begin{cases} b^2/(2L), & b \le L, \\ b - L/2 \ge b/2, & b > L. \end{cases}$$

Thus $b \le \sqrt{2LE_N} + 2E_N$, proving (351).

Under (352), $6^N\eta \le \eta^{1/2}$ and $(1-\beta)^N = O(\eta^{a_\beta})$, where $a_\beta = -\log(1-\beta)/(2\log 6) > 0$. Therefore $E_N = O(1/\log(1/\eta))$, while D_N decays as a positive power of η. This proves (353). □

Remark 58.3 (Interpretation of noisy observations). If measured normalized jets $\widehat{J}_k$ have error at most η_0, any two admissible models fitting the data to that tolerance satisfy (348) with $\eta = 2\eta_0$. The theorem bounds the diameter of this feasible set. It does not estimate derivatives from ordinary noisy values of G; that step is supplied in Section 60. It does not assert that a large gate jet can be acquired at a cost independent of its order. With exact jets, $\eta = 0$, the same proof gives profile diameter $O(N^{-1/2})$ as $N \to \infty$.

Remark 58.4 (A tail-free source of the Lipschitz and positivity bounds). Suppose $X_i \ge 0$, $Z = \sum_i X_i < \infty$ almost surely, $a_i \ge a_0 > 0$, and $1 \le q_i \le v < \infty$. For

$$f(t) = 1 - \mathbb{E}\exp\left(-\sum_i (a_i + t^{q_i})X_i\right), \qquad 0 \le t \le 1,$$

finite or countably many coordinates are allowed, provided the defining exponent is meaningful. The estimate

$$0 \le f'(t) \le v\,\mathbb{E}[Ze^{-a_0 Z}] \le \frac{v}{ea_0}$$

gives a common Lipschitz constant without any ordinary moment assumption. One may justify the inequality first for finite sums and then pass to the limit by monotone convergence of the Laplace integrands; the Lipschitz bound is preserved. If $Z \ge z_0 > 0$ almost surely, then $f(t) \ge 1 - e^{-a_0 z_0}$, providing an admissible β. If Z has no positive deterministic lower bound, a separately specified uniform lower bound on $f(0)$ suffices instead.

59 From finite gate jets to stable hidden structure

The preceding finite-profile theorem can be combined with the gate-jet estimate to obtain a guarantee at the gate observation stage itself. The observations here are normalized derivatives of the gate

curve at zero, not finitely many values of that curve. These are derivatives in the gate parameter θ, rather than the endpoint derivatives in the profile parameter t considered in the infinite-jet blindness example. Section 60 then controls these gate derivatives from ordinary noisy gate values.

Corollary 59.1 (Finite gate jets resolve dimension and powers)**.** *Fix an integer $D \ge 1$, $1 \le u \le v < 2u$, $0 < \Delta \le v$, $m > 0$, $c_* > 0$, $0 < \beta < 1$, $0 < z_* \le 1 - \beta$, and $0 < C_- \le C_+ < \infty$. Consider models consisting of a finite nonnegative random vector X, powers q_i, a baseline c, and a gain C, satisfying*

$$1 \le d \le D, \qquad u \le q_i \le v, \qquad |q_i - q_j| \ge \Delta \quad (i \ne j), \qquad c \ge c_*, \qquad C_- \le C \le C_+, \tag{356}$$

$$T = \sum_i X_i, \qquad Z = \mathbb{E}e^{-cT} \in [z_*, 1-\beta], \qquad \frac{\mathbb{E}[X_i e^{-cT}]}{Z} \ge m. \tag{357}$$

No positive moment of the original law is assumed finite. Define

$$A(t) = \mathbb{E}\exp\left(-cT - \sum_i X_i t^{q_i}\right), \qquad f(t) = 1 - A(t), \qquad G(\theta) = C\int_0^1 -\log(1 - \theta f(t))\,dt.$$

Let $N \ge 2$ be an integer and let $\eta \ge 0$. Suppose two models in this class have normalized gate jets satisfying

$$\left|\frac{G_1^{(k)}(0) - G_2^{(k)}(0)}{(k-1)!}\right| \le \eta, \qquad 1 \le k \le N. \tag{358}$$

Subscripts $1, 2$ distinguish the models. Set

$$L_* = \frac{v}{ec_*}, \qquad S_* = \frac{4}{e^2 c_*^2 z_*}, \qquad \kappa = \frac{1}{D}\left(\frac{\Delta}{2ev}\right)^{2D-2}, \qquad \alpha = 1 - \frac{v}{2u} > 0.$$

Let E_N be the bound in (350) of Theorem 58.1, with intensity bounds C_-, C_+, floor β, and moment error η. Put

$$h_N = 2E_N + \sqrt{2L_* E_N}, \qquad \delta_N = \frac{2h_N}{z_*}. \tag{359}$$

If

$$0 < \delta_N \le S_*, \qquad \delta_N^\alpha < \frac{m\kappa}{16eDS_*^{v/(2u)}}, \tag{360}$$

then the dimensions of the two models are equal, and their powers obey

$$\min_\pi \max_i |q_{1,i} - q_{2,\pi(i)}| \le \frac{32evDS_*^{v/(2u)}}{m\kappa}\,\delta_N^\alpha. \tag{361}$$

For fixed class parameters and $N = \lfloor \log(1/\eta)/(2\log 6)\rfloor$, all sufficiently small $\eta > 0$ satisfy these conditions, and

$$\min_\pi \max_i |q_{1,i} - q_{2,\pi(i)}| = O\left((\log(1/\eta))^{-\alpha/2}\right). \tag{362}$$

Thus sufficiently accurate finite gate jets determine the hidden dimension throughout this quantitative class and constrain its powers with the stated modulus.

Proof. For $0 \le t \le 1$, the function f is nondecreasing and takes values in $[\beta, 1]$. For $0 < t \le 1$, differentiation under the expectation is justified by the bounded envelope vTe^{-cT} and gives

$$0 \le f'(t) = \mathbb{E}\left[e^{-cT - \sum_i X_i t^{q_i}} \sum_i q_i X_i t^{q_i - 1}\right] \le v\mathbb{E}[Te^{-cT}] \le \frac{v}{ec_*} = L_*.$$

Continuity at zero extends this Lipschitz bound to the closed interval. Theorem 58.1 therefore gives $\|f_1 - f_2\|_\infty \le h_N$. Consequently $\|A_1 - A_2\|_\infty \le h_N$ and $|Z_1 - Z_2| = |A_1(0) - A_2(0)| \le h_N$. Since $0 \le A_2(t) \le Z_2$ and $Z_1, Z_2 \ge z_*$,

$$\left\| \frac{A_1}{Z_1} - \frac{A_2}{Z_2} \right\|_\infty \le \frac{2h_N}{z_*} = \delta_N.$$

Both normalized profiles are Laplace profiles of the exponentially tilted laws. These laws have coordinate means at least m, and the elementary bound on $T^2 e^{-cT}$ gives their common second-moment bound S_*. Apply Theorem 57.1 and Corollary 57.2, with $(a, b, S, \delta) = (u, v, S_*, \delta_N)$ and $\tau = (\delta_N/S_*)^{1/(2u)}$. Choose as common sample data the exact values of the first normalized profile at the prescribed $2D$ sample sites. Their error is zero for the first model and at most δ_N for the second. The common Hankel rank estimate must therefore equal the dimension of both models.

For completeness, the ratio of Hankel perturbation size to its signal lower bound in this application is at most

$$\frac{2DS_*^{v/(2u)}}{m\kappa}\delta_N^\alpha < \frac{1}{8e}.$$

It is below $1/2$, and the node-error bound is less than $1/(2e)$. Thus the power estimates are real, and each of the two true power lists is within $16evDS_*^{v/(2u)}\delta_N^\alpha/(m\kappa)$ of the same estimated list, after sorting. The triangle inequality proves (361).

With the stated choice of N, the gate-jet estimate is $E_N = O((\log(1/\eta))^{-1})$. Therefore $\delta_N = O((\log(1/\eta))^{-1/2})$, which proves the asymptotic claim and eventually satisfies the two strict small-error conditions. □

The class assumptions are quantitative visibility assumptions. In particular, the gain floor, the tilted mass floor, the tilted mean floor, and the separation of powers exclude distinct ways in which hidden components can become arbitrarily difficult to see. The corollary gives stability of dimension and powers; it does not assert recovery of the whole law when the powers have rational relations. It is a consistency statement: if each of two models fits the same observed gate jets to accuracy $\eta/2$, their mutual jet discrepancy is at most η, so the conclusion applies. No minimax optimality of the logarithmic exponent is asserted.

60 Ordinary gate values give a finite-data guarantee

The gate derivatives in the preceding section can themselves be controlled from finitely many ordinary gate values. A deliberately conservative forward-difference construction suffices. Its sampling locations depend on the noise tolerance and cluster near zero; no numerical efficiency or optimality is asserted.

Theorem 60.1 (A finite-value modulus for gate inversion)**.** *Retain the admissible class of Theorem 58.1: $C \in [C_-, C_+]$ and $f : [0,1] \to [\beta, 1]$ is nondecreasing and L-Lipschitz. Let $N \ge 2$ be an integer, $0 < h \le N^{-2}$, and $\delta \ge 0$. For two admissible gate curves G_1, G_2, suppose*

$$|G_1(jh) - G_2(jh)| \le \delta \qquad (1 \le j \le N). \tag{363}$$

The zeroth value $G_1(0) = G_2(0) = 0$ is known exactly. Define

$$\eta_{N,h,\delta} = 16C_+N^2h + 2^N\delta h^{-N}. \tag{364}$$

Then the normalized gate jets satisfy

$$\left|\frac{G_1^{(k)}(0)-G_2^{(k)}(0)}{(k-1)!}\right|\le \eta_{N,h,\delta}\qquad (1\le k\le N). \tag{365}$$

Consequently, with D_N, E_N defined by (349)–(350) using $\eta=\eta_{N,h,\delta}$,

$$|C_1-C_2|\le D_N,\qquad \|f_1-f_2\|_\infty\le 2E_N+\sqrt{2LE_N}. \tag{366}$$

In particular, if $0<\delta\le e^{-64}$, choose

$$N=\left\lfloor\frac{\sqrt{\log(1/\delta)}}{4}\right\rfloor,\qquad h=e^{-4N}. \tag{367}$$

For fixed class parameters, these N positive gate settings give

$$|C_1-C_2|+\|f_1-f_2\|_\infty=O\left((\log(1/\delta))^{-1/4}\right)\qquad(\delta\downarrow 0). \tag{368}$$

Proof. Write

$$\Delta_h^k G(0)=\sum_{j=0}^{k}(-1)^{k-j}\binom{k}{j}G(jh).$$

Repeated use of the fundamental theorem of calculus gives

$$\frac{\Delta_h^k G(0)}{h^k}=\int_{[0,1]^k} G^{(k)}\big(h(u_1+\cdots+u_k)\big)\,du_1\cdots du_k. \tag{369}$$

Every gate curve in the class is smooth on $[0,Nh]$, since $Nh\le 1/2$. Differentiation of its defining integral gives

$$\frac{G^{(k+1)}(\theta)}{(k-1)!}=Ck\int_0^1\frac{f(t)^{k+1}}{(1-\theta f(t))^{k+1}}\,dt\le C_+k(1-Nh)^{-(k+1)}\quad(0\le\theta\le Nh).$$

The mean value theorem in (369) therefore bounds the forward-difference bias by

$$\left|\frac{\Delta_h^k G(0)}{h^k(k-1)!}-\frac{G^{(k)}(0)}{(k-1)!}\right|\le C_+k^2h(1-Nh)^{-(k+1)}\le 8C_+N^2h. \tag{370}$$

For the last inequality, use $(1-Nh)^{-(k+1)}\le(1-1/N)^{-(N+1)}\le 8$ for $N\ge 2$. For example, the logarithm of the middle expression, viewed as a function of real $N\ge 2$, is decreasing: its derivative is $\log(1+1/(N-1))-(N+1)/(N(N-1))<0$.

The discrepancy between the two normalized forward differences is at most

$$\frac{2^k\delta}{h^k(k-1)!}\le 2^N\delta h^{-N}.$$

Combining this with the two bias bounds proves (365). Theorem 58.1 then proves (366).

For (367), one has $N\ge 2$ and $h\le N^{-2}$. Also $\log(1/\delta)\ge 16N^2$, and hence

$$\begin{aligned}6^N\eta_{N,h,\delta}&=16C_+N^2e^{-(4-\log 6)N}+12^N\delta e^{4N^2}\\&\le 16C_+N^2e^{-(4-\log 6)N}+e^{N\log 12-12N^2}=o(N^{-1}).\end{aligned}$$

Since $(1-\beta)^N=o(N^{-1})$, it follows that $E_N=O(N^{-1})$; the gain bound D_N is smaller. Thus the profile error is $O(N^{-1/2})$. Finally $N\asymp\sqrt{\log(1/\delta)}$, proving (368). □

Corollary 60.2 (Stable hidden structure from ordinary gate values)**.** *Consider two models in the class (356)–(357), with $\alpha = 1 - v/(2u) > 0$. Suppose their ordinary gate values satisfy (363). In Corollary 59.1, use $\eta = \eta_{N,h,\delta}$ from (364) to form E_N, h_N, and δ_N. Whenever (360) holds, the dimensions agree and (361) bounds their powers.*

In particular, under (367), all sufficiently small $\delta > 0$ force equality of the dimensions and give

$$\min_{\pi} \max_{i} |q_{1,i} - q_{2,\pi(i)}| = O\left((\log(1/\delta))^{-\alpha/4}\right). \tag{371}$$

No ordinary moment assumption on the original laws is needed, and no rational independence of their powers is required.

Proof. Theorem 60.1 supplies precisely the finite-jet hypothesis of Corollary 59.1. For the stated design, $E_N = O(N^{-1})$ and $\delta_N = O(N^{-1/2}) = O((\log(1/\delta))^{-1/4})$. The strict threshold in that corollary is therefore satisfied eventually, and its power bound is $O(\delta_N^{\alpha})$, as claimed. □

If measured gate values have individual error at most δ_0, any two models fitting those same data to that tolerance satisfy (363) with $\delta = 2\delta_0$. The conclusions therefore bound the diameter of the compatible model class from finitely many ordinary noisy observations. The proof does not require derivative observations. It does require the specified quantitative class bounds and noise-dependent sampling design. The powers may have rational relations, so stable dimension and power recovery still does not imply recovery of the full joint law. The logarithmic rate is a sufficient bound for this construction, not a minimax or computational-efficiency claim.

61 Sharp noise cost of radial window design

The finite-window identity of Proposition 43.7 is an exact-data inverse. We now quantify how a radial window changes the conditioning of recovery of the local Laplace exponent. We specify the target and the noise norm carefully: the result below concerns the function $h = -\log L$, not the subsequent inversion of L to a probability distribution. It is a deterministic noise theorem, not a statement about the sample size needed to estimate a rare-event rate.

Fix $p > \rho > 0$, $L > 0$, $p_0 \in (1/2, 1)$, and $M > 0$. Let

$$\mathcal{P} = \{P \text{ on } [0,\infty) : P\{0\} \geq p_0,\ \int x\, P(dx) \leq M\}, \qquad h_P(s) = -\log \int e^{-sx}\, P(dx).$$

Choose $0 < \gamma < p$, with $\gamma \neq \rho$, and put

$$f_P(t) = e^{-\gamma t} h_P(e^{pt}), \qquad g_P(t) = e^{-\gamma t} \rho \int_0^L e^{\rho u} h_P(e^{p(t-u)})\, du. \tag{372}$$

Thus $g_P = K f_P$, where

$$(Kf)(t) = \int k(u) f(t-u)\, du, \qquad k(u) = \rho e^{(\rho-\gamma)u} \mathbf{1}_{[0,L]}(u). \tag{373}$$

All norms in this section are $L^2(\mathbb{R})$ norms unless indicated otherwise. The observed function is $g_P + e$, where $\|e\|_2 \leq \delta$. For an observation operator T, write

$$\mathcal{E}_T(\delta) = \inf_{\mathcal{A}} \sup_{P \in \mathcal{P}} \sup_{\|e\|_2 \leq \delta} \|\mathcal{A}(T f_P + e) - f_P\|_2. \tag{374}$$

For several channels the observation and noise belong to the Hilbert direct sum of their L^2 spaces.

Theorem 61.1 (Optimal radial-window noise rate)**.** *There are constants $0 < c < C < \infty$, depending only on the fixed parameters above, such that, for all sufficiently small $\delta > 0$,*

$$c\delta \log(1/\delta) \le \mathcal{E}_K(\delta) \le C\delta \log(1/\delta). \tag{375}$$

The lower bound continues to hold if the exact atom at zero and the entire sequence of ordinary moments of P are supplied to the reconstruction procedure. More precisely, there is one fixed atom and one fixed moment sequence such that the lower bound already holds on the subclass of $\mathcal{P}$ with those data.

Moreover, let $k_1, \dots, k_J \in L^1(\mathbb{R}) \cap BV(\mathbb{R})$, extended by zero wherever a compactly supported window vanishes, and suppose

$$\left(\sum_{j=1}^{J} (\|k_j\|_1 + \mathrm{TV}(k_j))^2 \right)^{1/2} \le B. \tag{376}$$

For $Tf = (k_1 * f, \dots, k_J * f)$, the lower bound

$$\mathcal{E}_T(\delta) \ge c_B \delta \log(1/\delta) \tag{377}$$

holds uniformly over all such designs. The constants do not depend on J or on the chosen kernels. Hence one tilted boxcar attains the best order possible under this window-variation budget.

The lower bound allows a design to depend on δ, provided its budget remains bounded. The statement does not include atomic kernels: an ideal Dirac observation has no integration loss and is outside $L^1 \cap BV$.

61.1 The analytic property of genuine Laplace exponents

Lemma 61.2. *Put $b_* = \pi/(2p)$. There is $R < \infty$, depending only on p_0, M, p, γ, such that every f_P extends holomorphically to $|\operatorname{Im} t| < b_*$ and satisfies*

$$\int_{\mathbb{R}} e^{2b_*|\xi|} |\widehat{f}_P(\xi)|^2 \, d\xi \le 2R^2. \tag{378}$$

Here the Fourier transform is unitary. In particular this is a uniform property of the probability class, not an additional Sobolev or analyticity assumption on the unknown target.

Proof. For $\operatorname{Re} s \ge 0$, the Laplace transform $L_P(s)$ belongs to the disk with center $q = P\{0\}$ and radius $1 - q$. Consequently $\operatorname{Re} L_P(s) \ge 2p_0 - 1 > 0$, and the principal logarithm defines h_P without zeros or branch ambiguity. The logarithmic series about q gives

$$|h_P(s)| \le C_0 := -\log(2p_0 - 1).$$

Integrating the derivative of the logarithm on the segment joining 1 to $L_P(s)$, and using $|1 - e^{-sx}| \le |s|x$, gives the second bound

$$|h_P(s)| \le C_1 |s|, \qquad C_1 = M/(2p_0 - 1).$$

If $|v| \le b_*$, then $\operatorname{Re} e^{p(t+iv)} \ge 0$. It follows that

$$|f_P(t + iv)| \le e^{-\gamma t} \min(C_0, C_1 e^{pt}).$$

The right side belongs to both L^1 and L^2, uniformly in v. For example one may take

$$R^2 = \frac{C_1^2}{2(p-\gamma)} + \frac{C_0^2}{2\gamma}.$$

For $|v| < b_*$, contour shifting, justified by the same exponential bounds at the two ends of a horizontal strip, gives $\widehat{f_P(\cdot + iv)}(\xi) = e^{-v\xi}\widehat{f}_P(\xi)$. Apply Plancherel at v and $-v$, let $v \uparrow b_*$ using Fatou's lemma, and add the two boundary inequalities. □

61.2 An estimator and its uniform error bound

Set $a = \rho - \gamma \neq 0$. The Fourier multiplier of K is

$$m(\xi) = \rho \frac{e^{(a-i\xi)L} - 1}{a - i\xi}. \tag{379}$$

In particular,

$$\frac{\rho|e^{aL} - 1|}{\sqrt{a^2 + \xi^2}} \leq |m(\xi)| \leq \frac{\rho(e^{aL} + 1)}{\sqrt{a^2 + \xi^2}}. \tag{380}$$

The off-critical choice $\gamma \neq \rho$ removes the zeros of the ordinary boxcar multiplier. For observed g^δ, define the spectral-cutoff estimator by

$$\widehat{f}_T^\delta(\xi) = \mathbf{1}_{\{|\xi| \leq T\}} \frac{\widehat{g}^\delta(\xi)}{m(\xi)}.$$

Equations (378) and (380) imply the explicit estimate

$$\|f_T^\delta - f_P\|_2 \leq \frac{\sqrt{a^2 + T^2}}{\rho|e^{aL} - 1|}\delta + \sqrt{2}Re^{-b_*T}. \tag{381}$$

Taking $T = b_*^{-1}\log(\sqrt{2}R/\delta)$ when δ is sufficiently small proves the upper bound in Theorem 61.1. This estimator needs no numerical derivative of the noisy data.

61.3 A lower bound inside an actual probability class

The following construction uses the classical oscillatory lognormal Stieltjes class [47, 42]; its moment cancellation is not a new moment-indeterminacy result. Its role here is to make the noise lower bound hold for genuine local laws, even after every ordinary moment has been specified.

Choose

$$0 < \lambda < \min\{-\log p_0, Me^{-1/2}\}, \qquad 0 < \varepsilon \leq 1/2, \qquad \omega_n = 2\pi n.$$

Let $P_{n,\pm,\varepsilon}$ be the compound-Poisson laws with Levy measures obtained by pushing forward

$$\lambda \frac{e^{-u^2/2}}{\sqrt{2\pi}}(1 \pm \varepsilon \sin(\omega_n u))\, du \tag{382}$$

under $u \mapsto e^u$. These are positive finite measures. For every integer $j \geq 0$,

$$\int_{\mathbb{R}} e^{ju} \frac{e^{-u^2/2}}{\sqrt{2\pi}} \sin(2\pi n u)\, du = e^{(j^2 - 4\pi^2 n^2)/2} \sin(2\pi n j) = 0. \tag{383}$$

All measures in (382) therefore have mass λ and the same moments $\lambda e^{j^2/2}$. Their compound-Poisson laws have atom $e^{-\lambda}$ at zero and mean $\lambda e^{1/2}$, and lie in $\mathcal{P}$. They also have one common full sequence of ordinary moments: the jth cumulant is $\lambda e^{j^2/2}$, and each moment is the corresponding complete Bell polynomial in the cumulants. In particular these side data do not distinguish any of the alternatives.

Write $\Delta f_{n,\varepsilon} = f_{P_{n,+,\varepsilon}} - f_{P_{n,-,\varepsilon}}$, and set $b = \gamma/p \in (0,1)$. The Mellin integral

$$\int_0^\infty x^{-z-1}(1-e^{-x})\,dx = -\Gamma(-z), \qquad 0 < \operatorname{Re} z < 1,$$

and Fubini give, with $z = b + i\xi/p$,

$$\widehat{\Delta f}_{n,\varepsilon}(\xi) = \frac{\lambda\varepsilon}{ip\sqrt{2\pi}}[-\Gamma(-z)]\left(e^{(z+i\omega_n)^2/2} - e^{(z-i\omega_n)^2/2}\right). \tag{384}$$

Fubini is justified by the integrability of $x^{-b-1}(1-e^{-x})$ and the finite bth moment of the lognormal jump measure.

Lemma 61.3 (Frequency-packet estimates)**.** *Put $A_n = \omega_n^{-b-1/2}e^{-\pi\omega_n/2}$. There are positive constants c, C, C_r, independent of n and ε, such that, for all sufficiently large n,*

$$c\lambda\varepsilon A_n \le \|\Delta f_{n,\varepsilon}\|_2 \le C\lambda\varepsilon A_n, \tag{385}$$

$$\int_{\mathbb{R}} (1+\xi^2)^{-r}|\widehat{\Delta f}_{n,\varepsilon}(\xi)|^2\,d\xi \le C_r(\lambda\varepsilon)^2\omega_n^{-2r}A_n^2, \qquad r \ge 0. \tag{386}$$

Proof. For $b \in (0,1)$, Stirling's estimate, together with continuity and the absence of poles on the line $-b + i\mathbb{R}$, gives two-sided bounds

$$|\Gamma(-b-iv)| \asymp (1+|v|)^{-b-1/2}e^{-\pi|v|/2}.$$

The two Gaussian factors in (384) have absolute values $\exp\{[b^2-(\xi/p\pm\omega_n)^2]/2\}$. On $|\xi/p-\omega_n| \le 1$, one Gaussian is bounded below by a positive constant and the other is exponentially small in ω_n^2. This proves the lower bound in (385).

For the upper bounds use $|A-B|^2 \le 2|A|^2 + 2|B|^2$ and symmetry. Writing $v = \xi/p$ reduces the relevant integral, up to constants, to

$$\int_{\mathbb{R}} (1+|v|)^{-2b-1-2r}e^{-\pi|v|}e^{-(v-\omega_n)^2}\,dv.$$

On $|v-\omega_n| \le \omega_n/2$, divide by $\omega_n^{-2b-1-2r}e^{-\pi\omega_n}$ and use the integrable majorant $Ce^{-w^2-\pi w}$, $w = v-\omega_n$. On the complement the Gaussian is at most exponentially small in ω_n^2, after its product with $e^{-\pi|v|}$ is integrated; this is negligible compared with the displayed divisor. These estimates establish (385) and (386). □

For the general design in (376), integration by parts against the finite signed derivative of a BV function yields

$$|\widehat{k}_j^{\mathrm{mult}}(\xi)| \le \min\{\|k_j\|_1, \mathrm{TV}(k_j)/|\xi|\},$$

where $\widehat{k}_j^{\mathrm{mult}}(\xi) = \int k_j(u)e^{-i\xi u}\,du$ is the convolution multiplier. Hence

$$\sum_j |\widehat{k}_j^{\mathrm{mult}}(\xi)|^2 \le \frac{2B^2}{1+\xi^2}.$$

Lemma 61.3 implies

$$\|T\Delta f_{n,\varepsilon}\|_2 \le C_B\lambda\varepsilon A_n/\omega_n, \qquad \|\Delta f_{n,\varepsilon}\|_2 \ge c\lambda\varepsilon A_n. \tag{387}$$

Fix $\varepsilon_0 = 1/2$. For small δ, take the largest n for which $C_B\lambda\varepsilon_0 A_n/\omega_n \ge \delta$, and then choose $\varepsilon \le \varepsilon_0$ so the right side of the first inequality in (387) is δ. Because $\omega_n = 2\pi n$,

$$n \sim \pi^{-2}\log(1/\delta), \qquad \|\Delta f_{n,\varepsilon}\|_2 \ge c'_B\delta\log(1/\delta).$$

The midpoint of the two noiseless observations can be explained by either law with noise norm at most $\delta/2$. Every reconstruction makes an error of at least half their target separation for one of the laws. This proves both lower bounds in Theorem 61.1, including the assertion about full moment side information.

Corollary 61.4 (Cost of smoothing the edges of a window)**.** *For any fixed integer $r \ge 1$, observe $K^r f_P$ instead of Kf_P. Then*

$$\mathcal{E}_{K^r}(\delta) \asymp \delta[\log(1/\delta)]^r.$$

The log-kernel of K^r is $\rho^r e^{(\rho-\gamma)u}\mathbf{1}^{*r}_{[0,L]}(u)$, so this corresponds to a nonnegative B-spline radial mask, supported on $[0, rL]$ in logarithmic radius. Multiplication by a fixed normalizing constant does not alter the asserted order.

Proof. The multiplier is $m(\xi)^r$ and thus comparable to $(1+|\xi|)^{-r}$. The cutoff argument gives error $O(\delta T^r + e^{-b_* T})$. For the lower bound use (386) with exponent r and repeat the two-point argument with A_n/ω_n^r. □

Proposition 61.5 (No uniform total-variation inversion)**.** *On the same class $\mathcal{P}$, the local distribution cannot be recovered uniformly consistently in total variation from the noisy window data of Theorem 61.1. This remains true if the exact atom at zero and all ordinary moments are known. The claim also holds for any design satisfying a fixed budget (376).*

Proof. Keep $\varepsilon \in (0, 1/2]$ fixed in (382). Write $\nu_\pm$ for the two finite Levy measures, each of mass λ, and write $\|\cdot\|_{\mathrm{var}}$ for the total variation norm of a signed measure. The compound-Poisson expansion and the telescoping convolution identity imply

$$P_+ - P_- = e^{-\lambda}(\nu_+ - \nu_-) + R, \qquad \|\nu_+^{*j} - \nu_-^{*j}\|_{\mathrm{var}} \le j\lambda^{j-1}\|\nu_+ - \nu_-\|_{\mathrm{var}}.$$

Consequently

$$\|R\|_{\mathrm{var}} \le e^{-\lambda}\sum_{j\ge 2}\frac{j\lambda^{j-1}}{j!}\|\nu_+ - \nu_-\|_{\mathrm{var}} = (1-e^{-\lambda})\|\nu_+ - \nu_-\|_{\mathrm{var}}.$$

Since $e^{-\lambda} > p_0 > 1/2$, this yields

$$d_{\mathrm{TV}}(P_+, P_-) \ge (2e^{-\lambda} - 1)\lambda\varepsilon\mathbb{E}|\sin(2\pi nU)|, \qquad U \sim N(0,1),$$

where $d_{\mathrm{TV}}(P,Q) = \frac{1}{2}\|P - Q\|_{\mathrm{var}}$. The last expectation tends to $2/\pi$, by the Fourier series of $|\sin|$ and the Gaussian characteristic function. Thus the distributional separation stays bounded away from zero, while (387) makes the observation separation tend to zero. The midpoint argument proves the asserted impossibility. The common moment sequence and common atom were already verified in (383). □

Remark 61.6 (Scope and relation to classical inverse problems). Mellin convolution, spectral cutoff, ordinary-smooth deconvolution, and lognormal moment-indeterminacy are classical; see, in particular, the boxcar channel-design theory of [45] and the Stieltjes classes in [47, 42]. The role of the present result is to connect an implementable radial design to a sharp noise modulus on the actual class of local log-Laplace exponents. The uniform BV-budget lower bound also rules out improvement by merely adding finitely many ordinary radial windows. Neither the moment construction nor the Fourier cutoff method is claimed as a new general technique. The sharp lower bounds here use unbounded compound-Poisson laws; they do not assert the same lower bound under a fixed compact support restriction. Recovery of the distribution from its real Laplace transform remains a separate inverse problem.

62 The dimension cost of adaptive linear readings

An adaptive linear experiment with m readings sequentially chooses bounded Borel functions $\varphi_1, \dots, \varphi_m$ on $Q = [0,1]^d$, with each choice allowed to depend arbitrarily on the previous exact answers, and receives $\int_Q \varphi_i \, d\mu$. Neither continuity of the adaptive rule nor a bound on the magnitude of the query functions is imposed. Laplace values, compactly supported moment data, and Laplace derivatives are examples. The result below is a direct information-width argument, included to prevent an interpretation of curve uniqueness as finite-data compression. Its method is part of the established theory of adaptive information; see [51].

Theorem 62.1 (Sharp order for finite exact linear information). *For every fixed $d \ge 1$, there is a constant $b_d > 0$ with the following property. For any deterministic adaptive experiment using $m \ge 1$ linear readings, there are two smooth probability densities f_+, f_- on Q, with $1/2 \le f_\pm \le 3/2$, producing exactly the same complete transcript and satisfying*

$$W_1(f_+ \, dx, f_- \, dx) \ge b_d m^{-1/d}. \tag{388}$$

Consequently its worst-case reconstruction error is at least $\frac{1}{2} b_d m^{-1/d}$. If q additional prescribed bounded linear calibrations are supplied, the same assertion holds with m replaced by $m+q$. Conversely, arbitrary linear information admits a nonadaptive reconstruction of every probability law on Q with m readings and error at most $B_d m^{-1/d}$, for a constant B_d depending only on d.

Proof. Choose smooth functions ψ, ϕ supported in the interior of Q such that

$$\int_Q \psi = 0, \qquad \|\psi\|_\infty \le 1, \qquad \mathrm{Lip}(\phi) \le 1, \qquad A := \int_Q \phi\psi > 0.$$

For example, take a nonnegative smooth interior bump η, let ψ be a positive normalization of $-\partial_1 \eta$, and let ϕ agree with a sufficiently small positive multiple of x_1 on its support and vanish near the boundary of Q.

Let $k = \lceil (2m)^{1/d} \rceil$, $N = k^d$, and $h = 1/k$. Subdivide Q into N cubes of side length h, with lower corners z_j, and set

$$\psi_j(x) = \psi((x - z_j)/h), \qquad \phi_j(x) = h\phi((x - z_j)/h)$$

inside the corresponding cube and zero outside. Run the proposed algorithm on the uniform density. This fixes a particular list of query functions $\varphi_1, \dots, \varphi_m$. Let $V \subset \mathbb{R}^N$ be the nullspace of the matrix with entries $\int_Q \varphi_i \psi_j$. Then $\dim V \ge N - m$.

At an extreme point v of the nontrivial compact polytope $V \cap [-1,1]^N$, at least $\dim V$ coordinates have absolute value one. Indeed, if fewer coordinate constraints are active, a nonzero direction in V

annihilating the active coordinates allows small displacements of both signs within the polytope, contradicting extremality. Thus we can choose v with $|v_j| \le 1$ and $\sum_j |v_j| \ge N - m \ge N/2$.

The densities

$$f_\pm = 1 \pm \tfrac{1}{2} \sum_{j=1}^{N} v_j \psi_j$$

are smooth probability densities in $[1/2, 3/2]$. They give the same answer as the uniform density to every query in the fixed list. Induction over the query number therefore shows that the adaptive algorithm follows this same list and produces exactly the same transcript for both inputs.

The function $\Phi_v = \sum_j \operatorname{sgn}(v_j)\phi_j$ is 1-Lipschitz on Q: on each cube it is 1-Lipschitz and vanishes near the boundary, and the same bound follows across cubes by subdividing a straight line at its intersections with their boundaries. Hence

$$\begin{aligned} W_1(f_+\,dx, f_-\,dx) &\ge \int_Q \Phi_v (f_+ - f_-)\,dx = A h^{d+1} \sum_j |v_j| \\ &\ge \frac{A}{2} h \ge \frac{A}{4\,2^{1/d}} m^{-1/d}. \end{aligned}$$

This proves the lower bound. Appending the q calibration functions to the nullspace constraints gives the calibrated version, by the same proof. The densities then share the calibrations of the uniform law; no assertion is made for every possible prescribed calibration value.

For the upper bound, choose $k = \lfloor (m+1)^{1/d} \rfloor \ge 1$ and query the masses of all but one of the k^d grid cubes. The remaining mass follows from total mass one. Move each cube's mass to its center. This costs at most $\sqrt{d}/(2k) \le \sqrt{d}\, m^{-1/d}$ in W_1. □

The upper bound permits arbitrary bounded linear query functions, including indicators of cubes. It does not claim a matching upper bound for Laplace queries alone. The theorem is deterministic; no randomized complexity claim is made. Its lower bound also permits derivatives of unbounded order to be queried exactly, provided each derivative value counts as a separate linear reading. Smoothness here is qualitative: no uniform derivative bound on the densities is imposed.

63 Quantitative questions for controlled observations

The finite-value guarantees resolve one part of the quantitative problem left by exact blind identification. The sharp modulus from ordinary gate values, the best placement of finitely many settings, and recovery on a fixed interior gate interval remain open in this framework. In particular, the bound in Corollary 60.2 is a sufficient rate, and the linear-information lower bound does not establish its optimality for the nonlinear gate functional.

Recovering the entire joint law requires additional control of tails, regularity, or the topology of error, as well as quantitative separation of higher mixed-moment exponents. Pairwise separation of the original powers only addresses the first exponent band. A classification of the models representing a fixed resonant gate curve would further distinguish power collisions from vanishing mixed moments. The smooth null densities above rule out an unrestricted universal positive statement.

Empirical estimation of the rare-event response and propagation of that error through the inverse form a separate sampling problem. The deterministic window rate is not a sample-complexity result. In countable dimension, a quantitative bound on $\sum_{j>m} X_j$ may support an effective dimension notion, but a detection threshold must be specified because arbitrarily small added coordinates destroy uniform cardinality recovery.

The finite-window clock is known throughout the interior-probe theorem. An unknown clock changes the parameterization while preserving the weighted distribution of profile values, so additional clock calibration would be needed. Likewise, the proof with countably many powers uses $\inf_i q_i > 0$ to choose a common convergence radius for the signed exponent measure; exponents tending to zero require a different argument.

Part V
Infinite mixtures, moment coincidences, and statistical recovery

The preceding parts concern what arithmetic and analytic observations reveal about hidden structure. Here the unknown object is a probability measure. We study three questions separately: whether an exact collection of observed laws determines that measure; whether replication removes otherwise invisible directions; and whether a finite sample permits uniform recovery. These are different properties, even for strictly positive observation kernels that extend to entire functions.

For a known component family P_θ and an unknown mixing probability μ, the law of q conditionally independent observations sharing one latent parameter is

$$A_q\mu = \int P_\theta^{\otimes q}\, d\mu(\theta).$$

The replication threshold is the least positive integer q for which A_q is injective on all Borel mixing probabilities, with threshold ∞ if no such integer exists. This counts observations within one group, not the number N of independent groups available for estimation. Separately redrawing the latent parameter for every observation produces the product of the one-observation law and does not repair its ambiguities.

The results develop in three stages. First, a bounded-holomorphic Fourier–Laplace transform gives the sharp reciprocal-sum criterion for Gaussian mixtures observed through partial-sum marginals, and classifies fixed-shape stable families. Second, a missing Legendre mode separates finite parameter-derivative tests from global mixture identification. Finite-dimensional ambiguity becomes atomic after two observations, whereas infinite-dimensional ambiguity supports arbitrary replication thresholds on both the circle and an ordinary interval. Finally, multiplicative smoothing gives localized Gevrey annihilators and exact moment-coincidence sets; their interaction with compactness distinguishes analytic identification from uniform statistical recovery. The factorial Legendre example then supplies fixed-replication rates, growing-replication budget bounds, and a contrasting construction with arbitrarily slow recovery.

Literature and scope. Lin's normal-mixture question is used in the form stated in [54, Section 2.6]. The differential-barrier question concerns precisely the finite operator classes in [55, Definitions 1 and 4]; no absence of nonlocal or infinite-order obstructions is asserted. Vandermeulen and Scott [62] obtain the sharp $2m-1$ bound within the corresponding class of mixtures with m components, and Vandermeulen and Saitenmacher [63] refine finite-component bounds using independence assumptions. Here a fixed component family is allowed arbitrary Borel mixing laws, so finite-support identification and unrestricted identification must be distinguished.

The analytic ingredients are classical. Fourier descriptions of characteristic kernels appear in Fukumizu et al. [59]; arithmetic-progression criteria for strict positive definiteness and Skolem–Mahler–Lech arguments appear in Pinkus [61, Section 3] and Bell–Lagarias [57, Theorem 1.3]. Identification by the full tensor-moment hierarchy is an existing principle [58]. The Müntz and Clarkson–Erdős theorems are used in the forms of Borwein and Erdélyi [64, Theorems 3.1–3.2]; sparse moments have also been used in mixture identification by Stoyanov and Lin [66]. The localized smooth-annihilator theorem below proves the mathematical assertion in Bürgin–Epperlein–Wirth's Conjecture 17 [65], with stronger smoothness, support, and exact-coincidence conclusions. These source-specific claims do not assert historical priority or external acceptance. In particular, the rate $\log\log N/\log N$ is already known in other mixture models, including Poisson mixtures [60].

64 Normal mixtures from partial-sum marginals

64.1 The question being answered

Lin's preprint [54, Section 2.6] asks whether arbitrary mixtures of iid normal sequences are determined by the marginal laws of their partial sums. Its Proposition 2.6.4 proves this under an exponential-moment assumption on the random variance. We remove that assumption. The observation remains exactly the one in the question: the individual law of each partial sum, with no joint observations of different partial sums.

Let (A, V) have an arbitrary Borel probability law on $\mathbb{R}\times[0,\infty)$, and let $Z_1, Z_2, \dots$ be independent standard normal random variables independent of (A, V). Define

$$X_j = A + \sqrt{V}\,Z_j, \qquad S_n = \sum_{j=1}^{n} X_j. \tag{389}$$

The parameters A and V may be dependent, and neither is subject to any moment condition. Every mixture of iid normal sequences, including degenerate normal components, has this representation in distribution.

Theorem 64.1 (Normal mixtures from partial-sum marginals)**.** *The collection $(\mathcal{L}(S_n))_{n\ge1}$ determines the entire joint law of (A, V) in (389), and hence the law of the exchangeable sequence $(X_j)_{j\ge1}$.*

More generally, it is enough to know the marginal laws of S_n for n in any infinite set $\mathcal{N}\subset\mathbb{N}$ satisfying

$$\sum_{n\in\mathcal{N}} \frac{1}{n} = \infty. \tag{390}$$

The key is not analytic continuation across the boundary point zero. The relevant transform is bounded and holomorphic on the open right half-plane. Its prescribed zeros approach zero too densely to be the zeros of a nonzero bounded holomorphic function.

64.2 Two transform lemmas

We include the required half-plane uniqueness argument. It is the classical necessary condition on the zeros of a bounded holomorphic function, often expressed through Blaschke products.

Lemma 64.2 (A boundary uniqueness criterion)**.** *Let H be a bounded holomorphic function on $\{z : \operatorname{Re} z > 0\}$. Suppose $r_k > 0$ are distinct, $H(r_k) = 0$, and*

$$\sum_k \frac{r_k}{1+r_k^2} = \infty. \tag{391}$$

Then H is identically zero.

Proof. Put $h(w) = H((1+w)/(1-w))$ for $|w|<1$. If h is not identically zero, factor out its finite-order zero at zero, if present, to obtain a bounded holomorphic function g with $g(0)\neq0$. Boundedness of the quotient follows from the maximum principle on disks and passage of their radii to one. Jensen's formula implies that its nonzero zeros w_j, counted with multiplicity, satisfy

$$\sum_{|w_j|<r} \log\frac{r}{|w_j|} \le \log\|g\|_\infty - \log|g(0)| \qquad (0<r<1),$$

with radii meeting zeros omitted if needed. Monotone convergence as $r \uparrow 1$ gives $\sum_j (1 - |w_j|) < \infty$, since $1 - u \le -\log u$ for $0 < u \le 1$.

Our zeros correspond to $w_k = (r_k - 1)/(r_k + 1)$. Omitting the possible single zero $r_k = 1$, we have

$$1 - |w_k| = \frac{1 - |w_k|^2}{1 + |w_k|} \ge \frac{2r_k}{(1 + r_k)^2} \ge \frac{r_k}{1 + r_k^2}.$$

This contradicts (391). □

Lemma 64.3 (Fourier–Laplace uniqueness)**.** *A finite complex Borel measure ρ on $\mathbb{R} \times [0, \infty)$ is zero if*

$$\int e^{ita - sb} \rho(da, db) = 0 \qquad (t \in \mathbb{R},\ s > 0). \tag{392}$$

Proof. For a fixed $s > 0$, the finite measure $D \mapsto \int_{D \times [0,\infty)} e^{-sb} \rho(da, db)$ on $\mathbb{R}$ has zero Fourier transform and is therefore zero. Consequently, for every Borel $D \subset \mathbb{R}$, the finite measure $\rho_D(E) = \rho(D \times E)$ on $[0, \infty)$ has zero Laplace transform at every $s > 0$. Dominated convergence gives its vanishing total mass as well. Push it forward by $b \mapsto e^{-b}$ and regard the resulting measure as a measure on $[0, 1]$. All its nonnegative integer moments vanish. Density of polynomials in $C([0, 1])$ and finite-measure uniqueness show that $\rho_D = 0$. Thus ρ vanishes on every Borel rectangle and hence is zero. □

Proof of Theorem 64.1. Let μ_1, μ_2 be two candidate laws for (A, V) that yield the same stated observations, and set $\rho = \mu_1 - \mu_2$. Conditionally on (A, V), the average S_n/n is normal with mean A and variance V/n. Equality of its characteristic functions therefore gives

$$\int e^{ita - t^2 v/(2n)} \rho(da, dv) = 0 \qquad (t \in \mathbb{R},\ n \in \mathcal{N}). \tag{393}$$

Fix $t \neq 0$ and define

$$H_t(z) = \int e^{ita - zv} \rho(da, dv), \qquad \operatorname{Re} z > 0.$$

Its absolute value is at most $|\rho|(\mathbb{R} \times [0, \infty))$. It is holomorphic: on $\operatorname{Re} z \ge \varepsilon > 0$, differentiation under the integral is justified by boundedness of $v^m e^{-\varepsilon v}$ for every fixed m. Its distinct positive zeros include $r_n = t^2/(2n)$ for $n \in \mathcal{N}$. Condition (390) implies

$$\sum_{n \in \mathcal{N}} \frac{r_n}{1 + r_n^2} = \infty.$$

Lemma 64.2 yields $H_t(s) = 0$ for every $s > 0$. Continuity in t also gives this equality at $t = 0$. Lemma 64.3 now gives $\rho = 0$. The common mixing law determines every finite-dimensional distribution of the sequence by conditional independence, completing the proof. □

Remark 64.4 (What changes in the earlier argument). In the cited preprint the variance exponential moment lets the transform extend across zero, after which the usual identity theorem applies to $t^2/(2n) \to 0$. The present argument instead uses bounded holomorphy on the existing half-plane. No extension across zero and no variance exponential moment are needed. This also explains why an arbitrary sequence of boundary-approaching observations cannot simply be substituted: the divergence condition is the step that forces uniqueness.

65 Sharp sparse identifiability for Gaussian location–scale mixtures

Let $\mathcal{N} \subset \{1, 2, \dots\}$ and, conditionally on a random pair $(A, B) \in \mathbb{R} \times [0, \infty)$, let $X_j = A + \sqrt{B} Z_j$, where the Z_j are independent standard normal variables, independent of (A, B). Write $S_n = X_1 + \cdots + X_n$. The preceding theorem proves the positive direction. We now establish the converse with one joint density realizing all the required transform zeros simultaneously.

Theorem 65.1 (Sharp sampling criterion)**.** *The map $\mu \mapsto (\mathcal{L}_\mu(S_n))_{n \in \mathcal{N}}$ is injective on the Borel probability measures on $\mathbb{R} \times [0, \infty)$ if and only if*

$$\sum_{n \in \mathcal{N}} \frac{1}{n} = \infty.$$

If the sum converges, there are two distinct, absolutely continuous mixing laws with all polynomial moments finite and with identical selected sum laws. In particular the obstruction persists even if $B > 0$ almost surely.

Sufficiency. This is Theorem 64.1, whose proof applies to every set $\mathcal{N}$ with divergent reciprocal sum and imposes no moment condition. □

Necessity, including all regularity details. Put $\lambda_n = (2n)^{-1}$ and suppose $\Lambda := \sum_{n \in \mathcal{N}} \lambda_n < \infty$. The finite or infinite product

$$P(z) = \prod_{n \in \mathcal{N}} \frac{z - \lambda_n}{z + \lambda_n} \qquad (\operatorname{Re} z > 0)$$

converges locally uniformly, is not identically zero, has zeros at all λ_n, and satisfies $|P(z)| \le 1$. These assertions follow from $\sum_n \lambda_n < \infty$ and $(z - \lambda)/(z + \lambda) = 1 - 2\lambda/(z + \lambda)$; no additional phase normalization is needed. Its boundary values satisfy $|P(i\xi)| = 1$ for $\xi \neq 0$.

Use the principal square root and define

$$F(z) = P(z) \exp\{-\sqrt{z} - 1/\sqrt{z}\}, \qquad \operatorname{Re} z > 0.$$

This is a bounded, nonzero holomorphic function with $F(\lambda_n) = 0$ and $F(\overline{z}) = \overline{F(z)}$. We verify that its boundary function $H(\xi) = F(i\xi)$, extended by $H(0) = 0$, is Schwartz. On the imaginary axis away from zero, logarithmic differentiation gives

$$Q(z) := \frac{P'(z)}{P(z)} = \sum_n \left(\frac{1}{z - \lambda_n} - \frac{1}{z + \lambda_n} \right).$$

For each integer $r \ge 0$,

$$|Q^{(r)}(i\xi)| \le 2(r + 1)! \Lambda |\xi|^{-r-2}.$$

Indeed, express the difference of the two $(r + 1)$st inverse powers as an integral over the real interval $[-\lambda_n, \lambda_n]$; throughout that interval $|i\xi - x| \ge |\xi|$. The usual recursion $P^{(j+1)}/P = (P^{(j)}/P)' + Q(P^{(j)}/P)$ therefore gives, for each $j \ge 1$,

$$|P^{(j)}(i\xi)| \le C_j |\xi|^{-2j} \quad (0 < |\xi| \le 1), \qquad |P^{(j)}(i\xi)| \le C_j \quad (|\xi| \ge 1).$$

Here the product and all derivatives converge uniformly on compact subsets of the imaginary axis avoiding zero. Since

$$\operatorname{Re} \sqrt{i\xi} = \sqrt{|\xi|/2}, \qquad \operatorname{Re} \frac{1}{\sqrt{i\xi}} = \frac{1}{\sqrt{2|\xi|}},$$

the two exponential factors dominate every polynomial derivative bound both at zero and at infinity. Thus $H \in \mathcal{S}(\mathbb{R})$, with all derivatives zero at zero.

Set

$$f(b) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{i\xi b} H(\xi)\, d\xi.$$

Then $f \in \mathcal{S}(\mathbb{R})$, it is real, and it is nonzero. An elementary contour shift proves that $f(b) = 0$ for $b < 0$. For fixed $R > 0$ shift the integration line from $i\mathbb{R}$ to $R + i\mathbb{R}$ in the integral of $e^{bz}F(z)$. The horizontal integrals tend to zero, since for $0 \le x \le R$, $\operatorname{Re}\sqrt{x + iT} \ge \sqrt{|T|/2}$. The boundary value at zero causes no difficulty because F extends continuously there by zero. It follows that

$$|f(b)| \le \frac{e^{bR}}{2\pi} \int_{\mathbb{R}} |F(R + i\xi)|\, d\xi \le \frac{e^{bR}}{2\pi} \int_{\mathbb{R}} e^{-\sqrt{|\xi|/2}}\, d\xi.$$

The last integral is independent of R, and $R \to \infty$ proves the support assertion for $b < 0$. More formally one may first shift from $\varepsilon + i\mathbb{R}$ and let $\varepsilon \downarrow 0$ by dominated convergence, using the same integrable bound.

For $s > 0$, Fubini and the Cauchy integral formula on a right half-disk give

$$\int_0^\infty e^{-sb} f(b)\, db = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{H(\xi)}{s - i\xi}\, d\xi = F(s).$$

The large semicircle contributes zero because $|F(z)| \le e^{-\sqrt{|z|/2}}$ there. A small semicircle at zero can also be deleted, or treated by its continuous zero boundary value.

Choose any nonzero, real, even function $h \in C_c^\infty(\{t : 1 < |t| < 2\})$ and define

$$g(a, b) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-ita} h(t) t^2 f(bt^2)\, dt.$$

This is real. The function $h(t)t^2 f(bt^2)$ is smooth with compact t-support bounded away from zero; all of its t-derivatives decay faster than every power of $|b|$, uniformly in t. Repeated integration by parts in t proves, for all $M, N \ge 0$,

$$|g(a, b)| \le C_{M,N}(1 + |a|)^{-N}(1 + |b|)^{-M}.$$

The same estimates hold for all (a, b)-derivatives. Hence $g \in \mathcal{S}(\mathbb{R}^2)$, and $g = 0$ when $b < 0$. Fourier inversion in a and the Laplace identity for f give

$$\int_{\mathbb{R}} \int_0^\infty e^{ita - sb} g(a, b)\, db\, da = h(t)F(s/t^2) \quad (t \ne 0,\ s > 0).$$

The left side is zero when $t = 0$, because the partial Fourier transform in a is zero there. In particular $\int g = 0$. It is nonzero because h and f are nonzero. Consequently, if $g = g_+ - g_-$ and $m = \int g_+ = \int g_- > 0$, the two measures

$$\mu_+(da, db) = m^{-1} g_+(a, b)\, da\, db, \qquad \mu_-(da, db) = m^{-1} g_-(a, b)\, da\, db$$

are distinct probabilities on $\mathbb{R} \times [0, \infty)$, with all polynomial moments finite. They give no mass to $b = 0$.

Finally, for $n \in \mathcal{N}$ and a characteristic-function argument $v \ne 0$, set $t = nv$ and $s = nv^2/2$. Then $s/t^2 = 1/(2n) = \lambda_n$, so the difference of the two characteristic functions of S_n is

$$m^{-1} h(nv) F(\lambda_n) = 0.$$

It is also zero at $v = 0$. Thus all selected sum laws agree, completing the necessity proof. □

Corollary 65.2 (Equal marginals and all moments do not repair sparse sampling)**.** *If $\sum_{n\in\mathcal{N}} 1/n < \infty$, the two counterexample laws can be chosen to have the same A marginal, the same B marginal, and identical mixed polynomial moments of every order, all finite. They may additionally satisfy $B \geq b_0$ almost surely for any prescribed $b_0 > 0$.*

Proof. Use the signed Schwartz density g constructed above. Its partial Fourier transform is

$$G(t,b) := \int_{\mathbb{R}} e^{ita} g(a,b)\, da = h(t)t^2 f(bt^2).$$

It vanishes on a neighborhood of $t = 0$. Consequently, for each b,

$$\int_{\mathbb{R}} a^j g(a,b)\, da = i^{-j}\partial_t^j G(0,b) = 0 \qquad (j \geq 0).$$

The case $j = 0$ gives a zero signed B marginal. Multiplying by b^k and integrating gives zero for every mixed moment, with all interchanges justified by Schwartz decay.

Also $\int_0^\infty f(b)\, db = \lim_{s\downarrow 0} F(s) = 0$, since $|P(s)| \leq 1$ and the regularizing factor tends to zero. The Fourier transform of the signed A marginal is therefore

$$\int_0^\infty G(t,b)\, db = h(t) \int_0^\infty f(u)\, du = 0$$

for $t \neq 0$, and is zero at $t = 0$ as well. Fourier uniqueness proves that this marginal vanishes. The normalized positive and negative parts of g thus have the asserted equal marginals and moments.

Finally replace B by $B + b_0$ in both laws. This multiplies their signed Fourier–Laplace transform by $e^{-b_0 s}$, preserving every required zero. It preserves equality of the coordinate marginals and, by the binomial formula, equality of all mixed moments. □

Remark 65.3. All polynomial moments in the counterexamples are finite, but no finite exponential moment is asserted. Thus agreement of all mixed moments here does not contradict any moment-determinacy theorem.

65.1 Sampling examples and random drift–variance mixtures

For normal mixtures, Theorem 65.1 gives a complete classification of the sets of observed integer lengths:

Observed lengths	Reciprocal sum	Every mixing law identified?
All positive integers	Divergent	Yes
An infinite arithmetic progression	Divergent	Yes
Squares $1, 4, 9, \ldots$	Convergent	No
Powers of two $1, 2, 4, \ldots$	Convergent	No
Any finite set	Finite	No

A negative entry means that distinct mixing laws can give identical observations. It does not mean that every mixing law has a competitor. By Corollary 65.2, neither knowing both parameter marginals nor knowing every mixed polynomial moment removes these counterexamples. A common strictly positive variance lower bound also does not remove them.

The Gaussian theorem applies as well to mixtures of Brownian motions: if $Y_t = At + \sqrt{V}W_t$, with W independent of (A, V), then Y_n and S_n have the same law. Thus integer-time marginal laws identify the random drift and variance under exactly the same sampling condition. No joint observation at different times is needed.

66 A classification of fixed-shape stable families

The same boundary uniqueness mechanism extends beyond the Gaussian family. In this section the stability index and skewness are fixed; only the location and scale are mixed. The distinction is essential: the theorem does not permit the shape parameters themselves to vary inside the mixing measure.

Fix $0 < \alpha \le 2$ and $-1 \le \beta \le 1$. For $t \neq 0$ put

$$\psi_{\alpha,\beta}(t) = \begin{cases} |t|^{\alpha}(1 - i\beta \tan(\pi\alpha/2) \operatorname{sgn} t), & \alpha \neq 1, \\ |t|(1 + i(2\beta/\pi) \operatorname{sgn} t \log |t|), & \alpha = 1, \end{cases} \tag{394}$$

and put $\psi_{\alpha,\beta}(0) = 0$. Let $K_{a,b}^{\alpha,\beta}$ be the probability measure with characteristic function

$$\widehat{K_{a,b}^{\alpha,\beta}}(t) = \exp\{ita - b\psi_{\alpha,\beta}(t)\}, \qquad a \in \mathbb{R}, \quad b \in [0, \infty). \tag{395}$$

This is the standard 1-parameterization of stable laws: in the usual scale notation γ, $b = \gamma^{\alpha}$ for $\alpha \neq 1$, and $b = \gamma$ for $\alpha = 1$; see [56, Definition 1.8]. At $b = 0$ the measure is δ_a. At $\alpha = 2$ the choice of β has no effect, and the variance is $2b$. The parameter a is a location parameter, not in general an expectation.

We first record the analytic criterion in a form which covers both real and nonreal sampling points.

Lemma 66.1 (A Fourier–Laplace sampling criterion)**.** *Let μ, ν be probability measures on $\mathbb{R} \times [0, \infty)$. Suppose that for every $t \in \mathbb{R} \setminus \{0\}$ there are distinct points $z_j(t)$ in the open right half-plane such that*

$$\int e^{ita - z_j(t)b} \, d\mu(a, b) = \int e^{ita - z_j(t)b} \, d\nu(a, b) \tag{396}$$

and

$$\sum_j \frac{\operatorname{Re} z_j(t)}{1 + |z_j(t)|^2} = \infty. \tag{397}$$

Then $\mu = \nu$.

Proof. Write $\rho = \mu - \nu$ and, for fixed real t, define

$$H_t(z) = \int e^{ita - zb} \, d\rho(a, b), \qquad \operatorname{Re} z > 0.$$

This is bounded by 2 and is holomorphic. Indeed, on a compact subset of the right half-plane there is $\varepsilon > 0$ with $\operatorname{Re} z \ge \varepsilon$, and each integrand derivative is bounded in absolute value by $b^k e^{-\varepsilon b}$, a bounded function of $b \ge 0$. Thus differentiation under the finite signed measure is justified without a moment assumption.

A nonzero bounded holomorphic function on the right half-plane has zeros satisfying the Blaschke condition $\sum_j \operatorname{Re} z_j/(1 + |z_j|^2) < \infty$. For completeness, map the half-plane to the disk by $w = (z - 1)/(z + 1)$. Jensen's formula for a nonzero bounded holomorphic function on the disk gives $\sum_j (1 - |w_j|) < \infty$, after removing a possible zero at the origin. Moreover,

$$1 - |w|^2 = \frac{4 \operatorname{Re} z}{|z + 1|^2}, \qquad 1 + |z|^2 \le |z + 1|^2 \le 2(1 + |z|^2),$$

so the disk and half-plane conditions are equivalent. Equations (396)–(397) therefore imply $H_t \equiv 0$ for every $t \neq 0$. Dominated convergence in t gives $H_0 \equiv 0$ as well.

The finite signed measure $d\rho_1(a,b) = e^{-b}\,d\rho(a,b)$ now has Fourier transform

$$\int e^{ita+iub}\,d\rho_1(a,b) = H_t(1-iu) = 0 \qquad (t,u\in\mathbb{R}).$$

Fourier uniqueness yields $\rho_1 = 0$. Multiplication by e^b on each strip $0 \le b \le m$, followed by $m\to\infty$, yields $\rho = 0$. □

Theorem 66.2 (Stable-family classification)**.** *Fix $0 < \alpha \le 2$ and $-1 \le \beta \le 1$. Consider exchangeable sequences obtained by first drawing an arbitrary random pair $(A,B)\in\mathbb{R}\times[0,\infty)$ and, conditionally on this pair, drawing independent variables with law $K^{\alpha,\beta}_{A,B}$. The one-dimensional laws of all partial sums determine the law of (A,B), and hence the entire exchangeable law, if and only if*

$$(\alpha,\beta)\neq(1,0).$$

No independence between A and B, and no moment or exponential-moment condition on their joint law, is required.

More generally, let $\mathcal{I}\subset\mathbb{N}$ be a set of observed partial-sum lengths. The following are sufficient for determination:

$$\sum_{n\in\mathcal{I}} n^{-|1-\alpha|} = \infty, \qquad \alpha\neq 1; \tag{398}$$

$$\sum_{n\in\mathcal{I}} \frac{1}{1+(\log n)^2} = \infty, \qquad \alpha = 1,\quad \beta\neq 0. \tag{399}$$

The sparse sampling conditions are asserted only as sufficient conditions.

Proof. Let μ and ν be two mixing laws with identical observed partial-sum distributions, and write S_n for a partial sum. Conditional independence gives

$$\mathbb{E}e^{itS_n/n} = \int \exp\{ita - nb\psi_{\alpha,\beta}(t/n)\}\,d\mu(a,b). \tag{400}$$

The same formula holds with ν, and the two quantities agree for $n\in\mathcal{I}$.

Suppose first that $\alpha\neq 1$. For fixed $t\neq 0$, set

$$c_t = |t|^\alpha(1 - i\beta\tan(\pi\alpha/2)\operatorname{sgn} t).$$

The sampling points in (400) are $z_n(t) = c_t n^{1-\alpha}$. They are distinct and have positive real part. For fixed t,

$$\frac{\operatorname{Re} z_n(t)}{1+|z_n(t)|^2} = \frac{(\operatorname{Re} c_t)n^{1-\alpha}}{1+|c_t|^2 n^{2(1-\alpha)}} \asymp n^{-|1-\alpha|} \qquad (n\to\infty).$$

Consequently (398) allows application of Lemma 66.1 and gives $\mu=\nu$. For the full set $\mathcal{I}=\mathbb{N}$, this condition holds because $0<|1-\alpha|\le 1$.

Now suppose that $\alpha = 1$ and $\beta\neq 0$. Put $k = 2\beta/\pi$. Directly from (394),

$$z_n(t) = n\psi_{1,\beta}(t/n) = |t| + ikt\log|t| - ikt\log n.$$

These are distinct points on the vertical line $\operatorname{Re} z = |t| > 0$. The logarithmic location shift must be retained; omitting it would incorrectly identify this case with the symmetric Cauchy case. Since $kt\neq 0$,

$$\frac{\operatorname{Re} z_n(t)}{1+|z_n(t)|^2} = \frac{|t|}{1+t^2+k^2t^2(\log|t|-\log n)^2} \asymp \frac{1}{1+(\log n)^2}.$$

Condition (399) and the same lemma give $\mu = \nu$. In particular the condition holds on $\mathbb{N}$.

It remains to exhibit failure for $\alpha = 1$, $\beta = 0$. Fix $b, c > 0$ and let $Z_0, Z_1, Z_2, \dots$ and $Z_1', Z_2', \dots$ be independent standard symmetric Cauchy variables. Define

$$X_j = cZ_0 + bZ_j, \qquad Y_j = (b + c)Z_j'.$$

The first sequence has random location $A = cZ_0$ and fixed scale $B = b$; the second has fixed parameters $(A, B) = (0, b + c)$. For every $n \geq 1$, both partial sums are centered Cauchy variables with scale $n(b+c)$. Nevertheless $X_1 - X_2$ has Cauchy scale $2b$, whereas $Y_1 - Y_2$ has scale $2(b+c)$, so the exchangeable laws differ. All conditional distributions in this counterexample are nondegenerate. □

Remark 66.3 (What the classification says). For each fixed (α, β), the parameters (a, b) specify distinct laws in (395): the modulus of the characteristic function determines b, after which its phase determines a. The parameterization is continuous into the space of probability measures with its weak topology. Its injectivity and the Lusin–Souslin theorem give a Borel inverse on its Borel image. Thus the preceding theorem is also a statement about the goodness of the corresponding measurable class of directing distributions. It does not address mixtures over different values of α or β. The analytic ingredients and the stable scaling formulas are classical; the conclusion here is their application to identification from marginal partial-sum distributions.

67 Non-identifiability without finite differential barriers

The remark following Proposition 6 and the first question in the discussion of [55] ask whether absence of parameter differential structure suffices for identifiability of an infinite mixture. We give a negative answer for both classes explicitly defined there: finite-order parameter differential operators and finite shift, finite-order difference–differential parameter operators. The obstruction in the example is a missing global integral mode. No claim is made about exclusion of arbitrary infinite-order or nonlocal operators.

Let P_n be the ordinary Legendre polynomial with $P_n(1) = 1$, and set

$$L_n(\theta) = P_n(2\theta - 1), \qquad 0 \leq \theta \leq 1.$$

We use the standard elementary identities

$$L_0 = 1, \qquad L_1 = 2\theta - 1, \qquad |L_n(\theta)| \leq 1, \qquad \int_0^1 L_n(\theta) L_m(\theta)\, d\theta = \frac{\mathbf{1}_{\{n=m\}}}{2n+1}. \tag{401}$$

The sequence $(L_0, \dots, L_N)$ is a basis for polynomials of degree at most N.

Lemma 67.1 (A missing integral mode has no finite jet representation). *Let $y_1, \dots, y_J$ be distinct points of $[0, 1]$, let $r_1, \dots, r_J \in \mathbb{N}_0$, and define, on polynomials,*

$$T(p) = \sum_{j=1}^{J} \sum_{k=0}^{r_j} b_{j,k} p^{(k)}(y_j).$$

If $T(1) = 0$ and $T(L_n) = 0$ for every $n \geq 2$, then every coefficient $b_{j,k}$ is zero.

Proof. The Legendre expansion of a polynomial and (401) imply

$$T(p) = 3T(L_1) \int_0^1 p(\theta) L_1(\theta)\, d\theta \quad \text{for every polynomial } p. \tag{402}$$

Put $Q(\theta)=\prod_{j=1}^{J}(\theta-y_j)^{r_j+1}$. All jets entering T vanish on Q^2L_1, hence

$$0=T(Q^2L_1)=3T(L_1)\int_0^1 Q(\theta)^2L_1(\theta)^2\,d\theta.$$

The integral is strictly positive, so $T(L_1)=0$. Consequently T annihilates every polynomial. Hermite interpolation at the distinct points y_j provides, for each pair (j,k), a polynomial whose listed jets are zero except for $p^{(k)}(y_j)=1$. Applying T to it gives $b_{j,k}=0$. □

Theorem 67.2 (A strictly positive polynomial kernel with an invisible mode)**.** *Let $\Theta=[0,1]$, let $\mathcal{X}=\{\sigma n:\sigma\in\{-1,1\},\ n\ge 2\}\subset\mathbb{R}$ carry counting measure, and put $w_n=2^{1-n}$. Define*

$$f(\sigma n\mid\theta)=\frac{w_n}{2}\left(1+\frac{\sigma}{2}L_n(\theta)\right). \tag{403}$$

Then all the following hold.

1. *For every θ, $f(\cdot\mid\theta)$ is a strictly positive probability mass function. Each parameter section is a polynomial, and $\theta\mapsto f(\cdot\mid\theta)$ is injective.*

2. *For the mixture operator $A\nu=\int f(\cdot\mid\theta)\nu(d\theta)$ on finite signed measures,*

$$\ker A=\{bL_1(\theta)\,d\theta:b\in\mathbb{R}\}. \tag{404}$$

   In particular, for every $0<c<1$ the distinct probabilities

$$G_\pm(d\theta)=(1\pm cL_1(\theta))\,d\theta \tag{405}$$

   have strictly positive analytic densities and satisfy $AG_+=AG_-=(w_n/2)_{\sigma n\in\mathcal{X}}$.

3. *No nonzero finite combination of parameter evaluations and derivatives at finitely many distinct points annihilates every section $f(x\mid\cdot)$. Consequently the kernel satisfies no nontrivial parameter PDE of finite order on any nonempty open subset of $(0,1)$, and no nontrivial finite-shift, finite-order difference–differential parameter relation on any allowed open set. The latter statements hold even with arbitrary coefficient functions, provided the asserted identities hold pointwise.*

Proof. Since $\sum_{n\ge2}w_n=1$ and $|L_n|\le1$, the values in (403) lie in $[w_n/4,3w_n/4]$ and sum to one after summing the two signs and then n. Measurability is immediate.

Let ν be a finite signed measure. Adding the two values $(A\nu)(n)$ and $(A\nu)(-n)$ shows that $A\nu=0$ implies $\nu([0,1])=0$. Subtracting them then gives $\int L_n\,d\nu=0$ for every $n\ge2$. For any polynomial p, expansion in the Legendre basis therefore yields

$$\int_0^1 p\,d\nu=3\left(\int_0^1 L_1\,d\nu\right)\int_0^1 p(\theta)L_1(\theta)\,d\theta.$$

Polynomials are uniformly dense in $C([0,1])$, so finite-measure uniqueness gives $\nu=bL_1(\theta)d\theta$, with $b=3\int L_1\,d\nu$. Conversely every such measure is annihilated by A, by orthogonality. This proves (404) and the assertions about $G_\pm$. If $f(\cdot\mid\theta)=f(\cdot\mid\theta')$, then $\delta_\theta-\delta_{\theta'}$ is in this null space. A measure that is both purely atomic and absolutely continuous must be zero, so $\theta=\theta'$.

Now let T be a finite jet functional as in Lemma 67.1, and suppose $T(f(\sigma n\mid\cdot))=0$ for every sign and every $n\ge2$. Adding the two identities gives $T(1)=0$, and subtracting them gives $T(L_n)=0$. The lemma forces $T=0$ coefficientwise.

For a parameter differential operator $\mathcal{L}_\theta = \sum_{k=0}^{r} c_k(\theta)\partial_\theta^k$, fix any parameter value where the annihilation identity holds and apply the preceding conclusion with one node $y_1 = \theta$. All coefficients vanish there. For a difference–differential operator as defined in [55, Definition 4], fix θ and use the distinct nodes $y_j = \theta + v_j$ and coefficients $b_{j,k} = c_{j,k}(\theta)$. Every coefficient again vanishes. With counting measure the almost-everywhere statement in the sample variable excludes no sample points, so these pointwise arguments apply exactly to the operator definitions in that paper. □

Corollary 67.3 (Singular mixing measures remain identifiable)**.** *For the kernel in Theorem 67.2, every mixing probability singular with respect to Lebesgue measure is identifiable among all Borel mixing probabilities. This includes every finite or countably discrete mixing probability.*

Proof. If $AG = AH$ and G is singular, then $H = G + bL_1(\theta)d\theta$ by (404). The absolutely continuous part of the nonnegative measure H is $bL_1(\theta)d\theta$. Because L_1 has both signs on sets of positive Lebesgue measure, this can be nonnegative only when $b = 0$. □

Corollary 67.4 (Two observations sharing a latent parameter restore identification)**.** *The mixture model with kernel $f(x \mid \theta)f(y \mid \theta)$ on $\mathcal{X}^2$ is identifiable over all Borel probability measures on $[0, 1]$. The two observations are conditionally independent given the same latent parameter; two independent draws with separately redrawn latent parameters do not have this property.*

Proof. Let a finite signed measure ν annihilate the two-observation kernel. Marginalizing over y shows that $A\nu = 0$, so $\nu = bL_1(\theta)d\theta$. The marginalization is legitimate because the sums of absolute values are bounded by $\|\nu\|_{\mathrm{TV}}$. Take the signed sum over the four pairs $(x, y) = (\sigma 2, \tau 3)$:

$$\sum_{\sigma,\tau\in\{-1,1\}} \sigma\tau f(\sigma 2 \mid \theta)f(\tau 3 \mid \theta) = \frac{w_2 w_3}{4} L_2(\theta)L_3(\theta).$$

It follows that $b\int_0^1 L_1L_2L_3 = 0$. The Legendre recurrence gives $L_1L_2 = (3/5)L_3 + (2/5)L_1$, so

$$\int_0^1 L_1L_2L_3\, d\theta = \frac{3}{35} > 0.$$

Hence $b = 0$ and $\nu = 0$. □

Remark 67.5 (Exact separation and the scope of the counterexample). For (405), direct integration of their distribution functions gives $W_1(G_+, G_-) = c/3$, while their induced observation laws coincide. Thus the failure is exact, despite distinct component distributions, positive smooth mixing densities, and absence of both finite operator classes. It answers the general sufficiency question negatively for the two precisely specified differential structures in [55]. It does not contradict their theorems, which assert sufficiency of such structures for non-identifiability. An arbitrary nonlocal obstruction is present: integration against $L_1(\theta)d\theta$ annihilates the kernel by construction.

Corollary 67.6 (A jointly entire continuous-density counterexample)**.** *Let both the sample space and the parameter space be $[0, 1]$, with Lebesgue measure as the dominating sample measure, and define*

$$k(x \mid \theta) = 1 + \sum_{n=2}^{\infty} \frac{L_n(x)L_n(\theta)}{n!}. \tag{406}$$

This is a probability density in x for every θ, with $k(x \mid \theta) \geq 3 - e > 0$ on $[0, 1]^2$, and extends to an entire function of the two complex variables (x, θ). Its mixture operator has exactly the null space $\{bL_1(\theta)d\theta : b \in \mathbb{R}\}$. Its component distributions are distinct, every finite family of parameter jets at

distinct parameter points is linearly independent as a family of functions of x, and it admits neither of the finite differential operator classes excluded in Theorem 67.2. The probabilities $G_{\pm}$ in (405) both give the uniform sample density. Every singular mixing probability remains identifiable, and two conditionally independent observations sharing one latent parameter restore identifiability over all mixing probabilities.

Proof. On the real unit square, absolute uniform convergence and $\sum_{n\ge 2} 1/n! = e-2$ give the stated lower bound. Orthogonality allows termwise integration in x and gives integral one. For completeness, the elementary polynomial formula

$$P_n(z) = 2^{-n} \sum_{j=0}^{n} \binom{n}{j}^2 (z-1)^{n-j}(z+1)^j$$

shows that on every complex disc $|z| \le R$ one has $|P_n(z)| \le [2(R+1)]^n$, using $\sum_j \binom{n}{j}^2 = \binom{2n}{n} \le 4^n$. After replacing z by $2z-1$, this gives an exponential-in-n bound for L_n on every compact complex set. The factorial denominator therefore makes (406) locally uniformly convergent on $\mathbb{C}^2$. It defines an entire function, and the series can be differentiated termwise to every fixed order, locally uniformly. These observations justify all subsequent coefficient extraction and parameter differentiation.

If $B\nu(x) = \int k(x \mid \theta)\nu(d\theta)$, then

$$\int_0^1 B\nu(x)\,dx = \nu([0,1]), \qquad \int_0^1 B\nu(x)L_m(x)\,dx = \frac{\int_0^1 L_m(\theta)\nu(d\theta)}{m!(2m+1)} \quad (m \ge 2). \tag{407}$$

Thus $B\nu = 0$ if and only if ν has zero mass and zero Legendre moments of degrees $m \ge 2$. The proof of (404) applies without change, and also proves distinctness of component distributions, the two uniform mixtures, and identifiability of singular mixing probabilities.

Let T be any finite jet functional as in Lemma 67.1. If $T(k(x \mid \cdot)) = 0$ for almost every x, integrate against 1 and against $L_m(x)$ in (407). Parameter differentiation commutes with these integrals, so $T(1) = T(L_m) = 0$ for all $m \ge 2$. The lemma forces every coefficient of T to vanish. In particular every finite family of parameter jets is linearly independent. Applying this argument at each fixed parameter point excludes both finite differential and finite difference–differential operators, exactly as in the discrete example.

Finally suppose a finite signed measure ν annihilates the two-observation kernel $k(x \mid \theta)k(y \mid \theta)$. Marginalization gives $\nu = bL_1(\theta)d\theta$. Integrating the joint mixture against $L_2(x)L_3(y)$ and using (407) gives

$$0 = \frac{1}{2!\,5\,3!\,7} \int_0^1 L_2(\theta)L_3(\theta)\nu(d\theta) = \frac{b}{2!\,5\,3!\,7}\frac{3}{35}.$$

Therefore $b = 0$ and $\nu = 0$. □

68 Finite observational defects and replicated observations

The one-dimensional invisible space above is a special case of a general finite-dimensional phenomenon. Two replicated observations force every remaining invisible measure onto one fixed finite set.

Let Θ and $\mathcal{X}$ be compact metric spaces, and let $\theta \mapsto P_\theta$ be a weakly continuous family of Borel probability measures on $\mathcal{X}$. Write $\mathcal{M}(\Theta)$ for the real vector space of finite signed Borel measures on Θ. The observation operator with m conditionally independent observations sharing the same latent parameter is

$$A_m\mu = \int_\Theta P_\theta^{\otimes m}\,d\mu(\theta), \qquad K_m = \ker A_m.$$

Thus A_1 describes one observation from each independently selected latent parameter, whereas A_m describes a group of m observations from one common parameter. These are different statistical experiments.

For $f \in C(\mathcal{X};\mathbb{R})$, put

$$Tf(\theta) = \int f\, dP_\theta, \qquad V = \overline{T(C(\mathcal{X};\mathbb{R}))}^{\|\cdot\|_\infty} \subset C(\Theta;\mathbb{R}).$$

The function 1 belongs to V. The family P_θ is injectively parameterized if and only if V separates the points of Θ. Products of functions of the separate observation coordinates are dense in $C(\mathcal{X}^m)$, by the real Stone–Weierstrass theorem. Consequently,

$$K_m = \left\{ \nu : \int v_1 \cdots v_m \, d\nu = 0 \text{ for all } v_1, \dots, v_m \in V \right\}. \tag{408}$$

In particular, $K_1 = V^\perp$. If $d = \dim K_1 < \infty$, then V has codimension d in $C(\Theta;\mathbb{R})$: the dual of $C(\Theta)/V$ is $V^\perp$, and a normed space whose dual is finite dimensional is itself finite dimensional, with the same dimension.

Theorem 68.1 (Two observations reduce every finite defect to atoms)**.** *Suppose $d = \dim K_1 < \infty$. Every $\nu \in K_2$ is supported on at most $2d$ points. More precisely, its positive and negative parts are each supported on at most d points. There is one fixed set $E \subset \Theta$ of at most $2d$ points supporting every measure in K_2. Therefore, if K_1 contains no nonzero finitely supported signed measure, then $K_2 = \{0\}$.*

In particular, two observations identify every mixing probability measure whenever all measures in K_1 are absolutely continuous with respect to a fixed atomless measure. The same conclusion follows if every finite subfamily of $\{P_\theta : \theta \in \Theta\}$ is linearly independent.

Proof. For $\nu \in K_2$, consider the multiplication operator

$$M_\nu : C(\Theta;\mathbb{R}) \longrightarrow \mathcal{M}(\Theta), \qquad M_\nu(g) = g\nu.$$

Equation (408) shows that $M_\nu(V) \subset K_1$. Since V has codimension d, a vector space complement F of dimension d gives

$$\operatorname{rank} M_\nu \le \dim M_\nu(V) + \dim M_\nu(F) \le 2d.$$

If the support of $|\nu|$ contained r distinct points, choose pairwise disjoint neighborhoods of those points and continuous functions g_j supported in those neighborhoods, each nonzero on a set of positive $|\nu|$-measure. The measures $g_j\nu$ would be nonzero and have disjoint supports, hence be linearly independent. Thus a multiplication operator of rank at most $2d$ forces ν to have at most $2d$ atoms.

For the more precise assertion, let $\nu = \sum_{i=1}^{r} c_i \delta_{\theta_i}$ with all $c_i \neq 0$, and let p and q be the numbers of positive and negative coefficients. The restriction space

$$W = \{(v(\theta_1), \dots, v(\theta_r)) : v \in V\} \subset \mathbb{R}^r$$

has codimension at most d: its annihilator identifies with the subspace of K_1 supported on these r points. Moreover,

$$\sum_{i=1}^{r} c_i w_i z_i = 0 \qquad (w, z \in W).$$

A totally isotropic subspace for a nondegenerate real form of signature (p, q) has dimension at most $\min(p, q)$. This follows, for example, because its projections onto the positive and negative coordinate spaces are both injective. Hence

$$p + q - d \le \dim W \le \min(p, q), \qquad \max(p, q) \le d.$$

Since $1 \in V$, we have $K_2 \subset K_1$. To obtain a fixed set, choose a basis of the finite-dimensional space K_2 and let E be the union of its members' finite supports. A linear combination of the basis vectors whose coefficients avoid finitely many proper hyperplanes has nonzero mass at every point of E. Its support is therefore E, so the already established bound gives $|E| \le 2d$. Finally, if K_1 contains no nonzero finitely supported measure, the first part forces $K_2 = \{0\}$. The two stated sufficient conditions each exclude such a measure. □

The theorem concerns exact identifiability, not uniform statistical stability. It also states more than the special case $K_2 = 0$: for an arbitrary finite-dimensional defect, every non-atomic part of the mixing measure is determined by the two-observation law, and any remaining difference is finite atomic.

Theorem 68.2 (A finite bound on replication)**.** *Suppose the component map $\theta \mapsto P_\theta$ is injective and $d = \dim K_1 < \infty$. Then $K_{d+1} = \{0\}$. More precisely, until the nullspace becomes zero, each additional shared-parameter observation strictly decreases its dimension. The bound $d + 1$ cannot be decreased in general.*

Proof. Let U_m be the uniform closure of the span of products of m elements of V. Since $1 \in V$, the spaces U_m increase, their annihilators K_m decrease, and $\dim K_m \le d$. If $K_m = K_{m+1}$, the Hahn–Banach theorem gives $U_m = U_{m+1}$. For $v \in V$ and $u \in U_m$, approximation by finite linear combinations of products shows that $vu \in U_{m+1} = U_m$. Thus U_m contains the unital algebra generated by V. This algebra separates points, so Stone–Weierstrass gives $U_m = C(\Theta)$ and $K_m = \{0\}$. Every nonzero step therefore decreases a nonnegative integer dimension, giving $K_{d+1} = \{0\}$.

For sharpness, take $d + 2$ distinct parameters $0 < t_1 < \cdots < t_{d+2} < 1$ and $P_{t_i} = \text{Bernoulli}(t_i)$. The one-observation operator has rank 2 and nullity d. The m-observation law determines exactly the moments of the mixing measure through degree m. The corresponding Vandermonde matrix has rank $\min(m + 1, d + 2)$. Hence $K_d \neq 0$ when $d \ge 1$, whereas $K_{d+1} = 0$. Every nonzero null measure has total mass zero, so its normalized Jordan parts are distinct mixing probability measures with the same observation law. This proves sharpness for probability mixtures as well as for signed measures. When $d = 0$, one observation already identifies the mixture. □

69 Analytic kernels with an arbitrary finite polynomial defect

Continue to use the shifted Legendre polynomials and normalization in (401). We now prescribe an arbitrary finite collection of missing modes, rather than only L_1. Fix any nonempty finite set $S \subset \{1, 2, \ldots\}$ and put $d = |S|$, $M = \max S$, and $a_n = 2^{-n^2-2}$. Consider the probability density

$$k_S(x, \theta) = 1 + \sum_{\substack{n \ge 1 \\ n \notin S}} a_n L_n(x) L_n(\theta), \qquad 0 \le x, \theta \le 1. \tag{409}$$

The series converges locally uniformly on $\mathbb{C}^2$, since Legendre polynomials grow at most exponentially in n on each compact set, while a_n decays quadratically exponentially. It therefore defines an entire

function of the two variables. On the real unit square,

$$\frac{3}{4} < k_S(x,\theta) < \frac{5}{4}, \qquad \int_0^1 k_S(x,\theta)\,dx = 1.$$

We use Lebesgue measure on $[0,1]$ as the observation dominating measure.

Theorem 69.1 (An exactly prescribed global ambiguity)**.** *For the kernel* (409)*,*

$$K_1 = \{h(\theta)\,d\theta : h \in \operatorname{span}\{L_s : s \in S\}\}\,. \tag{410}$$

Consequently:

1. *The defect has exactly dimension d; positive analytic mixing densities can be nonidentifiable from one observation.*
2. *The singular part, relative to Lebesgue measure, of an arbitrary mixing measure is determined by its one-observation law. In particular, the operator is injective on all singular mixing measures, including all discrete and all singular continuous mixing measures.*
3. *Every finite set of component densities is linearly independent.*
4. *Two shared-parameter observations identify every mixing probability measure.*

Proof. For $\mu \in \mathcal{M}([0,1])$, termwise integration and orthogonality give

$$\int A_1\mu(x)\,dx = \mu([0,1]), \qquad \int L_n(x)A_1\mu(x)\,dx = \frac{a_n}{2n+1}\int L_n(\theta)\,d\mu(\theta) \quad (n \geq 1,\ n \notin S).$$

Thus $A_1\mu = 0$ implies that the mass and all moments indexed outside S vanish. Set

$$h(\theta) = \sum_{s\in S}(2s+1)\left(\int L_s\,d\mu\right)L_s(\theta).$$

The measure $\mu - h(\theta)d\theta$ annihilates every Legendre polynomial, hence every polynomial, and hence every continuous function by Weierstrass. It is zero. Conversely, every measure on the right side of (410) is annihilated by the kernel.

Small nonzero polynomial perturbations $1+h$ and $1-h$ of the uniform mixing density establish the first conclusion. Equality of two one-observation laws forces the difference of the mixing measures to be absolutely continuous, giving the second conclusion by uniqueness of the Lebesgue decomposition. No nonzero finitely supported measure is absolutely continuous with respect to Lebesgue measure, giving the third conclusion. The fourth follows from Theorem 68.1. □

Theorem 69.2 (Every finite parameter-jet test is independent)**.** *Let $\theta_1,\dots,\theta_q$ be distinct points of $[0,1]$ and let $r_1,\dots,r_q$ be nonnegative integers. The functions of x*

$$\left\{\partial_\theta^j k_S(x,\theta_i) : 1 \leq i \leq q,\ 0 \leq j \leq r_i\right\}$$

are linearly independent. In particular, no nonzero finite-order parameter differential operator, or finite sum of shifted finite-order parameter differential operators, annihilates this kernel wherever its evaluation points are admissible.

Proof. It suffices to treat real coefficients, since the kernel is real. Suppose a finite jet functional

$$T(p) = \sum_{i=1}^{q} \sum_{j=0}^{r_i} c_{ij} p^{(j)}(\theta_i)$$

annihilates $k_S(x, \cdot)$ for every x. Orthogonality in x, with termwise differentiation justified by local uniform holomorphic convergence, gives $T(1) = 0$ and $T(L_n) = 0$ for $n \notin S$. Hence for every polynomial p,

$$T(p) = \int_0^1 p(t)h(t)\,dt, \qquad h(t) = \sum_{s \in S} (2s+1) T(L_s) L_s(t).$$

Set $Q(t) = \prod_{i=1}^{q} (t - \theta_i)^{r_i+1}$. Every prescribed jet of $Q^2 h$ vanishes, so

$$0 = T(Q^2 h) = \int_0^1 Q(t)^2 h(t)^2\,dt.$$

Thus $h = 0$, and T annihilates all polynomials. Hermite interpolation at the distinct nodes θ_i then gives $c_{ij} = 0$ for all i, j. For complex coefficients, apply the real argument separately to their real and imaginary parts. For a shifted differential operator, fix the base parameter, collect coincident evaluation nodes, and apply the same finite-jet assertion; only the zero functional can remain. □

The assertion concerns finite differential and finite shift-differential operators. The kernel does have the nonlocal integral annihilators $h(t)dt$ in (410); no absence of nonlocal or infinite-order obstructions is claimed.

69.1 Exactly d scalar correlations repair the defect

The two-observation conclusion admits an explicit finite reconstruction. Let

$$u_r = \int_0^1 L_r(\theta)\,d\mu(\theta).$$

The one-observation law gives $u_0 = 1$ and every u_r for $r \notin S$. Fix any integer $N > M$. For $s \in S$, the joint law of two observations (X_1, X_2) sharing the same latent parameter gives

$$b_s = \frac{(2N+1)(2N+2s+1)}{a_N a_{N+s}} \mathbb{E}[L_N(X_1) L_{N+s}(X_2)] = \int_0^1 L_N(\theta) L_{N+s}(\theta)\,d\mu(\theta). \tag{411}$$

The scaling follows from $\mathbb{E}[L_n(X_i) \mid \theta] = a_n L_n(\theta)/(2n+1)$ for $n \notin S$.

Proposition 69.3 (A minimal number of scalar repairs)**.** *The one-observation law together with the d correlations in (411) determines μ uniquely. Among procedures that add expectations of a fixed finite list of bounded two-observation statistics to the one-observation law, fewer than d scalar expectations cannot identify every mixing probability measure.*

Proof. Write the finite polynomial identity

$$L_N L_{N+s} = \sum_{r=0}^{2N+s} c_{sr} L_r, \qquad c_{sr} = (2r+1) \int_0^1 L_N(t) L_{N+s}(t) L_r(t)\,dt.$$

If $r < s$, the polynomial $L_N L_r$ has degree less than $N + s$, so $c_{sr} = 0$. Let $\kappa_j = \binom{2j}{j}$ be the leading coefficient of L_j. Comparing leading coefficients in $L_N L_s$ and using orthogonality yields

$$c_{ss} = \frac{(2s+1)\kappa_N \kappa_s}{(2N+2s+1)\kappa_{N+s}} > 0.$$

Subtracting the already known terms $c_{sr}u_r$ with $r \notin S$ from b_s leaves a triangular linear system for the d missing moments u_s. It is solved by descending through S, since its diagonal entries are nonzero. All Legendre moments are then known, and polynomial density on $[0,1]$ determines μ.

For minimality, consider the open family of mixing densities

$$p_c(\theta) = 1 + \sum_{s\in S} c_s L_s(\theta), \qquad \sum_{s\in S} |c_s| < \frac{1}{2}.$$

All these densities have the same one-observation law. The expectation of any fixed bounded statistic of the two observations is an affine function of $c \in \mathbb{R}^d$, by linearity of mixing. Fewer than d such expectations define an affine map with nontrivial linear kernel, which is not injective on this open parameter set. This proves the claimed lower bound for scalar expectation repairs. □

The reconstruction is exact but may be ill-conditioned: the factors $1/(a_N a_{N+s})$ can be very large. No finite-sample accuracy guarantee is implied by this identifiability statement.

70 Arbitrary replication thresholds on the analytic circle

When the one-observation nullspace is infinite dimensional, the sharp finite-defect bound does not apply. The following construction realizes every threshold while keeping all finite parameter jets independent. Its Fourier mechanism differs from the interval construction in Section 72: here the indistinguishable mixing densities themselves can be analytic.

Let $\mathbb{T} = \mathbb{R}/(2\pi\mathbb{Z})$ and write $m(d\theta) = d\theta/(2\pi)$ for its Haar probability measure. The parameter space in this section is the analytic circle itself: the representatives 0 and 2π denote the same parameter, not two distinct points of a closed interval. Both the observations and the parameter take values in $\mathbb{T}$.

For a conditional density $k(x \mid \theta)$ and a Borel probability measure μ on $\mathbb{T}$, let

$$M_q^\mu(dx_1 \cdots dx_q) = \left[\int_{\mathbb{T}} \prod_{j=1}^{q} k(x_j \mid \theta)\, \mu(d\theta)\right] m(dx_1)\cdots m(dx_q).$$

Thus the q observations share one latent parameter and are conditionally independent. Identifiability at order q means injectivity of $\mu \mapsto M_q^\mu$ on all Borel probability measures.

Theorem 70.1 (Every finite replication threshold, and an infinite one)**.** *For every $r \in \{1,2,3,\dots\} \cup \{\infty\}$ there is a strictly positive conditional density k_r on $\mathbb{T}\times\mathbb{T}$ with the following properties.*

1. *It extends to an entire, 2π-periodic function of each complex variable jointly, and its real values are bounded below by a positive absolute constant.*
2. *For any distinct $\theta_1,\dots,\theta_J \in \mathbb{T}$ and any nonnegative integers $d_1,\dots,d_J$, the functions*

$$\left\{\partial_\theta^\ell k_r(\,\cdot \mid \theta_j) : 1 \le j \le J,\ 0 \le \ell \le d_j\right\}$$

   are linearly independent over $\mathbb{C}$.
3. *If $r < \infty$, identifiability at order q holds exactly when $q \ge r$. If $r = \infty$, it fails for every finite q, although the collection $(M_q^\mu)_{q\ge1}$ identifies μ.*

4. *Every finitely supported mixing measure is identifiable among all Borel probability measures from one observation. At every order where unrestricted identifiability fails, the failure can be witnessed by two distinct, strictly positive analytic mixing densities. For the assertion about unrestricted competitors, see Corollary 70.6.*

In particular, absence of finite-order differential or finite-shift differential annihilators supplies no finite universal bound on the number of shared-latent observations required for unrestricted mixture identification.

We prove the theorem using a Fourier criterion and an arithmetic independence lemma. For a symmetric subset $S \subseteq \mathbb{Z}$ containing zero, put

$$k_S(x \mid \theta) = 1 + \sum_{n \in S \cap \mathbb{N}} e^{-n^2} \cos(n(x-\theta)). \tag{412}$$

Here $\mathbb{N} = \{1, 2, \dots\}$. The series and all its derivatives converge uniformly on compact subsets of $\mathbb{C}^2$, since on every such compact set the summands are bounded by e^{-n^2+Cn} times a polynomial in n. Its integral in x is one, and

$$k_S(x \mid \theta) \geq 1 - \sum_{n \geq 1} e^{-n^2} \geq 1 - \frac{1}{e-1} > 0.$$

For $q \geq 1$ let $qS = \{s_1 + \cdots + s_q : s_1, \dots, s_q \in S\}$. The inclusion $0 \in S$ makes these sets increasing in q.

Lemma 70.2 (Replication is addition of spectral sets)**.** *For the kernel (412), $M_q^\mu = M_q^\nu$ if and only if $\widehat{\mu}(n) = \widehat{\nu}(n)$ for every $n \in qS$, where $\widehat{\mu}(n) = \int e^{-in\theta}\, \mu(d\theta)$. Consequently, identifiability at order q holds if and only if $qS = \mathbb{Z}$.*

Proof. The Fourier coefficient of $k_S(\cdot \mid \theta)$ at n is $a_n e^{-in\theta}$, where $a_0 = 1$, $a_n = e^{-n^2}/2$ for $n \in S \setminus \{0\}$, and $a_n = 0$ otherwise. Thus the Fourier coefficient of M_q^μ at $(n_1, \dots, n_q) \in \mathbb{Z}^q$ is

$$\left(\prod_{j=1}^{q} a_{n_j} \right) \widehat{\mu}(n_1 + \cdots + n_q).$$

Uniqueness of Fourier coefficients for finite measures proves the first assertion and the sufficient condition. If $b \notin qS$, then $b \neq 0$ and symmetry also gives $-b \notin qS$. For any $0 < c < 1$ the densities $1 + c\cos(b\theta)$ and $1 - c\cos(b\theta)$ relative to m are distinct and strictly positive, but their Fourier coefficients agree on qS. □

Lemma 70.3 (Arithmetic richness implies independence of all finite jets)**.** *Suppose that $S \cap \mathbb{N}$ meets every residue class modulo every positive integer infinitely often. Then k_S has the independence property in part (2) of Theorem 70.1.*

Proof. Suppose a linear combination of the displayed functions vanishes. Taking its Fourier coefficients at $n \in S \cap \mathbb{N}$ gives

$$E(n) = \sum_{j=1}^{J} P_j(n) \lambda_j^n = 0, \qquad \lambda_j = e^{-i\theta_j}, \quad P_j(n) = \sum_{\ell=0}^{d_j} c_{j\ell} (-in)^\ell.$$

The numbers λ_j are distinct and nonzero. The sequence $E(n)$ satisfies a constant-coefficient linear recurrence over $\mathbb{C}$. By the Skolem–Mahler–Lech theorem, its zero set in the nonnegative integers is a

finite union of arithmetic progressions together with a finite set; see [57, Theorem 1.3]. In particular its zero set is eventually periodic, with some period $Q \geq 1$. If any residue class modulo Q failed eventually to consist of zeros, its infinitely many intersections with $S \cap \mathbb{N}$ would give a contradiction. Hence $E(n) = 0$ for all sufficiently large n.

For completeness, an exponential polynomial with distinct nonzero bases cannot be eventually zero unless every P_j is zero. If $P_j \neq 0$, apply to the sequence the shift operator $\prod_{h \neq j} (\mathcal{E} - \lambda_h)^{d_h+1}$, where $(\mathcal{E}u)(n) = u(n+1)$. It kills every term except the jth and leaves $\lambda_j^n Q_j(n)$ for a polynomial Q_j of the same degree as P_j, with nonzero leading coefficient. Such a sequence is not eventually zero. This contradiction proves $P_j = 0$ for every j, and hence all the original coefficients vanish. □

This lemma also excludes operators with finitely many shifted parameter evaluations and derivatives, even if their coefficients vary arbitrarily with the base point: at each base point, combine coincident nodes and apply the lemma to the resulting distinct nodes. Only the zero functional remains. It does not exclude infinite-order or genuinely nonlocal annihilators.

70.1 A finite threshold prescribed in advance

Fix $r \geq 2$ and choose

$$\alpha_r = \frac{1}{2} - \frac{\sqrt{2}}{10r}, \qquad \delta_r = \frac{2r-1}{4r(r-1)}, \qquad S_r = \{n \in \mathbb{Z} : \|n\alpha_r\|_{\mathbb{R}/\mathbb{Z}} < \delta_r\}. \tag{413}$$

These choices satisfy

$$r\delta_r > \frac{1}{2}, \qquad (r-1)\delta_r = \frac{1}{2} - \frac{1}{4r} < \alpha_r < \frac{1}{2}. \tag{414}$$

Lemma 70.4. *The set S_r meets every residue class modulo every positive integer infinitely often, and*

$$rS_r = \mathbb{Z}, \qquad 1 \notin (r-1)S_r.$$

Proof. For a fixed residue class a modulo Q, the numbers $(a + Qj)\alpha_r$ are dense modulo one as $j \to \infty$, because $Q\alpha_r$ is irrational. They therefore enter the open interval $(-\delta_r, \delta_r)$ modulo one infinitely often. The triangle inequality for distance modulo one and (414) give $1 \notin (r-1)S_r$.

Now fix $b \in \mathbb{Z}$, and let $y \in [-1/2, 1/2]$ represent $b\alpha_r$ modulo one. We have $|y/r| < \delta_r$. Choose integers $s_1, \ldots, s_{r-1}$ such that their respective representatives $y_1, \ldots, y_{r-1}$ modulo one are sufficiently close to y/r. Irrationality supplies these integers, and the choices can be made so that all $|y_j| < \delta_r$ and $|y - \sum_{j<r} y_j| < \delta_r$. Set $s_r = b - \sum_{j<r} s_j$. Its phase modulo one is represented by $y - \sum_{j<r} y_j$, so $s_r \in S_r$ as well. This represents $b \in rS_r$. □

Taking $k_r = k_{S_r}$ proves all the assertions for finite $r \geq 2$. The same pair of mixing densities $1 \pm c\cos\theta$ witnesses nonidentifiability at every order $q < r$. For $r = 1$, use $S_1 = \mathbb{Z}$.

70.2 An analytic family with no finite replication threshold

Enumerate the pairs (Q, a) with $Q \geq 1$ and $0 \leq a < Q$ in a sequence $(Q_j, a_j)_{j \geq 2}$ in which each pair appears infinitely often. Set $n_1 = 1$ and recursively choose positive integers n_j such that

$$n_j \equiv a_j \pmod{Q_j}, \qquad n_j > 2^j \sum_{i<j} i n_i \quad (j \geq 2). \tag{415}$$

Each choice is possible in the prescribed infinite arithmetic progression. Set

$$S_\infty = \{0\} \cup \bigcup_{j \geq 1} \{\pm n_j, \pm 2n_j, \ldots, \pm j n_j\}.$$

The positive blocks are disjoint, by the growth condition. Since each n_j is present, this set meets every residue class infinitely often. The finite blocks will also ensure determination of finite mixing laws among unrestricted competitors.

We claim that $qS_\infty \neq \mathbb{Z}$ for each fixed $q \geq 1$. Choose $K \geq 2$ large enough that $2^j > 4q$ for every $j \geq K$ and $n_K > 4$. Put $b = \lfloor n_K/2 \rfloor$. In any representation

$$z = s_1 + \cdots + s_q = \sum_j c_j n_j, \qquad s_h \in S_\infty,$$

combine all terms from the same block and cancel their integer coefficients. The resulting coefficients satisfy $|c_j| \leq qj$. If every nonzero coefficient has index below K, then $|z| \leq q \sum_{j<K} j n_j < n_K/4 < b$. Otherwise let $J \geq K$ be the largest index with nonzero coefficient. Then

$$|z| \geq n_J - q \sum_{j<J} j n_j > \frac{3}{4} n_J \geq \frac{3}{4} n_K > b.$$

Thus $b \notin qS_\infty$. Lemma 70.2 gives nonidentifiability at every finite order, and its cosine perturbations give the asserted positive analytic witnesses (which may depend on q).

On the other hand, $1 \in S_\infty$, so every integer belongs to qS_∞ for some finite q. Equality of M_q^μ and M_q^ν for every q therefore gives equality of all Fourier coefficients of μ and ν, and hence $\mu = \nu$. This is also identification from the law of the entire conditionally i.i.d. sequence.

Finally, for any of the constructed kernels, equality of two finite mixtures is a finite linear dependence among the functions $k_S(\cdot \mid \theta_j)$. Lemma 70.3 with all derivative orders zero shows that their signed atomic difference vanishes. Corollary 70.6 strengthens this to identification among unrestricted competitors and completes the proof of Theorem 70.1.

Remark 70.5 (What the threshold measures). The theorem concerns exact probability laws of grouped observations, not the number of independently sampled groups needed for statistical estimation. In the infinite-threshold construction no single pair of distinct mixing measures is indistinguishable at every order: each finite order has a nontrivial kernel, while the intersection of all these kernels is zero. All finite-node derivative tests nevertheless remain independent already at order one.

Corollary 70.6 (Finite support remains determined without a support bound)**.** *For each $r \in \mathbb{N} \cup \{\infty\}$, let k_r be the kernel constructed above. If μ has finite support and ν is any Borel probability measure on $\mathbb{T}$, then $M_1^\mu = M_1^\nu$ implies $\mu = \nu$.*

Proof. The assertion is immediate for $r = 1$. Otherwise, write $\operatorname{supp} \mu = \{\theta_1, \ldots, \theta_J\}$. For finite $r \geq 2$, choose a positive integer n with $\|n\alpha_r\| < \delta_r/J$; density of the irrational rotation supplies one. The triangle inequality gives $\{-Jn, -(J-1)n, \ldots, Jn\} \subseteq S_r$. For $r = \infty$, choose any block index $h \geq J$ and set $n = n_h$; the same progression then lies in S_∞ by construction. The nonnegative trigonometric polynomial

$$p(\theta) = \prod_{j=1}^{J} |e^{in\theta} - e^{in\theta_j}|^2$$

has its Fourier spectrum in that finite subset of S_r. Consequently equality of the one-view laws gives $\int p \, d\nu = \int p \, d\mu = 0$. Since $p \geq 0$, the measure ν is supported on the zero set of p, which consists of at most nJ points of the circle. The already proved independence of the component densities then identifies the two finite mixing measures. □

Remark 70.7 (Uniform analytic bounds across the entire hierarchy). The freedom in the threshold does not come from deteriorating smoothness or positivity constants. Every kernel above has the same positive lower bound $1 - 1/(e-1)$. On any complex set with $|\operatorname{Im} x| + |\operatorname{Im} \theta| \le C$, each mixed derivative of total order d is bounded by the common finite majorant

$$\mathbf{1}_{\{d=0\}} + \sum_{n \ge 1} n^d e^{-n^2 + Cn},$$

independently of the chosen finite or infinite threshold.

71 Localized invisible perturbations and exact moment coincidences

To construct unrestricted thresholds on an ordinary interval, we need smooth signed measures annihilating prescribed sparse powers. Ordinary additive smoothing need not preserve those moments. Multiplicative smoothing does preserve them, because moments factor under multiplication. This also resolves the specified prescribed-moment conjecture and permits an exact classification of coincidence sets.

For $\sigma \ge 1$ and an interval J, a smooth function belongs to $G^\sigma(J)$ if, for some constants $C, B > 0$,

$$\sup_{\theta \in J} |f^{(k)}(\theta)| \le C B^k (k!)^\sigma \qquad (k \ge 0).$$

On a closed interval we require smoothness up to its endpoints. For $J = \mathbb{R}$, the displayed bounds are global. In particular $G^1([1,2])$ consists of the restrictions of functions real analytic on a neighborhood of the interval. The constants may depend on the function. The distinction between this union of classes with varying constants and one fixed bounded class matters for statistical conclusions.

Theorem 71.1 (Localized smooth Müntz annihilators). *Let $0 < a < b$, and let $T \subset \mathbb{N}_0$ contain 0. The following conditions are equivalent:*

1. $\displaystyle\sum_{s \in T \setminus \{0\}} s^{-1} < \infty.$

2. *For every nonempty open interval I with $\overline{I} \subset (a,b)$ and every $\sigma > 1$, there exists a nonzero real $h \in G^\sigma(\mathbb{R})$ supported in I such that*

$$\int_a^b x^s h(x)\, dx = 0 \qquad (s \in T). \tag{416}$$

3. *There is a nonzero continuous real function h supported in $[a,b]$ satisfying (416).*

For each I and σ in the second assertion, the space of such annihilators is infinite dimensional. Consequently every strictly positive G^σ probability density g_0 on $[a,b]$ has distinct strictly positive G^σ probability perturbations $g_0 \pm \varepsilon h$ with the same T-moments, agreeing with g_0 outside I.

Proof. Assume the reciprocal sum converges, fix I, and choose a nondegenerate closed interval $[c,d] \subset I$. The classical Müntz theorem on an interval away from zero says that

$$V = \operatorname{span}\{t^s : s \in T\}$$

is not uniformly dense in $C([c,d];\mathbb{R})$ (this is immediate if T is finite); see, for example, [64, Theorem 3.2]. Hahn–Banach and the Riesz representation theorem therefore supply a nonzero finite real signed measure ν, supported in $[c,d]$, such that

$$\int t^s \, d\nu(t) = 0 \qquad (s \in T).$$

This includes $\nu([c,d]) = 0$, because $0 \in T$.

Choose $\delta \in (0,1)$ so small that $[c(1-\delta), d(1+\delta)] \subset I$. By Lemma 71.4, there exists a nonnegative, nonzero G^σ function φ supported in $[1-\delta, 1+\delta]$. Define its multiplicative smoothing of ν by

$$h(x) = \int_{[c,d]} t^{-1}\varphi(x/t)\, d\nu(t), \qquad x \in \mathbb{R}. \tag{417}$$

It is supported in $[c(1-\delta), d(1+\delta)]$, hence in I. Differentiation under the finite measure gives

$$\|h^{(k)}\|_\infty \le \|\nu\|_{\mathrm{TV}} c^{-k-1} \|\varphi^{(k)}\|_\infty \le C'(B/c)^k (k!)^\sigma,$$

so $h \in G^\sigma(\mathbb{R})$. For every real $\lambda \ge 0$, Fubini and $x = tu$ yield the factorization

$$\int_0^\infty x^\lambda h(x)\, dx = \left(\int_{[c,d]} t^\lambda \, d\nu(t) \right) \left(\int_0^\infty u^\lambda \varphi(u)\, du \right). \tag{418}$$

Thus all moments in T vanish.

It remains essential to check that smoothing has not killed ν. If every nonnegative integer moment of ν were zero, the Weierstrass approximation theorem would imply $\nu = 0$. Hence there is an integer $m \ge 0$ with $\int t^m \, d\nu(t) \ne 0$. The second factor in (418) at $\lambda = m$ is strictly positive, since $\varphi \ge 0$ is nonzero and supported on positive real numbers. Therefore $\int x^m h(x)\, dx \ne 0$, and h is nonzero. This proves the second assertion.

For necessity, if the reciprocal sum diverges, the same Müntz theorem makes V dense in $C[a,b]$. A continuous h annihilating V then annihilates every continuous function; taking that function to be h gives $\int_a^b h^2 = 0$, so $h = 0$. The second assertion plainly implies the third.

To obtain infinite dimensionality, apply the construction on infinitely many pairwise disjoint nonempty open subintervals of I with closures inside I. The resulting nonzero functions have disjoint supports and are linearly independent. Finally $0 \in T$ gives $\int h = 0$, and boundedness of h and the positive minimum of g_0 show that $g_0 \pm \varepsilon h$ have all the asserted properties for sufficiently small $\varepsilon > 0$. □

Corollary 71.2 (The prescribed-moment conjecture)**.** *Let $0 < a < b$ and let (n_k) be a strictly increasing sequence of positive integers with $\sum_k n_k^{-1} < \infty$. There exists a nonzero signed measure with a smooth density supported strictly inside (a,b) whose n_kth moments vanish for every k. Its total mass can also be required to vanish, its support can be localized in any prescribed interior interval, and its density can be chosen in any prescribed Gevrey class G^σ, $\sigma > 1$.*

Proof. Apply Theorem 71.1 to $T = \{0, n_1, n_2, \dots\}$. □

Corollary 71.3 (Strictly log-concave moment twins)**.** *Under the summability assumption in Theorem 71.1, for every $\sigma > 1$ there are distinct probability densities g_+ and g_- in $G^\sigma(\mathbb{R})$, each supported on $[a,b]$, positive on (a,b) and strictly log-concave there, with*

$$\int x^s g_+(x)\, dx = \int x^s g_-(x)\, dx \qquad (s \in T).$$

Their difference can be supported in any prescribed interior interval. Any finite list of additional nonnegative integer moments can also be required to agree.

Proof. Put $p = (\sigma - 1)^{-1}$ and normalize

$$g_0(x) = C \exp\bigl(-(x-a)^{-p} - (b-x)^{-p}\bigr), \quad a < x < b, \qquad g_0(x) = 0 \quad \text{otherwise.}$$

The proof of Lemma 71.4 shows that $g_0 \in G^\sigma(\mathbb{R})$. On (a, b),

$$(\log g_0)''(x) = -p(p+1)\bigl((x-a)^{-p-2} + (b-x)^{-p-2}\bigr) < 0.$$

Choose the nonzero h of Theorem 71.1 with compact support K in the prescribed interval. On K, g_0 has a positive minimum and $g_0 g_0'' - (g_0')^2$ has a strictly negative maximum. Consequently, for sufficiently small $\varepsilon > 0$, $g_\pm = g_0 \pm \varepsilon h$ remain positive on K and satisfy $g_\pm g_\pm'' - (g_\pm')^2 < 0$ there. Outside K they equal g_0. Thus they are strictly log-concave on (a, b), remain probability densities, and have the required common moments. Adding finitely many exponents to T preserves the convergent reciprocal sum. □

This proves the mathematical assertion of Conjecture 17 in Bürgin–Epperlein–Wirth [65], with stronger smoothness and localization conclusions. The proof combines classical Müntz non-density with the elementary moment factorization (418). A claim about historical priority requires checking subsequent literature independently.

Lemma 71.4 (Compact Gevrey functions)**.** *For every $\sigma > 1$ and $\delta > 0$ there is a nonnegative, nonzero $\varphi \in G^\sigma(\mathbb{R})$ supported in $[-\delta, \delta]$.*

Proof. Put $p = (\sigma - 1)^{-1}$ and $f(t) = e^{-t^{-p}}$ for $t > 0$, $f(t) = 0$ for $t \le 0$. Choose $\epsilon_p > 0$ small enough that, on the complex disk $|z - t| \le \epsilon_p t$ about any positive real t, the principal branch satisfies $\operatorname{Re}(z^{-p}) \ge c_p t^{-p}$ for a constant $c_p > 0$. Cauchy's estimate yields

$$|f^{(k)}(t)| \le k!(\epsilon_p t)^{-k} e^{-c_p t^{-p}}.$$

For $k \ge 1$, maximizing $t^{-k} e^{-c_p t^{-p}}$ and using $(k/e)^k \le k!$ shows

$$\sup_{t>0} |f^{(k)}(t)| \le B_p^k (k!)^{1+1/p}.$$

For each fixed k the preceding pointwise estimate tends to zero as $t \downarrow 0$, so extension by zero is smooth. Now take $\varphi(t) = f(t + \delta) f(\delta - t)$. It is positive on $(-\delta, \delta)$, supported in the required interval, and Leibniz's rule gives the claimed Gevrey estimates. □

Theorem 71.5 (Endpoint rigidity)**.** *Let $E \subset [0, \infty)$ be unbounded. If $h \in C^\infty([1, 2])$ satisfies*

$$\int_1^2 \theta^s h(\theta)\, d\theta = 0 \qquad (s \in E),$$

then $h^{(k)}(2) = 0$ for every $k \ge 0$. In particular, if h is real analytic on a neighborhood of $[1, 2]$, it is identically zero. The same conclusion holds in any linear smooth class on a neighborhood of $[1, 2]$ for which the Taylor jet at 2 determines the function.

Proof. If the endpoint jet is not zero, let m be its least nonzero order. Taylor's formula and the substitution $t = s(1 - \theta/2)$ give

$$\int_1^2 \theta^s h(\theta)\, d\theta \sim (-1)^m h^{(m)}(2) \frac{2^{s+m+1}}{s^{m+1}} \qquad (s \to +\infty). \tag{419}$$

Here is a direct justification. On the whole interval, $|h(\theta)| \le C(2-\theta)^m$, since all lower endpoint derivatives vanish. After multiplying the integral by $s^{m+1}2^{-s-m-1}$, the transformed integrand converges pointwise to $(-1)^m h^{(m)}(2)t^m e^{-t}/m!$ and is dominated by $Ct^m e^{-t}$; we used $(1-t/s)^s \le e^{-t}$ on the transformed domain $0 \le t \le s/2$. Dominated convergence proves (419). Its nonzero leading term excludes zeros on an unbounded set of positive exponents. Thus the endpoint jet vanishes. Analyticity, or the stated jet uniqueness property, now implies $h = 0$. □

Corollary 71.6 (A sharp regularity distinction)**.** *Suppose a grouped-observation model determines exactly the moments with exponents in qS at replication order q, where $S \subset \mathbb{N}_0$ is unbounded and contains 0. If $\sum_{s\in qS\setminus\{0\}} s^{-1} < \infty$, then its q-observation law is not injective even on strictly positive G^σ densities, for any $\sigma > 1$, and the ambiguity can be localized in any interior interval. Nevertheless its one-observation law is injective on the class of real-analytic mixing densities.*

Proof. Apply Theorem 71.1 with $T = qS$ for the smooth ambiguity. For two real-analytic densities with equal one-observation laws, their difference has vanishing moments on the unbounded set S; Theorem 71.5 forces that difference to vanish. □

The last injectivity statement compares two analytic densities. It does not say that an analytic mixing density is identifiable among arbitrary Borel competitors: even the uniform density has the invisible smooth perturbations supplied by Theorem 71.1. Likewise the endpoint rigidity statement permits nonzero smooth annihilators, but forces every such annihilator to have zero Taylor jet at the right endpoint. These statements concern the regularity of the unknown mixing law, whereas the observation kernels in the following constructions are entire throughout.

71.1 Exact prescribed coincidence sets

The preceding existence theorem can be strengthened: one can exclude all unrequested coincidences. This refinement uses the classical Clarkson–Erdős description of the closure of a non-dense integer Müntz space, rather than only its non-density.

Theorem 71.7 (Exact moment-zero patterns)**.** *Let $T \subset \mathbb{N}_0$ contain 0 and satisfy $\sum_{s\in T\setminus\{0\}} s^{-1} < \infty$. Fix $0 < a < b$, an open interval I with $\overline{I} \subset (a,b)$, and $\sigma > 1$. There is a real function $h \in G^\sigma(\mathbb{R})$ supported in I such that*

$$\int_a^b x^m h(x)\,dx = 0 \quad \Longleftrightarrow \quad m \in T \qquad (m \in \mathbb{N}_0).$$

Consequently the moment coincidence set of two distinct strictly log-concave, compactly supported G^σ probability densities can be prescribed to be exactly any such set T.

Proof. Choose a nondegenerate interval $[c,d] \subset I$ and let $V = \overline{\operatorname{span}\{x^s : s \in T\}}^{C[c,d]}$. For every $m \notin T$, one has $x^m \notin V$. For finite T this follows from independence of distinct monomials. For infinite T, the Clarkson–Erdős theorem, in the form of [64, Theorem 3.1], states that every element of V has a representation

$$f(x) = \sum_{s\in T} a_s x^s \qquad (c \le x < d).$$

The positive gap condition required by that theorem holds because the exponents are distinct integers. Convergence at each $x < d$ implies that this is a power series of radius at least d. If $f(x) = x^m$ on $[c,d]$, the identity theorem and uniqueness of power-series coefficients would give the coefficient 1 at degree m, impossible when $m \notin T$.

Enumerate $\mathbb{N}_0 \setminus T = \{m_1, m_2, \dots\}$. Hahn–Banach and Riesz now give real signed measures ν_j, supported on $[c, d]$, of total variation at most 1, satisfying

$$\int x^s \, d\nu_j(x) = 0 \quad (s \in T), \qquad \int x^{m_j} \, d\nu_j(x) \neq 0.$$

Fix one nonnegative, nonzero Gevrey function φ supported in a sufficiently small interval about 1, and apply (417) to each ν_j. The resulting functions h_j have supports in one common compact subset of I and satisfy common estimates

$$\|h_j^{(k)}\|_\infty \leq CB^k (k!)^\sigma \quad \text{for all } j, k.$$

They annihilate every moment in T, while the m_jth moment of h_j is nonzero by (418).

For $0 < t < 1$, set

$$h_t(x) = \sum_{j=1}^{\infty} 2^{-j} t^j h_j(x).$$

All derivative series converge uniformly, with common Gevrey bounds, and the support stays in I. For each $m \notin T$, the scalar function

$$H_m(t) = \int_a^b x^m h_t(x) \, dx$$

is analytic on $|t| < 2$ and is not identically zero: if $m = m_j$, its coefficient of t^j is the nonzero number $2^{-j} \int x^{m_j} h_j(x) \, dx$. Its real zeros in $(0, 1)$ therefore form a countable set. Avoiding the union of these zero sets over the countably many $m \notin T$ gives a parameter t_* for which h_{t_*} has exactly the asserted vanishing moments. In particular it is nonzero. The final statement follows by using this h_{t_*} in the compact strictly log-concave baseline perturbation of Corollary 71.3; multiplication by a nonzero constant does not change its moment-zero set. □

Corollary 71.8 (Classification of exact coincidence sets)**.** *For a set $T \subset \mathbb{N}_0$, the following are equivalent:*

1. *$0 \in T$ and $\sum_{s \in T \setminus \{0\}} s^{-1} < \infty$.*
2. *For every $0 < a < b$ and every $\sigma > 1$, there exist distinct probability densities $g_+, g_- \in G^\sigma(\mathbb{R})$, each with support $[a, b]$ and strictly log-concave on (a, b), whose integer moment coincidence set is exactly T.*
3. *There exist distinct probability measures, both supported in some compact interval contained in $(0, \infty)$, whose integer moment coincidence set is exactly T.*

In the second assertion the difference of the two densities can be supported in any prescribed interior interval.

Proof. Theorem 71.7 gives the first implication, and the second is immediate. For the converse, probability normalization forces $0 \in T$. If the reciprocal sum diverged, the Müntz theorem would force any two compactly supported positive-interval measures agreeing on those moments to coincide. This contradicts distinctness. □

71.2 Examples and the constructive scope of the classification

The exact coincidence classification is stronger than the existence of two distributions sharing some infinite collection of moments. For example, there are two distinct strictly log-concave smooth densities on any chosen positive interval that agree at precisely the orders $0, 1, 4, 9, 16, \dots$ and disagree at every other integer order. In contrast, distinct compactly supported laws cannot agree at all prime orders, since the sum of reciprocal primes diverges. The first example follows by taking $T = \{0\} \cup \{n^2 : n \geq 1\}$; the second is the classical completeness direction. The proof of the first is an existence construction, using Hahn–Banach and avoidance of countably many analytic zero sets; it is not a numerical algorithm for evaluating the densities.

72 Arbitrary replication thresholds on an ordinary interval

Throughout this section the observation space is $[0, 1]$ with Lebesgue measure and the parameter space is $[1, 2]$. The observation polynomials L_s retain the normalization in (401); the parameter functions will now be monomials. The endpoints of the parameter interval are distinct, and no periodic identification is used. For an infinite set $S \subseteq \mathbb{Z}_{\geq 0}$ containing 0, define

$$a_s = 2^{-s^2-s-2}, \qquad k_S(x \mid \theta) = 1 + \sum_{s \in S \setminus \{0\}} a_s L_s(x) \theta^s. \tag{420}$$

For a Borel probability measure μ on $[1, 2]$, let

$$A_{q,S}\mu(dx_1 \cdots dx_q) = \left\{ \int_1^2 \prod_{j=1}^{q} k_S(x_j \mid \theta)\, d\mu(\theta) \right\} dx_1 \cdots dx_q.$$

The same latent parameter is used for the q observations in a group. Set $qS = \{s_1 + \cdots + s_q : s_i \in S\}$. The inclusion $0 \in S$ means that moments accessible with fewer observations remain accessible with more.

Lemma 72.1 (Regularity and the exact observable moments)**.** *The function in* (420) *is jointly entire in* $(x, \theta) \in \mathbb{C}^2$*, integrates to one in* x*, and satisfies*

$$\frac{3}{4} < k_S(x \mid \theta) < \frac{5}{4} \quad (x \in [0, 1],\ \theta \in [1, 2]).$$

On every fixed complex compact set, every fixed mixed derivative is bounded uniformly over the choice of S*. For any two finite signed Borel measures* μ, ν *on* $[1, 2]$*,*

$$A_{q,S}\mu = A_{q,S}\nu \quad \Longleftrightarrow \quad \int \theta^t \, d\mu(\theta) = \int \theta^t \, d\nu(\theta) \quad \text{for every } t \in qS. \tag{421}$$

Proof. On the real rectangle, the sum of the absolute values of the nonconstant terms is bounded by

$$\sum_{s \geq 1} 2^{-s^2-s-2} 2^s = \frac{1}{4} \sum_{s \geq 1} 2^{-s^2} < \frac{1}{4} \sum_{s \geq 1} 2^{-s} = \frac{1}{4}.$$

Orthogonality to L_0 gives normalization. On each compact subset of the complex x-plane, $|L_s(x)| \leq C^s$ for some $C \geq 1$: for example, this follows immediately by induction from the three-term Legendre recurrence. The Gaussian decrease in s in a_s therefore gives normal convergence on every complex

compact rectangle. The same majorant for all S, followed by Cauchy estimates on a slightly larger rectangle, bounds every mixed derivative uniformly in S.

For $n \in S\setminus\{0\}$, the conditional mean of $L_n(X)$ is $a_n\theta^n/(2n+1)$; for $n=0$ it is one. Consequently, for $n_1,\dots,n_q \in S$,

$$\int \prod_{j=1}^{q} L_{n_j}(x_j)\, d(A_{q,S}\mu) = \left\{ \prod_{j:n_j>0} \frac{a_{n_j}}{2n_j+1} \right\} \int \theta^{n_1+\cdots+n_q}\, d\mu(\theta). \tag{422}$$

This proves the forward implication in (421). For the reverse implication, expand the product of the q kernel series. It converges absolutely and uniformly on the real rectangle, and every resulting coefficient depends on a moment whose exponent belongs to qS. Termwise integration against a finite signed measure is therefore valid and gives equality of the two densities. □

Proposition 72.2 (Exact moment criterion for replication)**.** *For an arbitrary infinite $S \subseteq \mathbb{Z}_{\ge 0}$ containing 0, the operator $A_{q,S}$ is injective on all Borel probabilities on $[1,2]$ if and only if*

$$\sum_{t\in qS\setminus\{0\}} \frac{1}{t} = \infty. \tag{423}$$

If the sum converges, noninjectivity holds even for strictly positive Gevrey densities of any prescribed order $\sigma > 1$, with their difference supported in an arbitrarily small interior interval.

Proof. When the sum diverges, the Müntz theorem on a positive interval says that the real span of $\{\theta^t : t \in qS\}$ is uniformly dense in $C[1,2]$. Thus (421) implies equality of the measures. This use of Müntz's theorem is classical; the formulation needed here is also stated as [65, Theorem 2]. When the sum converges, Theorem 71.1 supplies the stated positive density perturbations, and (421) makes their grouped laws identical. □

Lemma 72.3 (All finite parameter jets are independent)**.** *For any distinct $\theta_1,\dots,\theta_J \in [1,2]$ and any nonnegative integers $m_1,\dots,m_J$, the functions*

$$\left\{ \partial_\theta^{\ell} k_S(\,\cdot \mid \theta_j) : 1 \le j \le J,\ 0 \le \ell \le m_j \right\}$$

are linearly independent over $\mathbb{C}$.

Proof. Suppose their linear combination with coefficients $c_{j\ell}$ is zero. Taking its L_s coefficient gives, for every positive $s \in S$,

$$0 = \sum_{j=1}^{J} \theta_j^s B_j(s), \qquad B_j(s) = \sum_{\ell=0}^{m_j} c_{j\ell}\, \theta_j^{-\ell}(s)_\ell,$$

where $(s)_\ell = s(s-1)\cdots(s-\ell+1)$ and $(s)_0 = 1$. Order the nodes so that $\theta_1 < \cdots < \theta_J$. If B_J is a nonzero polynomial, division by θ_J^s would give

$$|B_J(s)| \le C(1+s)^M \rho^s \quad (s \in S,\ s \longrightarrow \infty)$$

for some $\rho < 1$, whereas a nonzero polynomial is bounded below by a positive constant times a nonnegative power of s for all large positive s. This is impossible. Repeating the argument gives $B_j = 0$ for all j. The falling factorial polynomials are linearly independent, so all $c_{j\ell} = 0$. □

Lemma 72.4 (Sparse moments determine every finite atomic measure)**.** *If μ is a finitely supported probability measure on $[1,2]$ and ν is any Borel probability measure on $[1,2]$, then*

$$A_{1,S}\mu = A_{1,S}\nu \quad \Longrightarrow \quad \mu = \nu.$$

Proof. We first recall an elementary zero-counting fact, and include its proof. A nonzero real linear combination of $m+1$ distinct real powers of t has at most m zeros on $(0,\infty)$, counted with multiplicity. Divide by the smallest power. Differentiation leaves at most m powers, and induction together with Rolle's theorem proves the assertion. More explicitly, any finite collection of zeros of total multiplicity N gives at least $N-1$ zeros of the derivative, counted with multiplicity; division by a nonvanishing power does not alter the original zeros.

Let the support of μ be $\{\theta_1,\dots,\theta_J\}$. Choose $2J+1$ exponents $0 = s_0 < s_1 < \cdots < s_{2J}$ from S. The $2J$ homogeneous real linear equations

$$p(\theta_j) = p'(\theta_j) = 0 \quad (1 \le j \le J), \qquad p(t) = \sum_{i=0}^{2J} b_i t^{s_i},$$

have a nonzero solution. The zero-counting fact implies that these are all the positive zeros of p and that each has multiplicity exactly two. Thus, after multiplying p by -1 if necessary, $p \ge 0$ on $(0,\infty)$ and its zero set there is precisely $\{\theta_1,\dots,\theta_J\}$. This argument also applies when a node is an endpoint of $[1,2]$, since both endpoints are interior points of $(0,\infty)$.

Equality of the one-observation laws gives equality of all the S-moments. Hence $\int p\,d\nu = \int p\,d\mu = 0$. Positivity of ν forces its support to lie in the same finite set of nodes. Lemma 72.3, with derivative order zero, now gives equality of the weights, and hence $\mu = \nu$. □

Theorem 72.5 (Every replication threshold on an interval)**.** *For every $r \in \{1,2,\dots\} \cup \{\infty\}$ there is an infinite set $S_r \subseteq \mathbb{Z}_{\ge 0}$ containing 0 and 1 such that the kernel (420) has the following properties.*

1. *It is strictly positive and jointly entire, with the uniform bounds in Lemma 72.1.*
2. *Every finite family of parameter jets is linearly independent, and every finitely supported mixing probability is identified by one observation even among arbitrary Borel competitors.*
3. *If $r < \infty$, then A_{q,S_r} is injective on all Borel probabilities exactly when $q \ge r$.*
4. *If $r = \infty$, no finite q gives injectivity, but the collection $(A_{q,S_r}\mu)_{q\ge 1}$ determines μ.*
5. *Whenever injectivity fails, it already fails for distinct strictly positive C^∞ mixing densities. For finite $r \ge 2$, one pair can witness failure simultaneously for every $q < r$.*

The smooth densities in the last assertion may, more strongly, be chosen in any prescribed Gevrey class of order $\sigma > 1$.

Proof. For $r = 1$, take $S_1 = \mathbb{Z}_{\ge 0}$. All monomial moments are observed, so polynomial approximation on $[1,2]$ proves injectivity.

Fix $2 \le r < \infty$. For $j = 0,\dots,r-1$, let

$$D_{r,j} = \left\{ \sum_{k\ge 0} \varepsilon_k 2^{rk+j} : \varepsilon_k \in \{0,1\},\ \varepsilon_k = 0 \text{ for all but finitely many } k \right\}, \qquad S_r = \bigcup_{j=0}^{r-1} D_{r,j}.$$

Separating the binary digits of an arbitrary nonnegative integer according to their positions modulo r proves

$$rS_r = \mathbb{Z}_{\ge 0}. \tag{424}$$

Since $0 \in S_r$, the same is true with r replaced by any $q \geq r$. Lemma 72.1 and polynomial approximation therefore prove injectivity for $q \geq r$.

For $t \geq 1$, the number of free binary digits in each $D_{r,j}$ below t is at most $1 + (\log_2 t)/r$. Consequently

$$\#(S_r \cap [0,t]) \leq 2r\, t^{1/r}, \qquad \#(qS_r \cap [0,t]) \leq (2r)^q t^{q/r}.$$

The second inequality holds because all summands are nonnegative, so that a sum not exceeding t uses only elements not exceeding t. If $q < r$, the exponent q/r is strictly below one. More generally, if $T \subseteq \mathbb{Z}_{\geq 0}$ satisfies $\#(T \cap [0,t]) \leq Ct^\beta$ for $t \geq 1$ and $\beta < 1$, then

$$\sum_{t \in T \setminus \{0\}} \frac{1}{t} \leq \sum_{k \geq 0} 2^{-k} \#(T \cap [0, 2^{k+1}]) \leq C 2^\beta \sum_{k \geq 0} 2^{-k(1-\beta)} < \infty.$$

Apply Theorem 71.1 with $T = (r-1)S_r$: it provides distinct strictly positive smooth densities with the same moments at every exponent in T. Since $qS_r \subseteq T$ for every $q < r$, the same pair has identical q-observation laws for all such q.

For the infinite threshold take

$$S_\infty = \{0, 1\} \cup \{2^{2^j} : j = 0, 1, 2, \ldots\}.$$

For each fixed q, the counting function of qS_∞ is bounded by a constant times $(1 + \log\log(t+3))^q$. In particular it is $O_q(t^{1/2})$ as $t \longrightarrow \infty$. Applying Theorem 71.1 to $T = qS_\infty$ proves failure of injectivity for every fixed q. The witnessing pair may depend on q. On the other hand, $1 \in S_\infty$ implies $\bigcup_{q \geq 1} qS_\infty = \mathbb{Z}_{\geq 0}$; equality of all grouped laws thus gives equality of all moments and hence of the measures. The regularity, finite-jet and finite-atomic assertions follow from the preceding lemmas. □

Corollary 72.6 (Localized infinite-dimensional ambiguity)**.** *Fix one of the models in Theorem 72.5, a subcritical replication count $q < r$ (any finite q when $r = \infty$), and a nonempty open interval I with $\overline{I} \subset (1,2)$. For every $\sigma > 1$ the nullspace of A_{q,S_r} contains an infinite-dimensional real vector space of signed densities in $G^\sigma(\mathbb{R})$ supported in I. If g_0 is any strictly positive continuous probability density on $[1,2]$, then, for every finite collection of such null densities $h_1, \ldots, h_m$, all densities*

$$g_0 + \sum_{j=1}^{m} c_j h_j$$

with $\sum_j |c_j|$ sufficiently small are strictly positive probability densities with the same q-observation law as g_0. If g_0 belongs to $G^\sigma([1,2])$, then so do these perturbations.

Proof. The reciprocal sum over $qS_r \setminus \{0\}$ converges by the preceding proof. Choose countably many disjoint open subintervals whose closures lie in I. Apply Theorem 71.1 separately in each subinterval. The resulting nonzero compactly supported functions are linearly independent because their supports are disjoint, and (421) puts them in the nullspace. Each has integral zero. For a finite collection, boundedness of the h_j and the positive minimum of g_0 on $[1,2]$ prove the remaining assertions. □

Remark 72.7 (What produces the threshold). The mechanism is an additive covering problem for the degrees of observable monomials. Splitting binary digits makes r copies cover every degree, whereas fewer copies have a counting exponent strictly below one. Positivity and analyticity of the observation kernel, and independence of every finite collection of parameter jets, are unchanged as this global threshold varies. In particular, this construction takes place on an ordinary compact interval and uses neither an identification of endpoints nor a periodic parameter.

73 Analytic identifiability and a sharp boundary for uniform learning

Identifiability within a regularity class need not imply uniform statistical recovery on that class. Here this distinction survives every prescribed Gevrey bound of order strictly greater than one, even if each density under consideration extends to an entire function. The corresponding fixed analytic bounds behave differently.

We first state the compactness principle used to separate these issues. For a compact metric parameter space Θ, let W_1 denote the Wasserstein distance for its metric. Suppose X is also a compact metric space and $\theta \mapsto P_\theta \in \mathcal{P}(X)$ is continuous in total variation. Fix a positive integer q and put

$$A_q\mu = \int P_\theta^{\otimes q}\, d\mu(\theta).$$

For a nonempty class $\mathcal{D} \subseteq \mathcal{P}(\Theta)$, define

$$R_{N,q}(\mathcal{D}) = \inf_{\widehat{\mu}} \sup_{\mu\in\mathcal{D}} \mathbb{E}_{(A_q\mu)^{\otimes N}} W_1(\widehat{\mu},\mu). \tag{425}$$

There are N independent groups, each containing q observations; the estimator may take values in all of $\mathcal{P}(\Theta)$. We use $\mathrm{TV}(P,Q) = \sup_B |P(B) - Q(B)|$.

Proposition 73.1 (Uniform learning depends on the closure)**.** *Let $\overline{\mathcal{D}}$ denote weak closure in $\mathcal{P}(\Theta)$. Then*

$$R_{N,q}(\mathcal{D}) \longrightarrow 0 \quad\Longleftrightarrow\quad A_q \text{ is injective on } \overline{\mathcal{D}}. \tag{426}$$

More precisely, for every $N \geq 1$,

$$R_{N,q}(\mathcal{D}) \geq \frac{1}{2}\sup\left\{W_1(\mu,\nu) : \mu,\nu \in \overline{\mathcal{D}},\ A_q\mu = A_q\nu\right\}. \tag{427}$$

Proof. First, A_q is continuous from the weak topology on mixing measures to total variation on observation laws. Indeed, $\mathrm{TV}(P^{\otimes q}, Q^{\otimes q}) \leq q\,\mathrm{TV}(P,Q)$, so the map $\theta \mapsto P_\theta^{\otimes q}$ is norm continuous as a map into the Banach space of finite signed measures. A finite partition of unity on Θ approximates this map uniformly by maps of the form $\sum_j f_j(\theta)Q_j$, with each f_j continuous and each Q_j fixed. Integration of each approximant is weak-to-norm continuous; uniform approximation proves the assertion.

For arbitrary parameters μ,ν, arbitrary data laws P,Q, and an arbitrary estimator, the triangle inequality gives

$$\max\{\mathbb{E}_P W_1(\widehat{\mu},\mu), \mathbb{E}_Q W_1(\widehat{\mu},\nu)\} \geq \frac{W_1(\mu,\nu)}{2}(1 - \mathrm{TV}(P,Q)). \tag{428}$$

To verify the factor $1/2$, dominate P,Q by $P+Q$ and integrate the pointwise inequality $W_1(\widehat{\mu},\mu) + W_1(\widehat{\mu},\nu) \geq W_1(\mu,\nu)$ against the smaller of their two densities; this common measure has mass $1 - \mathrm{TV}(P,Q)$.

Now let $\mu,\nu \in \overline{\mathcal{D}}$ satisfy $A_q\mu = A_q\nu$. Choose $\mu_j,\nu_j \in \mathcal{D}$ converging weakly to them. Then $W_1(\mu_j,\nu_j) \to W_1(\mu,\nu)$ and, for each fixed N,

$$\mathrm{TV}\left((A_q\mu_j)^{\otimes N}, (A_q\nu_j)^{\otimes N}\right) \leq N\,\mathrm{TV}(A_q\mu_j, A_q\nu_j) \longrightarrow 0.$$

Apply (428) and let $j \to \infty$. The result holds for every estimator, proving (427).

For the converse put $K = \overline{\mathcal{D}}$, which is compact. Let d_{BL} be the bounded Lipschitz metric on $\mathcal{P}(X^q)$. If A_q is injective on K, its continuous inverse on the compact image is uniformly continuous. Equivalently,

$$\omega(t) = \sup\{W_1(\mu,\nu) : \mu,\nu \in K, d_{\mathrm{BL}}(A_q\mu, A_q\nu) \le t\} \longrightarrow 0 \qquad (t \downarrow 0).$$

Choose finite $1/N$-nets $F_N \subseteq \mathcal{D}$ for K in W_1. If $\widehat{Q}_N$ is the empirical law of the N groups, choose $\widehat{\mu}_N \in F_N$ minimizing $d_{\mathrm{BL}}(A_q\widehat{\mu}_N, \widehat{Q}_N)$, breaking ties by a fixed ordering. This is measurable. Uniform continuity of A_q gives numbers $b_N \to 0$ such that, for every $\mu \in \mathcal{D}$,

$$d_{\mathrm{BL}}(A_q\widehat{\mu}_N, A_q\mu) \le 2d_{\mathrm{BL}}(\widehat{Q}_N, A_q\mu) + b_N.$$

On a compact sample space,

$$\sup_{Q\in\mathcal{P}(X^q)} \mathbb{E}_Q d_{\mathrm{BL}}(\widehat{Q}_N, Q) \longrightarrow 0. \tag{429}$$

Indeed, the bounded Lipschitz unit ball has a finite uniform η-net by Arzelà–Ascoli. For fixed η, the expected maximum of the finitely many empirical integral errors tends to zero, while replacing an arbitrary function by a net element costs at most 2η. Let $N \to \infty$ and then $\eta \downarrow 0$. Uniform inverse continuity, (429), and the bounded diameter of Θ imply $\sup_{\mu\in\mathcal{D}} \mathbb{E}W_1(\widehat{\mu}_N, \mu) \to 0$. □

The lower-bound mechanism is a standard consequence of indistinguishable limits and two-point decision theory. In particular, the use of nonidentifiable limits to prove constant minimax risk over finite discrete mixing laws appears in Proposition 3 of Le, Bariletto, Rinaldo, and Ho [55]. The purpose of the next theorem is the regularity-restricted consequence and its exact replication threshold, not a claim of novelty for the compactness principle itself.

For $\sigma \ge 1$, $M > 1$, and $B > 0$, let $\mathcal{E}_{\sigma,M,B}$ consist of the probability measures $g(\theta)\,d\theta$ on $[1,2]$ such that g extends to an entire function of one complex variable, is real on $[1,2]$, and satisfies

$$\frac{1}{2} \le g(\theta) \le \frac{3}{2}, \qquad \|g^{(k)}\|_{L^\infty([1,2])} \le MB^k(k!)^\sigma \quad (k = 0,1,\ldots). \tag{430}$$

Thus the derivative constants are fixed for the whole class, and every one of its members is entire, whatever the value of σ. The uniform density belongs to every such class.

Theorem 73.2 (A sharp boundary inside classes of entire densities)**.** *Consider the interval kernel k_{S_r} of Theorem 72.5, where $r \in \{1,2,\ldots\} \cup \{\infty\}$. For every $\sigma \ge 1$, $M > 1$, and $B > 0$, A_{1,S_r} is injective on $\mathcal{E}_{\sigma,M,B}$. Nevertheless uniform estimation has the following sharp distinctions.*

1. *For $\sigma = 1$, every fixed $q \ge 1$ satisfies*
$$R_{N,q}(\mathcal{E}_{1,M,B}) \longrightarrow 0.$$

2. *For $\sigma > 1$ and every finite $q < r$, there exists $c = c(r,q,\sigma,M,B) > 0$ such that*
$$R_{N,q}(\mathcal{E}_{\sigma,M,B}) \ge c \qquad \text{for every } N \ge 1.$$
If $2 \le r < \infty$, the same constant $c > 0$ may be chosen for all $q < r$ simultaneously. If $r = \infty$, the constant may depend on q.

3. *If $r < \infty$ and $q \ge r$, then*
$$R_{N,q}(\mathcal{P}([1,2])) \longrightarrow 0,$$
and in particular uniform consistency holds on every $\mathcal{E}_{\sigma,M,B}$.

The assertion for $\sigma > 1$ holds for every prescribed $M > 1$ and $B > 0$, not merely for some sufficiently large derivative constants.

Proof. Any two elements of $\mathcal{E}_{\sigma,M,B}$ have analytic density difference. Their one-observation laws determine its moments on the unbounded set S_r. Theorem 71.5 therefore proves the stated injectivity within the class.

For the first assertion, the weak closure of $\mathcal{E}_{1,M,B}$ consists of measures having densities with the same derivative bounds. To see this, any sequence of these densities has, by the bounds on successive derivatives and diagonal Arzelà–Ascoli, a subsequence converging in $C^k([1,2])$ for every fixed k. The resulting smooth density g satisfies $\|g^{(k)}\|_\infty \le MB^k k!$. Taylor's remainder estimate shows that g is real analytic, with local radius at least $1/B$; the one-sided Taylor series at the two endpoints extend it analytically to a neighborhood of $[1,2]$. Although such a limit need not be entire, analytic moment rigidity still applies. Thus A_{1,S_r} is injective on this weak closure, and so is every A_{q,S_r}. Proposition 73.1 proves uniform consistency.

For the second assertion fix $\sigma > 1$, and choose $1 < \tau < \sigma$. If $2 \le r < \infty$, set $T = (r-1)S_r$; if $r = \infty$ and q is fixed, set $T = qS_\infty$. Theorem 71.1 supplies a nonzero real $h \in G^\tau(\mathbb{R})$ compactly supported inside $(1,2)$ with

$$\int_1^2 \theta^s h(\theta)\,d\theta = 0 \quad (s \in T), \qquad \|h^{(k)}\|_\infty \le AC^k (k!)^\tau$$

for some $A, C > 0$. In particular $\int h = 0$. For $t > 0$ let

$$\varphi_t(z) = \frac{1}{\sqrt{2\pi t}} e^{-z^2/(2t)}, \qquad h_t(z) = \int_{\mathbb{R}} \varphi_t(z-y)h(y)\,dy, \qquad c_t = \int_1^2 h_t(u)\,du.$$

Compact support of h makes h_t entire. On the real line, integration by parts gives $h_t^{(k)} = \varphi_t * h^{(k)}$, whence

$$\|h_t^{(k)}\|_\infty \le AC^k (k!)^\tau \quad (k \ge 0), \qquad \|h_t - c_t\|_{L^\infty([1,2])} \le 2A.$$

Furthermore $h_t \to h$ uniformly and $c_t \to 0$ as $t \downarrow 0$. Because $\sigma - \tau > 0$,

$$D = \max\left\{2A, A \sup_{k \ge 1} \frac{(C/B)^k}{(k!)^{\sigma-\tau}}\right\} < \infty.$$

Choose a fixed $\varepsilon > 0$ satisfying $\varepsilon D \le \min\{1/2, M-1\}$ and put

$$g_t^{\pm} = 1 \pm \varepsilon(h_t - c_t), \qquad g^{\pm} = 1 \pm \varepsilon h.$$

The interval has length one, so $\int g_t^{\pm} = 1$ exactly. Both g_t^+ and g_t^- are entire, between $1/2$ and $3/2$, and satisfy (430). Indeed the bound for $k = 0$ is at most $1 + \varepsilon D \le M$, and for $k \ge 1$ the derivative bound is at most $\varepsilon DB^k (k!)^\sigma \le MB^k (k!)^\sigma$. Thus $g_t^{\pm} d\theta \in \mathcal{E}_{\sigma,M,B}$ for every $t > 0$. Their distinct limits $\mu^{\pm} = g^{\pm} d\theta$ belong to its weak closure and have identical q-observation laws by (421). For finite r, this equality holds simultaneously for every $q < r$. The lower bound (427) gives the positive constant

$$c = \frac{1}{2} W_1(\mu^+, \mu^-) = \varepsilon \int_1^2 \left| \int_1^x h(u)\,du \right| dx > 0. \tag{431}$$

Positivity follows because a nonzero continuous h cannot have an identically zero indefinite integral.

For the final assertion, A_{q,S_r} is injective on all of the compact space $\mathcal{P}([1,2])$ when $q \ge r$. Proposition 73.1 gives uniform consistency. □

Remark 73.3 (Interpretation of the two boundaries). For a fixed bound with $\sigma = 1$, the closure retains analytic rigidity. When $\sigma > 1$, even if every member of the parameter class is entire, its closure contains nonanalytic invisible perturbations. No finite number of independent groups overcomes that uniform obstruction below the replication threshold. The identities between analytic densities are still unique: the lower bound concerns pairs of analytic densities whose data laws become arbitrarily close, not an equality between two distinct analytic densities. Above the threshold, compactness restores uniform consistency, without asserting any particular rate.

Corollary 73.4 (Strong log-concavity does not remove the obstruction). *Fix $\sigma > 1$, $M > 1$, $B > 0$, and a finite $q < r$ in Theorem 73.2. There exist $\kappa, c > 0$ such that the smaller class*

$$\mathcal{E}^{\kappa}_{\sigma,M,B} = \{g\,d\theta \in \mathcal{E}_{\sigma,M,B} : (\log g)'' \le -\kappa \text{ on } [1,2]\}$$

still satisfies $R_{N,q}(\mathcal{E}^{\kappa}_{\sigma,M,B}) \ge c$ for every $N \ge 1$. For finite r, one choice of $\kappa, c > 0$ works simultaneously for every $q < r$.

Proof. Choose $1 < M_0 < M$. For a sufficiently small $a > 0$, the entire probability density

$$g_0(z) = \frac{e^{-a(z-3/2)^2}}{\int_1^2 e^{-a(u-3/2)^2}\,du}$$

is between $3/4$ and $5/4$ on $[1,2]$, and its modulus is at most M_0 on the closed complex $2/B$-neighborhood of that interval. These assertions follow from uniform convergence to 1 on this complex compact set as $a \downarrow 0$. Cauchy's estimate gives

$$\|g_0^{(k)}\|_\infty \le M_0(B/2)^k k! \le M_0 B^k (k!)^\sigma \qquad (k \ge 0).$$

Use the same invisible h and entire centered approximants $b_t = h_t - c_t$ as in the preceding proof, and put $g_t^{\pm} = g_0 \pm \varepsilon b_t$. Their derivatives fit the prescribed bounds if $\varepsilon D \le M - M_0$. Their integrals are one exactly. The functions b_t have uniformly bounded $C^2([1,2])$ norms. Because $(\log g_0)'' = -2a$, taking $\varepsilon > 0$ still smaller ensures simultaneously that all $g_t^{\pm}$ lie between $1/2$ and $3/2$ and satisfy $(\log g_t^{\pm})'' \le -a$. This follows also directly from continuity of $g \mapsto g''/g - (g'/g)^2$ in the C^2 norm on densities bounded away from zero. The distinct limits $(g_0 \pm \varepsilon h)d\theta$ have equal q-observation laws. Proposition 73.1 supplies the positive all-N lower bound, with $\kappa = a$. □

73.1 Identification and consistency across mixing classes

For a fixed finite threshold $r \ge 2$, the different consequences of an exact observation law and of a finite statistical sample can be displayed together. In the table, "least group size" refers to identification within the stated class or to uniform W_1 consistency as the number of independent groups tends to infinity.

Mixing class	Exact identification	Uniform consistency
All Borel probability measures	r	r
All finite atomic probabilities, with no common bound on atom count	1	r
Entire densities with fixed bounds $\mathcal{E}_{1,M,B}$	1	1
Entire densities with fixed bounds $\mathcal{E}_{\sigma,M,B}$, $\sigma > 1$	1	r

The finite atomic row follows because finite atomic probabilities are weakly dense in all probability measures, together with Lemma 72.4 and Proposition 73.1. If a common finite upper bound on atom count is imposed, that subclass is compact and one-observation uniform consistency instead holds. No rate for that compact finite-atom problem is claimed here.

The distinction between local and global obstructions is also sharp. Lemma 72.3 excludes every nonzero finite point-derivative functional annihilating the component kernel. In particular there is no nontrivial finite-order differential relation at an interior parameter point. Nevertheless the global observation operator has localized, infinite-dimensional smooth nullspaces below the prescribed threshold. Smoothness of an invisible perturbation does not imply that it arises from a finite parameter differential identity.

74 Sharp quantitative recovery for the factorial kernel

The exact recovery supplied by replication need not be numerically well conditioned. We return to the factorial Legendre kernel (406), now denoted k_{fac}. Its qualitative identification properties were proved in Corollary 67.6. For an integer $r \geq 1$ and a mixing probability G, write

$$q_{r,G}(x_1, \dots, x_r) = \int_0^1 \prod_{i=1}^{r} k_{\mathrm{fac}}(x_i \mid \theta)\, G(d\theta), \qquad Q_r(G) = q_{r,G}\, dx_1 \cdots dx_r. \tag{432}$$

The r observations in a group share the same latent parameter. Different groups in the statistical experiment have independent latent draws. The density is normalized by Legendre orthogonality, is bounded below by $3 - e > 0$, and is jointly entire by locally uniform convergence on complex compact sets. We use $\mathrm{TV}(P, Q) = \frac{1}{2} \int |p - q|$ and the ordinary Euclidean Wasserstein distance W_1 on $[0, 1]$.

Theorem 74.1 (Sharp modulus and statistical rate)**.** *For every fixed integer $r \geq 2$, define*

$$\omega_r(\varepsilon) = \sup\{W_1(G, H) : \mathrm{TV}(Q_r(G), Q_r(H)) \leq \varepsilon\},$$

where the supremum ranges over all Borel probabilities on $[0, 1]$. There are constants $0 < c_r \leq C_r < \infty$ such that, for all sufficiently small $\varepsilon > 0$,

$$c_r \frac{\log\log(1/\varepsilon)}{\log(1/\varepsilon)} \leq \omega_r(\varepsilon) \leq C_r \frac{\log\log(1/\varepsilon)}{\log(1/\varepsilon)}. \tag{433}$$

If N independent groups with law $Q_r(G)$ are observed, then

$$\inf_{\widehat{G}} \sup_{G} \mathbb{E}_G W_1(\widehat{G}, G) \asymp_r \frac{\log\log N}{\log N} \qquad (N \longrightarrow \infty). \tag{434}$$

The infimum is over measurable probability-valued estimators. For $r = 1$, the minimax risk is at least $1/12$ for every N, and $\omega_1(0) \geq 1/6$.

The rate $\log\log N / \log N$ is already familiar in nonparametric mixture estimation, including compactly supported Poisson mixtures [60]. Our assertion is a quantitative description of the present replication transition, not a new universal mixture rate. In particular, any fixed number of further replicates leaves the rate order unchanged after identifiability has been restored. The theorem fixes the replicate count. Growing replication and an equal-group observation budget are treated in Section 75; those bounds leave a logarithmic gap.

74.1 Moment recovery and approximation

Let $\nu = G - H$ and $\delta_j = \int L_j \, d\nu$. Orthogonality gives

$$\int L_j(x) \, d\{Q_2(G) - Q_2(H)\}(x,y) = \frac{\delta_j}{j!(2j+1)}, \qquad j \geq 2, \tag{435}$$

$$\int L_2(x) L_3(y) \, d\{Q_2(G) - Q_2(H)\}(x,y) = \frac{1}{420} \int L_2 L_3 \, d\nu. \tag{436}$$

The elementary product identity

$$L_2 L_3 = \frac{10}{21} L_5 + \frac{4}{15} L_3 + \frac{9}{35} L_1 \tag{437}$$

therefore recovers the missing first moment from finitely many observable expectations. For $M \geq 5$, consider the M bounded test functions

$$\mathcal{F}_M = \{(x,y) \mapsto L_j(x) : 2 \leq j \leq M\} \cup \{(x,y) \mapsto L_2(x) L_3(y)\}.$$

If the absolute differences of their expectations under $Q_2(G)$ and $Q_2(H)$ are bounded by d, then

$$|\delta_j| \leq j!(2j+1)d \quad (2 \leq j \leq M), \qquad |\delta_1| \leq 4200d. \tag{438}$$

Indeed, the constant for the last inequality is bounded by

$$\frac{35}{9} \left(420 + \frac{10}{21} \, 1320 + \frac{4}{15} \, 42 \right) < 4200.$$

Lemma 74.2 (Polynomial approximation and moment comparison)**.** *There is an absolute constant C such that, for every integer $M \geq 5$ and every 1-Lipschitz function $f : [0,1] \to \mathbb{R}$ with $f(0) = 0$, there is a polynomial p of degree at most M satisfying*

$$\|p\|_\infty \leq 1, \qquad \|f - p\|_\infty \leq C/M.$$

Consequently, under the expectation discrepancy assumption above,

$$W_1(G,H) \leq \frac{C}{M} + CM^3 M! d. \tag{439}$$

Proof. We include the classical Jackson-kernel argument. Define the even 2π-periodic function $F(u) = f((1+\cos u)/2)$. It is $1/2$-Lipschitz for circular distance, and $\|F\|_\infty \leq 1$. Put $\ell = \lfloor M/2 \rfloor + 1$ and normalize

$$J_\ell(u) = c_\ell \left(\frac{\sin(\ell u/2)}{\sin(u/2)} \right)^4, \qquad \int_{-\pi}^{\pi} J_\ell(u) \, du = 1.$$

The quotient at zero is understood by continuity. This is a nonnegative trigonometric polynomial of degree $2(\ell - 1) \leq M$. On $[-\pi, \pi]$ the modulus of the quotient is bounded by $\min(\ell, \pi/|u|)$. On $|u| \leq 1/\ell$ it is at least $2\ell/\pi$. These inequalities show that its fourth-power integral is comparable to ℓ^3, whereas its fourth-power integral weighted by $|u|$ is $O(\ell^2)$. Hence

$$\int_{-\pi}^{\pi} |u| J_\ell(u) \, du \leq C/\ell.$$

The convolution $F * J_\ell$ is even, has sup norm at most one, and differs uniformly from F by at most C/ℓ. Every even trigonometric polynomial of degree at most M is a polynomial of degree at most M in $\cos u$. Substituting $\cos u = 2x - 1$ gives the required p.

Expand $p = \sum_{j=0}^{M} b_j L_j$. Orthogonality and $\|p\|_\infty \le 1$ give $|b_j| \le 2j+1$. Since $\nu([0,1]) = 0$, the Kantorovich–Rubinstein formula and the approximation just proved yield

$$W_1(G,H) \le \frac{C}{M} + \sum_{j=1}^{M} (2j+1)|\delta_j|.$$

Use (438) and the sufficient coarse bound $\sum_{j=2}^{M}(2j+1)^2 j! \le M(2M+1)^2 M!$. □

74.2 Explicit indistinguishable approximations

Lemma 74.3 (Moment matching with an exact transport gap)**.** *For every integer $m \ge 1$, set $n = m+1$ and define*

$$G_m = 2^{1-n} \sum_{\substack{0 \le j \le n \\ j \text{ even}}} \binom{n}{j} \delta_{j/n}, \qquad H_m = 2^{1-n} \sum_{\substack{0 \le j \le n \\ j \text{ odd}}} \binom{n}{j} \delta_{j/n}.$$

These are probabilities with equal polynomial moments through degree m, and

$$W_1(G_m, H_m) = \frac{1}{m+1}. \tag{440}$$

For every integer $r \ge 1$,

$$\mathrm{TV}(Q_r(G_m), Q_r(H_m)) \le T_{m,r} := \sum_{s=m+1}^{\infty} \frac{r^s}{s!}. \tag{441}$$

If $m+2 \ge 2r$, then $T_{m,r} \le 2r^{m+1}/(m+1)!$.

Proof. The even and odd binomial sums both equal 2^{n-1}, so the measures are probabilities. Their difference is a constant multiple of an nth finite difference and therefore annihilates every polynomial of degree at most $n-1 = m$. On the interval $[k/n, (k+1)/n)$, their cumulative distribution functions differ by

$$2^{1-n} \sum_{j=0}^{k} (-1)^j \binom{n}{j} = 2^{1-n}(-1)^k \binom{n-1}{k}.$$

Integrating the absolute difference proves (440).

For the kernel expansion, allow indices in $\{0, 2, 3, \dots\}$, with $L_0 = 1$ and coefficient $1/0! = 1$. A term in the r-fold product indexed by $(j_1, \dots, j_r)$ is a polynomial in θ of degree at most $s = j_1 + \cdots + j_r$. Its integral against $G_m - H_m$ vanishes for $s \le m$ and otherwise has absolute value at most two. Absolute uniform convergence justifies all interchanges. Enlarging the index sets to all nonnegative integers and using the multinomial identity gives

$$\|q_{r,G_m} - q_{r,H_m}\|_\infty \le 2 \sum_{s=m+1}^{\infty} \sum_{j_1 + \cdots + j_r = s} \frac{1}{j_1! \cdots j_r!} = 2 \sum_{s=m+1}^{\infty} \frac{r^s}{s!}.$$

The cube has volume one, proving (441). When $m+2 \ge 2r$, successive tail terms have ratio at most $1/2$, which proves the final bound. □

74.3 Proof of the sharp modulus and minimax theorem

Proof of Theorem 74.1. If $\mathrm{TV}(Q_2(G), Q_2(H)) \le \varepsilon$, all tests in $\mathcal{F}_M$ have expectation discrepancy at most 2ε. Write $L = \log(1/\varepsilon)$ and choose $M = \lfloor L/(2\log L) \rfloor$. For sufficiently large L this is at least five, and

$$\log(\varepsilon M^3 M!) \le -L + M\log M + 3\log M \le -L/3.$$

Lemma 74.2 gives the upper bound in (433). For $r > 2$, marginalization to the first two coordinates can only decrease total variation, so the same upper bound applies. For the lower bound, choose $m = \lceil 2L/\log L \rceil$. For every fixed r, the elementary factorial bound $(m+1)! \ge ((m+1)/e)^{m+1}$ shows that $T_{m,r} \le \varepsilon$ for all sufficiently large L. Lemma 74.3 then gives a transport gap $1/(m+1) \ge c_r \log L/L$.

For the statistical upper bound, retain the first two coordinates of each group. Let $\widehat{a}$ be the vector of empirical means of the M tests in $\mathcal{F}_M$, and let $a(G)$ denote their population expectations. Choose $\widehat{G}$ with

$$\|a(\widehat{G}) - \widehat{a}\|_\infty \le \inf_H \|a(H) - \widehat{a}\|_\infty + N^{-1}.$$

For completeness, a measurable choice is obtained by fixing a countable dense family of finitely supported probabilities with rational atoms and rational weights, taking the infimum over this family, and selecting its first qualifying element. Continuity of all feature expectations in the weak topology ensures that this infimum equals the displayed one. If $Z = \|\widehat{a} - a(G)\|_\infty$, then $\|a(\widehat{G}) - a(G)\|_\infty \le 2Z + N^{-1}$. Hoeffding's inequality for bounded tests, a union bound, and integration yield

$$\mathbb{E}_G Z \le C\sqrt{\frac{\log(2M)}{N}},$$

uniformly in G. Consequently

$$\sup_G \mathbb{E}_G W_1(\widehat{G}, G) \le \frac{C}{M} + CM^3 M! \left(\sqrt{\frac{\log(2M)}{N}} + \frac{1}{N}\right).$$

Take $M = \lfloor \log N/(4\log\log N) \rfloor$. Since $M! \le M^M \le N^{1/3}$ eventually, the second term is $o(1/M)$, proving the upper rate.

For the lower bound, set $m = \lceil 2\log N/\log\log N \rceil$. For fixed r, $NT_{m,r} \to 0$. The total variation distance between the two N-group sample laws is at most $NT_{m,r}$, and hence at most $1/2$ eventually. Integrating the triangle inequality for W_1 against the overlap of these sample laws shows that the larger of the two estimation risks at G_m, H_m is at least $1/[4(m+1)]$. This proves the lower rate.

Finally, the mixing densities $1 + \frac{1}{2}L_1$ and $1 - \frac{1}{2}L_1$ both give the uniform one-view observation law. Their signed density difference is $2\theta - 1$, so their cumulative distribution functions differ by $\theta^2 - \theta$. Their W_1 distance is $\int_0^1 (\theta - \theta^2)d\theta = 1/6$. The identical sample laws and the triangle inequality force the larger estimation risk to be at least $1/12$ for every N. □

75 Growing replication under an equal-group budget

We continue with the factorial kernel $k_* = k_{\mathrm{fac}}$ from (406). The preceding theorem determines the rate for each fixed group size; here the number of replicates may grow with the number of groups. Let $Q_q(G) = \int P_\theta^{\otimes q}\, dG(\theta)$, and write

$$R_{N,q} = \inf_{\widehat{G}} \sup_{G \in \mathcal{P}([0,1])} \mathbb{E}_{Q_q(G)^{\otimes N}} W_1(\widehat{G}, G).$$

Thus N independent groups each contain q observations sharing one latent parameter. All constants below are independent of N, q.

Theorem 75.1. *There are constants $c, C > 0$ such that, for $N \geq 1$ and $q \geq 2$,*

$$R_{N,q} \geq c \max\left\{ \frac{1}{\sqrt{N}}, \min\left(\frac{1}{\log(eN)}, \frac{1}{\sqrt{q \log(eN)}} \right) \right\}, \tag{442}$$

$$R_{N,q} \leq \min\left\{ 1, \frac{1}{2\sqrt{N}} + \frac{2100}{\sqrt{\lfloor q/2 \rfloor}} \right\}. \tag{443}$$

Consequently, for sufficiently large B,

$$c\{B \log(eB)\}^{-1/4} \leq \inf_{\substack{N,q \in \mathbb{N},\ q \geq 2 \\ Nq \leq B}} R_{N,q} \leq C B^{-1/4}. \tag{444}$$

In particular $q \geq N$ gives $R_{N,q} \asymp N^{-1/2}$, whereas $R_{N,q} = O(N^{-1/2})$ along a sequence $N \to \infty$ requires $q = \Omega(N/\log(eN))$.

These inequalities determine the equal-group budget exponent up to a logarithmic factor; they do not resolve the complete growing-q minimax curve. The infimum in (444) concerns equal, nonadaptively chosen group sizes, not arbitrary allocations. Related logarithmic gains and the limitations of empirical latent parameter estimates are classical in binomial-mixture estimation [67]; no novelty of those rate phenomena is asserted.

Proof of the upper bound. Put $A_j(x) = j!(2j+1)L_j(x)$, so $\mathbb{E}_\theta A_j(X) = L_j(\theta)$ for $j \geq 2$. The product identity (437) shows that the statistic

$$Y(x,y) = \frac{1}{2} + \frac{35}{18}\left\{ A_2(x)A_3(y) - \frac{10}{21}A_5(x) - \frac{4}{15}A_3(x) \right\}$$

has conditional expectation θ when X, Y are independent with common law P_θ. Moreover

$$|Y(x,y)| \leq \frac{1}{2} + \frac{35}{18}\left(420 + \frac{10}{21}\,1320 + \frac{4}{15}\,42 \right) < 2100.$$

Within each group, average this statistic over $M = \lfloor q/2 \rfloor$ disjoint pairs and clip the result to $[0,1]$, obtaining $\widehat{\theta}_i$. Conditional independence and clipping give $\mathbb{E}_\theta|\widehat{\theta}_i - \theta| \leq 2100/\sqrt{M}$. For $\widehat{G} = N^{-1}\sum_i \delta_{\widehat{\theta}_i}$, couple with the latent empirical measure $G_N = N^{-1}\sum_i \delta_{\theta_i}$. If F_N, F are the distribution functions of G_N, G, respectively,

$$\mathbb{E}W_1(\widehat{G}, G) \leq \frac{2100}{\sqrt{M}} + \int_0^1 \mathbb{E}|F_N(t) - F(t)|\,dt \leq \frac{2100}{\sqrt{M}} + \frac{1}{2\sqrt{N}}.$$

This proves (443). □

Proof of the lower bound. An oracle observing the N latent parameters is more informative than the grouped experiment. Consider $G_\pm = (\frac{1}{2} \pm a)\delta_1 + (\frac{1}{2} \mp a)\delta_0$ with $a = 1/(8\sqrt{N})$. Their W_1 distance is $2a$ and their N-sample relative entropy is at most $1/6$. Pinsker's inequality and the two-point bound therefore give $R_{N,q} \geq 1/(16\sqrt{N})$.

For the other bound, put $\theta_0 = 1/2$ and $\mathcal{H} = L^2(P_{\theta_0}; \mathbb{C})$. The normalized likelihood

$$f_z(x) = \frac{k_*(x \mid \theta_0 + z)}{k_*(x \mid \theta_0)}$$

is $\mathcal{H}$-holomorphic near $z=0$ and satisfies $\int f_z\,dP_{\theta_0}=1$. Since the denominator is bounded below and the numerator and its complex derivative are bounded on compact sets, there exist $0<R\le 1/4$ and $C_0\ge 1$ such that $\|f_z-1\|_{\mathcal{H}}\le\sqrt{C_0}|z|$ for $|z|\le R$. Normalization cancels the linear term in the squared norm, giving

$$\|f_z\|_{\mathcal{H}}^2=1+\|f_z-1\|_{\mathcal{H}}^2\le 1+C_0|z|^2.$$

The q-fold likelihood $F_z=f_z^{\otimes q}$ in $\mathcal{H}_q=L^2(P_{\theta_0}^{\otimes q};\mathbb{C})$ consequently satisfies $\|F_z\|_{\mathcal{H}_q}\le\exp(C_0q|z|^2/2)$. Writing $F_z=\sum_{j\ge 0}T_jz^j$, the Hilbert-space Cauchy estimate on $|z|=2h$, for $0<h\le R/2$, gives

$$\|T_j\|_{\mathcal{H}_q}\le e^{2C_0qh^2}(2h)^{-j}. \tag{445}$$

Let $m\ge 1$, $n=m+1$, and $\theta_j=\theta_0-h+2hj/n$. The probabilities

$$G=2^{1-n}\sum_{\substack{0\le j\le n\\ j\ \text{even}}}\binom{n}{j}\delta_{\theta_j},\qquad H=2^{1-n}\sum_{\substack{0\le j\le n\\ j\ \text{odd}}}\binom{n}{j}\delta_{\theta_j}$$

have matching moments through degree m, by the formula for an nth finite difference. Their transport distance is $W_1(G,H)=2h/(m+1)$: on the jth grid interval the difference of their cumulative distribution functions is $2^{1-n}(-1)^j\binom{n-1}{j}$, whose absolute values sum to one. Since $|\int(\theta-\theta_0)^j\,d(G-H)|\le 2h^j$, (445) yields

$$\left\|\frac{dQ_q(G)-dQ_q(H)}{dP_{\theta_0}^{\otimes q}}\right\|_{\mathcal{H}_q}\le 2e^{2C_0qh^2}\sum_{j>m}2^{-j}=2e^{2C_0qh^2}2^{-m}.$$

In particular their one-group total variation distance is at most $e^{2C_0qh^2}2^{-m}$. Choose

$$m=\left\lceil\frac{4}{3\log 2}\log(4N)\right\rceil,\qquad h=\min\left\{\frac{R}{2},\sqrt{\frac{m\log 2}{8C_0q}}\right\}.$$

Then the one-group distance is at most $1/(4N)$, so the N-group distance is at most $1/4$. The two-point inequality gives

$$R_{N,q}\ge\frac{3h}{4(m+1)}\ge c\min\left\{\frac{1}{\log(eN)},\frac{1}{\sqrt{q\log(eN)}}\right\}.$$

Together with the oracle argument this proves (442). Notice that this local lower-bound proof applies to any normalized analytic likelihood satisfying the displayed L^2 bound; it does not depend on the particular missing Legendre coefficient. □

Proof of the budget and replication conclusions. Choosing $q=\lfloor\sqrt{B}\rfloor$ and $N=\lfloor B/q\rfloor$ proves the upper bound in (444). Write $L=\log(eN)$. If $q\ge L$, the lower bound is at least a constant times

$$\max\{N^{-1/2},(qL)^{-1/2}\}\ge(NqL)^{-1/4}\ge\{B\log(eB)\}^{-1/4}.$$

If $q<L$, it is at least $c/L\ge c/\log(eB)$, which is larger than the asserted bound for all sufficiently large B. Finally, $q\ge N$ makes the upper bound $O(N^{-1/2})$. Conversely, $R_{N,q}=O(N^{-1/2})$ first forces $q\ge L$ eventually, since $1/L\gg N^{-1/2}$, and then forces $qL=\Omega(N)$ by (442). □

76 A fixed entire kernel with arbitrarily slow recovery

The preceding rate is a feature of the factorial coefficients, not a consequence of entire regularity or of the finite-jet independence property. The following statement makes this separation quantitative. It concerns worst-case estimation of the mixing probability in W_1; it does not concern estimation of the observation density.

Theorem 76.1 (No rate follows from the qualitative properties)**.** *Let $(\rho_N)_{N\ge1}$ be any positive sequence tending to zero. There is a single strictly positive, jointly entire probability kernel k_ρ on $[0,1]^2$ with the following properties. Its one-view mixture operator has precisely the signed-measure null space $\{bL_1(\theta)d\theta : b\in\mathbb{R}\}$; its two-view mixture operator is injective; and every finite family of parameter jets at distinct parameter points is linearly independent. If*

$$R_{N,r}(k_\rho)=\inf_{\widehat G}\sup_{G}\mathbb{E}_{G,k_\rho}W_1(\widehat G,G)$$

denotes the risk from N independent groups of r observations, then, for every fixed integer $r\ge2$,

$$R_{N,r}(k_\rho)\longrightarrow 0,\qquad \lim_{N\to\infty}\frac{R_{N,r}(k_\rho)}{\rho_N}=\infty. \tag{446}$$

The same kernel works for all fixed r, and the rate comparison holds for all sufficiently large sample sizes, not only along a subsequence.

Proof. Choose strictly increasing positive integers N_j such that $N_j\ge j$ and $\sup_{N\ge N_j}\rho_N\le(j+1)^{-2}$. Such a choice is possible even when (ρ_N) is not monotone. Set

$$a_j=\frac{2^{-j}}{j!N_j^2}\quad(j\ge2),\qquad k_\rho(x\mid\theta)=1+\sum_{j=2}^{\infty}a_jL_j(x)L_j(\theta).$$

All coefficients are positive, and $a_j\le1/j!$. The Legendre polynomials grow at most exponentially in degree on every fixed compact subset of $\mathbb{C}$, so the series converges locally uniformly on $\mathbb{C}^2$. It is jointly entire and is bounded below by $3-e>0$ on the real unit square. Orthogonality gives normalization in x.

For clarity, the qualitative properties can be checked directly. Coefficient extraction gives

$$\int_0^1 L_j(x)\int_0^1 k_\rho(x\mid\theta)\nu(d\theta)\,dx=\frac{a_j}{2j+1}\int_0^1 L_j\,d\nu\quad(j\ge2).$$

Thus a null measure has zero mass and zero Legendre moments except possibly its first moment. Polynomial density identifies it as $bL_1(\theta)d\theta$. Two views additionally give its L_2L_3 moment; the calculation in Corollary 67.4 gives $\int_0^1L_1L_2L_3=3/35\neq0$, so the two-view null measure is zero. For finite-jet independence, coefficient extraction gives $T(1)=0$ and $T(L_j)=0$ for $j\ge2$ for any purported finite-jet annihilator T. Lemma 67.1 then forces every coefficient of T to vanish.

Fix $r\ge2$. For every sufficiently large sample size N, take the unique integer $j\ge2$ with $N_{j-1}\le N<N_j$ and put $m=r(j-1)$. Apply the measures in Lemma 74.3 with this m. In an r-fold kernel product, every term of total polynomial degree exceeding $r(j-1)$ has at least one index at least j. Write $A=1+\sum_{k\ge2}a_k<\infty$. The same absolute-convergence argument as in that lemma, followed by a union bound on the large index, gives

$$\mathrm{TV}(Q_r(G_m),Q_r(H_m))\le rA^{r-1}\sum_{k\ge j}a_k\le\frac{rA^{r-1}}{N_j^2}.$$

The last inequality uses monotonicity of N_k and $\sum_{k\ge j} 2^{-k}/k! \le 1$. For the N-group sample laws, the total variation distance is at most $rA^{r-1}N/N_j^2 \le rA^{r-1}/N_j$, which tends to zero as $N \to \infty$. The two-point risk bound and the exact transport gap consequently imply

$$R_{N,r}(k_\rho) \ge \frac{1}{4\{r(j-1)+1\}} \quad \text{for all sufficiently large } N.$$

Our choice of N_{j-1} ensures that $\rho_N \le j^{-2}$. Hence the risk ratio is bounded below by $j^2/[4\{r(j-1)+1\}]$, which tends to infinity.

Finally, consistency is not lost. For each fixed approximation degree $M \ge 5$, all coefficients $a_2, \dots, a_M$ are nonzero. The moment-recovery and minimum-discrepancy arguments used above therefore give, with a finite constant C_M depending on this kernel and on M,

$$R_{N,r}(k_\rho) \le C/M + C_M/\sqrt{N} + C_M/N.$$

First let $N \to \infty$ and then $M \to \infty$. □

This construction is consistent with the general distinction between identifiability and stability in inverse problems. Its role here is to show that strict positivity, joint entire regularity, independence of all finite parameter jets, and recovery from two replicates can coexist with an arbitrarily slow statistical inverse.

77 Scope and remaining quantitative questions

Finite point and derivative independence, exact recovery from grouped laws, and statistical conditioning govern different questions. The circle examples retain analytic mixing-density ambiguity, whereas the interval examples distinguish analytic rigidity from localized smooth ambiguity. A finite-dimensional one-observation nullspace reduces to a fixed finite atomic set after two observations; arbitrary thresholds exceeding two therefore require an infinite-dimensional defect when all finite component families are independent. In the infinite-threshold constructions, the witnessing pair may vary with the finite group size: no two distinct mixing laws agree at every order, because the complete hierarchy identifies the law.

The stable-family classification fixes the index and skewness within each model. Its sparse conditions outside the Gaussian case are only sufficient, and mixtures over the shape parameters are not covered. The exact scalar repairs for finite Legendre defects may be severely ill-conditioned. The fixed analytic-bound learning theorem supplies consistency without a sharp rate. Section 75 leaves a logarithmic gap and treats only equal, nonadaptive group sizes; it does not settle unequal or adaptive replication or the full growing-replication minimax curve. The exact moment-coincidence construction does not control the size of the first unequal moment or give an efficient numerical representation of the densities. The threshold kernels are constructed examples; analogous conclusions for particular standard statistical families need additional arguments.

Source versions and verification status. The normal-mixture question is used as formulated in `arXiv:1507.00678v1`, Section 2.6. The finite differential and difference–differential classes are those in the author manuscript of Le et al. dated 8 August 2026, where the sufficiency question follows Proposition 6 and appears in Section 6. The localized moment theorem proves the assertion of Conjecture 17 in the cited Bürgin–Epperlein–Wirth source. Historical priority remains unverified; Lin's final publisher PDF was not inspected. The proofs and the exact Legendre identity received AI-assisted checks only.

78 Source correspondence and consolidation

This integrated manuscript incorporates the eleven English research drafts listed below, all authored under the name KEIO and prepared with AI assistance on 13–14 October 2026. The list records provenance rather than eleven independent external references. Earlier inspirational papers that were not generated in this sequence are cited in the bibliography where used.

No.	Source manuscript	Location and retained material
1	*Sharp Recurrence Exponents for Prime Flows and Finite-Height Zeros at a Frozen Boundary*	Part I. Power-weight recurrence, finite-height boundary zeros, and the explicit Fourier certificate.
2	*Entropy-Scale Recurrence and Sharp Quantitative Rigidity at Zeta Edges*	Part I. Logarithmic weights, Euler-edge locking, random perturbations, and positive-kernel bounds.
3	*Prime-Phase Localization of Simple Zeta Zeros and Quantitative Euler-Edge Repulsion*	Part II. Fixed prime-phase localization, edge triples, inverse principles, and spectral certificates.
4	*Shrinking Prime-Phase Targets, Gibbs Profiles, and Simultaneous Zeta-Edge Recurrence*	Part II. Shrinking and growing phase targets, simultaneous heights, Gibbs laws, and joint edge transitions.
5	*Two-Scale Identification from Joint Rare Events: Microscopic Laws, Spectral Fluctuations, and Singular Constraint Activation*	Part III. Microscopic-law inversion, discrete spectral fluctuations, hard constraints, and activation laws.
6	*Minimal Observation and Blind Calibration of Rare-Event Laws*	Part III. Minimal channels, blind power calibration, correlated laws, and finite-information obstructions.
7	*Arithmetic Observability through Controlled Thinning: Blind Reconstruction, Resonance, and Sharp Information Barriers*	Part IV. Controlled thinning, primitive exponent recovery, resonances, and optimal window noise.
8	*Blind Identification beyond Moment Assumptions: Interior Probes, Hidden Coordinates, and Finite Information*	Part IV. Interior and flat probes, countable coordinates, finite gate values, and information bounds.
9	*Mixture Identification beyond Moments and Differential Barriers*	Part V. Sparse partial sums, stable-family classification, and differential obstructions.
10	*Replication Thresholds and Finite-Dimensional Ambiguity in Infinite Mixtures*	Part V. Finite defects, analytic-circle replication thresholds, factorial rates, and slow recovery.
11	*Prescribed Moment Coincidences, Smooth Densities, and Replication Thresholds*	Part V. Prescribed moments, ordinary-interval thresholds, Gevrey boundaries, and growing replication.

Editorial treatment. Repeated foundations and identities are proved once and referenced internally. Distinct results, constructions, and proof arguments are retained. The later recurrence, shrinking-target, and interval-mixture theorems supersede the corresponding earlier restrictions or open questions. Conditional hypotheses and remaining quantitative gaps are unchanged; verification status is discussed in the introduction.

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